Architecting on AWS

Course Introduction

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Preparing for class

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Optimize your learning

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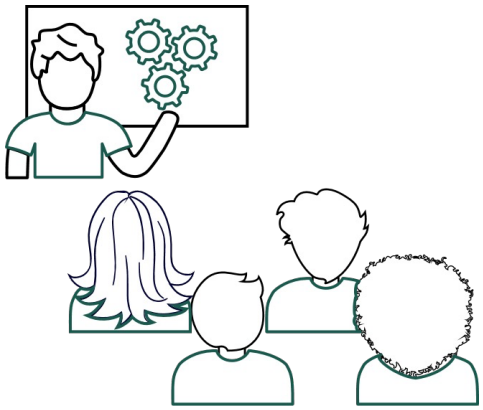
Logistics

- Breaks and lunch
- Security
- Cell phones
- Virtual classroom features
  - Audio
  - Chat
  - Raise hand



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Prerequisites

We recommend that course attendees complete one of the following prerequisites:

1. AWS Cloud Practitioner Essentials
2. AWS Technical Essentials
3. Build a working knowledge of:
    - Distributed systems
    - Networking concepts
    - IP addressing
    - Multi-tier architectures
    - Cloud computing concepts

AWS Cloud Practitioner Essentials course

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Register for access to guides and lab environments

Make sure you register for AWS Builder Labs.

- Refer to your welcome email for registration information.

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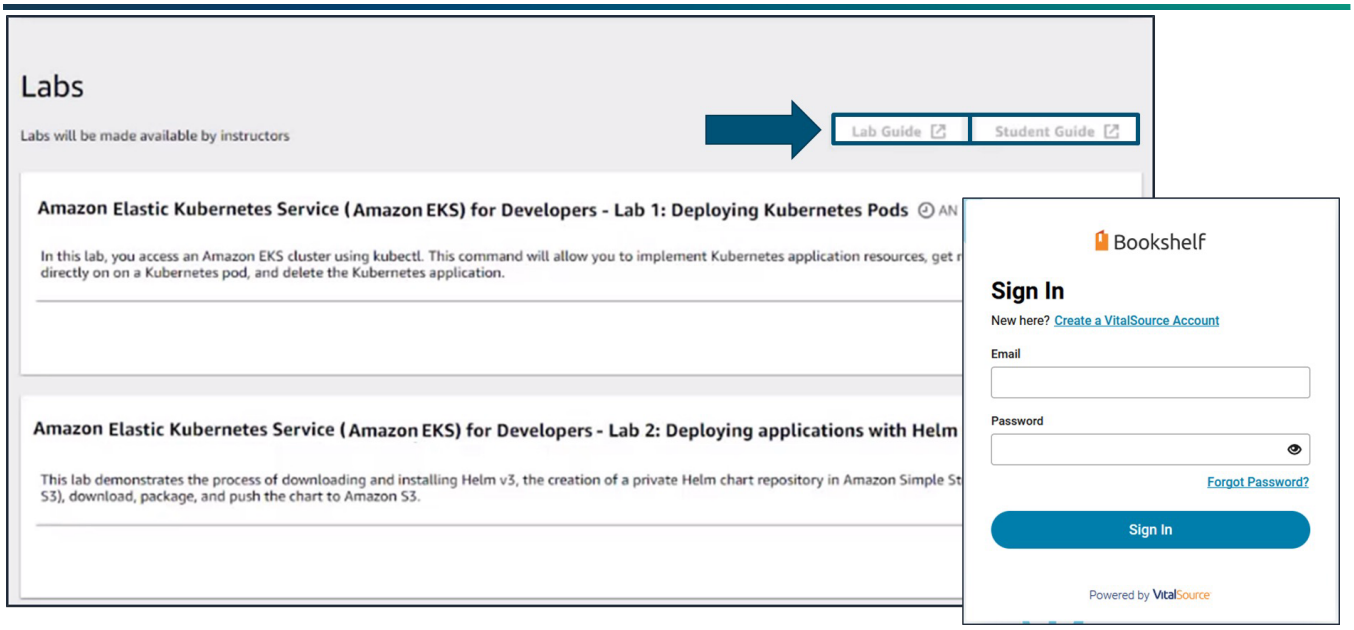
Student and lab guides



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Lab requirements

- Computer running:
  - Windows
  - macOS
  - Linux: Ubuntu, SUSE, or Red Hat
- Recommended web browsers:
  - Google Chrome
  - Mozilla Firefox
  - Microsoft Edge
- Reliable internet connection able to browse the internet by using HTTPS
- Register for AWS Builder Labs:
  - Turn off ad blockers and script blockers

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About you

Tell us the following:

- First name
- Organization and role
- What do you expect of this course?

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Course overview

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Course objective

- Identify Amazon Web Services (AWS) services, compare features, and explore best practices to architect resilient, secure, and highly available IT solutions on AWS.

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Day 1 agenda

Section title	Estimated time
Module 1: Architecting Fundamentals	45 minutes
Lab 1: Explore and Interact with the AWS Management Console and AWS Command Line Interface	35 minutes
Module 2: Account Security	60 minutes
Lunch	60 minutes
Module 3: Networking 1	60 minutes
Module 4: Compute	75 minutes
Lab 2: Build your Amazon VPC Infrastructure	45 minutes

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Day 2 agenda

Section title	Estimated time
Module 5: Storage	70 minutes
Module 6: Database Services	70 minutes
Lab 3: Create a Database Layer in Your Amazon VPC Infrastructure	45 minutes
Lunch	60 minutes
Module 7: Monitoring and Scaling	70 minutes
Lab 4: Configure High Availability in Your Amazon VPC	45 minutes
Module 8: Automation	30 minutes
Module 9: Containers	40 minutes

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Day 3 agenda

Section title	Estimated time
Module 10: Networking 2	45 minutes
Module 11: Serverless	45 minutes
Lab 5: Build a Serverless Architecture	45 minutes
Module 12: Edge Services	60 minutes
Lunch	60 minutes
Lab 6: Configure an Amazon CloudFront Distribution with an Amazon S3 Origin	60 minutes
Module 13: Backup and Recovery	40 minutes
Capstone Lab: Build an AWS Multi-Tier Architecture	90 minutes
Course Summary	10 minutes

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Capstone lab

- Review and analyze architectural solutions based on project data, best practices, and the AWS Well-Architected Framework.
- Design the architecture in a lab, without specific guidance.

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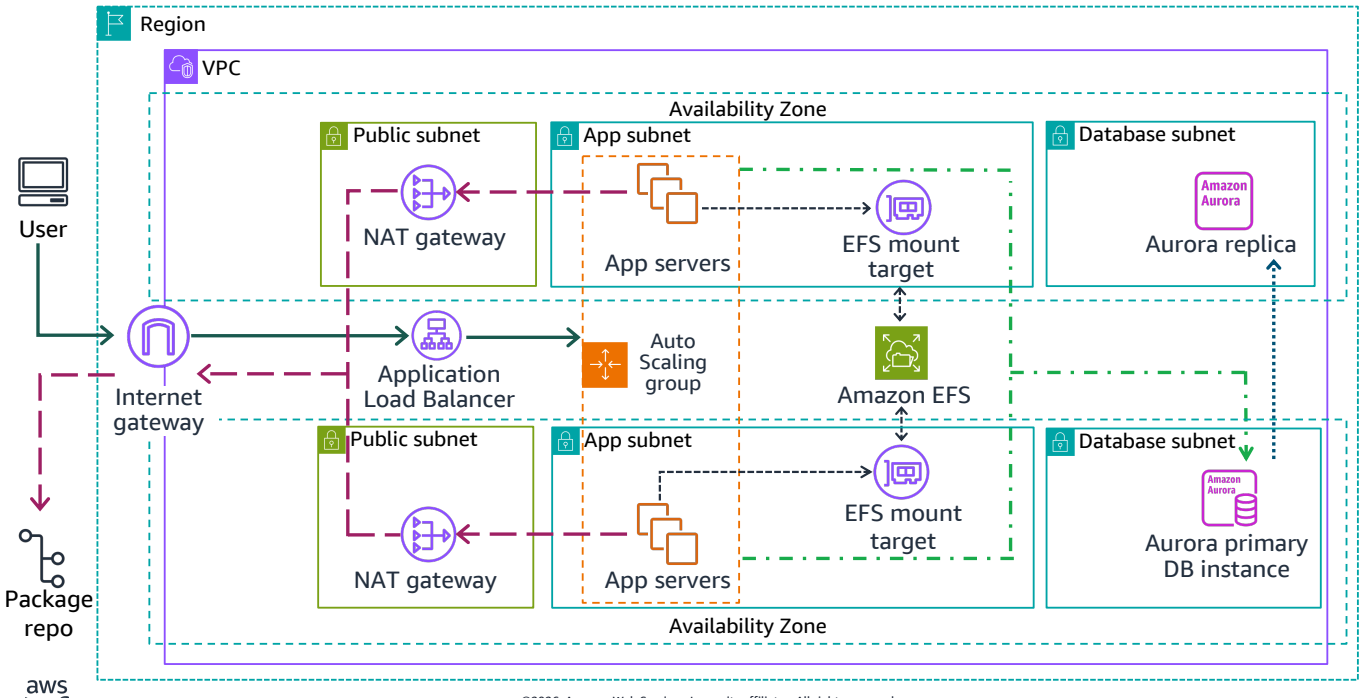
Capstone architecture



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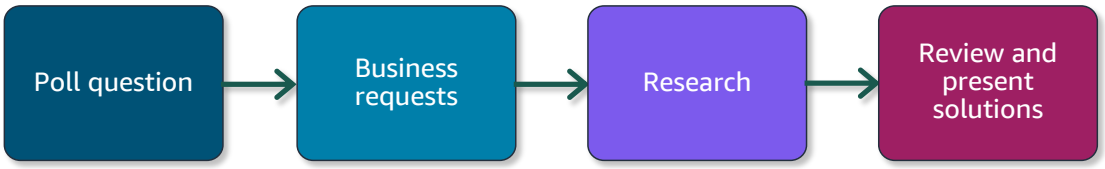
Module format



Live group chat

- Polls and knowledge checks
- Questions for the instructor

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Business request

Chief technology officer

The chief technology officer has hired you as a solutions architect to work on a number of projects for Example Corp.

Pay attention to what your stakeholder needs to know. It is your job to help them in their journey in the AWS Cloud.

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Supplemental learning

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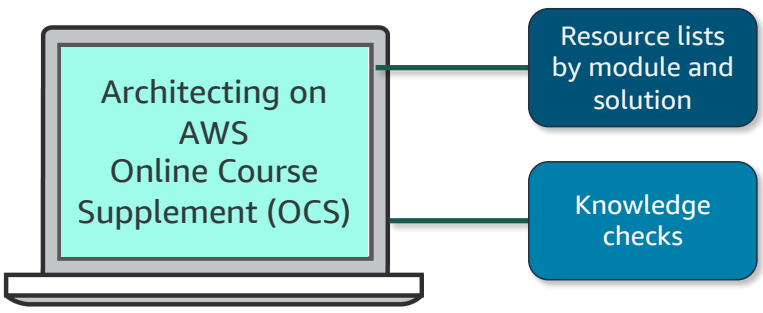
Online Course Supplement (OCS)



OCS

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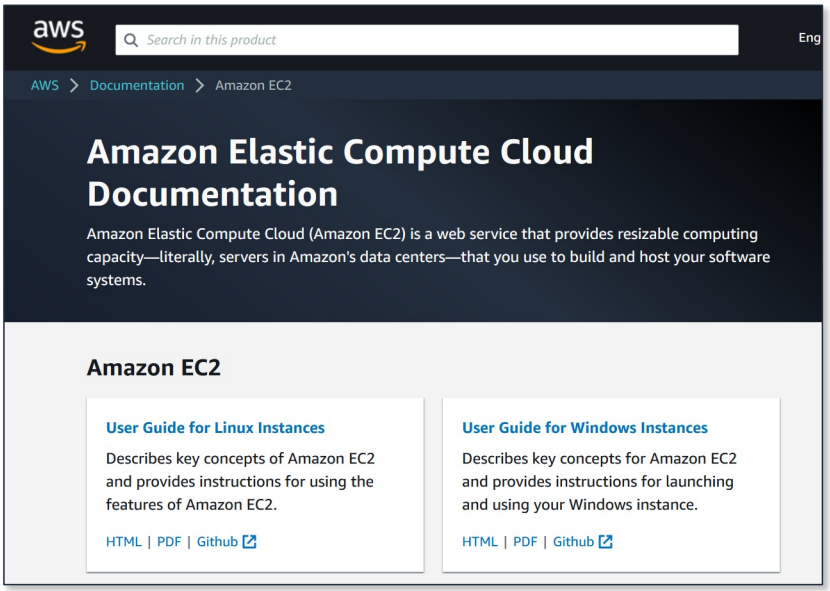
AWS Documentation resources



AWS Documentation

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aws

End of introduction

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Architecting on AWS

Module 1: Architecting Fundamentals

Lab 1

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Poll question

How far is your organization in its journey to the AWS Cloud?

A. We're just getting started.

B. We already have prototypes running.

C. We have production workloads running.

D. We run 100 percent of our operations in the AWS Cloud.

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Module overview

- Business request
- Amazon Web Services (AWS) services
- AWS infrastructure
- AWS Well-Architected Framework
- Present solutions
- Knowledge check
- Lab 1: Explore and Interact with the AWS Management Console and AWS Command Line Interface

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Business request

Chief technology officer

The chief technology officer (CTO) wants you to explore the following questions:

- What are the benefits of using AWS services?
- How is the AWS Global Infrastructure organized?
- How can you build your cloud infrastructure according to best practices?

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AWS services

"What are the benefits of using AWS services?"

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Amazon Web Services

- Global data centers
- More than 200 services
- Secure and robust
- Pay as you go
- Built for business needs

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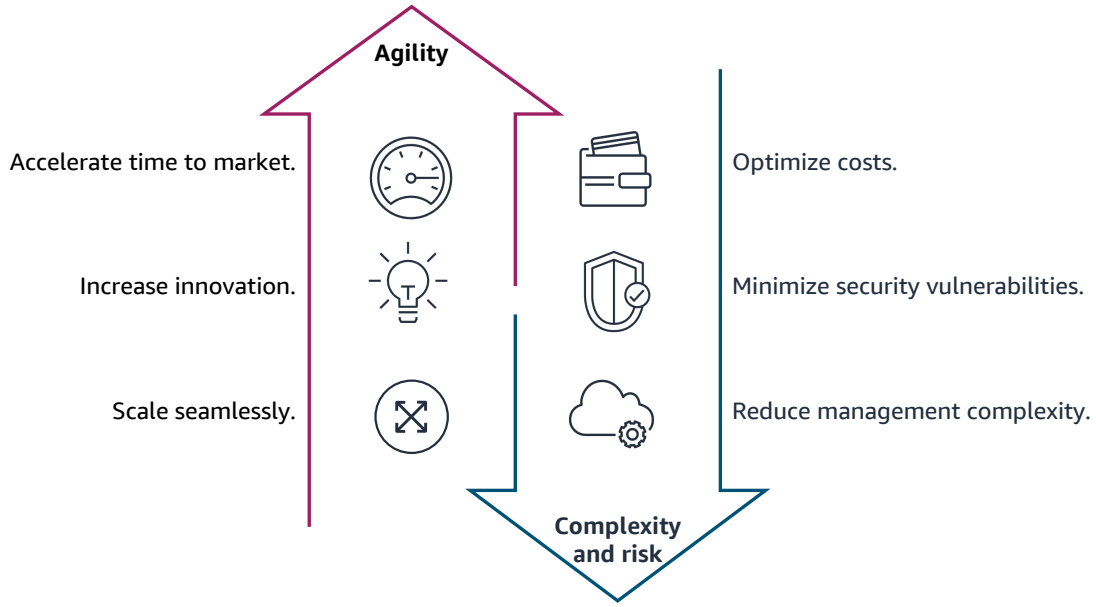
Why customers move to AWS



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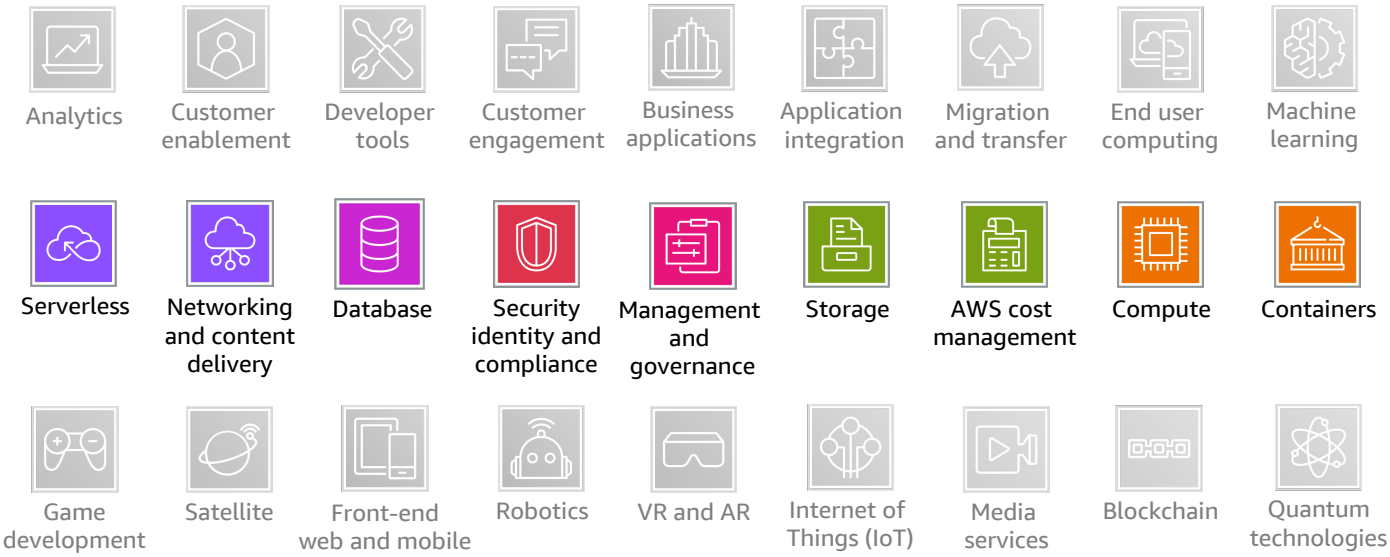
AWS service categories



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AWS infrastructure

"How is AWS global infrastructure organized?"

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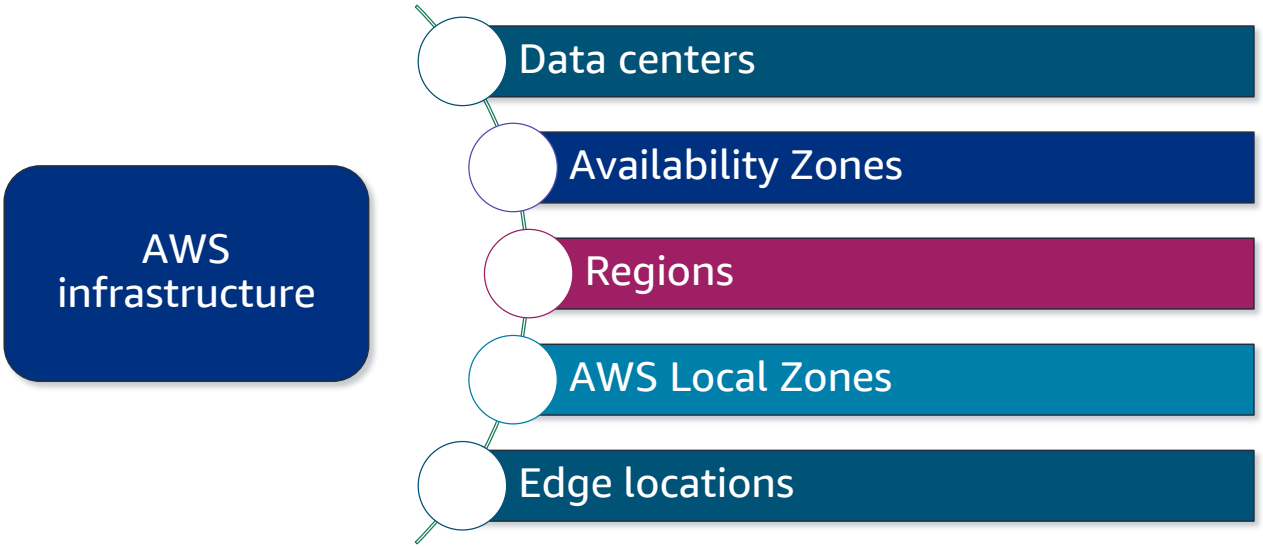
AWS infrastructure topics



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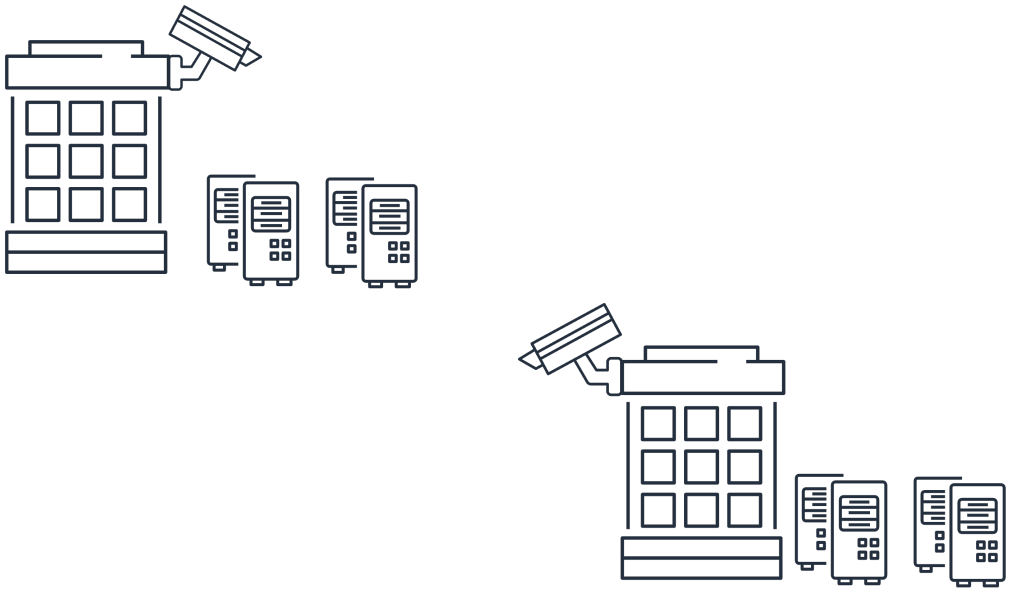
AWS data centers

- AWS services operate within AWS data centers.
- Data centers host thousands of servers.
- Each location uses AWS proprietary network equipment.
- Data centers are organized into Availability Zones.



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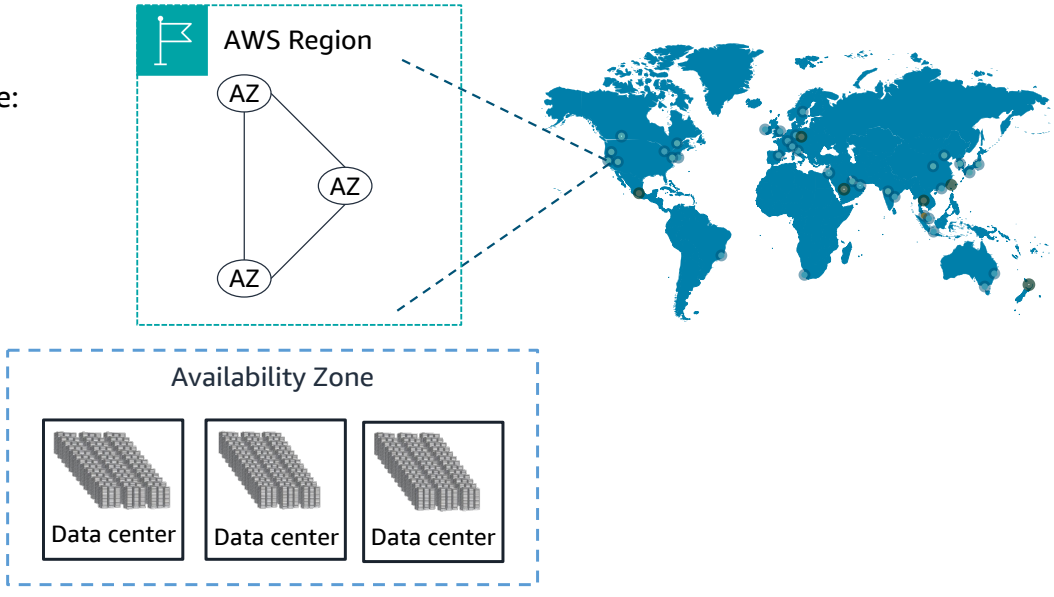
Availability Zones

Availability Zones are:

- Data centers in a Region
- Designed for fault isolation
- Interconnected by using high-speed private links
- Used to achieve high availability



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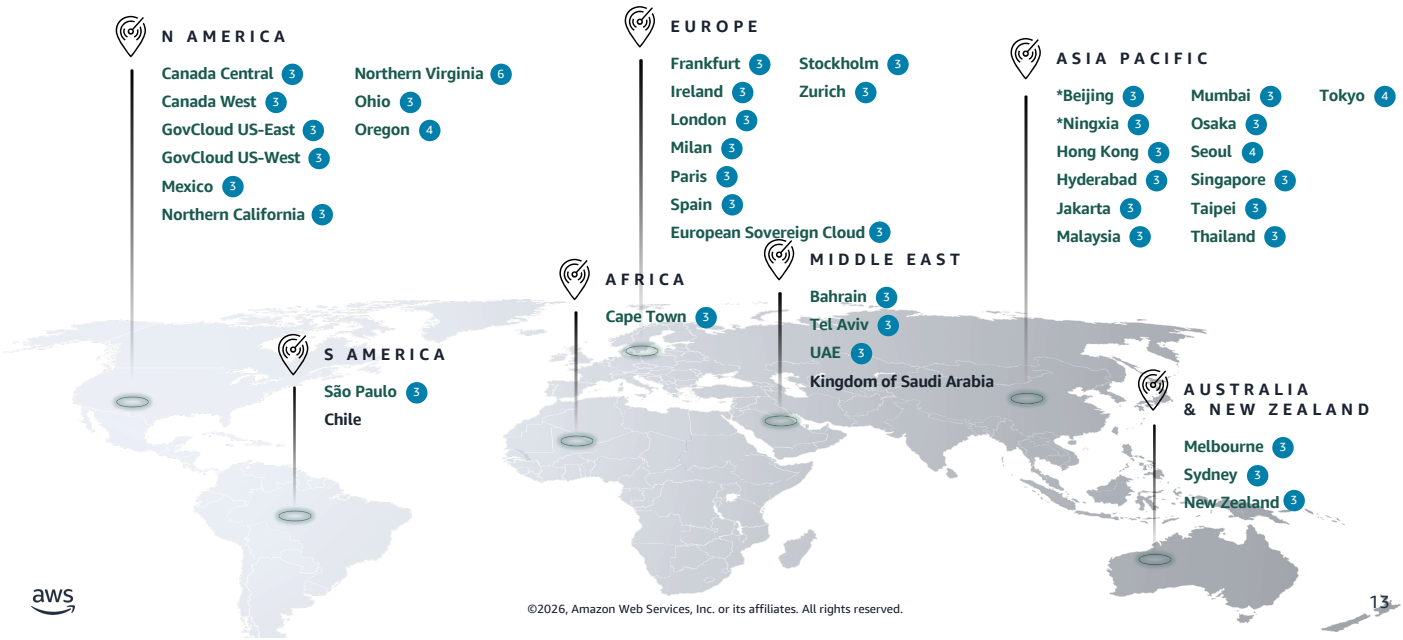
AWS Global Infrastructure Regions & AZs



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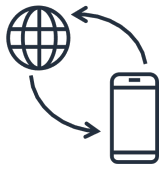
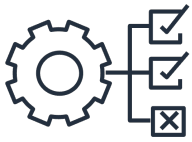
Factors that impact Region selection

Governance



Latency



Service availability

Cost

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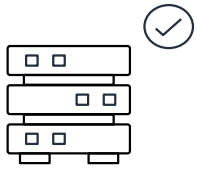
AWS Local Zones

Use cases:

- Media and entertainment content creation
- Real-time gaming
- Machine learning inference
- Live video streaming
- Augmented reality (AR) and virtual reality (VR)



AWS infrastructure at the edge

Local compute, storage, databases, and other services

Connecting to services in AWS Regions

Delivering new low-latency applications

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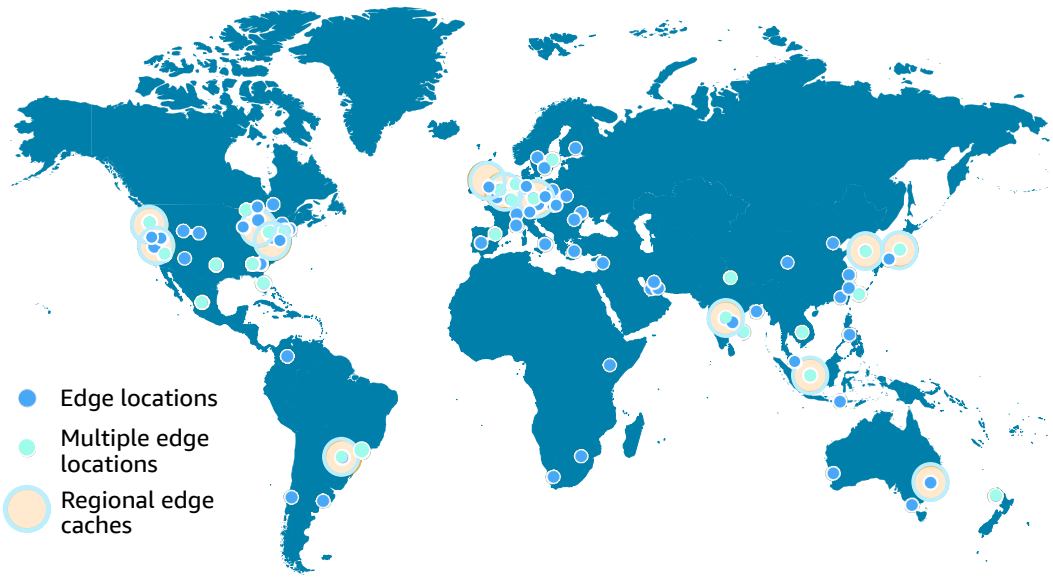
Edge locations

Edge locations:

- Run in major cities around the world
- Support AWS services such as Amazon Route 53 and Amazon CloudFront



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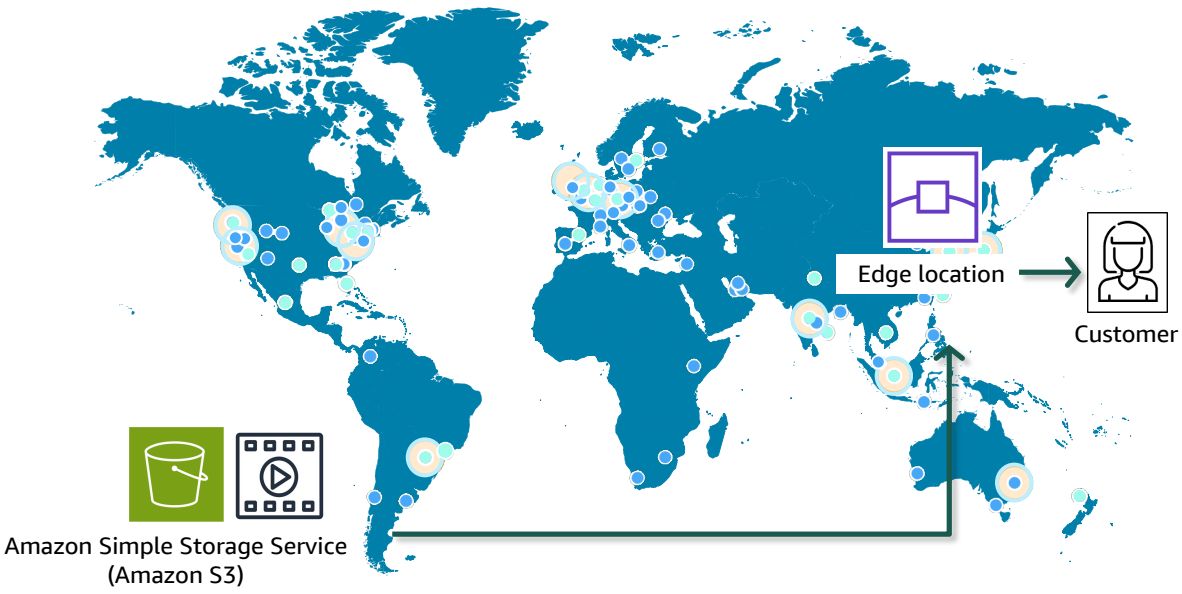
Edge location use case



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AWS Local Zone and edge location features

AWS Local Zones

- Low latency
- Local data processing
- Consistent AWS experience

Edge locations

- Caching of data
- Fast delivery of content
- Better user experience

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AWS Well-Architected Framework

"How can we build our cloud infrastructure according to best practices?"

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AWS architect responsibilities

Plan

- Set technical cloud strategy with business leads.
- Analyze solutions for business needs and requirements.

Research

- Investigate cloud services specifications and workload requirements.
- Review existing workload architectures.
- Design prototype solutions.

Build

- Design the transformation roadmap with milestones, work streams, and owners.
- Manage the adoption and migration.

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AWS Well-Architected Framework pillars

Security
- Apply at all layers
- Enforce the principle of least privilege
- Use multi-factor authentication (MFA)

Performance efficiency
- Reduce latency
- Use serverless architecture
- Incorporate monitoring

Cost optimization
- Analyze and attribute expenditures
- Use cost-effective resources
- Stop guessing

Operational excellence
- Perform operations with code
- Test response for unexpected events

Reliability
- Recover from failure
- Test recovery procedures
- Scale to increase availability

Sustainability
- Understand your impact
- Maximize utilization

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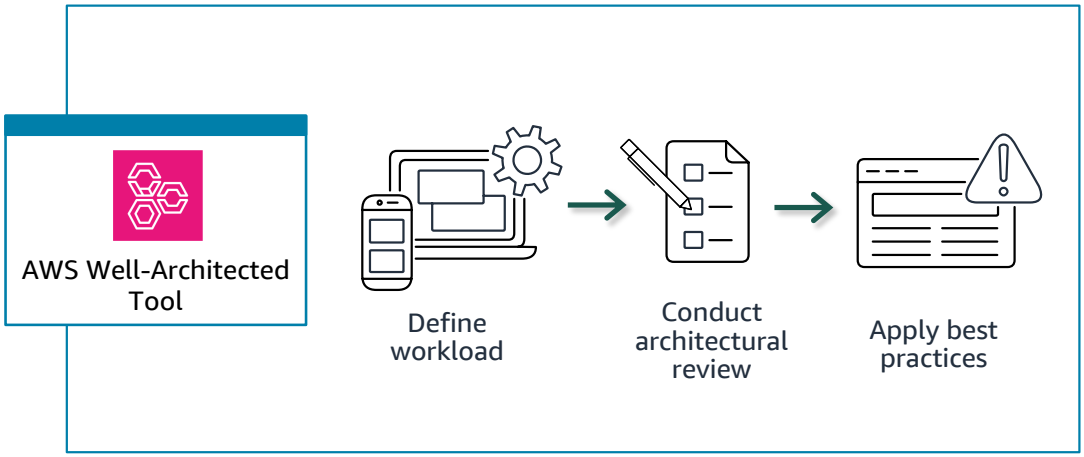
AWS Well-Architected Tool

- Based on the AWS Well-Architected Framework
- Can review your applications and workloads
- Central place for best practices and guidance
- Used in tens of thousands of workload reviews



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Review

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Present solutions

Chief technology officer

Consider how you would answer the following questions:

- What are the benefits of using AWS services?
- How is the AWS global infrastructure organized?
- How can you build your cloud infrastructure according to best practices?

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Module review

In this module, you learned about the following:

- ✓ AWS services
- ✓ AWS infrastructure
- ✓ AWS Well-Architected Framework

Next, you review the following:

Knowledge check

Lab introduction

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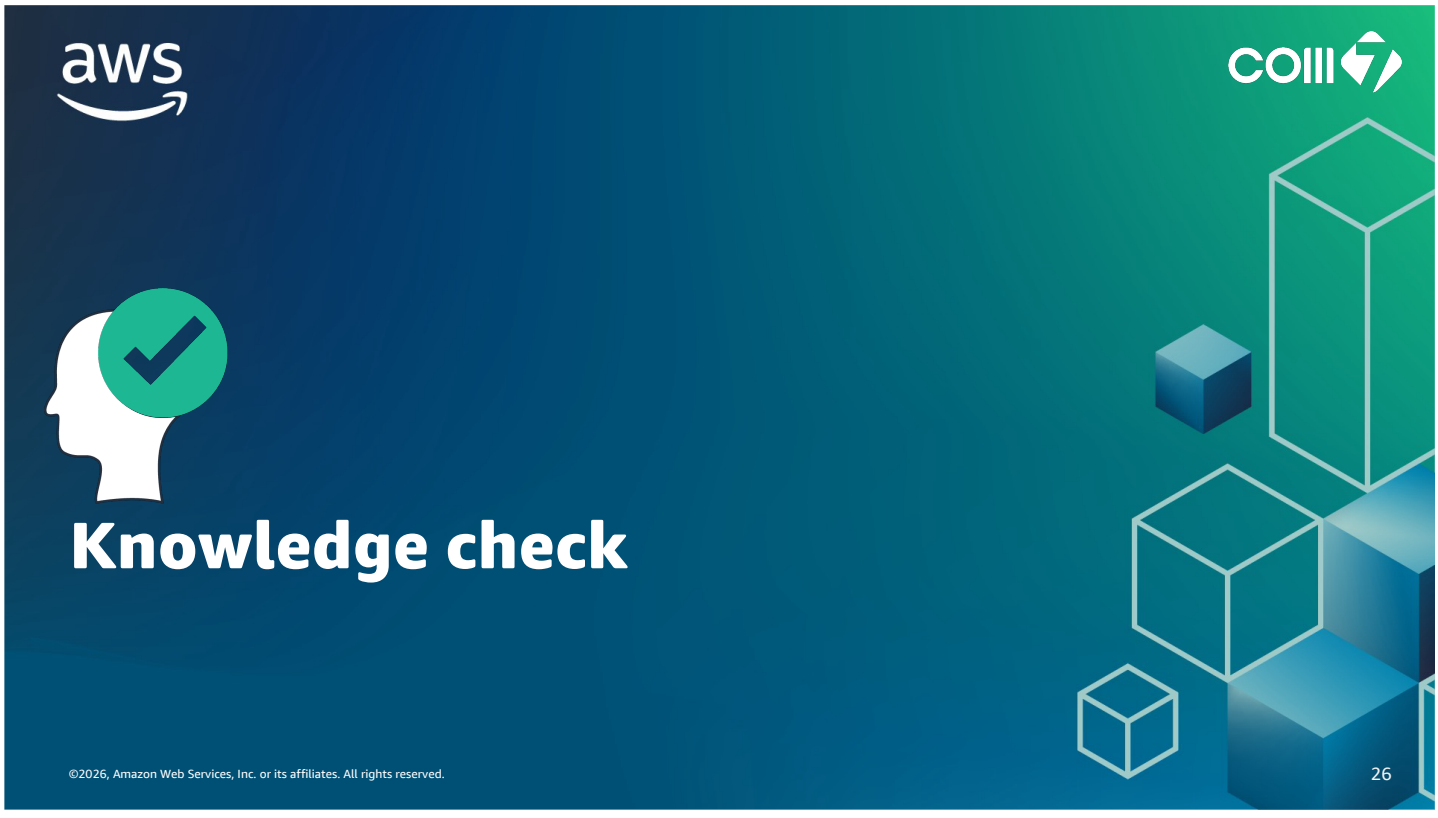

Knowledge check question 1

Which item is the best example of one responsibility of an AWS architect?

Choice	Response
A	Monitor alarms for disaster response.
B	Maintain application-level code in the AWS Cloud.
C	Manage access to a group of AWS accounts.
D	Analyze solutions for business needs and requirements.

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Knowledge check question 1 and answer

Which item is the best example of one responsibility of an AWS architect?

Choice	Response
A	Monitor alarms for disaster response.
B	Maintain application-level code in the AWS Cloud.
C	Manage access to a group of AWS accounts.
D correct	Analyze solutions for business needs and requirements.

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Knowledge check question 2

Which item is a cluster of data centers within a geographic location with low latency network connectivity?

Choice	Response
A	Availability Zone
B	Region
C	Edge location
D	Outposts

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Knowledge check question 2 and answer

Which item is a cluster of data centers within a geographic location with low latency network connectivity?

Choice	Response
A **correct**	Availability Zone
B	Region
C	Edge location
D	Outposts

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Knowledge check question 3

Which factors must be considered when picking an AWS Region? (Select TWO.)

Choice	Response
A	Local data regulations
B	Operating system requirements
C	Latency to end users
D	Support for hybrid networking
E	Programming language of the application

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Knowledge check question 3 and answer

Which factors must be considered when picking an AWS Region? (Select TWO.)

Choice	Response
A **correct**	Local data regulations
B	Operating system requirements
C **correct**	Latency to end users
D	Support for hybrid networking
E	Programming language of the application

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Knowledge check question 4

What is the primary benefit of deploying applications into multiple Availability Zones?

Choice	Response
A	Stronger security policies for resources
B	Decreased latency to resources
C	High availability for resources
D	There is no benefit to this design

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Knowledge check question 4 and answer

What is the primary benefit of deploying applications into multiple Availability Zones?

Choice	Response
A	Stronger security policies for resources
B	Decreased latency to resources
C correct	High availability for resources
D	There is no benefit to this design

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Knowledge check question 5

Which AWS Well-Architected Framework pillar contains the principle of least privilege?

Choice	Response
A	Operational excellence
B	Security
C	Reliability
D	Performance efficiency

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Knowledge check question 5 and answer

Which AWS Well-Architected Framework pillar contains the principle of least privilege?

Choice	Response
A	Operational excellence
B correct	Security
C	Reliability
D	Performance efficiency

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Lab 1:

Explore and Interact with the AWS Management Console and AWS Command Line Interface

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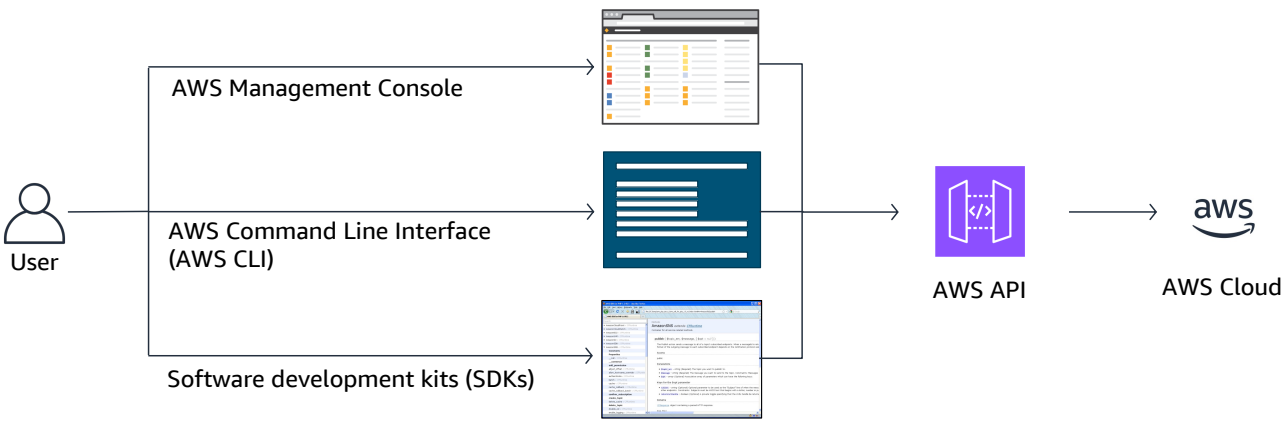
Connecting to an AWS service



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Lab tasks

Task 1: Explore and configure the AWS Management Console.

Task 2: Create an Amazon S3 bucket by using the AWS Management Console.

Task 3: Upload an object into the Amazon S3 bucket by using the S3 console.

Task 4: Create an Amazon S3 bucket and upload an object by using the AWS CLI.

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End of Module 1

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Architecting on AWS

Module 2: Account Security

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Poll question

How many Amazon Web Services (AWS) accounts does your organization use?

A. 1

B. 2–10

C. More than 10

D. I don't know

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Module overview

- Business requests
- Principals and identities
- Security policies
- Managing multiple accounts
- Module review
- Knowledge check

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Business request

Security specialist

The security specialist needs to know the following:

- What are the best practices to manage access to AWS accounts and resources?
- How can we give users access to only the resources that they need?
- What is the best way to manage multiple accounts?

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Principals and identities

"What are the best practices to manage access to AWS accounts and resources?"

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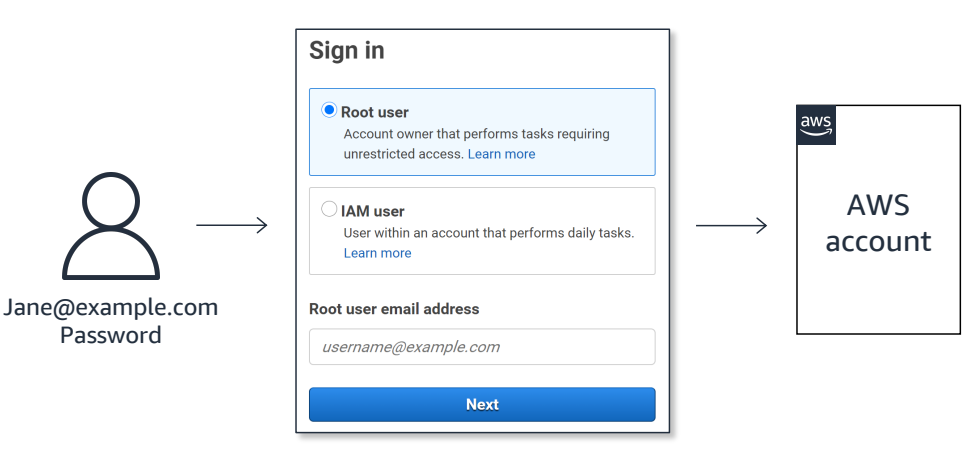
AWS account root user

A root user:

- Has full access to all AWS services
- Cannot be restricted in a single account model
- Should not be used for day-to-day interactions with AWS



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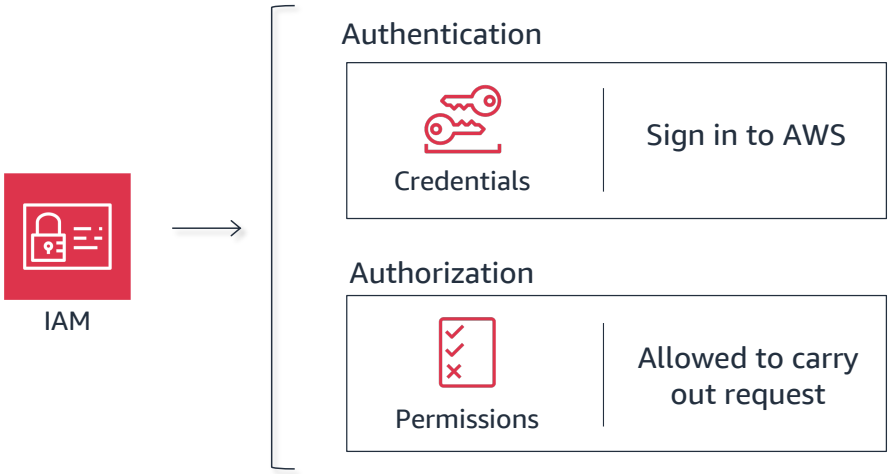
AWS Identity and Access Management (IAM)

Use IAM to:

- Create and manage users, groups, and roles.
- Manage access to AWS services and resources.
- Analyze access controls.



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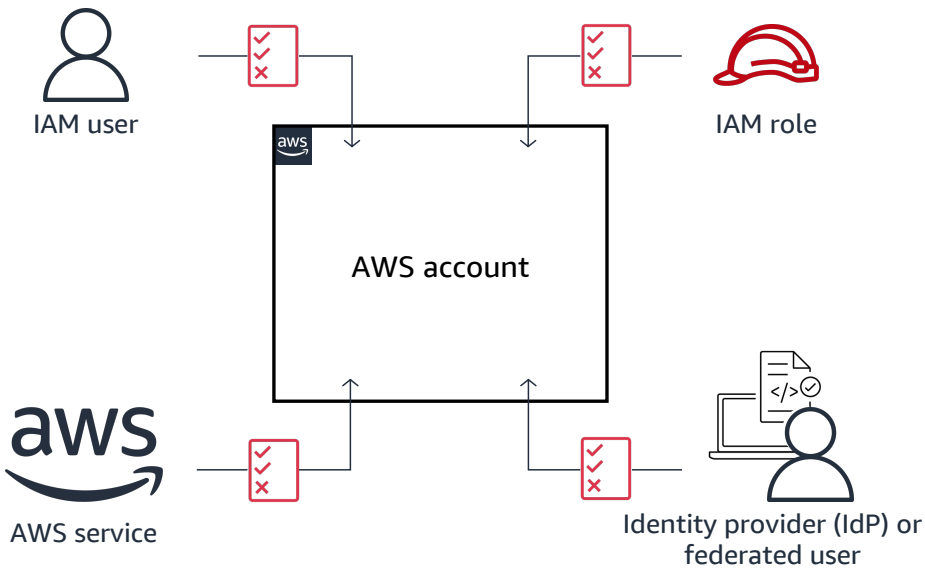
Principals

A principal:

- Can make a request for an action or operation on an AWS resource
- Can be a person, application, federated user, or assumed role



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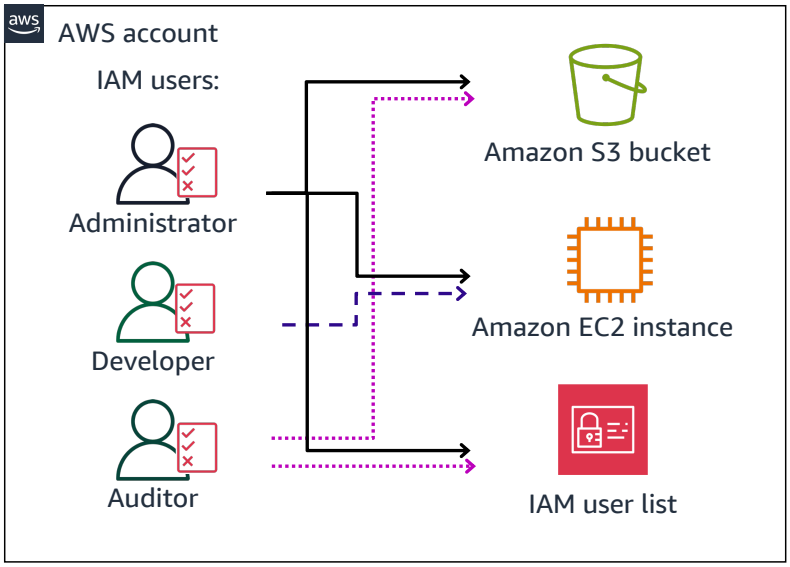
IAM users

IAM users are entities within an AWS account that represent human users or workloads.

- Each user has their own credentials.
- IAM users are authorized to perform specific AWS actions based on permissions.



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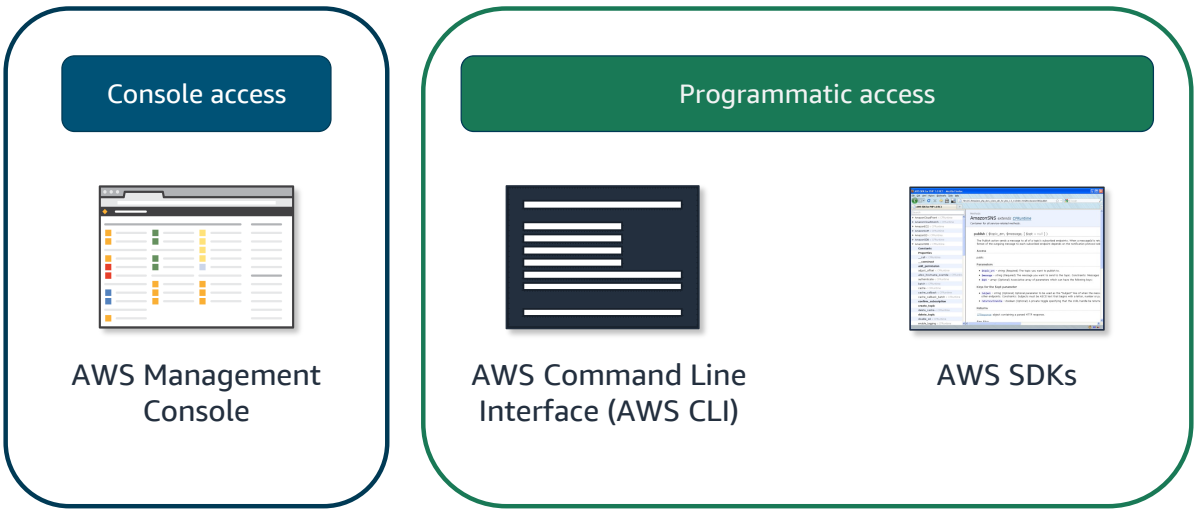
IAM users and AWS API calls



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Programmatic access

```
Access Key ID: AKIAIOSFODNN7EXAMPLE
Secret Access Key: wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY
```

AWS CLI

```
$ aws configure
AWS Access Key ID [****************MPLE]:
AWS Secret Access Key [************EKEY]:
Default region name [ap-southeast-1]:
Default output format [json]:
```

AWS SDK

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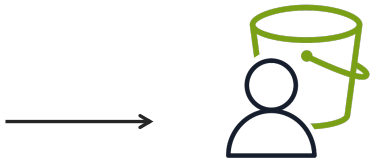
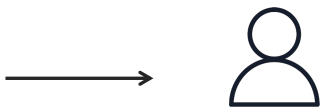
Setting permissions with IAM policies

Select	Policy name
	AdministratorAccess
	AmazonEC2ReadOnlyAccess
✓	AmazonS3FullAccess
	AmazonS3ReadOnlyAccess



Select	Policy name
	AdministratorAccess
✓	AmazonEC2ReadOnlyAccess
	AmazonS3FullAccess
✓	AmazonS3ReadOnlyAccess



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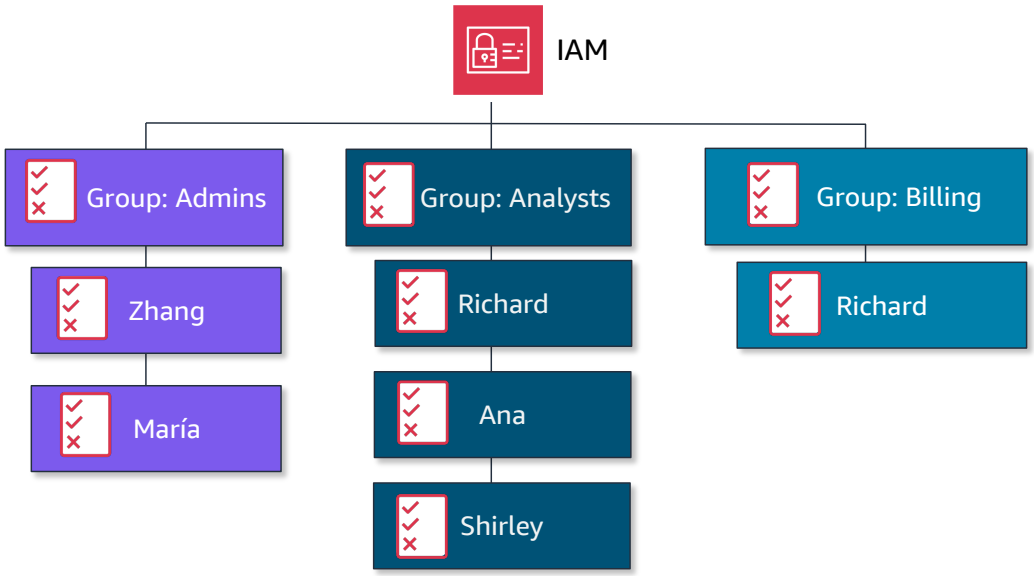
IAM user groups

- Assign IAM users to an IAM user group.
- Attach policies to an IAM user group to apply to all users within the group.



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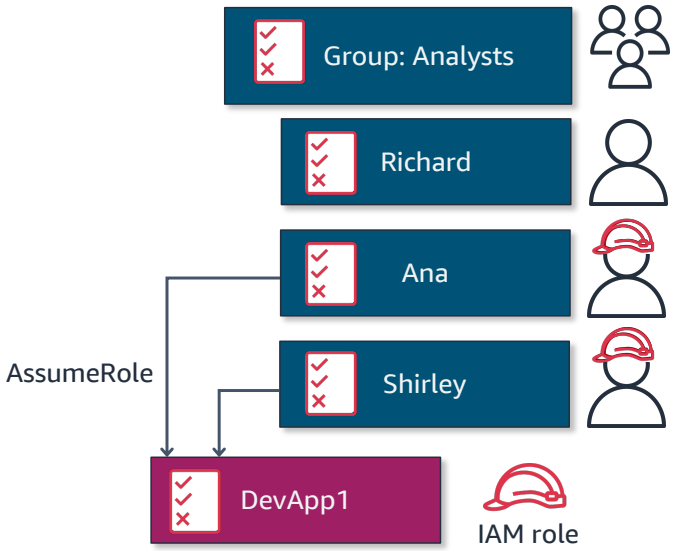
IAM roles

- Delegate a set of permissions to specific users or services using temporary credentials.
- Users assume a role without sharing credentials with others.
- Permissions are only valid while operating under the assumed role.



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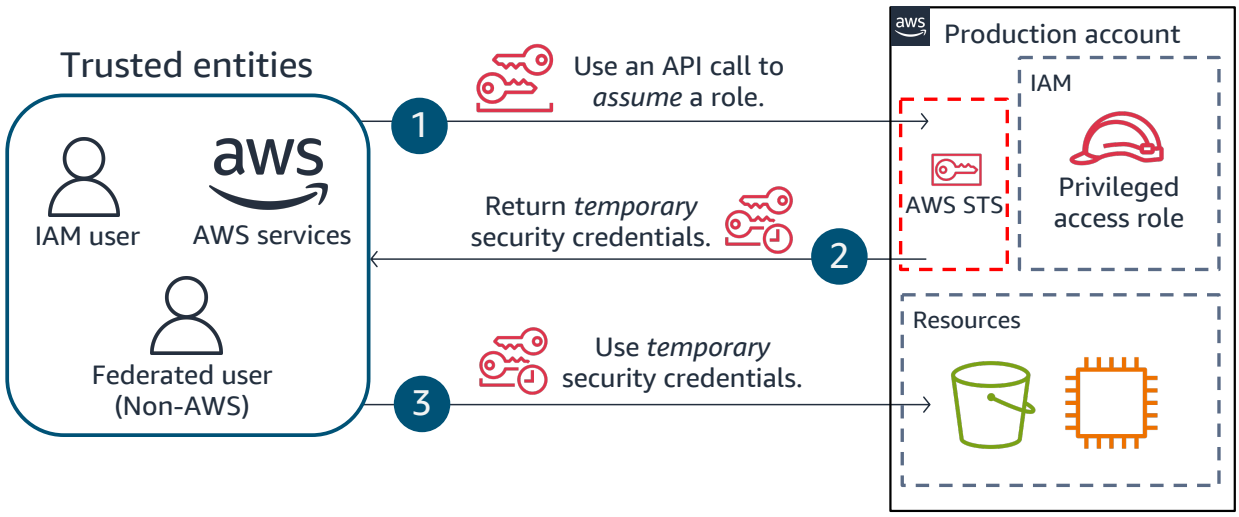
Assuming a role



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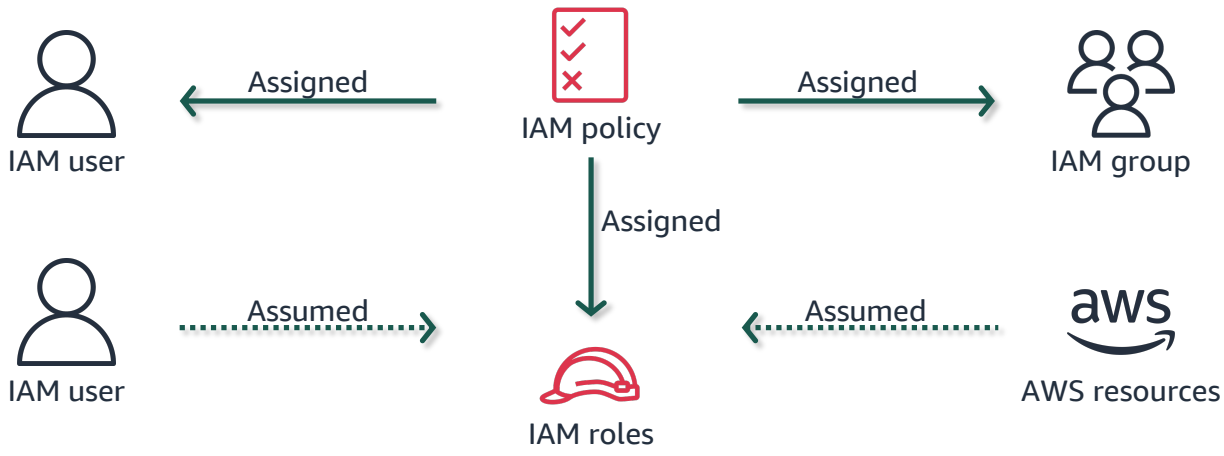
IAM policy assignments



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Security policies

"How can we give users access to only the resources they need?"

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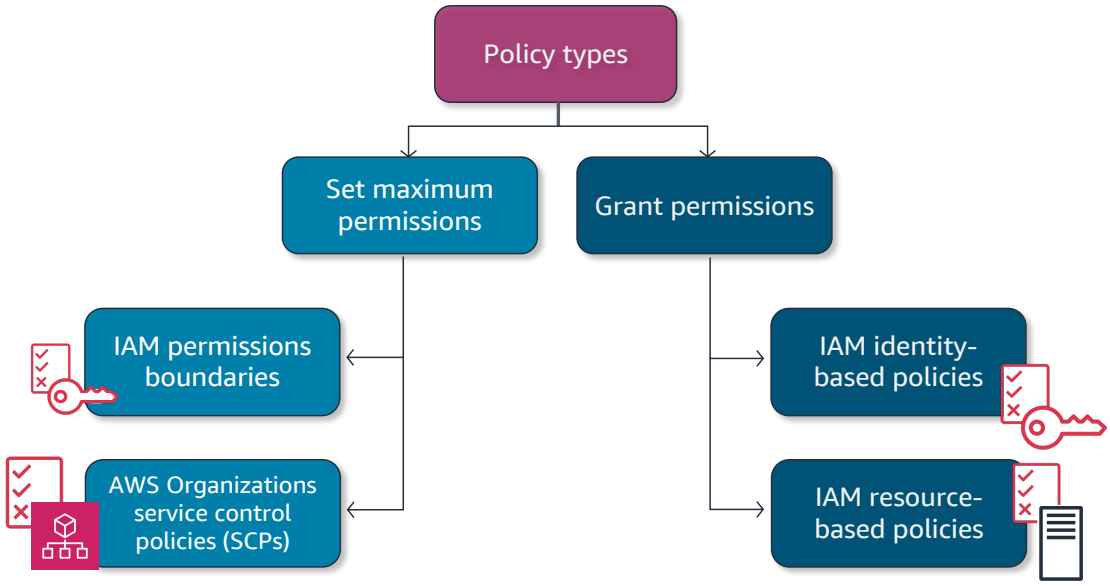
Security policy categories



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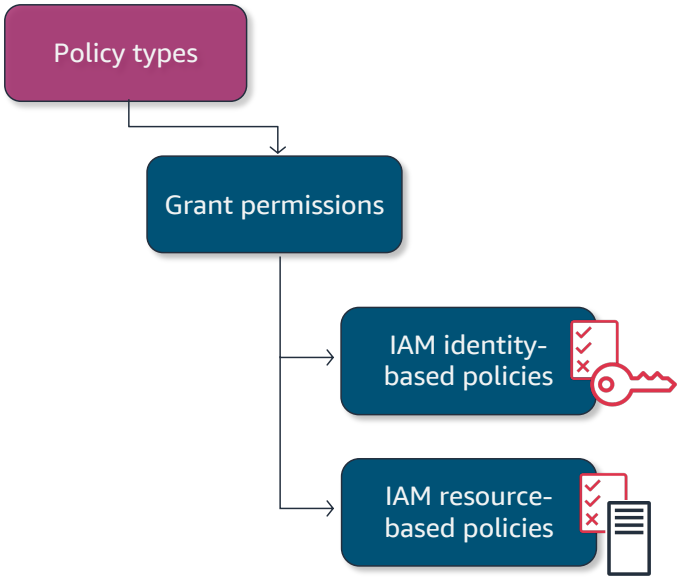
Granting permissions

- Identity-based policies are assigned to users, groups, and roles.
- Resource-based policies are assigned to resources.
- Resource-based policies are checked when someone tries to access the resource.



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Types of identity-based policies

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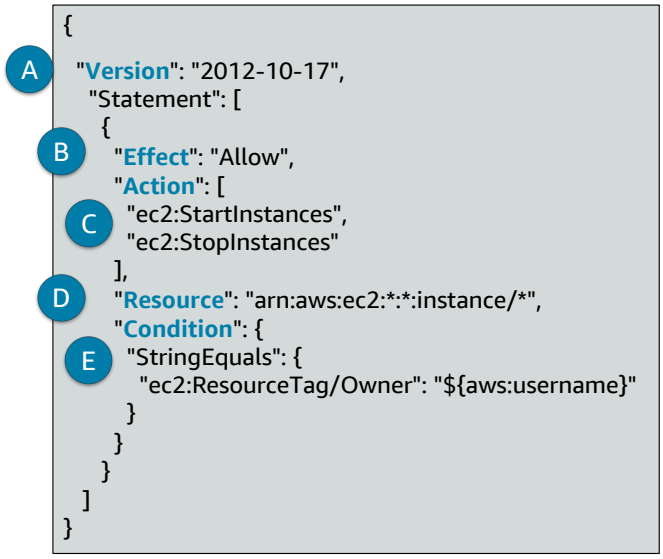
Identity-based policy example



A. Use this version date to use all of the available policy features.

B. Indicate whether the policy allows or denies an action.

C. Include a list of actions that the policy allows or denies.

D. Choose a list of resources that the effect applies to.

E. Optional: Specify the conditions under which the policy applies.

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Explicit allow and explicit deny

This section from a policy allows access.
It is called an *explicit allow.*

```
{
  "Effect": "Allow",
  "Action": [
    "s3:ListBucket",
    "s3:GetObject"
  ],
  "Resource": [
    "arn:aws:s3:::BUCKET-NAME",
    "arn:aws:s3:::BUCKET-NAME/*"
  ]
}
```

This section from a policy denies access.
It is called an *explicit deny*.

```
{
  "Effect": "Deny",
  "Action": [
    "ec2:*",
    "s3:*"
  ],
  "Resource": "*"
}
```

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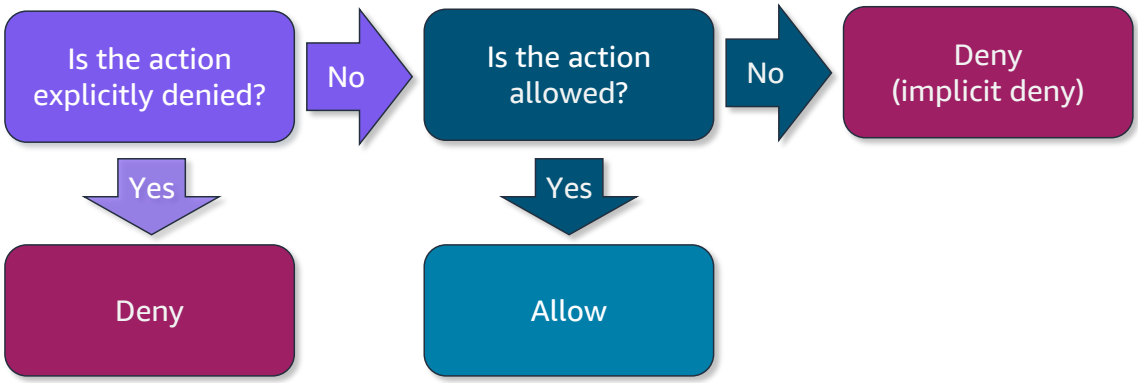
How IAM policies are evaluated



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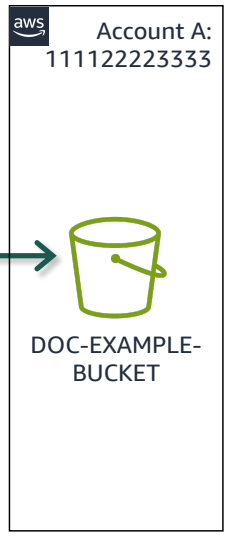
Using a resource-based policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AccountBAccess",
      "Effect": "Allow",
      "Principal": {"AWS": "444455556666"},
      "Action": "s3:PutObject",
      "Resource": [
        "arn:aws:s3:::DOC-EXAMPLE-BUCKET/folder123/*"
      ]
    }
  ]
}
```



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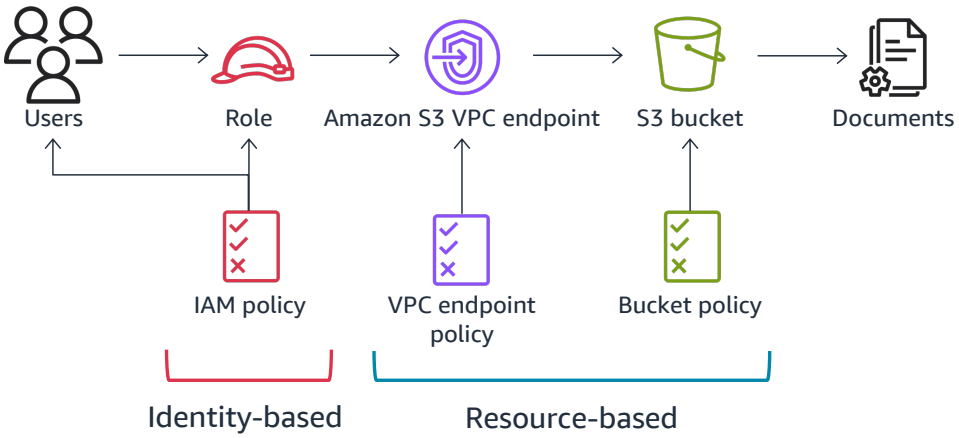
Defense in depth



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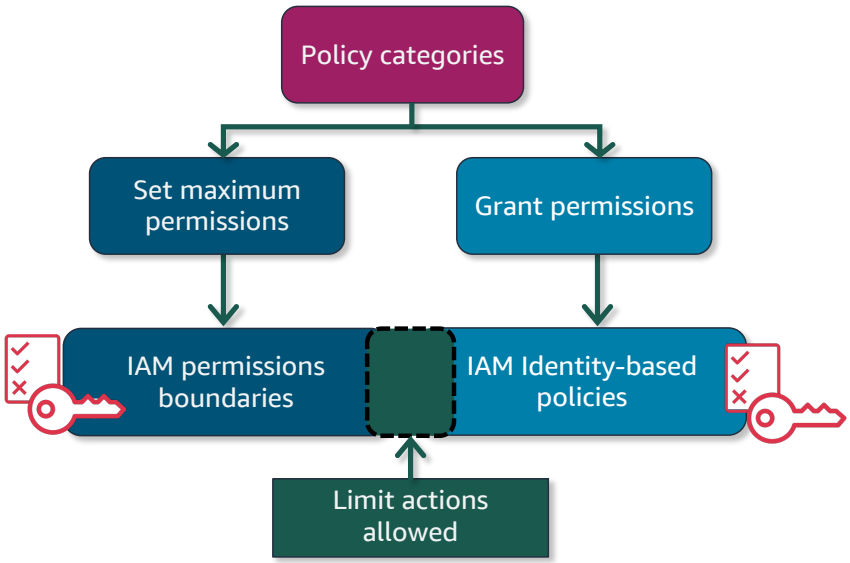
IAM permissions boundaries

IAM permissions boundaries:

- Limit the user's permissions
- Do not provide permissions on their own



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Managing multiple accounts

"What is the best way to manage multiple accounts?"

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Reasons to use multiple accounts

- Develop a multi-account strategy early.
- Refine as business needs evolve.

Many teams

Security and compliance controls

Billing

Isolation

Business process

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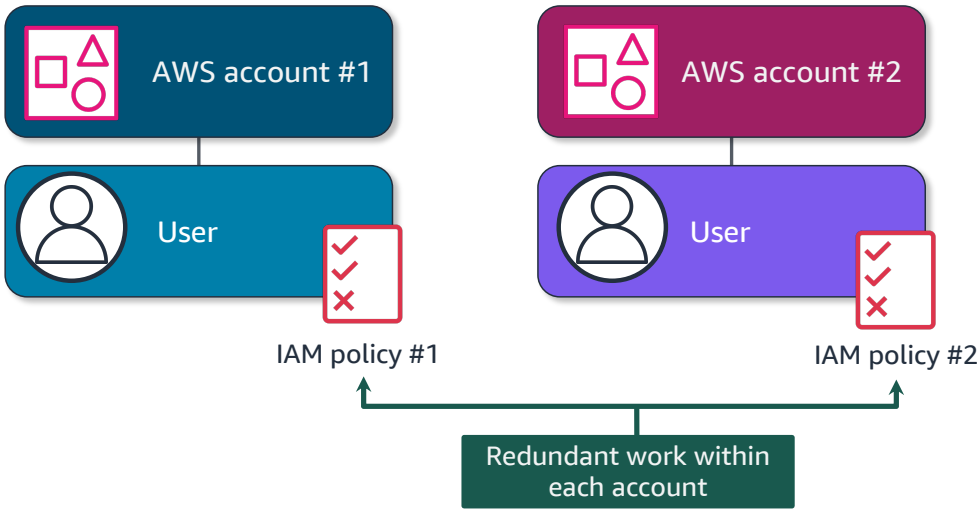
Without AWS Organizations

- IAM policies only apply to individual principals in a single account.
- You must manage policies within each account to enforce restrictions.
- You are required to generate multiple bills.



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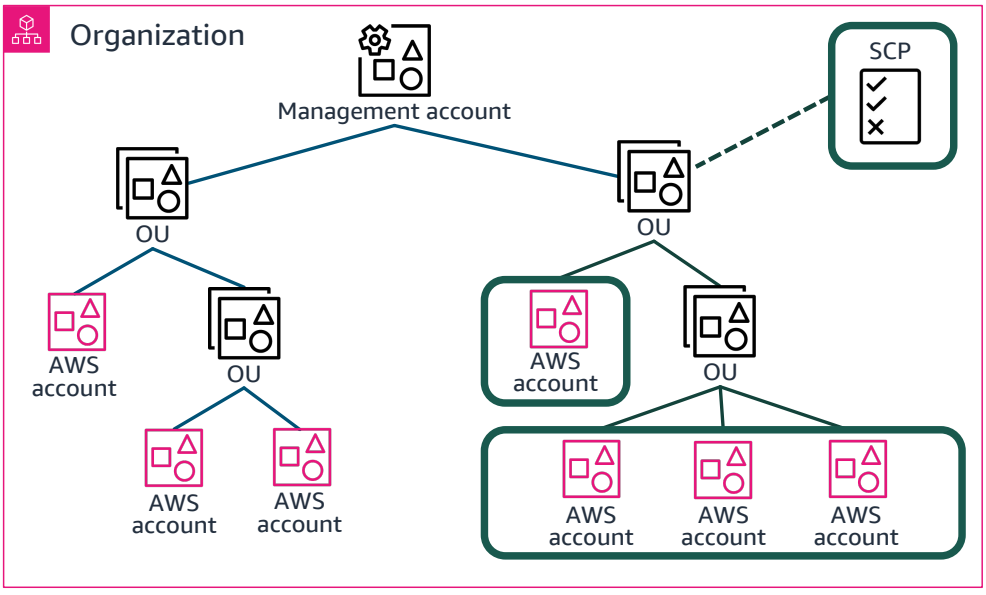
With AWS Organizations

- Create a hierarchy by grouping accounts into organizational units (OUs).
- Apply service control policies (SCPs) to control maximum permissions in every account under an OU.
- Take advantage of consolidated billing.



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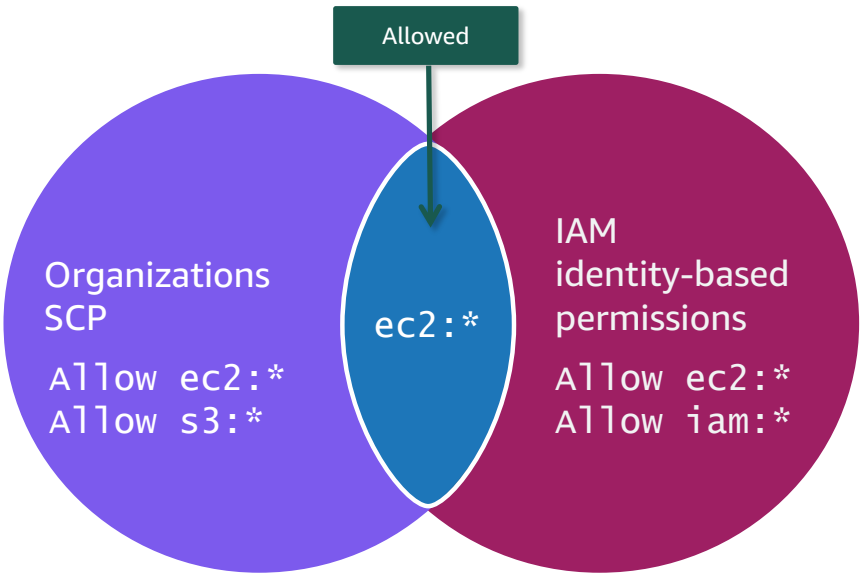
How IAM policies interact with SCPs

- SCPs allow only what is at the intersection of IAM permissions and SCPs.
- SCPs do not grant permissions. They act as a filter.



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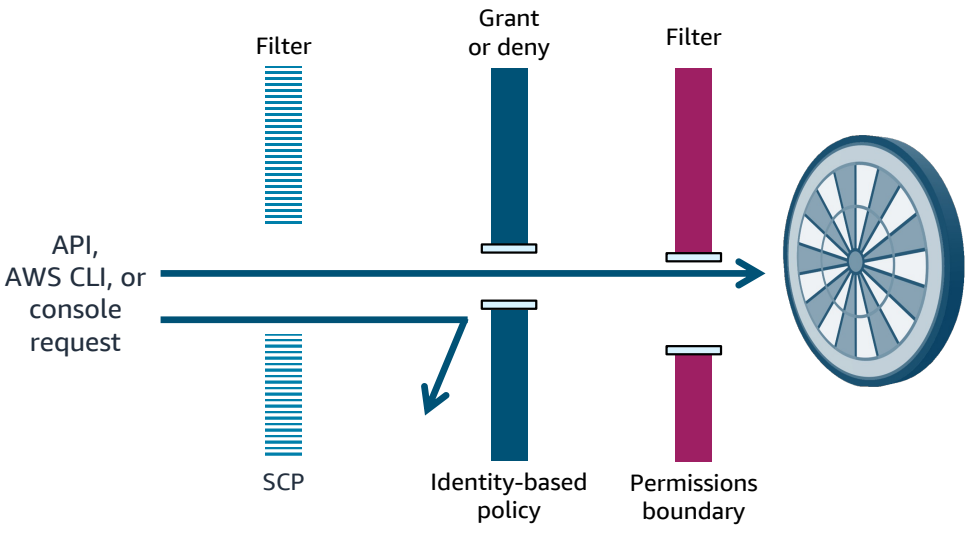
Using policies for a layered defense

- SCPs and permissions boundaries act as a filter to limit permissions.



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Review

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Present solutions

Security specialist

Consider how you would answer the following questions:

- What are the best practices to manage access to AWS accounts and resources?
- How can you give users access to only the resources they need?
- What is the best way to manage multiple accounts?

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Module review

In this module, you learned about the following:

- ✓ Principals and identities
- ✓ Security policies
- ✓ Managing multiple accounts

Next, you review the following:

Knowledge check

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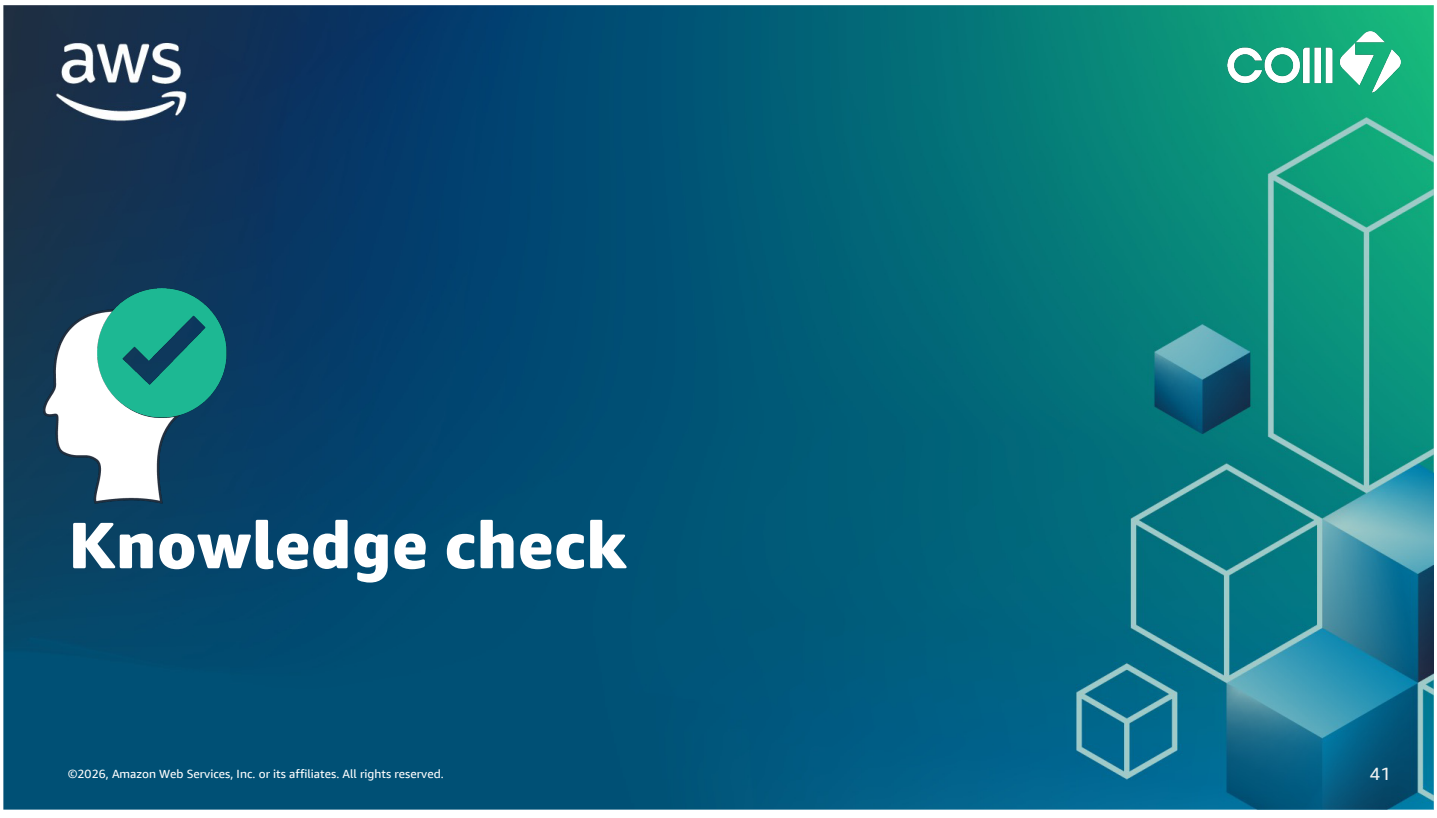

Knowledge check question 1

Which option can be attached to a user, group, or role?

Choice	Response
A	Resource-based policies
B	AWS Security Token Service (AWS STS)
C	Security groups
D	Identity-based policies

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Knowledge check question 1 and answer

Which option can be attached to a user, group, or role?

Choice	Response
A	Resource-based policies
B	AWS Security Token Service (AWS STS)
C	Security groups
D correct	Identity-based policies

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Knowledge check question 2

Which option sets permissions on a specific resource and requires a principal to be listed in the policy?

Choice	Response
A	Identity-based policies
B	Service control policies (SCPs)
C	Resource-based policies
D	Permissions boundaries

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Knowledge check question 2 and answer

Which option sets permissions on a specific resource and requires a principal to be listed in the policy?

Choice	Response
A	Identity-based policies
B	Service control policies (SCPs)
C correct	Resource-based policies
D	Permissions boundaries

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Knowledge check question 3

Which options are elements of an IAM user's long-term programmatic access? (Select TWO.)

Choice	Response
A	Username
B	Access key ID
C	Password
D	Secret access key
E	Multi-factor authentication (MFA) token

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Knowledge check question 3 and answer

Which options are elements of an IAM user's long-term programmatic access? (Select TWO.)

Choice	Response
A	Username
B correct	Access key ID
C	Password
D correct	Secret access key
E	Multi-factor authentication (MFA) token

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Knowledge check question 4

True or False: The root user should be used for daily administration of an AWS account.

Choice	Response
A	True
B	False

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Knowledge check question 4 and answer

The root user should be used for daily administration of an AWS account.

Choice	Response
A	True
B correct	False

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Knowledge check question 5

Which option can only be managed with AWS Organizations?

Choice	Response
A	Service control policies (SCPs)
B	Resource-based policies
C	Permissions boundaries
D	Identity-based policies

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Knowledge check question 5 and answer

Which option can only be managed with AWS Organizations?

Choice	Response
A correct	Service control policies (SCPs)
B	Resource-based policies
C	Permissions boundaries
D	Identity-based policies

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End of Module 2

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Architecting on AWS

Module 3: Networking 1

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Poll question

Which network components are you familiar with? Choose all that apply:

A. IP addressing and subnetting

B. Switching and routing

C. Network security

D. None of the above

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Module overview

- Business request
- IP addressing
- Amazon Virtual Private Cloud (VPC) fundamentals
- VPC traffic security
- Present solutions
- Capstone check-in
- Knowledge check

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Business request

Network engineer

The network engineer needs to know the following:

- How can we make sure that our network has enough IP addresses to support our workloads?
- How do we build a dynamic and secure network infrastructure in our Amazon Web Services (AWS) account?
- How can we filter inbound and outbound traffic to protect resources on our network?

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IP addressing

"How can we make sure that our network has enough IP addresses to support our workloads?"

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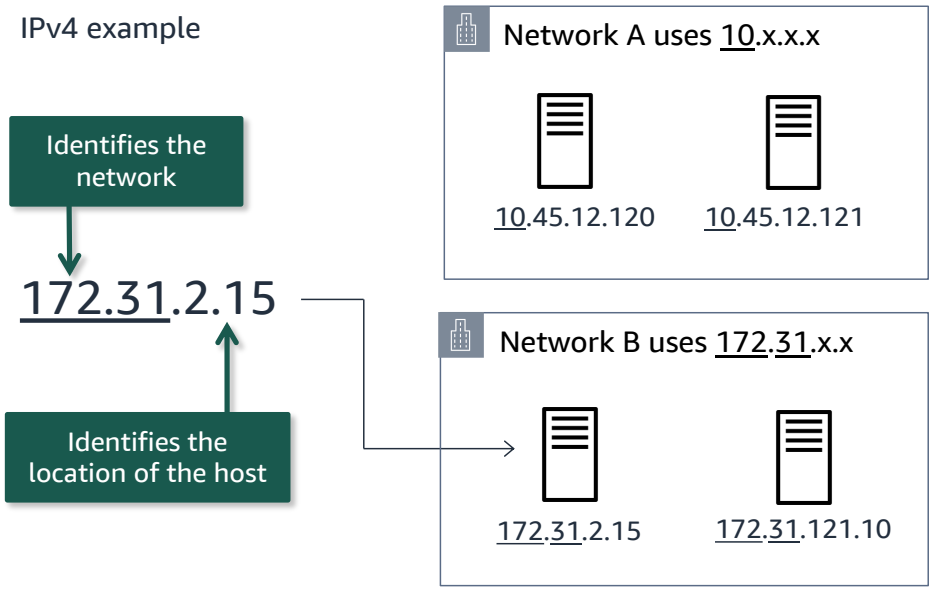
IP addresses

- An IP address identifies a location within a network.
- It identifies the network and the host.
- There are two types of IP addresses:
  - IPv4
  - IPv6



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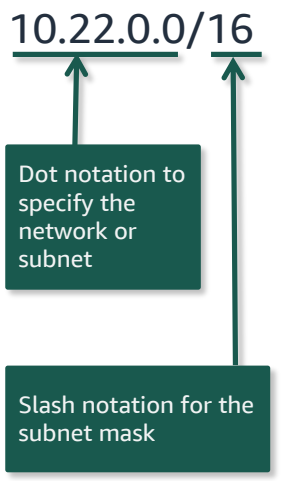
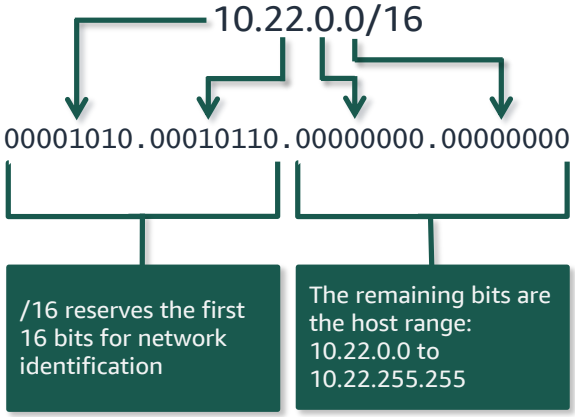
Classless Inter-Domain Routing (CIDR)

CIDR specifies a range of IP addresses called a *CIDR block*.



An IPv4 address is four groups of eight bits.



VPC supported CIDR blocks

CIDR	Total IPs
/28	16
…	…
/20	4,096
/19	8,192
/18	16,384
/17	32,768
/16	65,536

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VPC fundamentals

"How do we build a dynamic and secure network infrastructure in our AWS account?"

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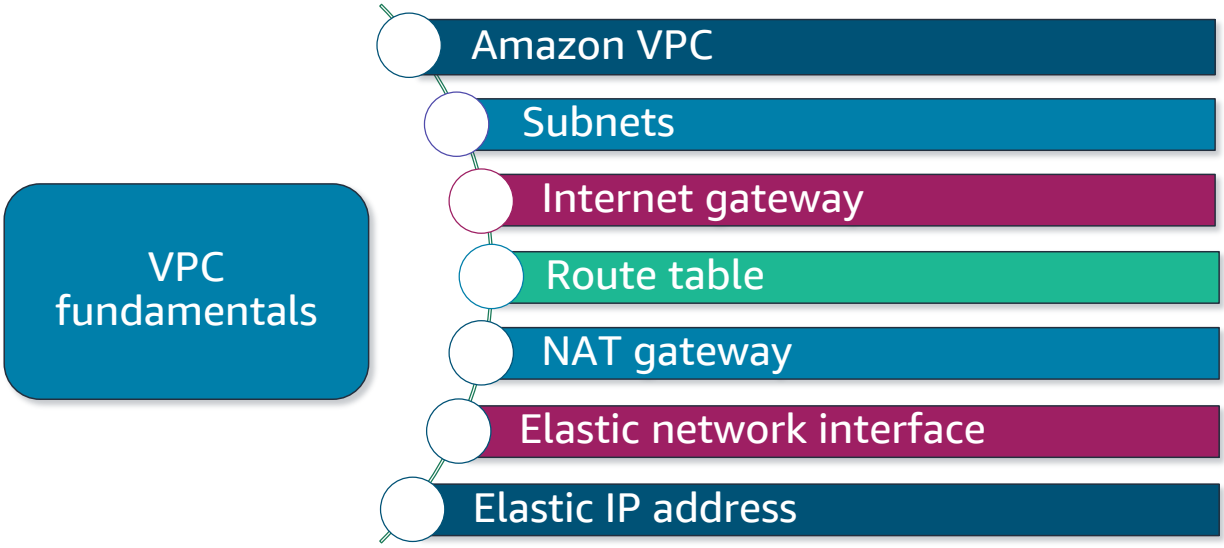
VPC fundamentals topics



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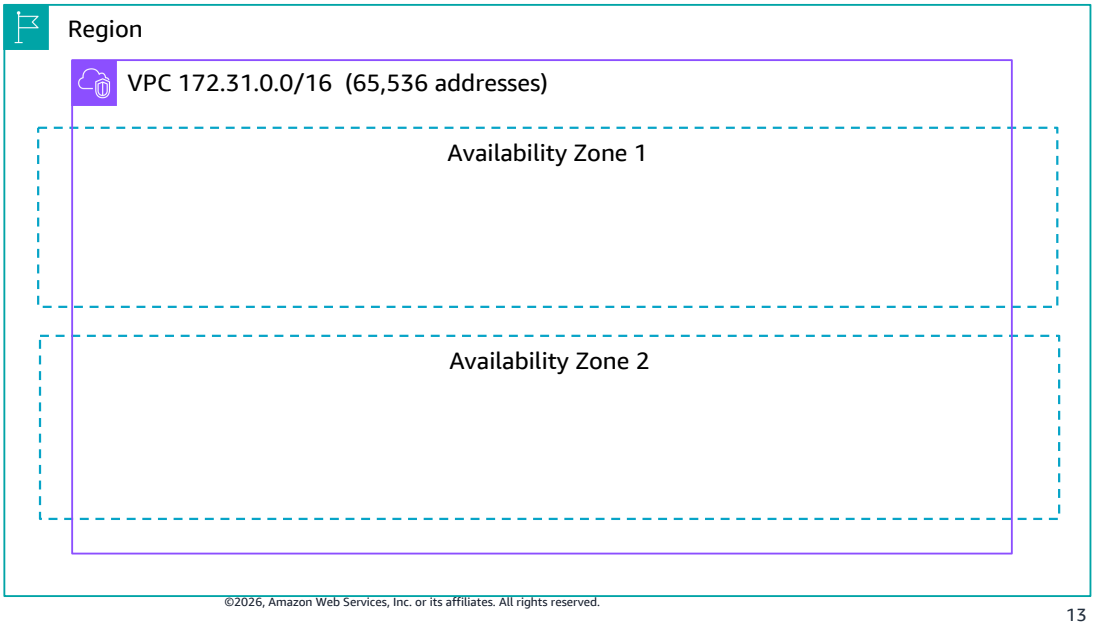
Amazon VPC

- Provides logical isolation for your workloads
- Permits custom access controls and security settings for your resources
- Is bound to a single AWS Region
- Default VPCs already created



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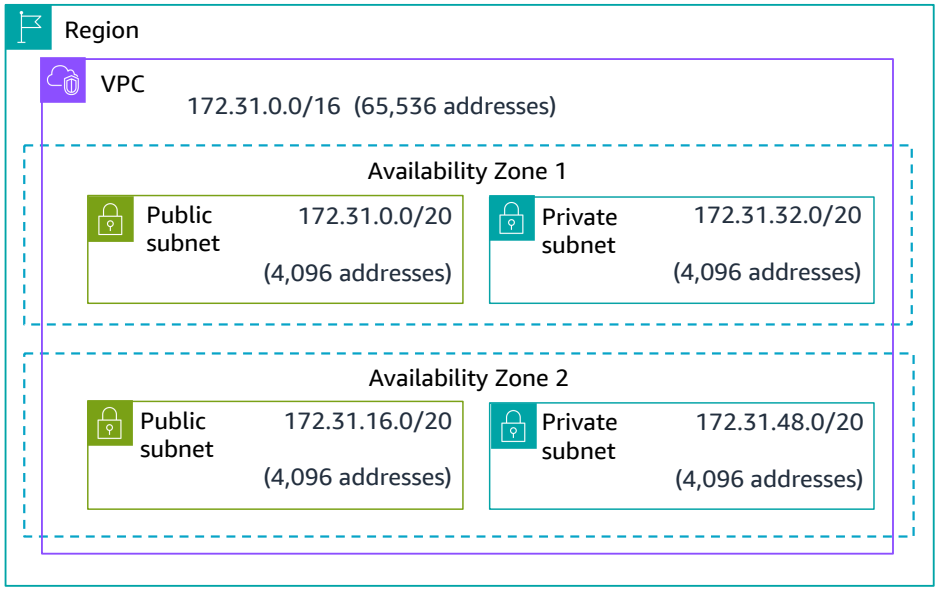
Subnets

- Subnets are a subset of the VPC CIDR block.
- Subnet CIDR blocks cannot overlap.
- Each subnet resides within one Availability Zone.
- An Availability Zone can contain multiple subnets.
- Five addresses are reserved.



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Public subnets

A public subnet holds resources that work with inbound and outbound internet traffic. It requires the following:

Route table

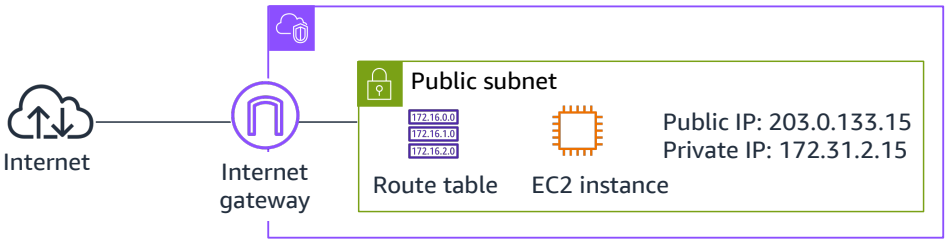
- A set of rules that the VPC uses to route network traffic
- Requires a route to the internet

Internet gateway

- Allows communication between resources in your VPC and the internet

Public IP addresses

- IP addresses that can be reached from the internet
- Protects the private IP addresses only reachable on the network



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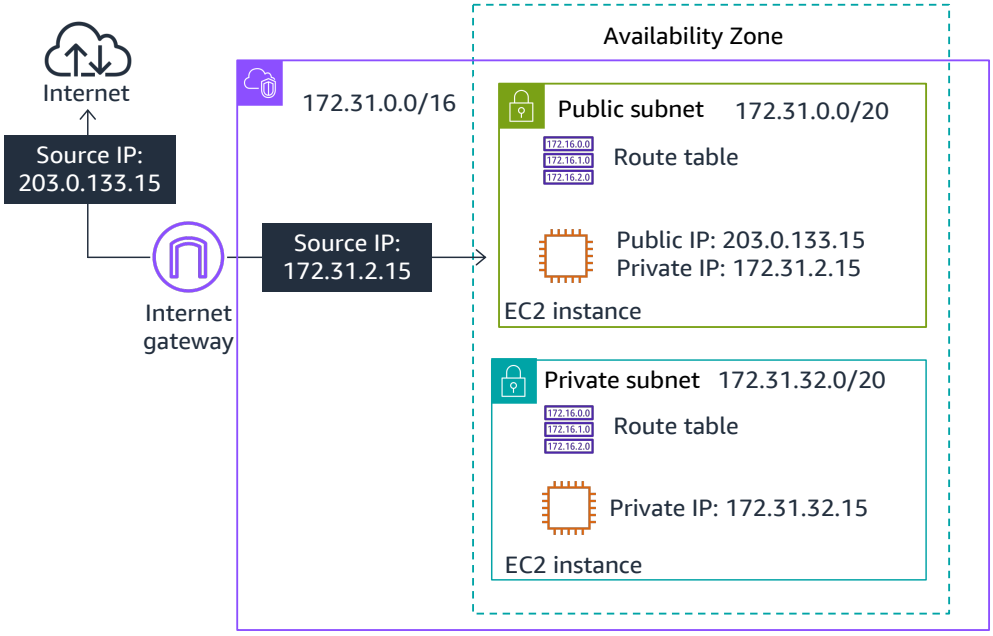
Internet gateways

- Internet gateways permit communication between instances in your VPC and the internet.
- They provide a target in your subnet route tables for internet-routable traffic.
- They protect IP addresses on your network by performing NAT.



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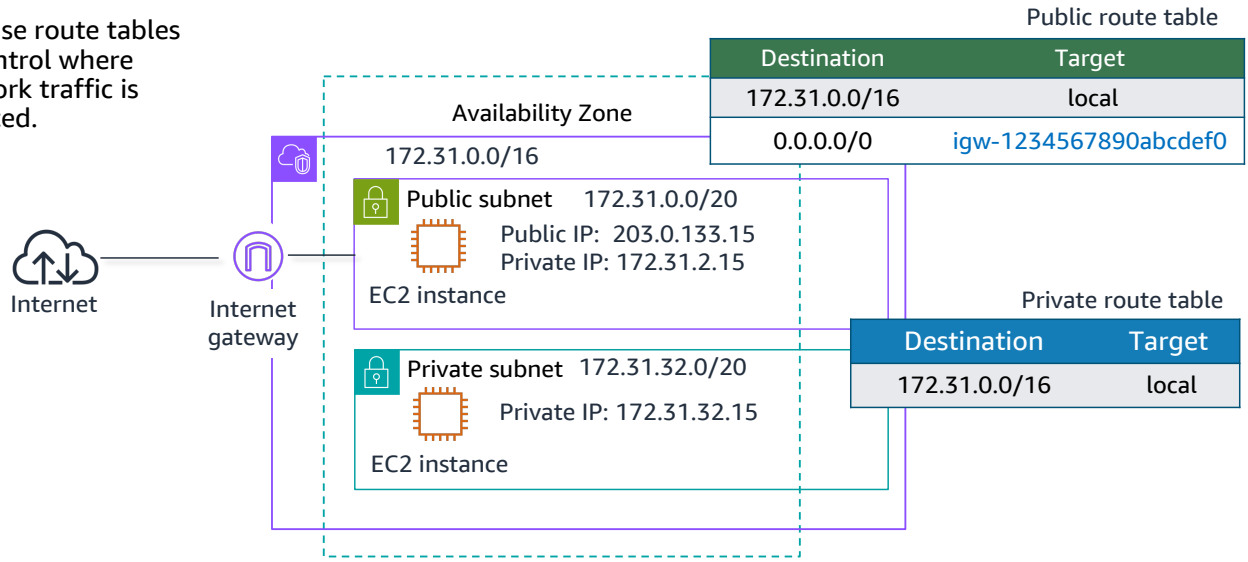
Route tables

- Your VPC has an *implicit router*.
- You use route tables to control where network traffic is directed.



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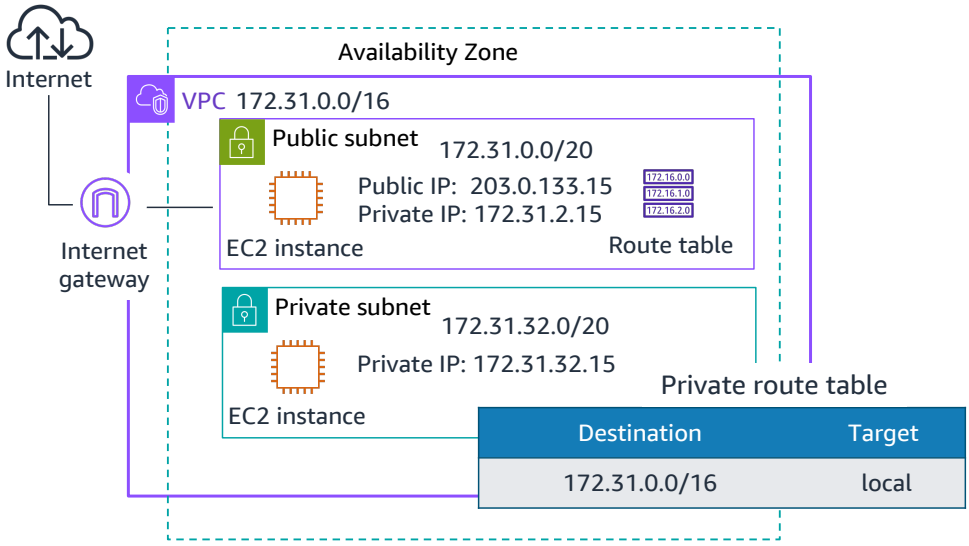
Private subnets

- Private subnets allow indirect access to the internet.
- The private IP address never changes.
- Traffic in the VPC stays local.



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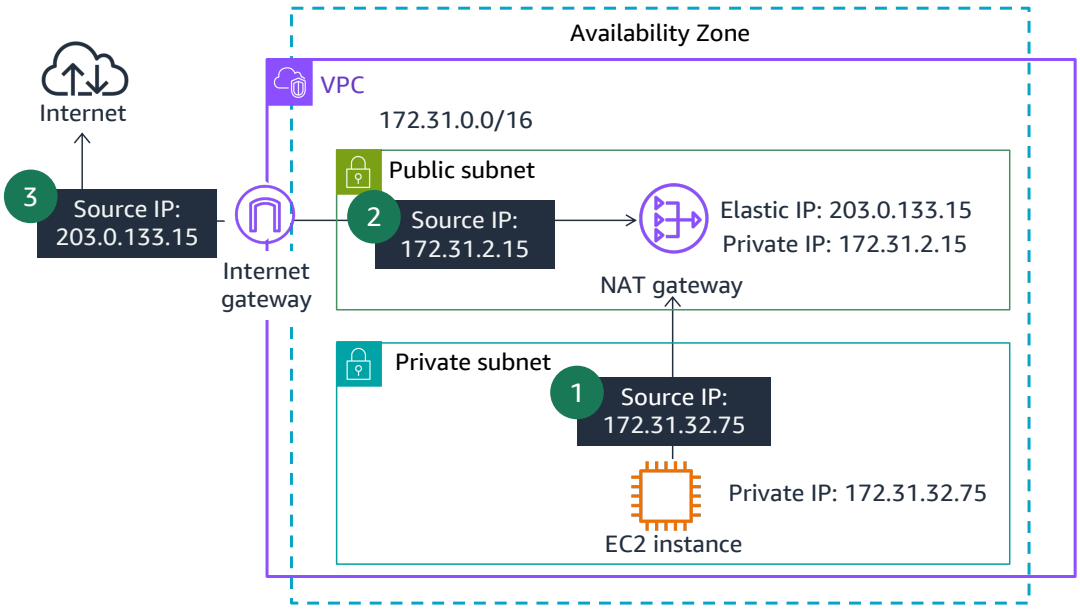
Network address translation with NAT gateways

- You use NAT to protect your private IP addresses.
- A NAT gateway uses an Elastic IP address as the source IP address for traffic from the private subnet.



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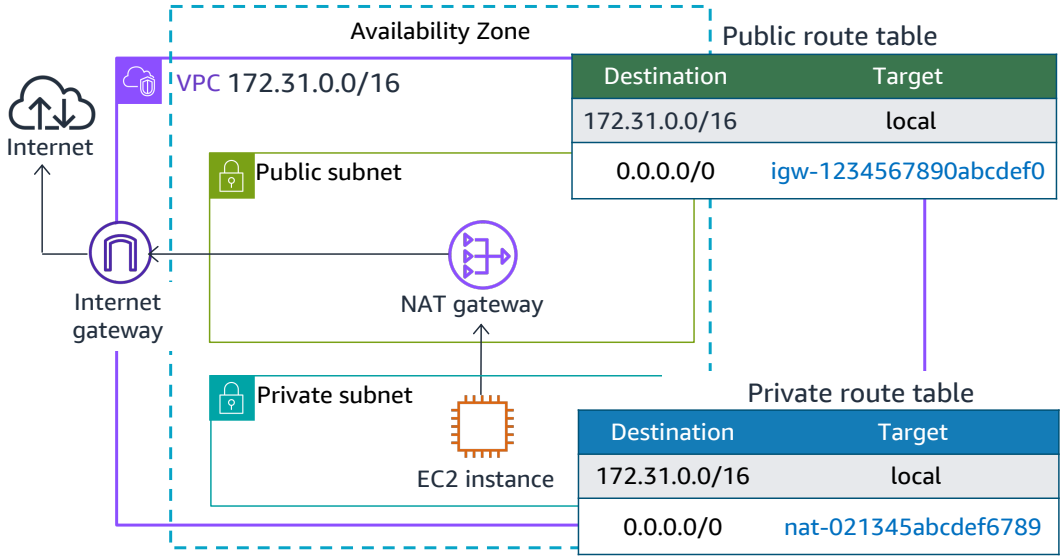
Connecting private subnets to the internet

- The route table for the private subnet sends all IPv4 internet traffic to the NAT gateway.
- The route table for the public subnet sends all internet traffic to the internet gateway.

NAT gateway use case: Connecting resources in a private subnet to the internet



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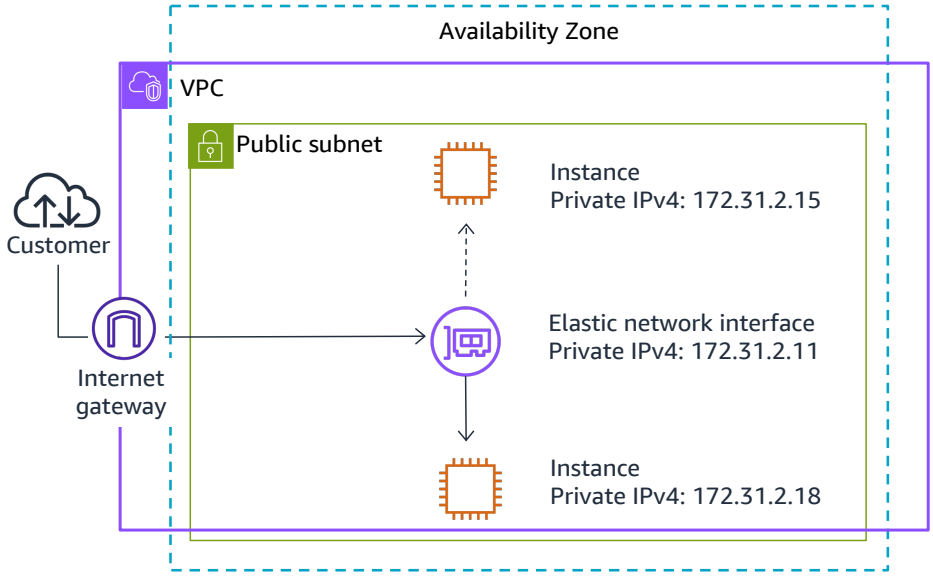
Elastic network interface

An *elastic network interface* is a logical networking component in a VPC that:

- Can be moved across resources in the same Availability Zone
- Maintains its private IP address, Elastic IP address, and media access control (MAC) address



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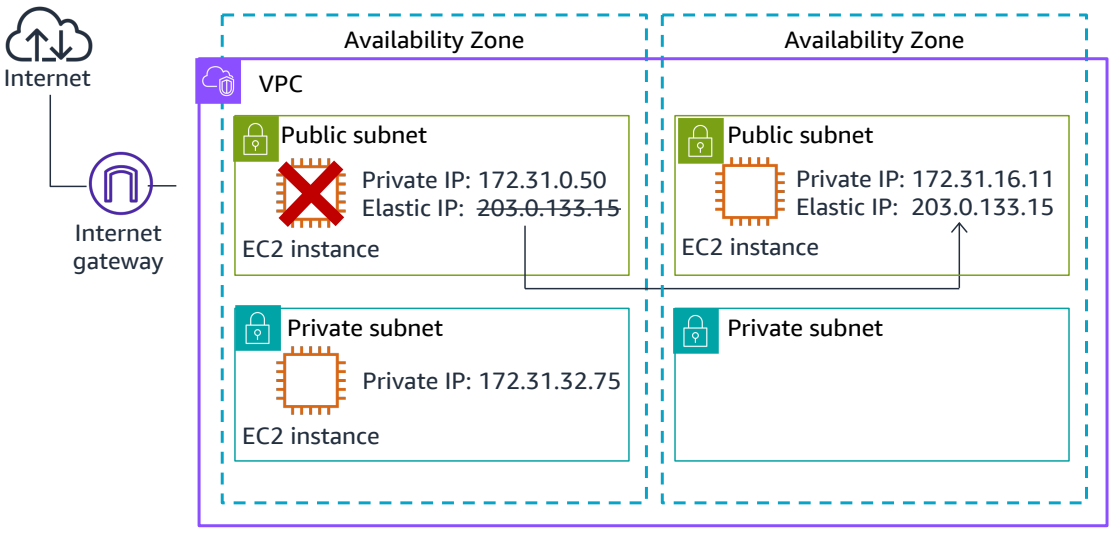
Elastic IP addresses

- Permit association with an instance or a network interface
- Can be reassociated and can direct new traffic immediately
- Have a default restriction of five per Region, per account
- Support Bring Your Own IP (BYOIP)



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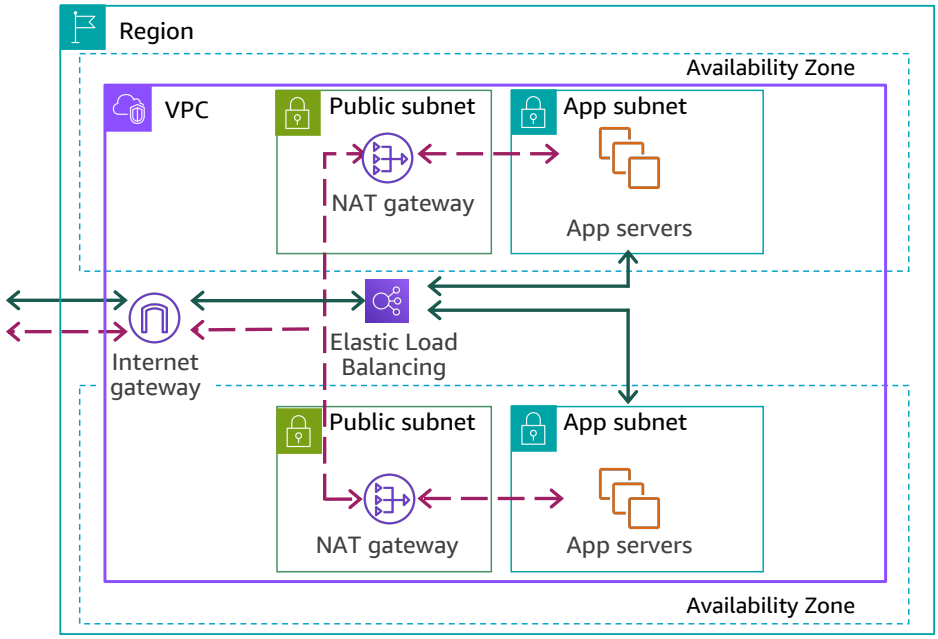
Deploy a VPC across multiple Availability Zones

- Deploy your VPCs across multiple Availability Zones to achieve high availability.
- Create subnets in each Availability Zone.
- Deploy resources in each Availability Zone.
- Distribute traffic between the Availability Zones using load balancers.



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VPC traffic security

"How can we filter inbound and outbound traffic to protect resources on our network?"

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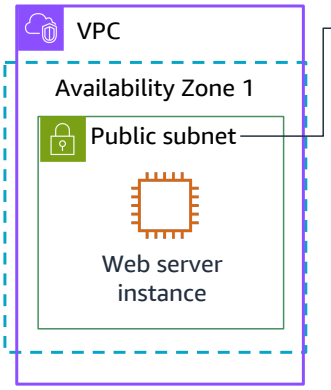
Network access control lists (network ACLs)

- A network ACL acts as a firewall at the subnet boundary.
- By default, it allows all inbound and outbound traffic.
- It is stateless, and requires explicit rules for all traffic.
- It evaluates rules, starting with the lowest numbered rule.



nacl-PublicSubnet

Inbound rules					
Rule #	Type	Protocol	Port Range	Source	Allow or Deny
100	HTTP	TCP	80	0.0.0.0/0	*Allow*
200	HTTPS	TCP	443	0.0.0.0/0	*Allow*
*	ALL Traffic	ALL	ALL	0.0.0.0/0	Deny

Outbound rules					
Rule #	Type	Protocol	Port Range	Source	Allow or Deny
100	Custom TCP Rule	TCP	1024-65535	0.0.0.0/0	*Allow*
*	ALL Traffic	ALL	ALL	0.0.0.0/0	Deny

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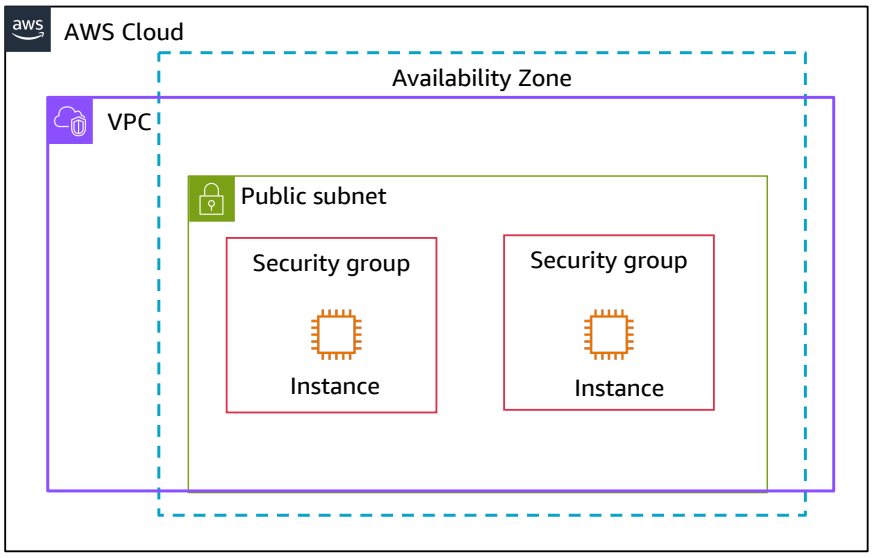
Security groups

- A security group is a virtual firewall that controls inbound and outbound traffic into AWS resources.
- It allows traffic based on IP protocol, port, or IP address.
- It uses stateful rules.



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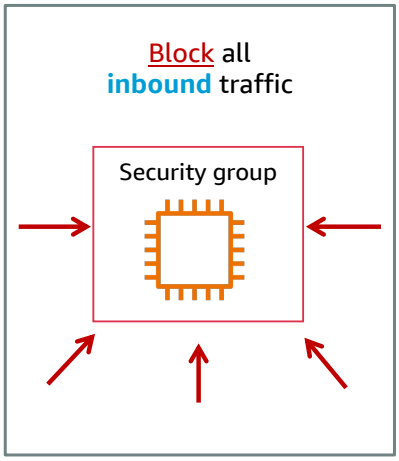
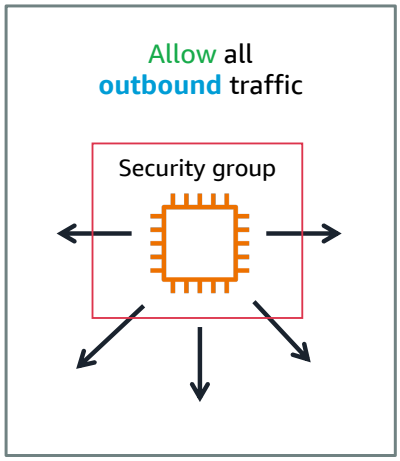
Default and new security groups

- Security groups in default VPCs allow all outbound traffic.
- Custom security groups have no inbound rules, and they allow all outbound traffic.





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Custom security group rules

Inbound

Source	Protocol	Port	Comments
0.0.0.0/0	TCP	80	Allows inbound HTTP access from all IPv4 addresses
0.0.0.0/0	TCP	443	Allows inbound HTTPS traffic from anywhere

Outbound

Destination	Protocol	Port	Comments
SG ID of DB servers	TCP	1433	Allows outbound Microsoft SQL Server access to instances in the specified security group
SG ID of MySQL servers	TCP	3306	Allows outbound MySQL access to instances in the specified security group

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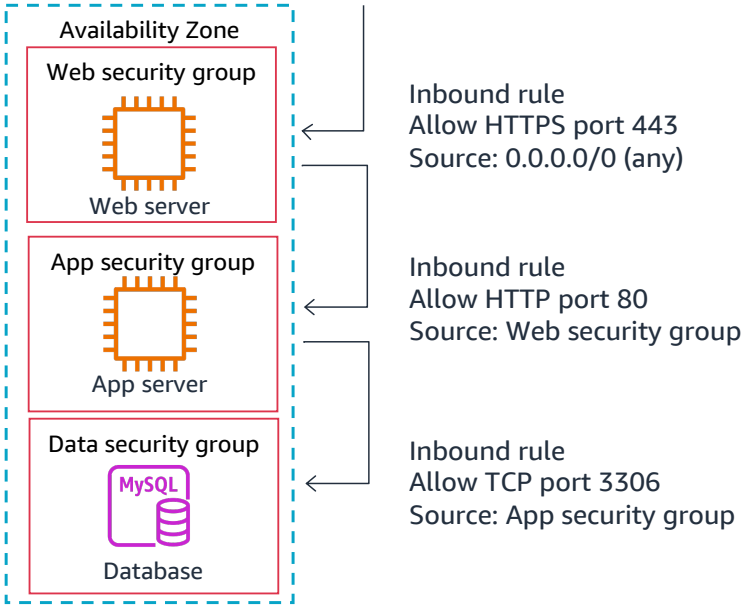
Security group chaining

- Inbound and outbound rules allow traffic flow from the top tier to the bottom tier.
- The security groups act as firewalls to prevent a subnet-wide security breach.



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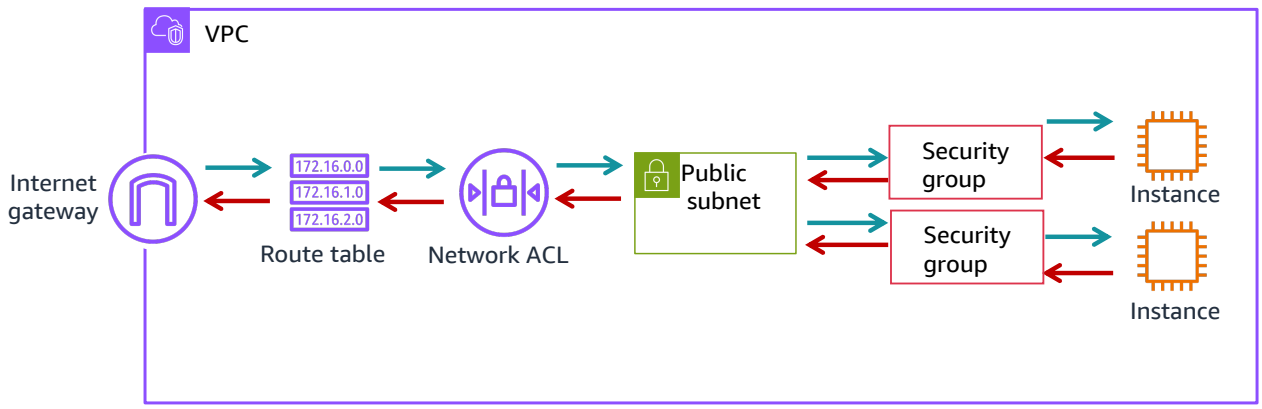
Design your infrastructure with multiple layers of defense



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Comparing security groups and network ACLs

Security Group	Network ACL
Associated to an elastic network interface and implemented in the hypervisor	Associated to a subnet and implemented in the network
Supports Allow rules only	Supports Allow rules and Deny rules
A stateful firewall	A stateless firewall
All rules evaluated before deciding whether to allow traffic	All rules processed in order when deciding whether to allow traffic
Applies to an instance only if it is associated with the instance	Applies to all instances deployed in the associated subnet

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Review

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Present solutions

Network engineer

Consider how you would answer the following:

- How can we make sure that our network has enough IP addresses to support our workloads?
- How do we build a dynamic and secure network infrastructure in our AWS account?
- How can we filter inbound and outbound traffic to protect resources on our network?

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Module review

In this module, you learned about the following:

- ✓ IP addresses
- ✓ VPC fundamentals
- ✓ VPC traffic security

Next, you review the following:

- Capstone check-in
- Knowledge check

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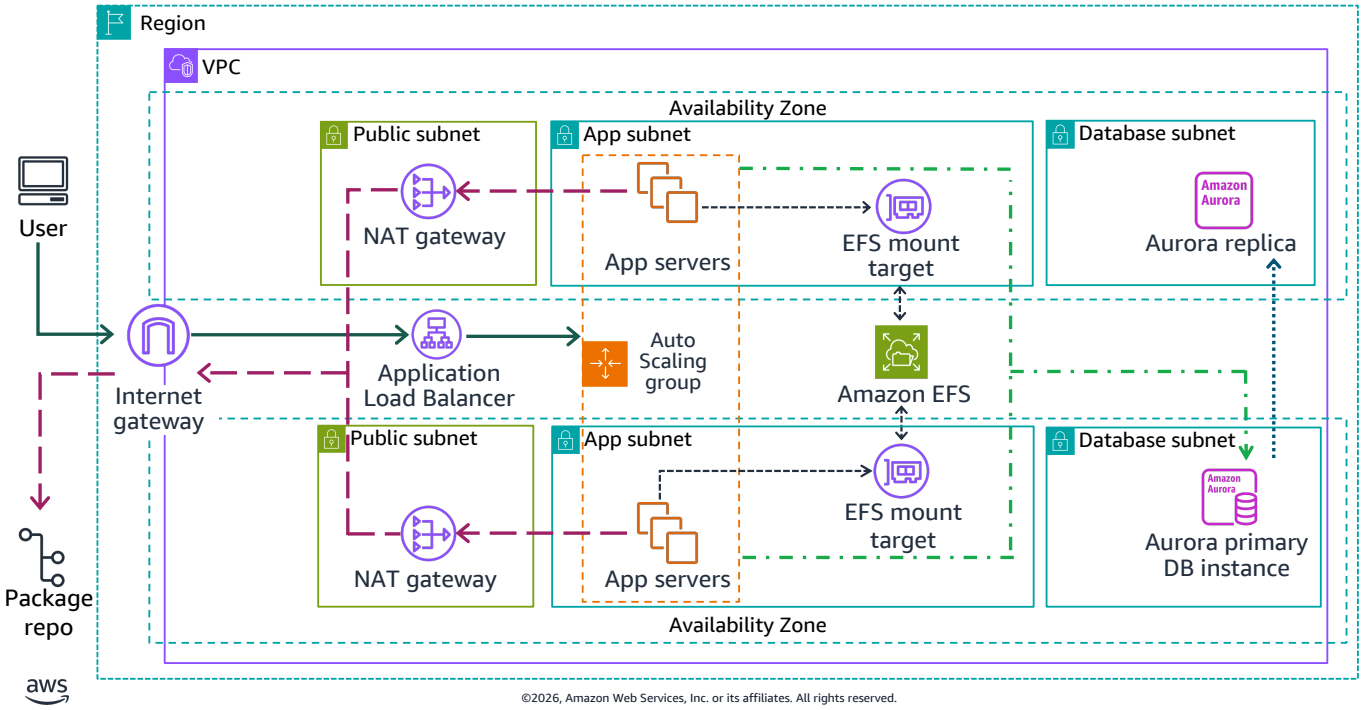
Capstone architecture



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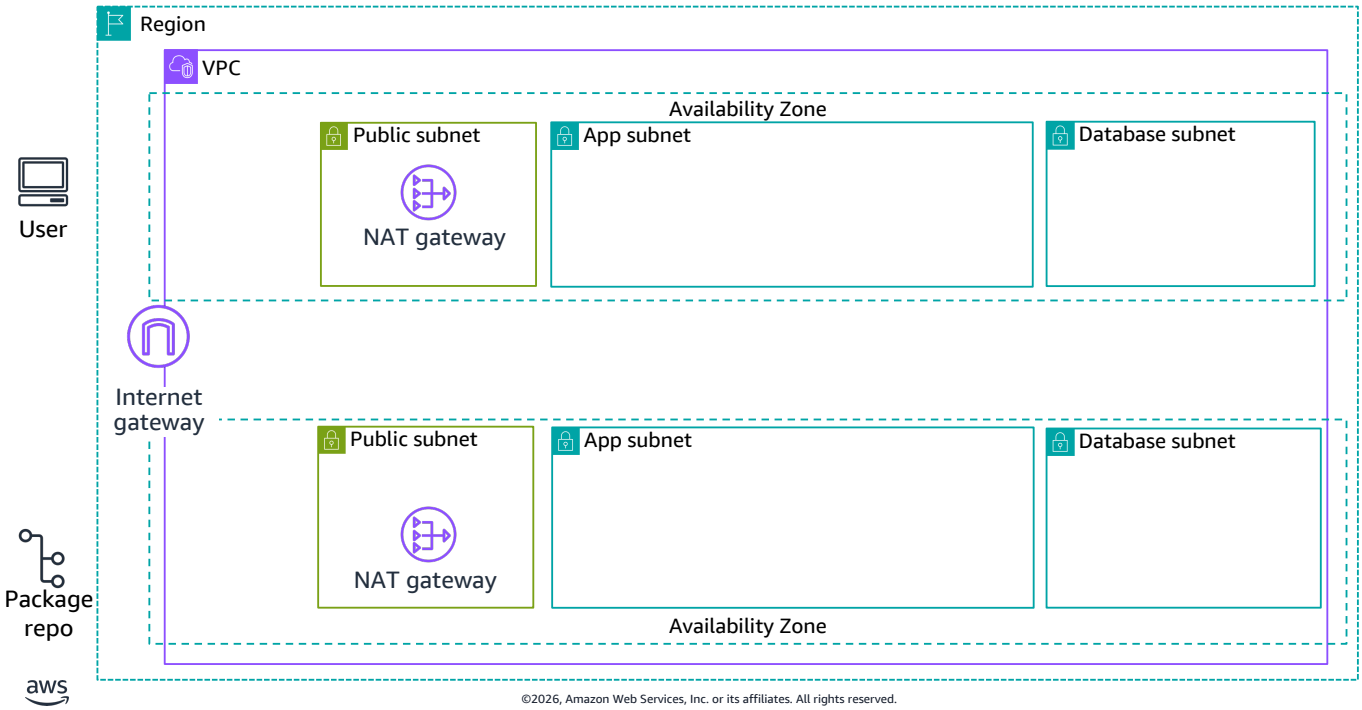
Capstone architecture check-in



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Knowledge check

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Knowledge check question 1

True or False: A single Amazon Virtual Private Cloud (Amazon VPC) can span multiple Regions.

Choice	Response
A	True
B	False

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Knowledge check question 1 and answer

True or False: A single Amazon Virtual Private Cloud (Amazon VPC) can span multiple Regions.

Choice	Response
A	True
B correct	False

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Knowledge check question 2

Which action must be taken to make a subnet public?

Choice	Response
A	Route outbound traffic from the subnet.
B	Route inbound traffic from the internet gateway.
C	Route outbound traffic to the internet gateway.
D	Subnets are public by default.

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Knowledge check question 2 and answer

Which action must be taken to make a subnet public?

Choice	Response
A	Route outbound traffic from the subnet.
B	Route inbound traffic from the internet gateway.
C correct	Route outbound traffic to the internet gateway.
D	Subnets are public by default.

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Knowledge check question 3

Which function does the NAT gateway serve?

Choice	Response
A	Load balances incoming traffic to multiple instances
B	Allows internet traffic initiated by private subnet instances
C	Allows instances to communicate between subnets
D	Increases security for instances in a public subnet

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Knowledge check question 3 and answer

Which function does the NAT gateway serve?

Choice	Response
A	Load balances incoming traffic to multiple instances
B correct	Allows internet traffic initiated by private subnet instances
C	Allows instances to communicate between subnets
D	Increases security for instances in a public subnet

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Knowledge check question 4

What should be used to create traffic filtering rules for a subnet?

Choice	Response
A	NAT gateway
B	Route table
C	Security group
D	Network access control list (network ACL)

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Knowledge check question 4 and answer

What should be used to create traffic filtering rules for a subnet?

Choice	Response
A	NAT gateway
B	Route table
C	Security group
D correct	Network access control list (network ACL)

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Knowledge check question 5

Which ports are open by default when a new security group is created? (Select TWO.)

Choice	Response
A	Nothing is allowed inbound.
B	Nothing is allowed outbound.
C	Anything is allowed inbound.
D	Anything is allowed outbound.
E	Inbound traffic is allowed on public subnets.

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Knowledge check question 5 and answer

Which ports are open by default when a new security group is created? (Select TWO.)

Choice	Response
A correct	Nothing is allowed inbound.
B	Nothing is allowed outbound.
C	Anything is allowed inbound.
D correct	Anything is allowed outbound.
E	Inbound traffic is allowed on public subnets.

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End of Module 3

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Architecting on AWS

Module 4: Compute

Lab 2

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Poll question

Where do you run the majority of your compute workloads?

A. On-premises physical or virtual servers

B. Cloud-based servers

C. On-premises containers

D. Cloud-based containers

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Module overview

- Business request
- Compute services
- Amazon Elastic Compute Cloud (Amazon EC2) instances
- EC2 instance storage
- Amazon EC2 pricing options
- AWS Lambda
- Present solutions
- Knowledge check
- Capstone check-in
- Lab 2: Build Your Amazon VPC Infrastructure

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Business request

Compute operations manager

The compute operations manager wants to know the following:

- What AWS compute services are available?
- What should the team consider when deploying new and existing servers to Amazon EC2?
- How do we know which volume type to attach to our EC2 instances?
- How can we optimize cost for compute resources?
- Where can we start with serverless compute options?

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Compute services

"What AWS compute services are available?"

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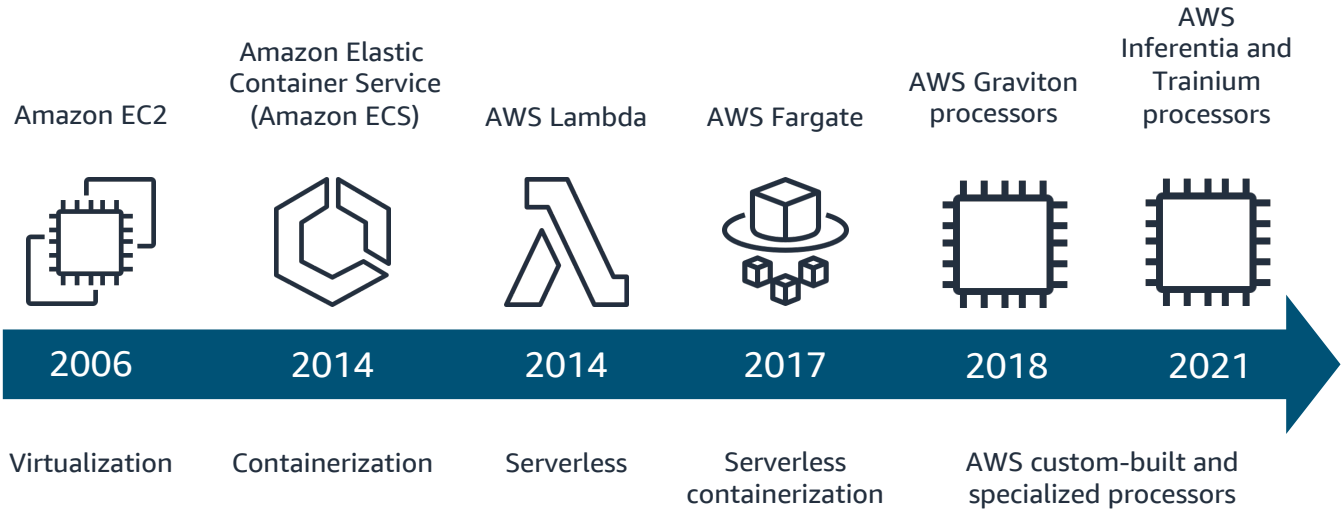
Evolution of AWS compute



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AWS services in this module

We cover other compute-related services later in this course.

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EC2 instances

"What should the team consider when deploying new and existing servers to Amazon EC2?"

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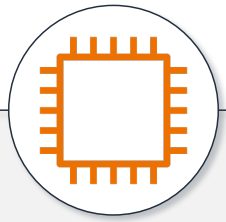
EC2 instances overview

Physical servers host EC2 instances in AWS Regions around the world.



EC2 instances give you secure and resizable compute capacity in the cloud.

You can add or remove compute capacity to meet changes in demand.

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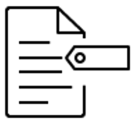
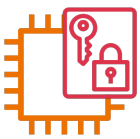
EC2 instance launch considerations



Name and tags

Application and OS image

Instance type and size

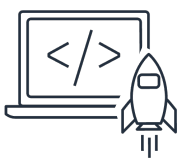


Key pair

Network and security

Storage

Placement and tenancy



Scripts and metadata

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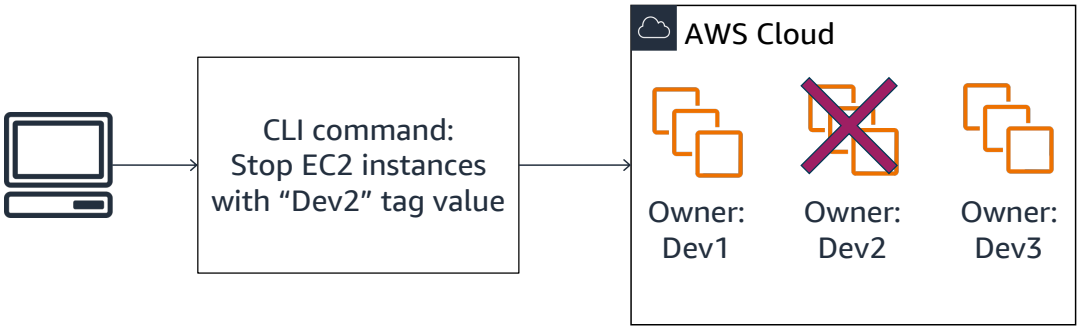
Tags in Amazon EC2

- Assign a name and other tags to your AWS resources.
- Manage, search, and filter resources.
- More tags are better than fewer.
- Tags are case sensitive.



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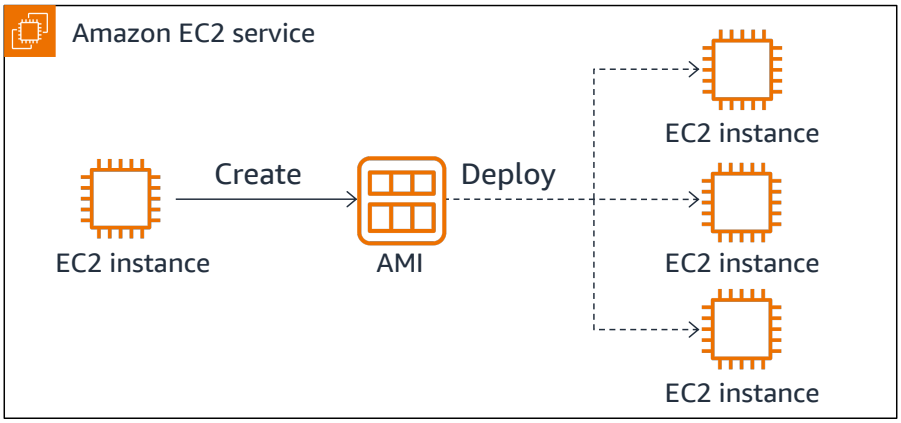
Amazon Machine Image (AMI)

AMI components:

- Template for instance volumes
- Launch permissions
- Block device mapping

Benefits:

- Repeatable
- Reusable
- Recoverable



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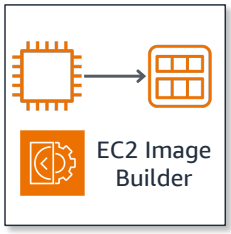
Where to get an AMI

Choose from the following options:

Use prebuilt AMIs offered by AWS.

Search the AWS Marketplace for a catalog with thousands of solutions.



Create your own AMIs.

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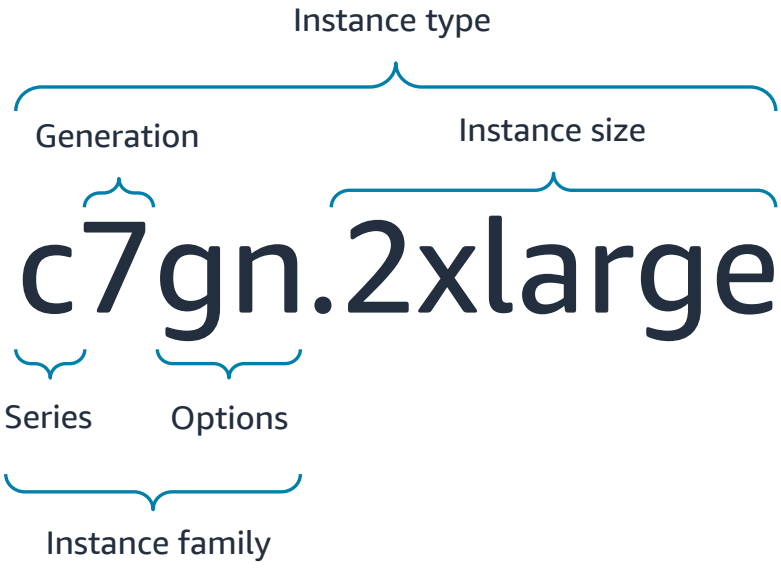
Understanding instance type names



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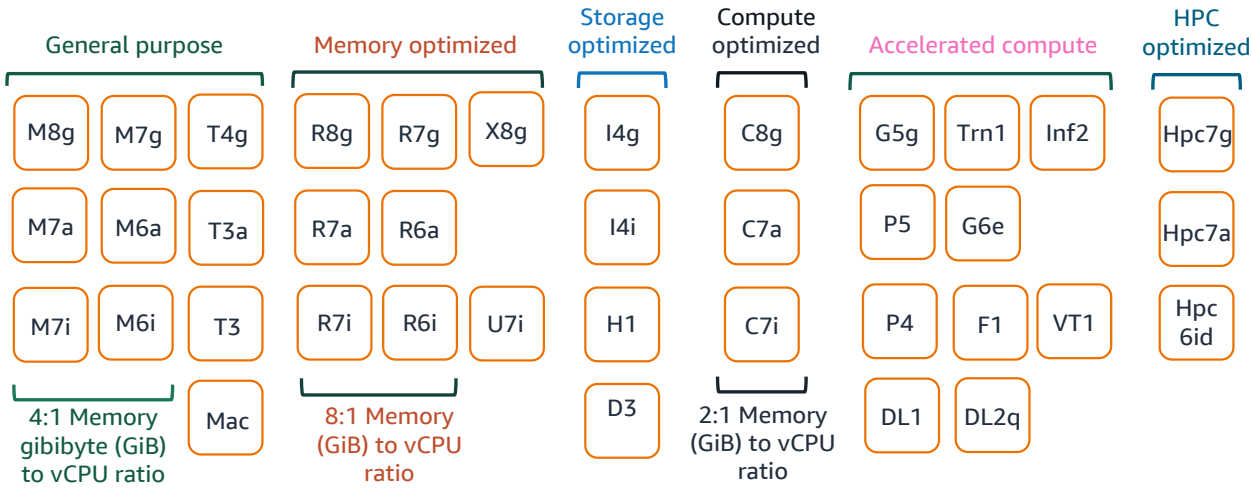
EC2 instance families



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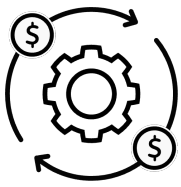
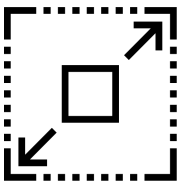
Benefits of newer-generation instance types

Improved performance



Cost efficiency



Scalability

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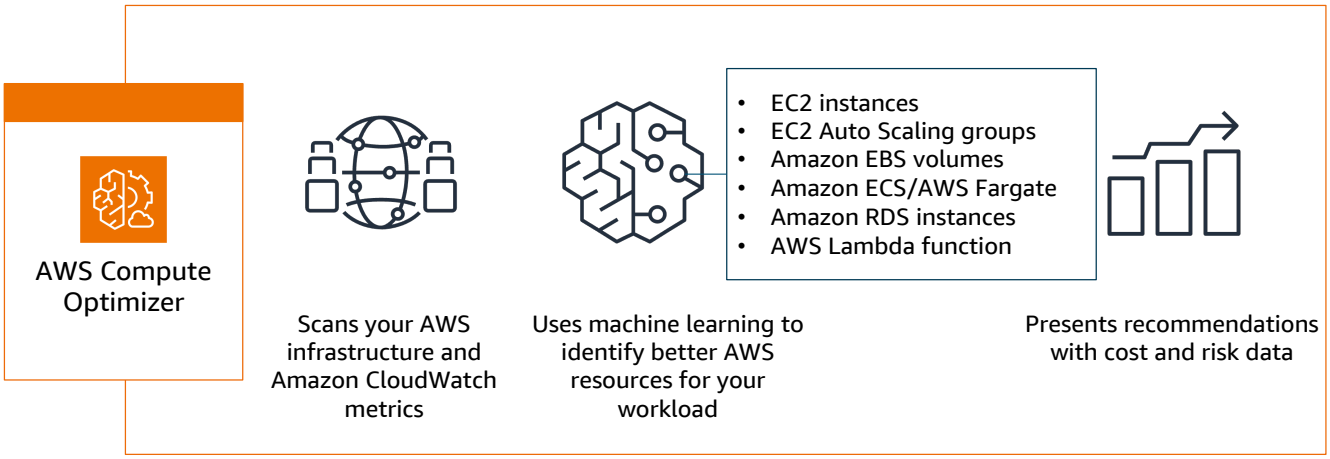
AWS Compute Optimizer



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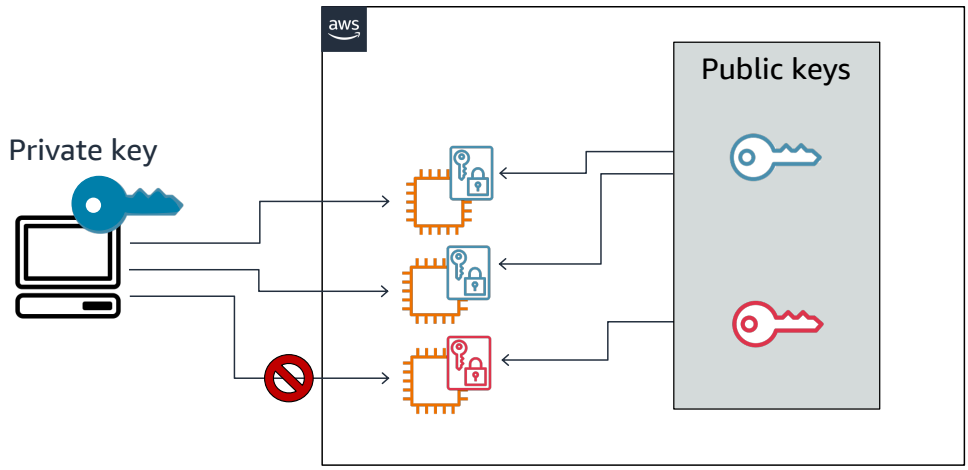
Amazon EC2 key pairs



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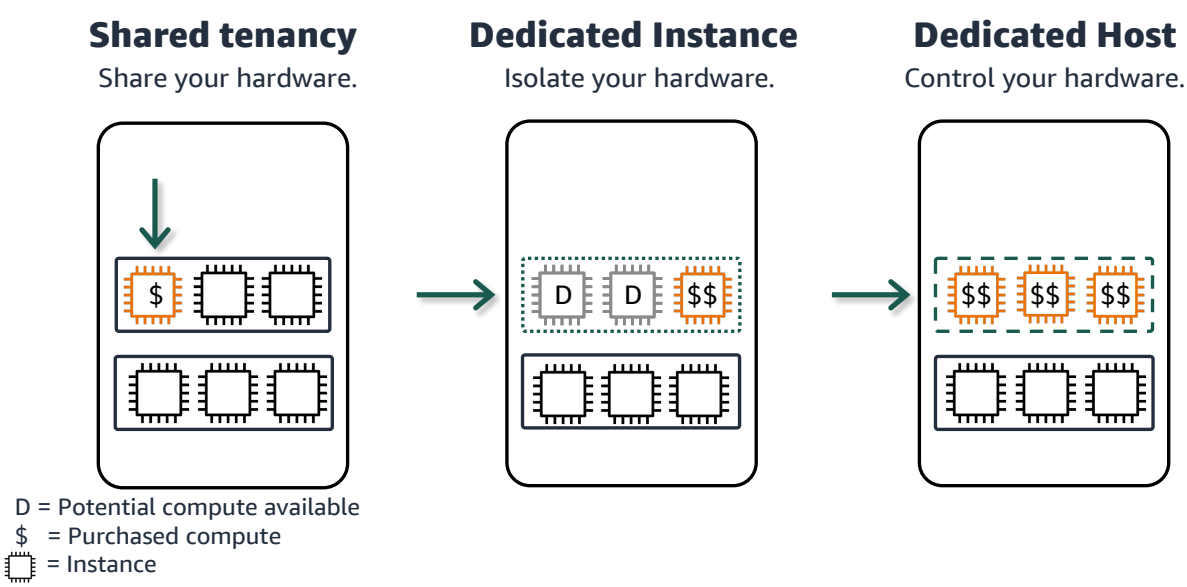
Tenancy



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Placement groups and use cases

With **placement groups**, you choose how close or far your instances are from each other.

Requirements	Solution	Example use case
Provide low network latency and high network throughput.	**Cluster** EC2 instances near each other.	High performance computing (HPC)
Critical instances must be fault-tolerant.	**Spread** across network segments and racks.	Medical health record system
Avoid correlated hardware failures.	**Partition** in logical groups on separate hardware.	Large distributed and replicated workloads such as Apache Kafka, Apache Hadoop, and Apache Cassandra

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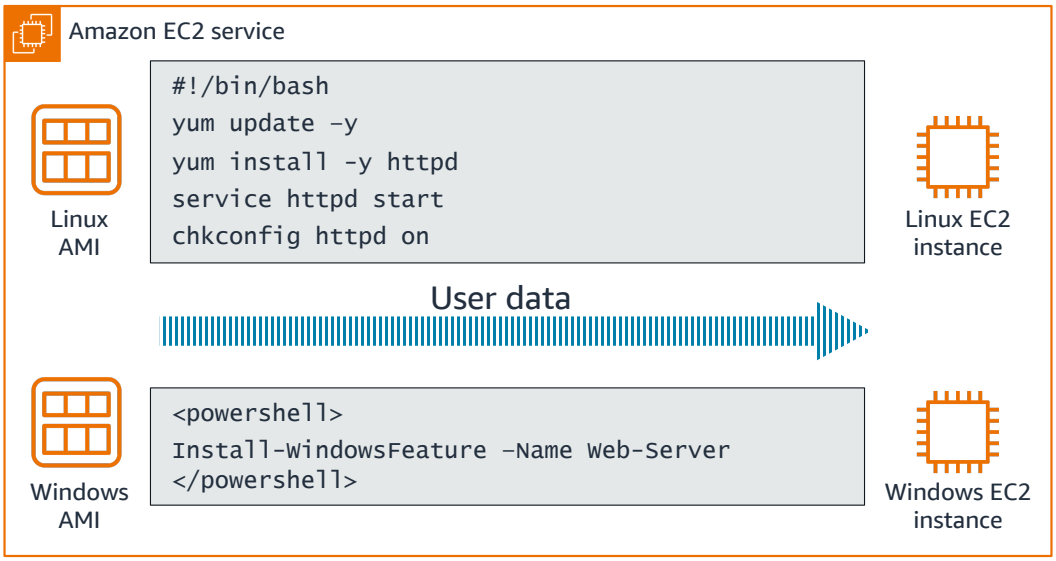
User data

- Runs scripts as root after the instance launches
- Can be used to perform common automated configuration tasks



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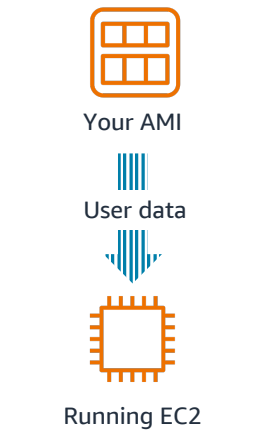
Instance metadata



Note: You can only get metadata with a request from your EC2 instance.

EC2 instance metadata can be used for automation.

```
TOKEN=`curl -X PUT "http://169.254.169.254/latest/api/token"
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600"`

curl http://169.254.169.254/latest/meta-data/public-hostname
-H "X-aws-ec2-metadata-token: $TOKEN"
```

Metadata	Value
instance-id	i-1234567890abcdef0
mac	00-1B-63-84-45-E6
public-hostname	ec2-203-0-113-25.compute-1.amazonaws.com
public-ipv4	203.0.113.25
local-hostname	ip-10-251-50-12.ec2.internal
local-ipv4	10.251.50.12

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Storage for EC2 instances

"How do we know which volume type to attach to our EC2 instances?"

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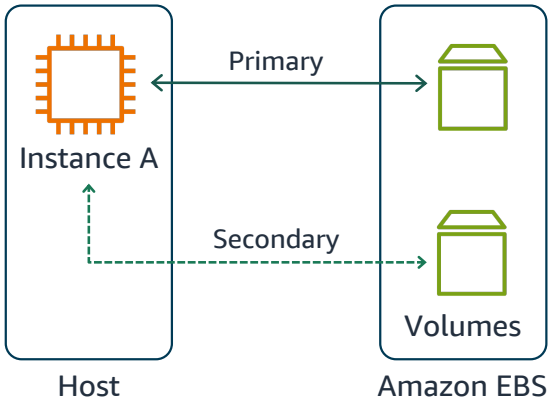
Amazon Elastic Block Store (Amazon EBS)

- Create block-level storage with automatic volume replication in your Availability Zone.
- Attach one or more EBS volumes to a single EC2 instance.
- Move EBS volumes between EC2 instances as needed.



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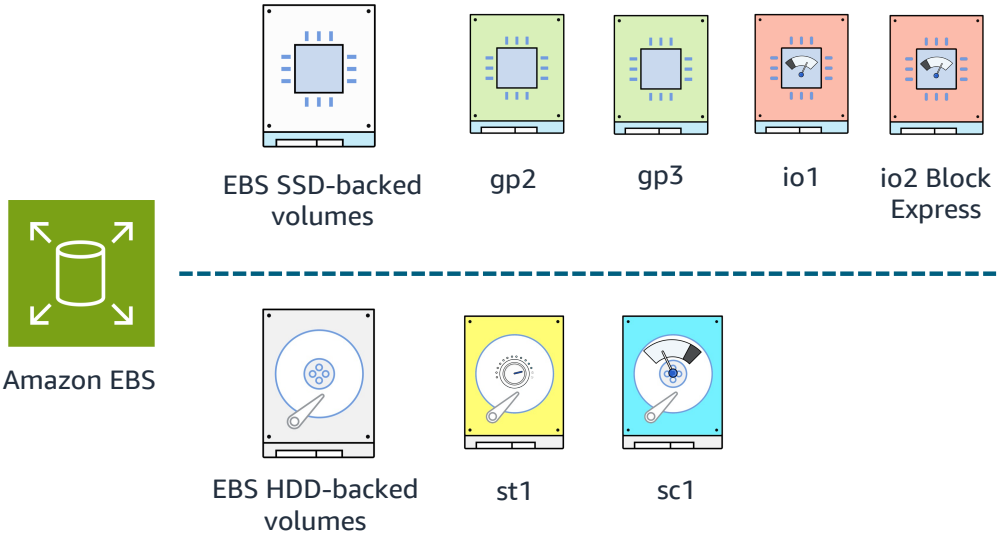
Amazon EBS volume types

- Solid state drive (SSD) is for high-performance and general-purpose workloads.
- Hard disk drive (HDD) is for big or infrequently accessed data.



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Amazon EBS volume characteristics (1 of 2)

	General Purpose SSD		Provisioned IOPS SSD	
Volume type	**gp2**	**gp3**	**io1**	**io2 Block Express**
Description	Volume that balances price and performance for a wide variety of workloads		Highest performance volume for critical, IOPS-intensive, and throughput-intensive workloads that require low latency	
Size	1 gibibyte (GiB) to 16 TiB (tebibytes)		4 GiB to 16 TiB	4 GiB to 64 TiB
Maximum IOPS	16,000 (burst)	16,000 (no burst)	64,000	256,000
Maximum throughput per volume	250 mebibytes per second (MiBps)	1,000 MiBps	1,000 MiBps	4,000 MiBps

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Amazon EBS volume characteristics (2 of 2)

	Throughput Optimized HDD	Cold HDD
Volume type	**st1**	**sc1**
Description	Low-cost HDD volume designed for frequently accessed, throughput-intensive workloads	Lowest-cost HDD volume designed for less frequently accessed workloads
Size	125 GiB to 16 TiB	125 GiB to 16 TiB
Maximum IOPS	500	250
Maximum throughput per volume	500 MiBps	250 MiBps

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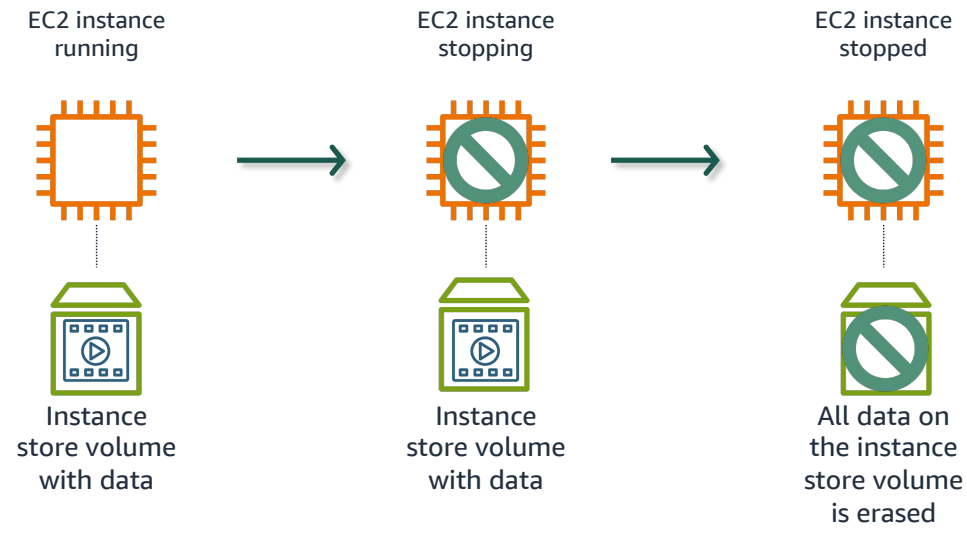
Instance store volumes

- Local to instance
- Non-persistent
- Doesn't support snapshots
- Available in HDD, SSD, and non-volatile memory express SSD (NVMe SSD) varieties



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Amazon EC2 pricing options

"How can we optimize cost for compute resources?"

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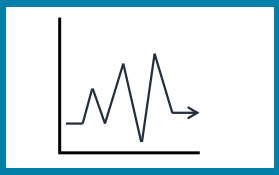
Amazon EC2 purchase options

On-Demand

Pay for compute capacity per second or hour, with no long-term commitments



Spiky workloads or temporary needs

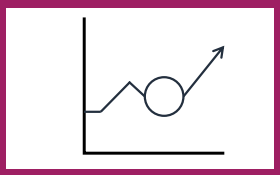
Savings Plans

1-year or 3-year commitment with varied flexibility based on type of Savings Plan

Committed flexible access to compute

Spot Instances

Spare Amazon EC2 capacity at savings of up to 90 percent off On-Demand costs



Fault-tolerant, flexible, stateless workloads

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Savings Plan types

Compute Savings Plans

Greatest flexibility, up to **66 percent** off On-Demand rates, and applies to AWS Fargate and AWS Lambda usage

Flexible across:

- Instance family
- Size
- OS
- Tenancy
- Availability Zone
- Region

EC2 Instance Savings Plans

Provide the lowest prices, up to **72 percent** off On-Demand rates on the selected instance family in a specific AWS Region

Flexible across:

- Availability Zone
- Size
- OS
- Tenancy

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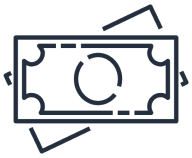
EC2 Spot Instances

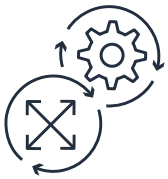
Use the same infrastructure

Run on the same hardware as On-Demand and Savings Plans.



Get the best value

Decide what you can pay for compute, and save up to 90 percent from the On-Demand price.



Plan for interruptions

Prepare for capacity changes in your Availability Zones.

Diversify your fleet

Choose different instance types, sizes, and Availability Zones.

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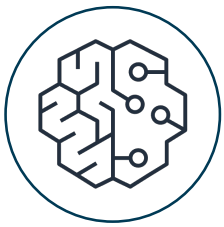
Use cases for Spot Instances

Image and media rendering

Manage rendering projects cost effectively to meet deadlines.

Web services

Launch Spot Instances to scale web services and applications at a lower cost.



Big data and analytics

Accelerate and scale time-critical, hyper-scale workloads.

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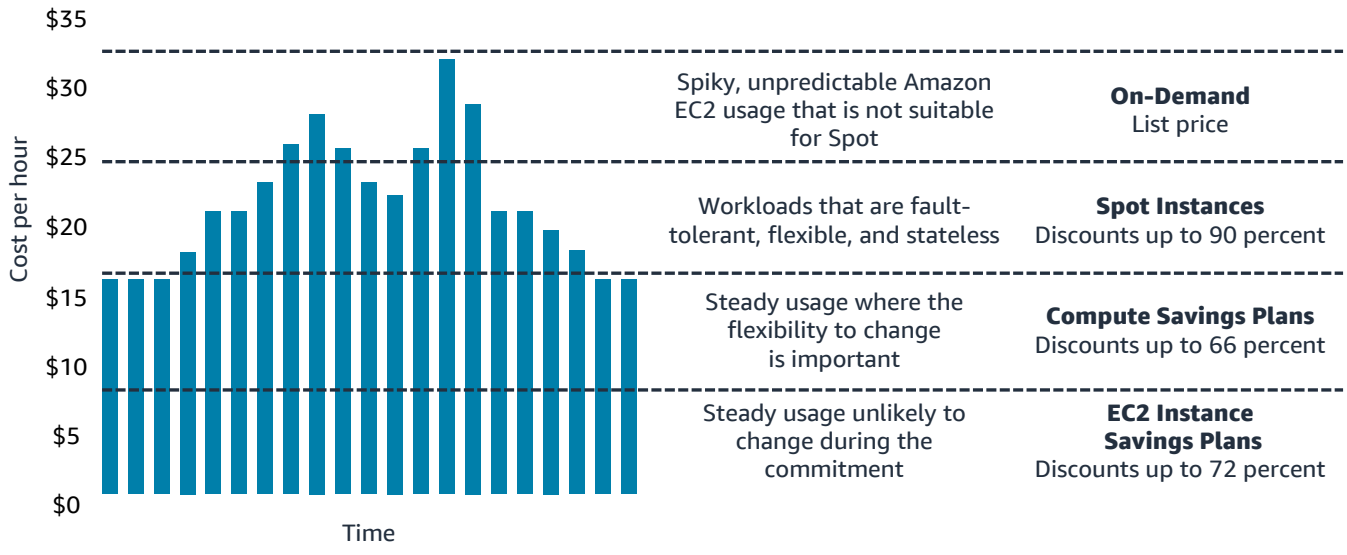
Combining purchase options



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AWS Lambda overview

"Where can we start with serverless compute options?"

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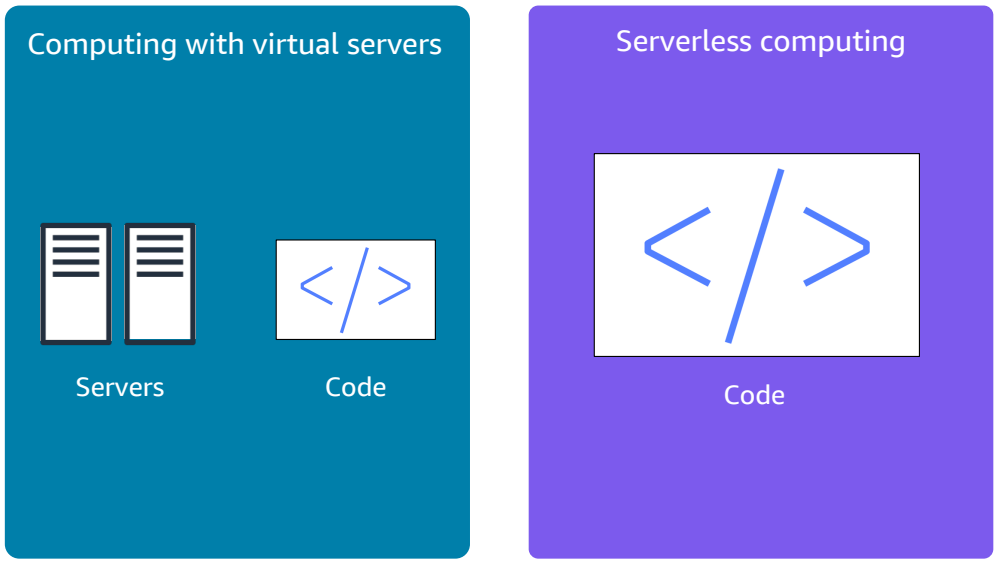
Serverless computing

- Highly available
- Fully managed by AWS



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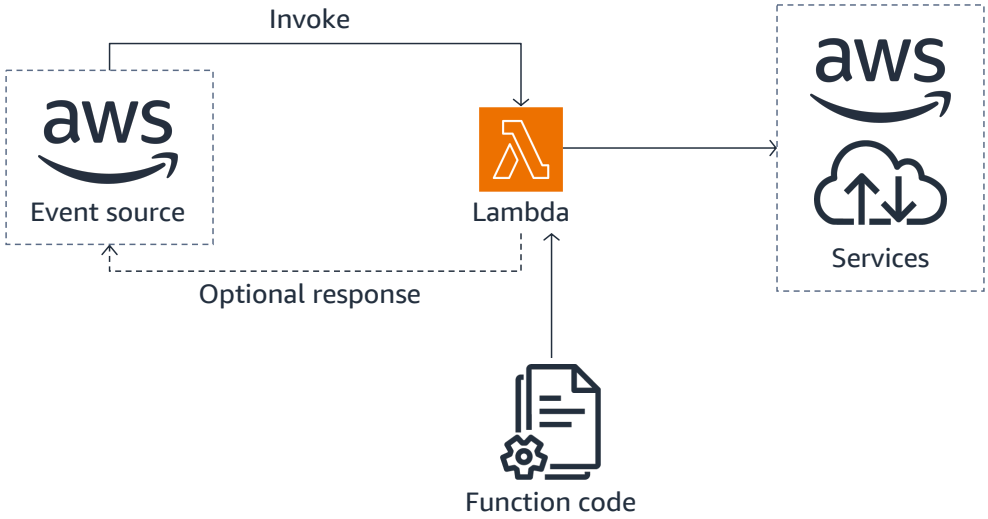
AWS Lambda

- Serverless compute
- Supports Node.js, Java, Python, C#, Go, PowerShell, Ruby, and more
- Runs for up to 15 minutes
- Supports up to 10 GB memory



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Event source examples

Amazon CloudWatch

AWS CloudFormation

Amazon Simple Queue Service (Amazon SQS)

Amazon S3

Amazon DynamoDB

AWS Config

Amazon EventBridge

AWS Step Functions

Application Load Balancer

Amazon API Gateway

Amazon Simple Notification Service (Amazon SNS)

Amazon Kinesis

Amazon Data Firehose

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Use cases

Web applications

Backends

Data processing

AI assistants

Amazon Alexa

IT automation

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Review

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Present solutions

Compute operations manager

Consider how you would answer the following questions:

- What AWS compute services are there?
- What should the team consider when deploying new and existing servers to Amazon EC2?
- How do we know which volume type to attach to our EC2 instances?
- How can we optimize cost for compute resources?
- Where can we start with serverless compute options?

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Module review

In this module, you learned about the following:

- ✓ Compute services
- ✓ EC2 instances
- ✓ Instance storage
- ✓ Amazon EC2 pricing options
- ✓ AWS Lambda

Next, you review the following:

- Capstone check-in
- Knowledge check
- Lab introduction

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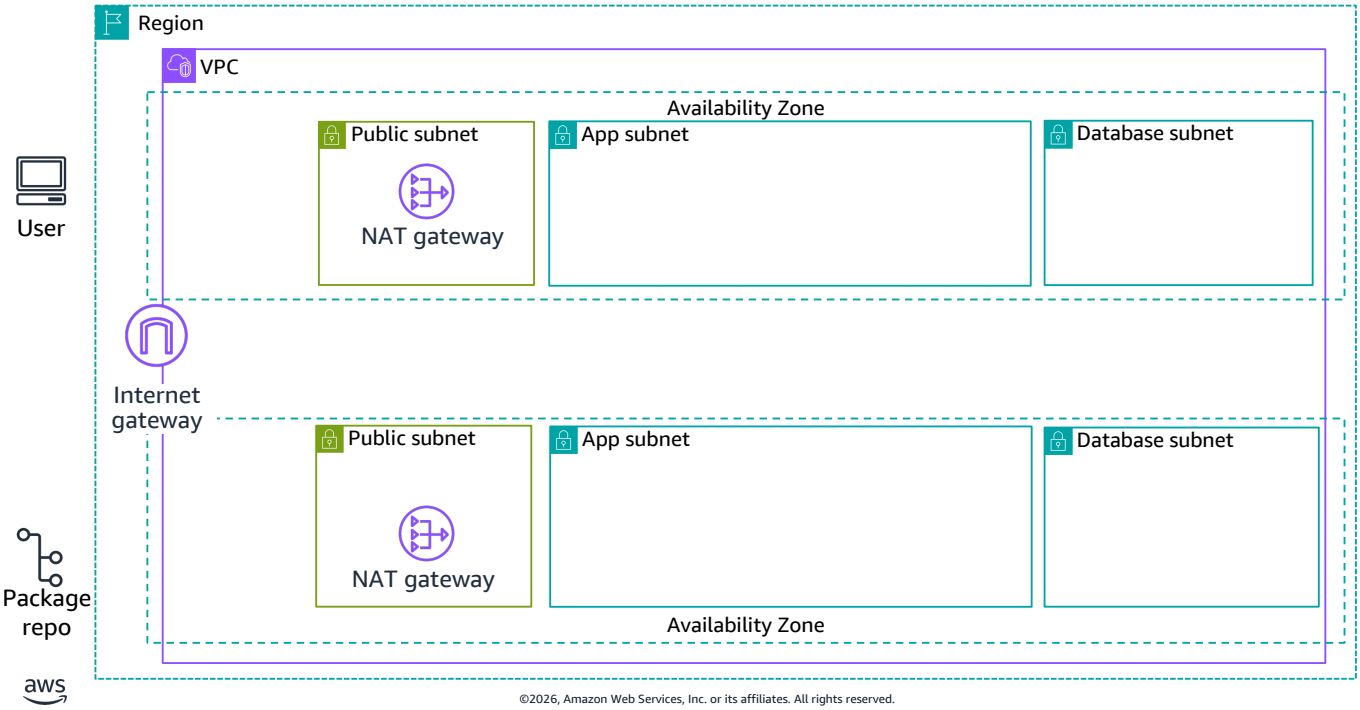
Capstone architecture previously



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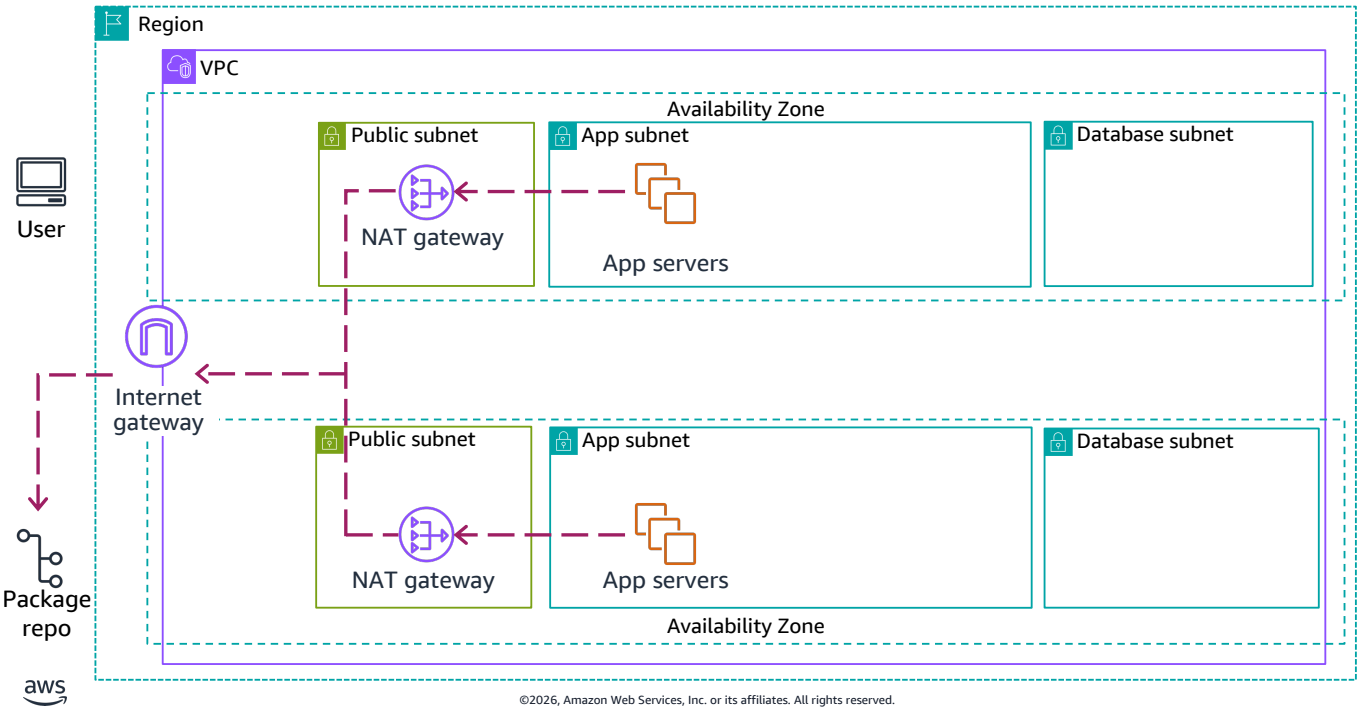
Capstone architecture check-in



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Knowledge check

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Knowledge check question 1

Which statements are true of Amazon Machine Images (AMIs)? (Select TWO.)

Choice	Response
A	AMIs can specify the subnets for launch.
B	AMIs can include block device mapping that specifies the volumes to attach to the Amazon EC2 instance when it is launched.
C	AMIs can only be obtained from the AWS Marketplace.
D	You can launch multiple instances from a single AMI.
E	AMIs can only be used by users within a single account.

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Knowledge check question 1 and answer

Which statements are true of Amazon Machine Images (AMIs)? (Select TWO.)

Choice	Response
A	AMIs can specify the subnets for launch.
B correct	AMIs can include block device mapping that specifies the volumes to attach to the Amazon EC2 instance when it is launched.
C	AMIs can only be obtained from the AWS Marketplace.
D correct	You can launch multiple instances from a single AMI.
E	AMIs can only be used by users within a single account.

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Knowledge check question 2

In the instance type name m8g.2xlarge, which aspect of the name indicates the generation of the instance family?

Choice	Response
A	m
B	g
C	2xlarge
D	8

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Knowledge check question 2 and answer

In the instance type name m8g.2xlarge, which aspect of the name indicates the generation of the instance family?

Choice	Response
A	m
B	g
C	2xlarge
D correct	8

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Knowledge check question 3

Which statements are true regarding Lambda? (Select TWO.)

Choice	Response
A	Functions currently only support Python.
B	You are responsible for updating and patching Lambda servers.
C	Functions can be allocated up to 10 GB of memory.
D	Functions can run for a maximum of 15 minutes.
E	Functions require a security group.

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Knowledge check question 3 and answer

Which statements are true regarding Lambda? (Select TWO.)

Choice	Response
A	Functions currently only support Python.
B	You are responsible for updating and patching Lambda servers.
C correct	Functions can be allocated up to 10 GB of memory.
D correct	Functions can run for a maximum of 15 minutes.
E	Functions require a security group.

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Lab 2:

Build Your Amazon VPC Infrastructure

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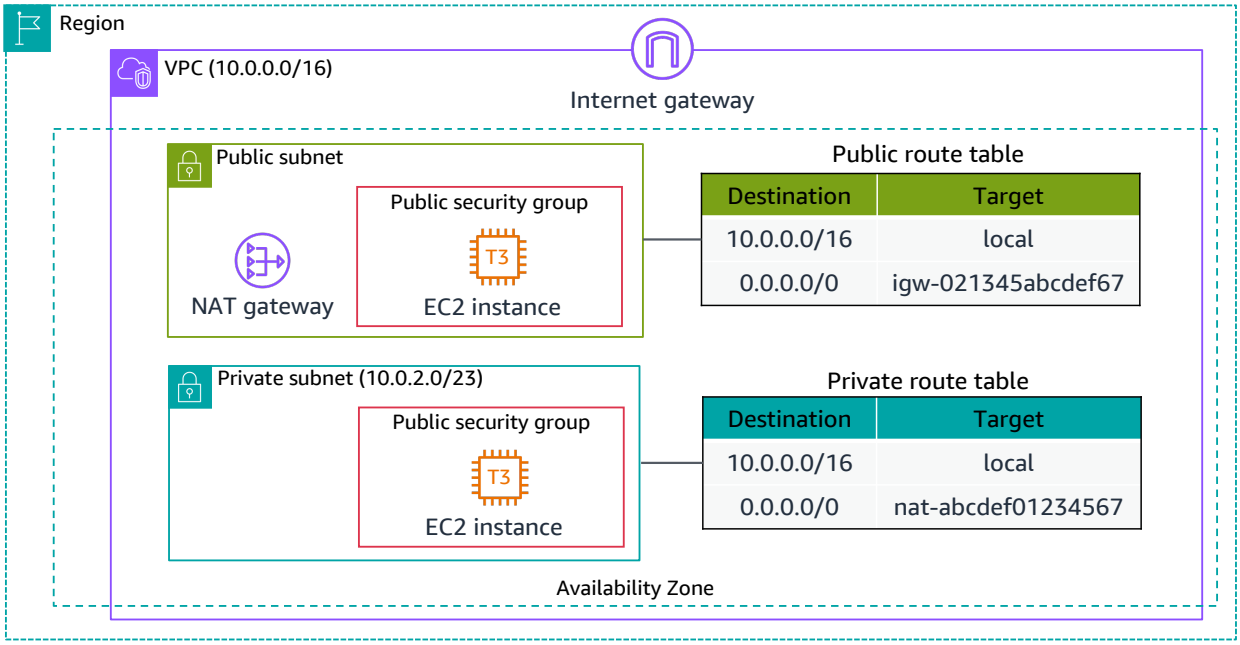
Lab 2 diagram



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Lab tasks

Task Step	Action
1	Create an Amazon VPC in a Region.
2	Create public and private subnets.
3	Create an internet gateway.
4	Route internet traffic in the public subnet to the internet gateway.
5	Create a public security group.
6	Launch an Amazon EC2 instance into a public subnet.
7	Connect to a public instance through HTTP.
8	Connect to the Amazon EC2 instance in the public subnet.
9	Create a NAT gateway and configure routing in the private subnet.
10	Create a security group for private resources.
11	Launch an Amazon EC2 instance into a private subnet.
12	Connect to the Amazon EC2 instance in the private subnet.
Optional 1	Test connectivity to the private instance from the public instance.
Optional 2	Retrieve instance metadata.

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End of Module 4

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Architecting on AWS

Module 5: Storage

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Poll question

What type of storage are you or your teams using in your environments? Choose all that apply.

A. Block storage

B. File storage

C. Object storage

D. I don't know

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Module overview

- Business request
- Storage services
- Amazon Simple Storage Service (Amazon S3)
- Shared file systems
- Data migration tools
- Present solutions
- Capstone check-in
- Knowledge check

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Business request

Storage team lead

The storage team lead needs to know the following:

- What are some services to consider when looking at block, file, and object storage?
- How do we choose the right object storage solution for my use case?
- What are some file-based options for building secure and scalable storage in the AWS Cloud?
- How can we move lots of data to the cloud in a relatively short time period?

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Storage services

"What are some services to consider when looking at block, file, and object storage?"

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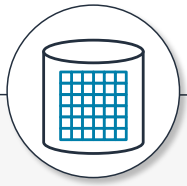
Cloud storage overview

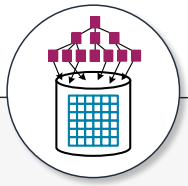
Block storage



Raw storage. Data organized as an array of unrelated blocks.

Examples: Hard disk, storage area network (SAN), storage arrays

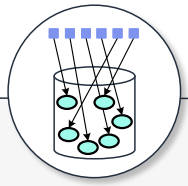
File storage



Unrelated data blocks managed by a file (serving) system. Native file system places data on disk.

Examples: Network attached storage (NAS) appliances, Windows file servers

Object storage



Stores virtual containers that encapsulate the data, data attributes, metadata, and object IDs.

Examples: Ceph, OpenStack Swift

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AWS data building blocks

Block storage

Amazon Elastic Block Store (Amazon EBS)

Instance store

File storage

Amazon Elastic File System (Amazon EFS)

Amazon FSx

Object storage

Amazon Simple Storage Service (Amazon S3)

Amazon S3 Glacier

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Amazon S3 overview

"How do we choose the right object storage solution for my use case?"

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Amazon S3

Amazon S3

Amazon S3 is a durable object storage solution.

Accelerate innovation

Increase agility

Reduce cost

Strengthen security

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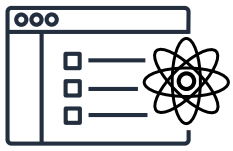
Amazon S3 use cases

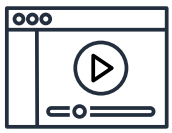
Use Amazon S3 when you have:

- A large number of users that access your content
- Growing data sets
- Data that you will write once and read many times

Backup and restore

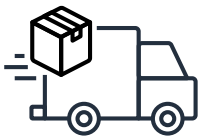


Data lakes for analytics



Media storage and streaming

Static website



Archiving and compliance

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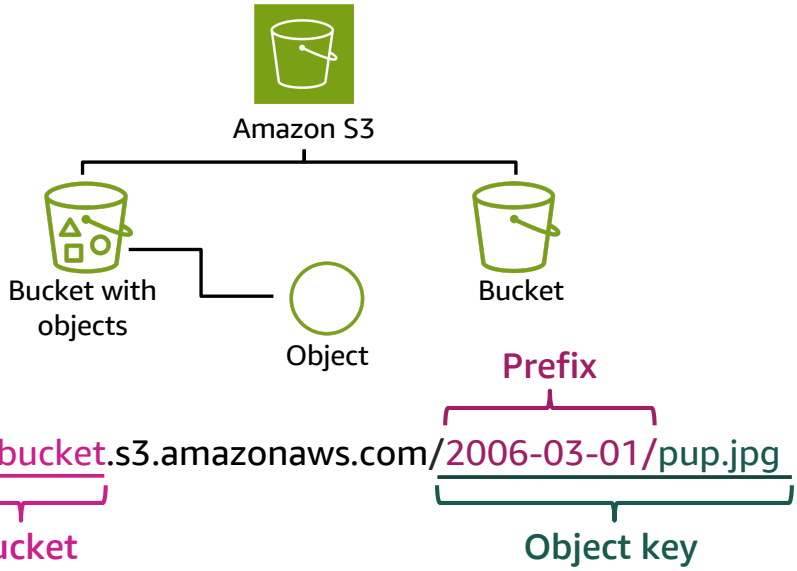
S3 buckets and objects

- Amazon S3 stores data as objects within buckets.
- An object includes a file and any metadata that describes the file.
- You can control access to the bucket and its objects.



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Securing objects

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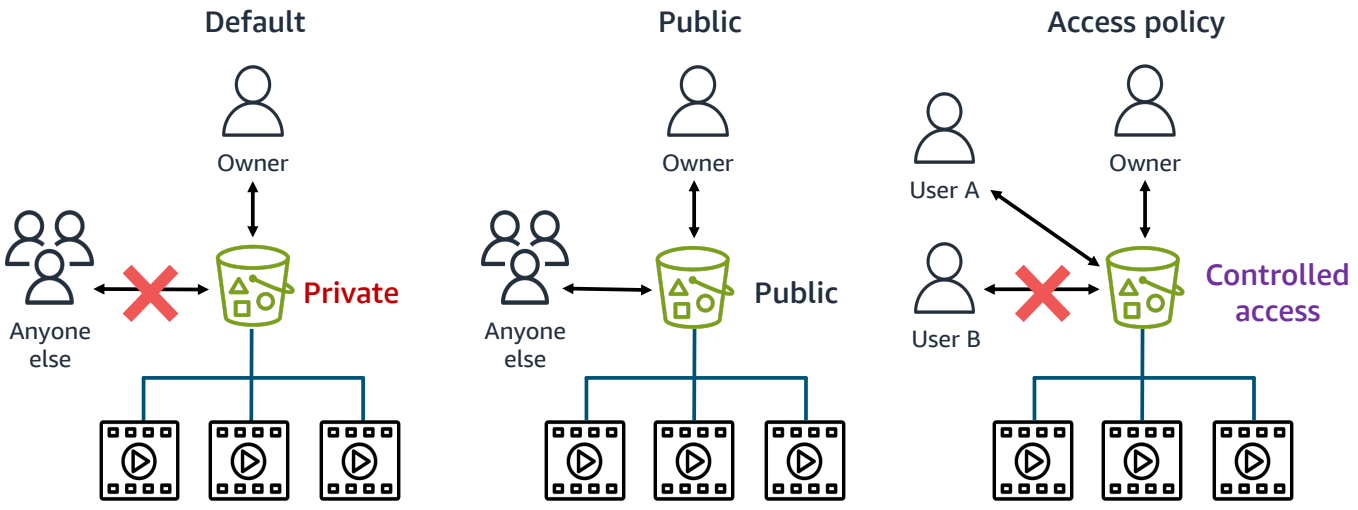
Amazon S3 access control



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Bucket policies

- Resource-based policy for an S3 bucket
- Controls access to a bucket without managing permissions in AWS Identity and Access Management (IAM)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
      "s3:ListBucket",
    "s3:GetObject"
    ],
    "Resource": [
     "arn:aws:s3:::doc-example-bucket",
     "arn:aws:s3:::doc-example-bucket/*"
      ]
    }
  ]
}
```

Bucket policy in JSON format

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Amazon S3 Block Public Access

Block public ...

- ❑ Access to buckets and objects granted through *new* ACLs
- ❑ Access to buckets and objects granted through *any* ACLs
- ❑ Access to buckets and objects granted through a *new* public bucket or access point policies
- ❑ *Cross-account access* to buckets and objects through *any* public bucket or access point policies

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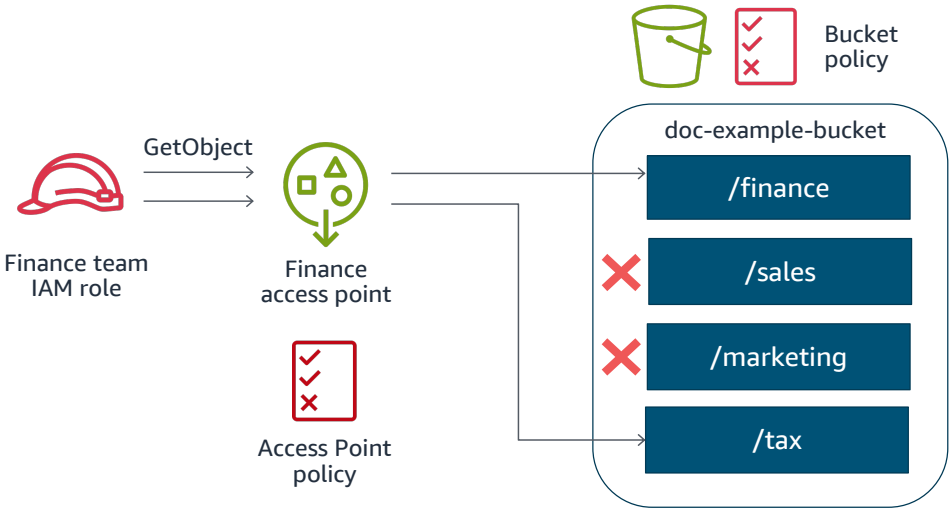
Amazon S3 Access Points

Each access point has:

- A unique Domain Name System (DNS) name and Amazon Resource Name (ARN)
- Distinct permissions and network controls



ARN: arn:aws:s3:us-west-2:123456789012:accesspoint/finance

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Server-side encryption key types

Choose how you encrypt objects in your S3 buckets:

- Amazon S3-managed keys (SSE-S3)
- AWS KMS keys (SSE-KMS)
- Dual-layer server-side encryption (DSSE-KMS)
- Customer-provided keys (SSE-C)

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Storing objects

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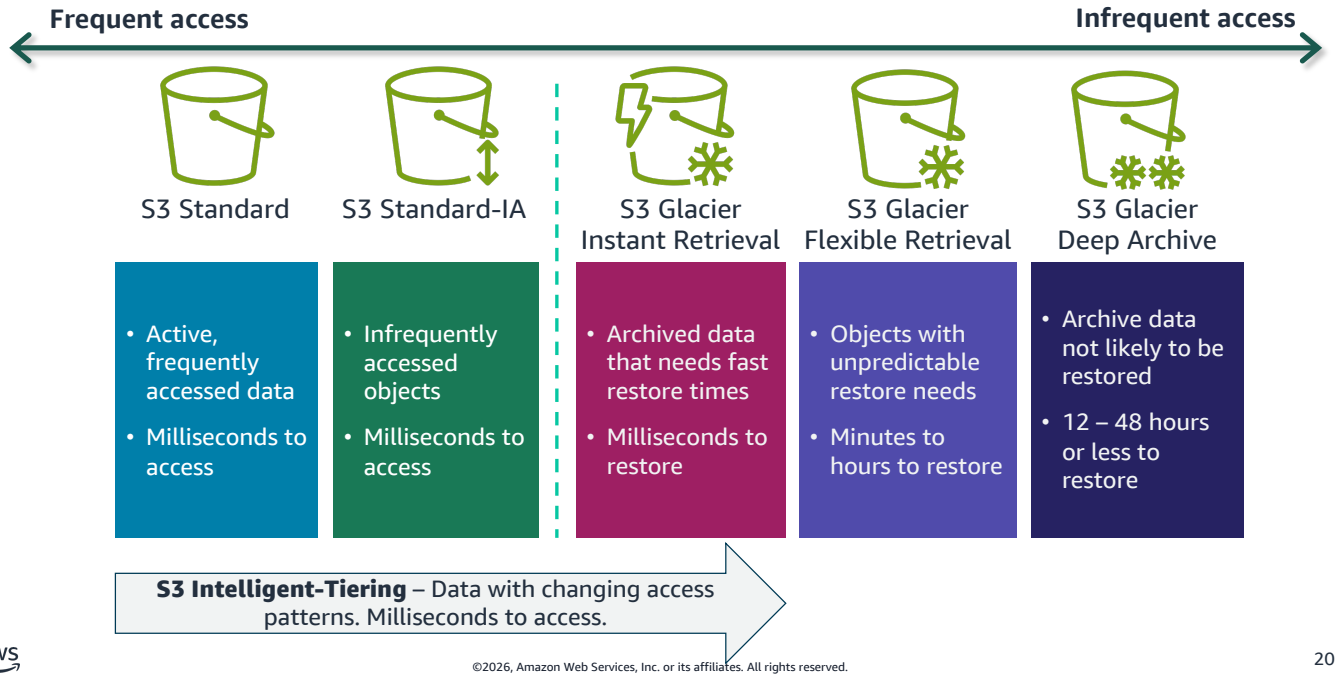
Amazon S3 storage classes



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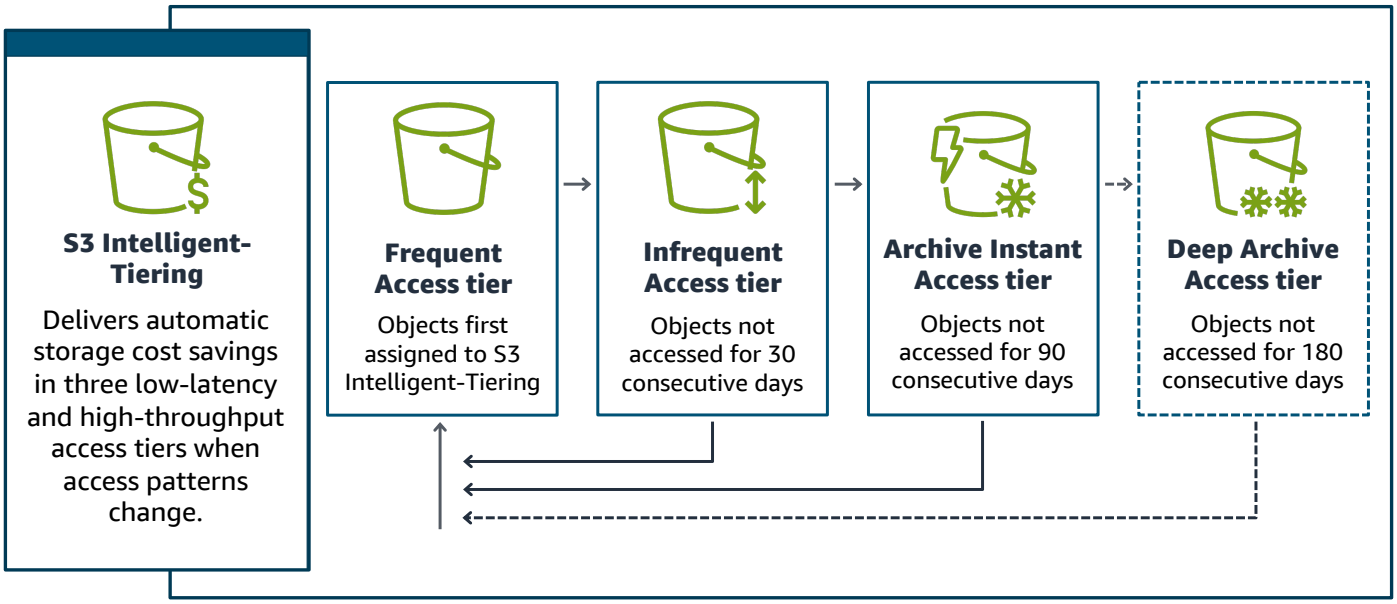
Amazon S3 Intelligent-Tiering



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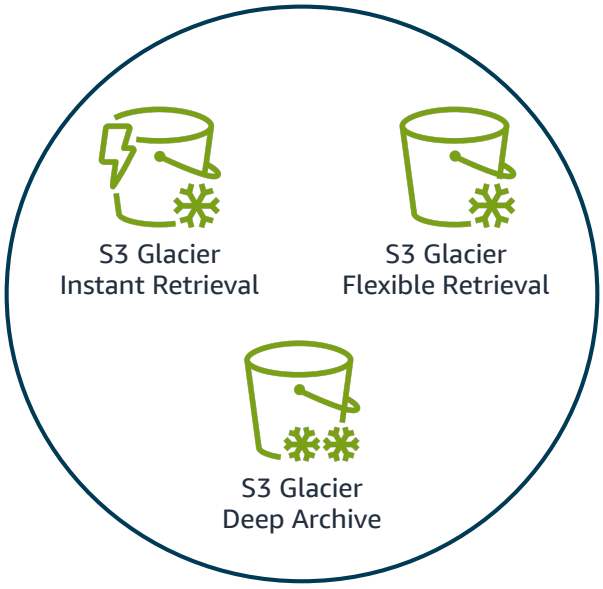
Amazon S3 Glacier storage class benefits

1. **Cost-effective storage**
   Lowest cost for specific data access patterns
2. **Flexible data retrieval**
   Three storage classes with variable access options
3. **Secure and compliant**
   Encryption at rest, AWS CloudTrail integration, and retrieval policies
4. **Scalable and durable**
   Meets needs from gigabytes to exabytes with 11 9s of durability



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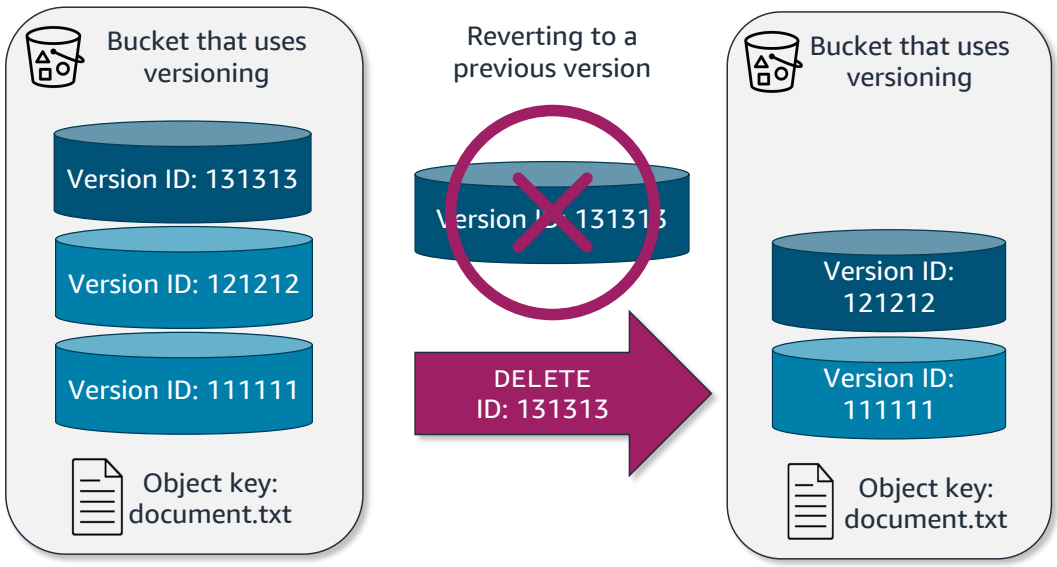
Amazon S3 Versioning

- Keep multiple variants of an object in the same bucket.
- Restore an object to a previous or specific version.
- Use S3 Object Lock for data retention or protection.



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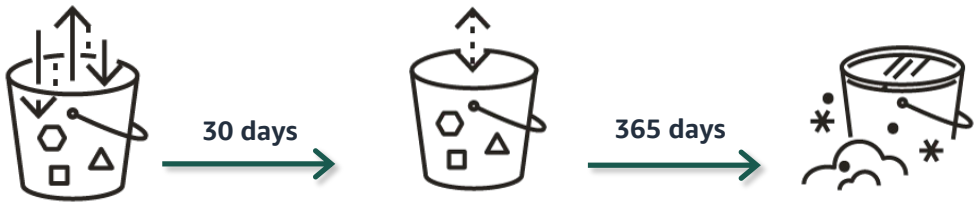
Lifecycle policies

Use S3 Lifecycle polices to transition objects to another storage class. S3 Lifecycle rules take action based on object age. Here's an example:

1. Move objects older than **30 days** to S3 Standard-Infrequent Access (S3 Standard-IA).
2. Move objects older than **365 days** to Amazon S3 Glacier Deep Archive.



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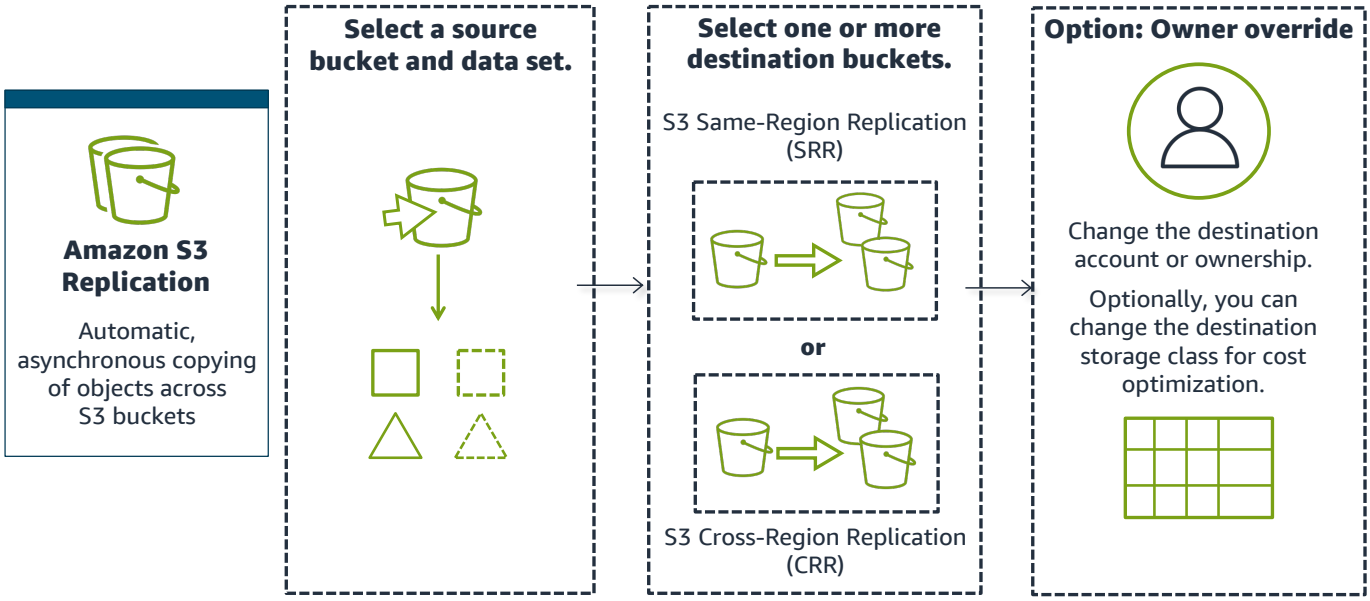
Replicating S3 objects



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Additional features of Amazon S3

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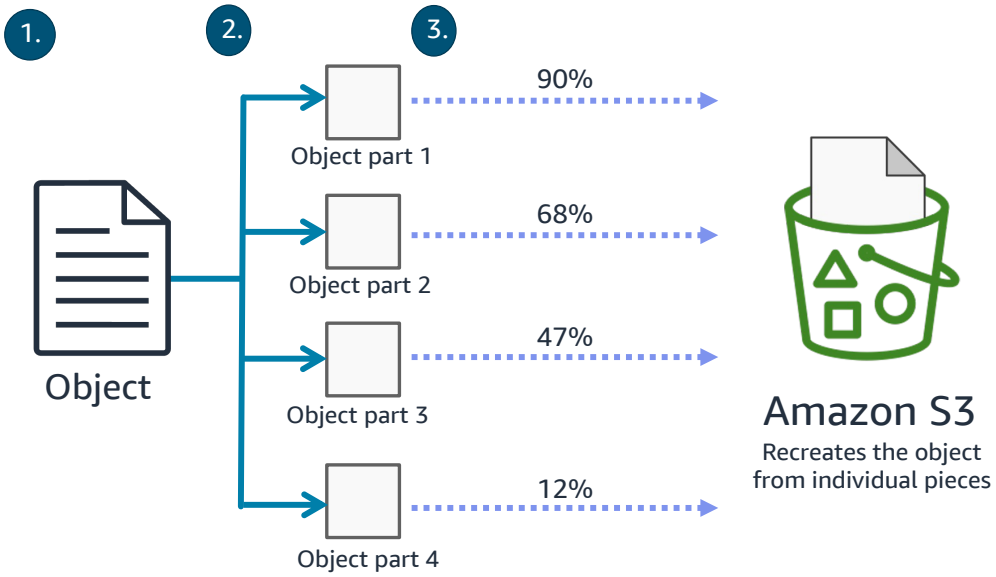
Amazon S3 multipart upload

1. Initiate the upload.
2. Upload the object parts.
3. Complete the multipart upload.

Note: You cannot perform multipart uploads manually by using the AWS Management Console.



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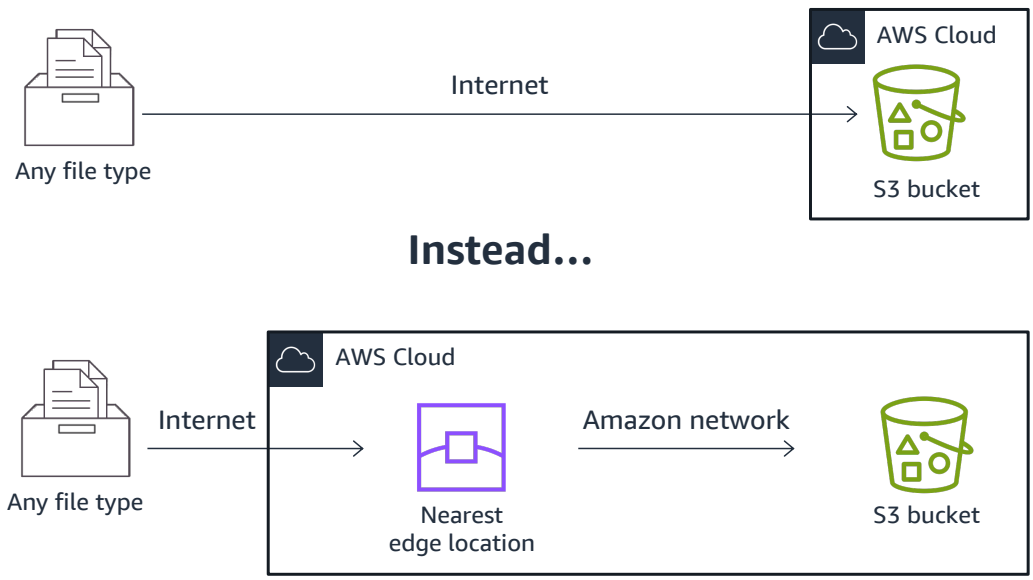
Amazon S3 Transfer Acceleration

- Move data faster over long distances.
- Reduce network variability.



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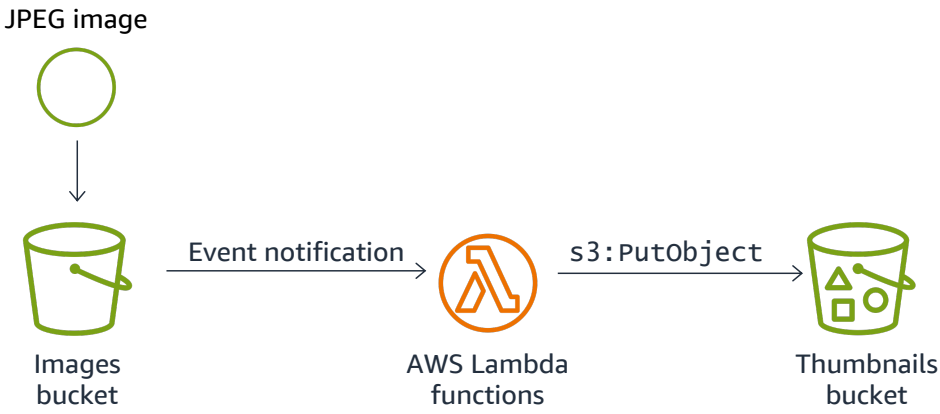
Amazon S3 Event Notifications

- Get notifications when events happen in your S3 bucket.
- AWS manages event monitoring for you. No polling is needed.

The following is an example event notification workflow to convert images to thumbnails:



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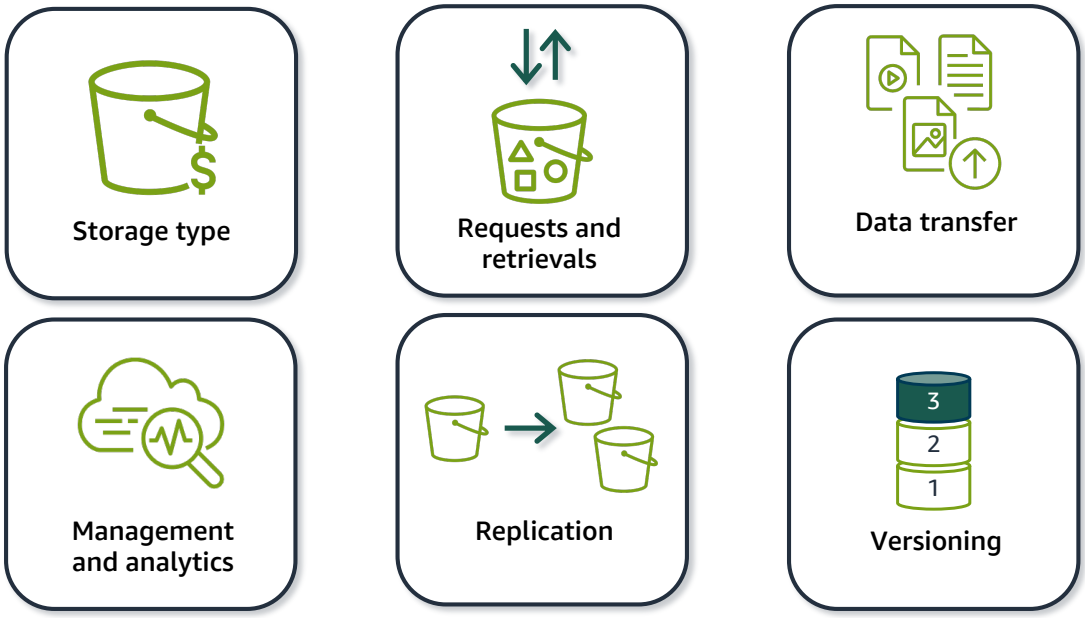
Amazon S3 cost factors



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S3 Express One Zone

- Specialized high-performance storage class with single-digit millisecond latency
- Requires directory buckets (not general purpose buckets)
- Organizes data hierarchically within a single Availability Zone
- Supports up to 2 million GET and 200,000 PUT transactions per second per directory bucket
- Ideal for performance-critical applications where data can be re-created if needed
- Stores data in only one Availability Zone (no multi-zone redundancy)

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Overview of shared file systems

"What are some file-based options for building secure and scalable storage in the AWS Cloud?"

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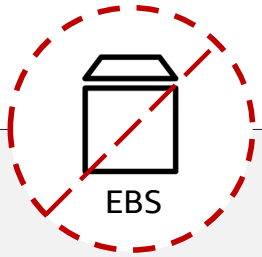
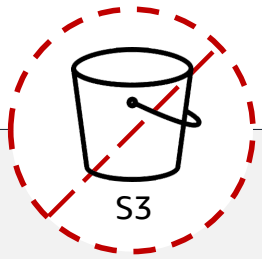
Shared file systems

What if you have multiple instances that need to use the same storage?



Amazon EBS is usually attached to one instance.



Object storage is not built for file systems.

Amazon EFS and Amazon FSx are ideal for this task.

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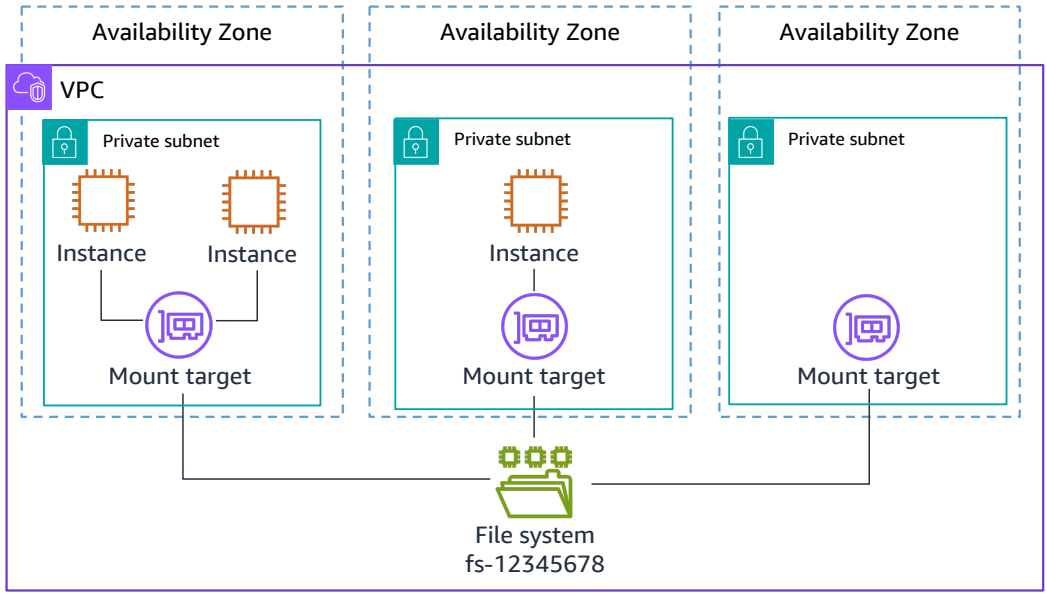
Amazon EFS

- Choose Amazon EFS for a scalable and elastic file system.
- Connect by using the NFSv4 protocol.
- Access file systems across EC2 instances at the same time.



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Amazon EFS benefits

Amazon EFS uses burst throughput mode to **scale throughput** based on your storage use.

Additionally, you can **provision throughput** independent of storage.

Amazon EFS **automatically grows and shrinks** file storage without provisioning.

Monitoring is not required to avoid storage limits.

Amazon EFS managed file systems **lower your total cost of ownership (TCO)**. Pay for only what you use.

Save on cost with EFS Infrequent Access or One Zone storage types.

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Amazon FSx

- Launch, run, and scale high-performing file systems on AWS.
- Use familiar and feature-rich products without managing hardware provisioning, patching, and backups.

Amazon FSx for Windows File Server

Amazon FSx for Lustre

Amazon FSx for NetApp ONTAP

Amazon FSx for OpenZFS

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Data migration tools

"How can we move lots of data to the cloud in a relatively short time period?"

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AWS data migration tools

Online

AWS Storage Gateway

Sync files with SMB, NFS, and iSCSI protocols from on-premises to AWS.

AWS DataSync

Sync files from on-premises file storage to Amazon EFS, Amazon FSx, and Amazon S3.

AWS Transfer Family

Transfer files into and out of Amazon S3 with SFTP, FTPS, FTP, and AS2.

Offline

AWS Snowball Edge

Move petabytes of data to AWS by using appliances that are designed for secure, physical transport.

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AWS Storage Gateway

AWS Storage Gateway

AWS Storage Gateway is a service that gives your applications seamless and secure integration between on-premises environments and AWS storage.

It provides you with low-latency access to cloud data with a Storage Gateway appliance.

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Storage Gateway types

Amazon S3 File Gateway

Native file access to Amazon S3 for backups, archives, and ingest for data lakes

Tape Gateway

Virtual tape library that uses Amazon S3 archive tiers for long-term retention

Volume Gateway

Block-level backups of volumes with Amazon EBS snapshots, AWS Backup integration, and cloud recovery

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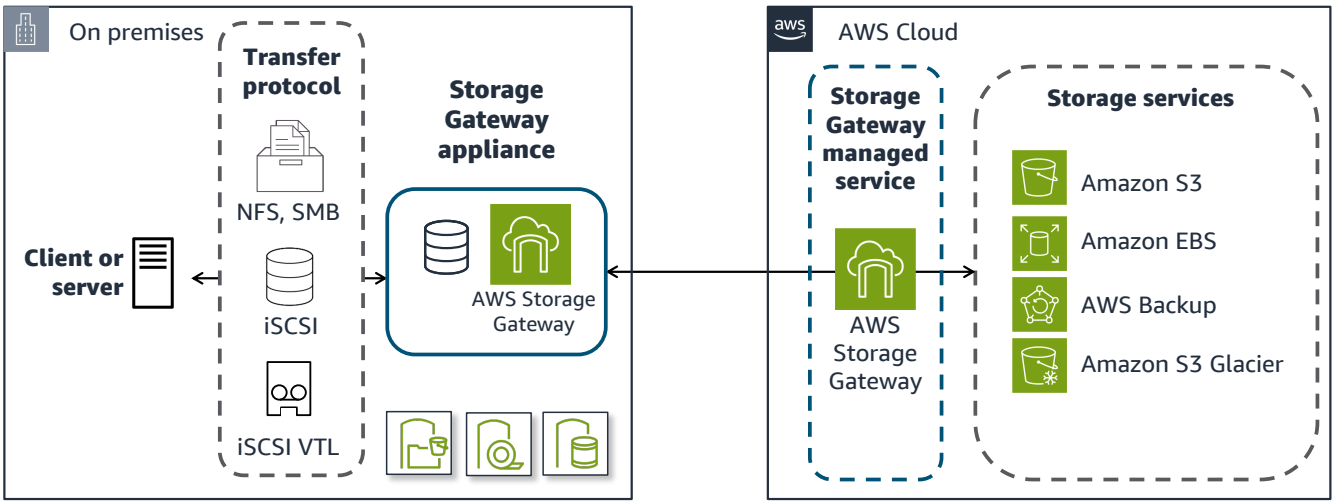
Storage Gateway architecture



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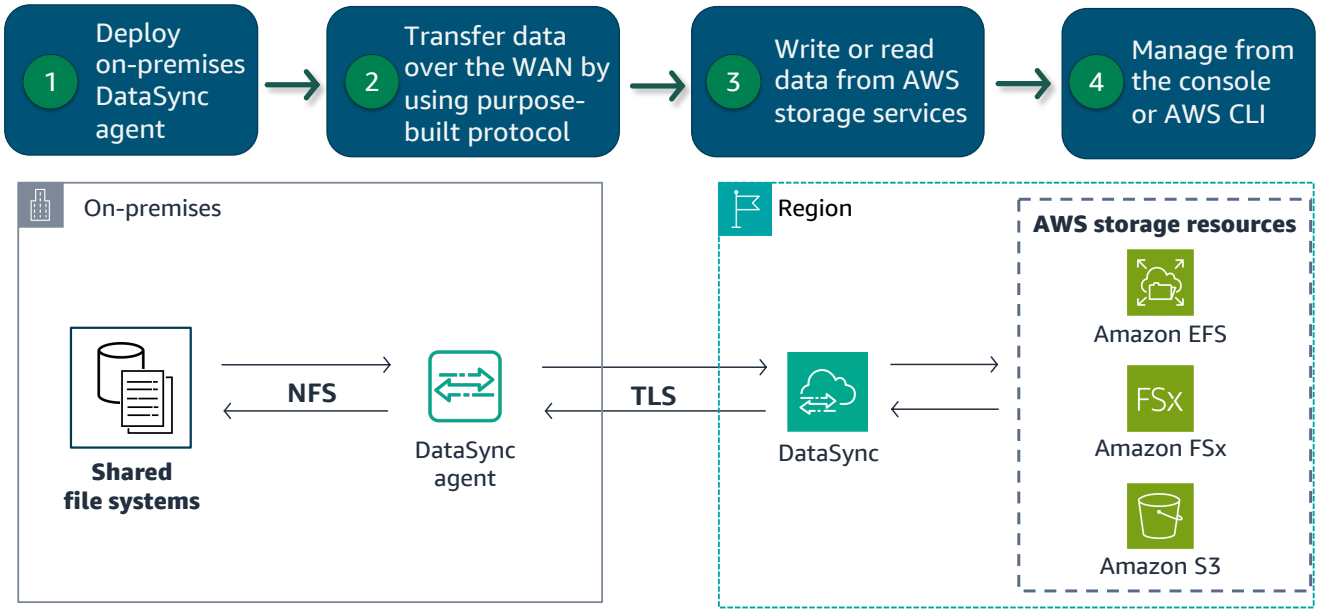
AWS DataSync



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AWS Snowball Edge

- An edge computing and data transfer device that the AWS Snowball service provides
- Includes Snowball Edge Storage Optimized and Compute Optimized

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Review

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Present solutions

Storage team lead

Consider how you would answer the following questions:

- What are some services to consider when looking at block, file, and object storage?
- How do you choose the right object storage solution for your use case?
- What are some file-based options for building secure and scalable storage in the AWS Cloud?
- How can you move lots of data to the cloud in a relatively short time period?

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Module review

In this module, you learned about the following:

- ✓ Storage services
- ✓ Amazon S3
- ✓ Shared file systems
- ✓ Data migration tools

Next, you review the following:

- Capstone check-in
- Knowledge check

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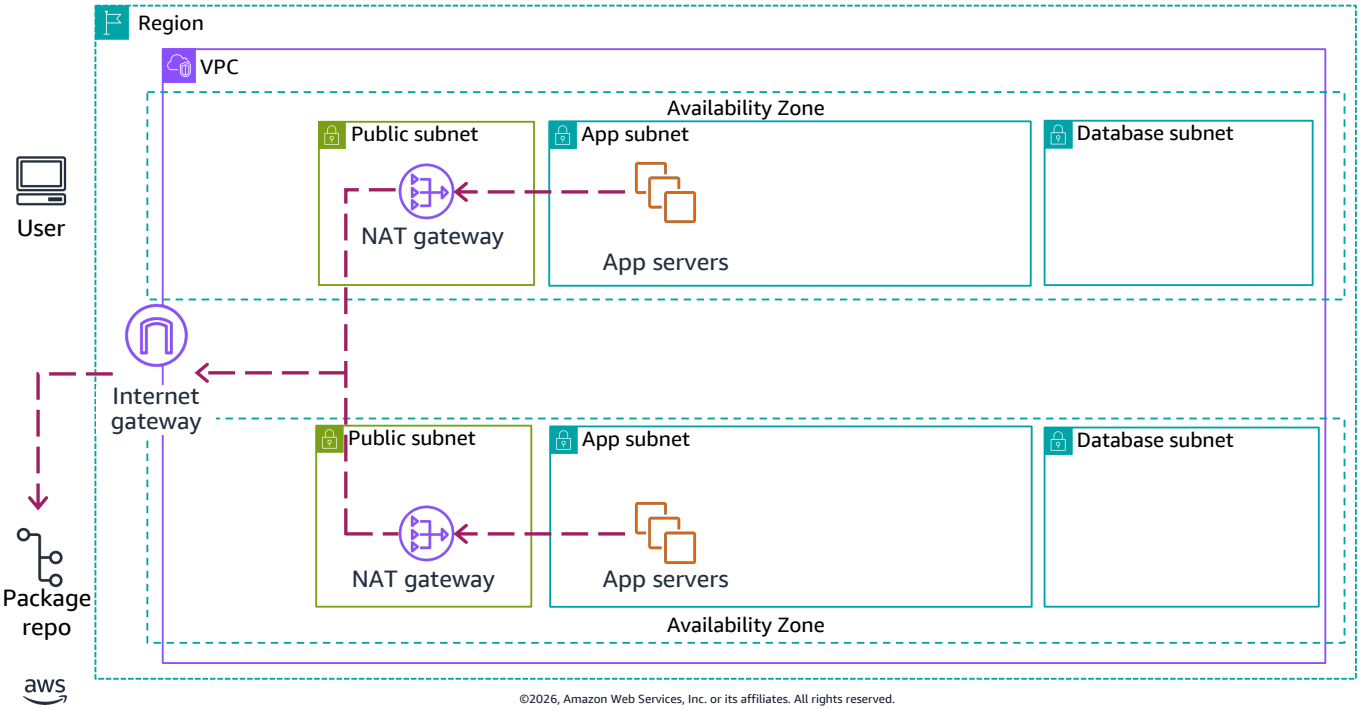
Capstone architecture previously



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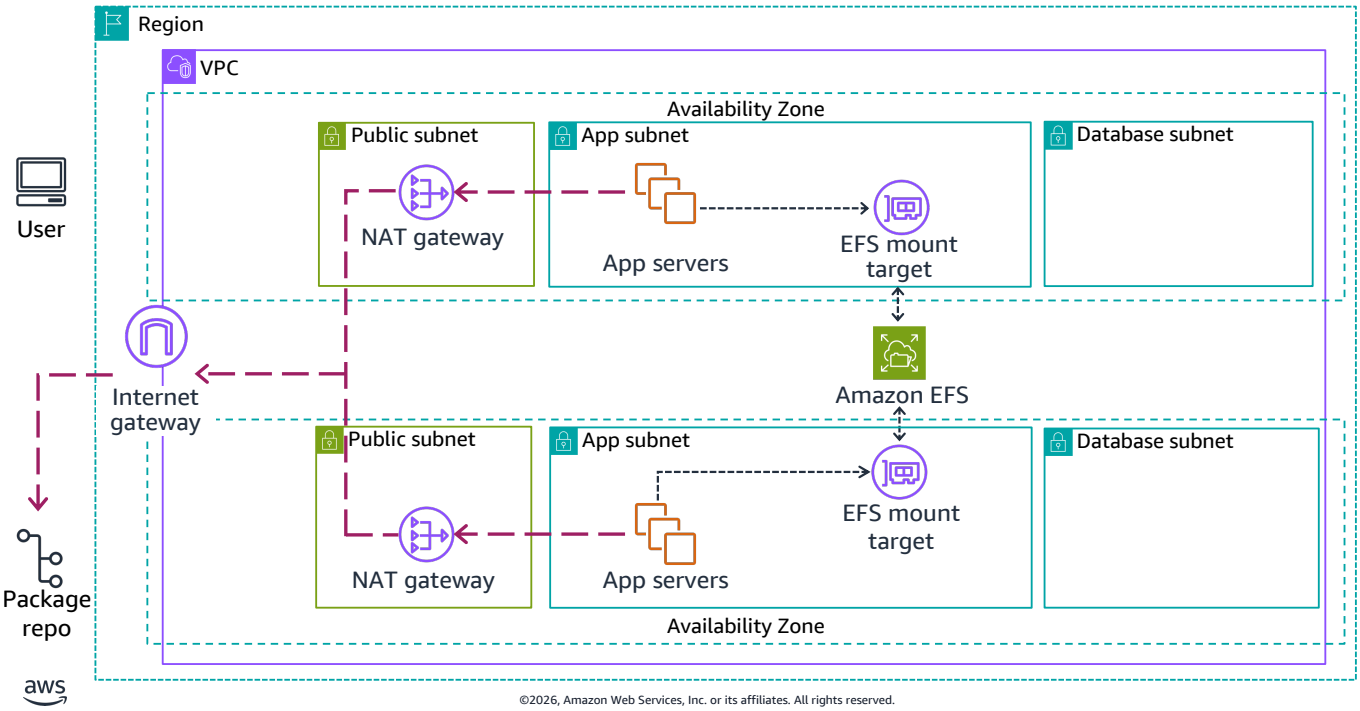
Capstone architecture check-in



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Knowledge check

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Knowledge check question 1

Which Amazon S3 feature would be used to automatically copy new objects to a bucket in a different AWS Region?

Choice	Response
A	Same-Region Replication (SRR)
B	Amazon S3 Versioning
C	AWS DataSync
D	Cross-Region Replication (CRR)

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Knowledge check question 1 and answer

Which Amazon S3 feature would be used to automatically copy new objects to a bucket in a different AWS Region?

Choice	Response
A	Same-Region Replication (SRR)
B	Amazon S3 Versioning
C	AWS DataSync
D correct	Cross-Region Replication (CRR)

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Knowledge check question 2

Which Amazon S3 feature can force an action to occur after an event takes place within a bucket?

Choice	Response
A	Invoking
B	Event Notifications
C	Lambda
D	Alarm

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Knowledge check question 2 and answer

Which Amazon S3 feature can force an action to occur after an event takes place within a bucket?

Choice	Response
A	Invoking
B correct	Event Notifications
C	Lambda
D	Alarm

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Knowledge check question 3

A developer has two Linux applications in different Availability Zones that must share a common file system. Which solution is best for this use case?

Choice	Response
A	AWS Storage Gateway
B	Amazon FSx for Windows File Server
C	Amazon S3
D	Amazon Elastic File System (Amazon EFS)

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Knowledge check question 3 and answer

A developer has two Linux applications in different Availability Zones that must share a common file system. Which solution is best for this use case?

Choice	Response
A	AWS Storage Gateway
B	Amazon FSx for Windows File Server
C	Amazon S3
D correct	Amazon Elastic File System (Amazon EFS)

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Knowledge check question 4

Which modes are available in the AWS Storage Gateway appliance? (Select THREE.)

Choice	Response
A	Memory Gateway
B	Tape Gateway
C	Volume Gateway
D	Amazon EBS File Gateway
E	Amazon S3 File Gateway
F	Amazon S3 Glacier File Gateway

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Knowledge check question 4 and answer

Which modes are available in the AWS Storage Gateway appliance? (Select THREE.)

Choice	Response
A	Memory Gateway
B correct	Tape Gateway
C correct	Volume Gateway
D	Amazon EBS File Gateway
E correct	Amazon S3 File Gateway
F	Amazon S3 Glacier File Gateway

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End of Module 5

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Architecting on AWS

Module 6: Database Services

Lab 3

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Poll question

Which database technologies have you used in your workloads? (Select all that apply.)

A. Relational databases

B. Nonrelational databases

C. Database caching

D. Database migration tools

E. None of these

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Module overview

- Business request
- Database services
- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Database caching
- Database migration tools
- Present solutions
- Capstone check-in
- Knowledge check
- Lab 3: Create a Database Layer in Your Amazon VPC Infrastructure

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Business request

Database services manager

The database services manager wants to know:

- What are the Amazon Web Services (AWS) database solutions?
- How can we more efficiently manage our relational databases in the cloud?
- How can we build a scalable key-value NoSQL database?
- How can we cache databases in the cloud to maximize performance?
- Which tools are available for migrating an existing database to the AWS Cloud?

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Database services

"What are the AWS database solutions?"

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AWS database services

Amazon Relational Database Service (Amazon RDS)

Amazon Aurora

Amazon Redshift

Amazon DocumentDB (with MongoDB compatibility)

Amazon DynamoDB

Amazon ElastiCache

Amazon MemoryDB

Amazon Keyspaces (for Apache Cassandra)

Amazon Timestream

Amazon Neptune

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Relational and nonrelational databases

Characteristic	Relational (SQL) databases	Nonrelational (NoSQL) databases
Data storage	Tables with rows and columns	Key-value, wide-column, graph, document, or other models
Schemas	Fixed	Dynamic
Example database services	Amazon RDS Aurora	DynamoDB ElastiCache

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Choosing the right database

Relational database	Nonrelational (NoSQL) database
You require strict schema rules and data quality enforcement.	You need your database to scale horizontally.
Your database doesn't need extreme read/write capacity.	Your data does not lend itself well to traditional schemas.
If you have a relational dataset that does not require extreme performance, a relational database management system can be the best, lowest effort solution.	Your read/write rates exceed the rates that can be economically supported through a traditional SQL database.

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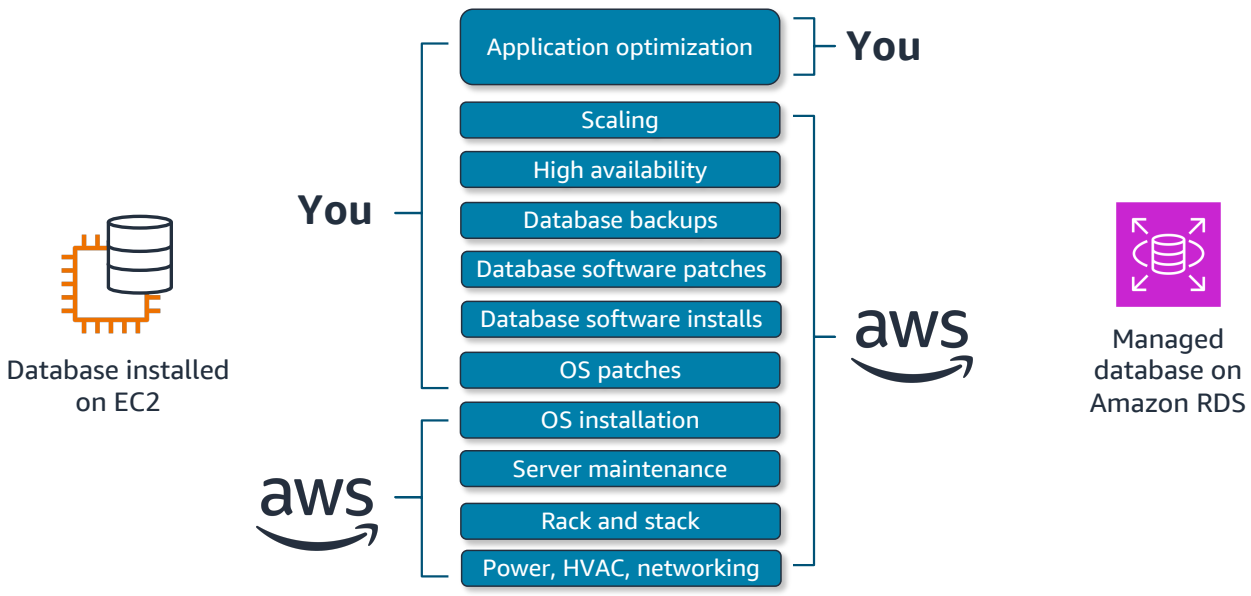
Managed and unmanaged services



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Amazon RDS

"How can we more efficiently manage our relational databases in the cloud?"

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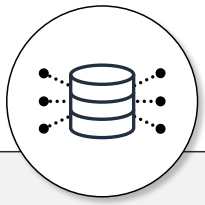
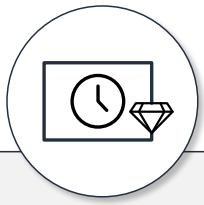
Amazon RDS features

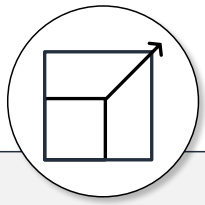


- Hardware, OS, and database software deployment and maintenance
- Built-in monitoring

- Data encryption at rest and in transit
- Industry compliance



- Automatic Multi-AZ data replication



- Compute and storage scaling
- Minimal application downtime

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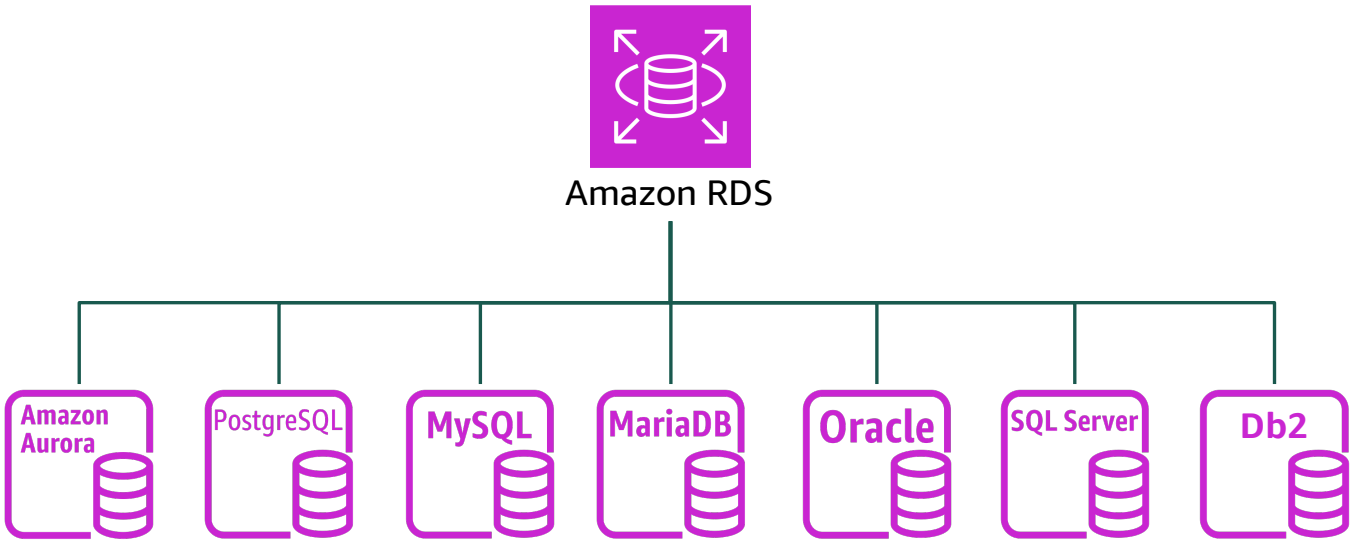
Amazon RDS database engines



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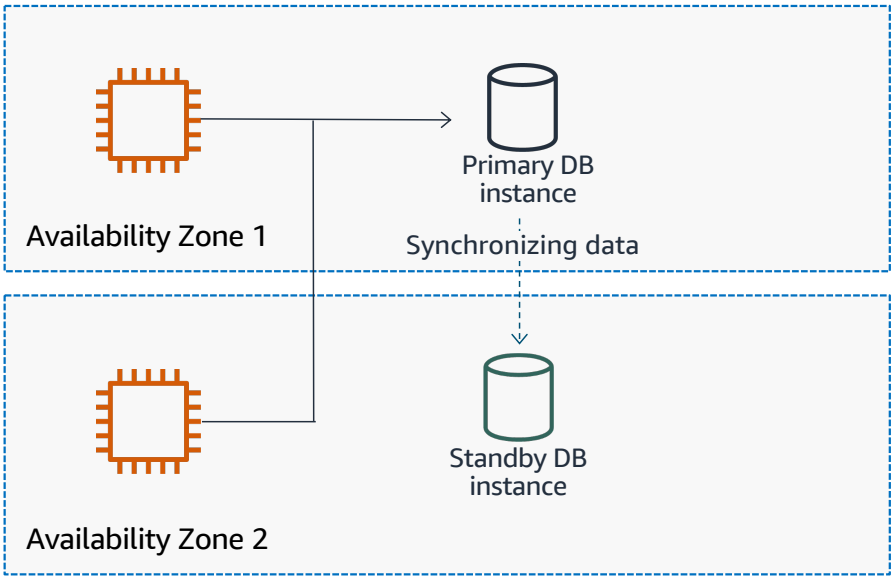
Amazon RDS Multi-AZ deployments

Multi-AZ deployments:

- Replicate data to a standby DB instance in another Availability Zone
- Are not used for read-only scenarios



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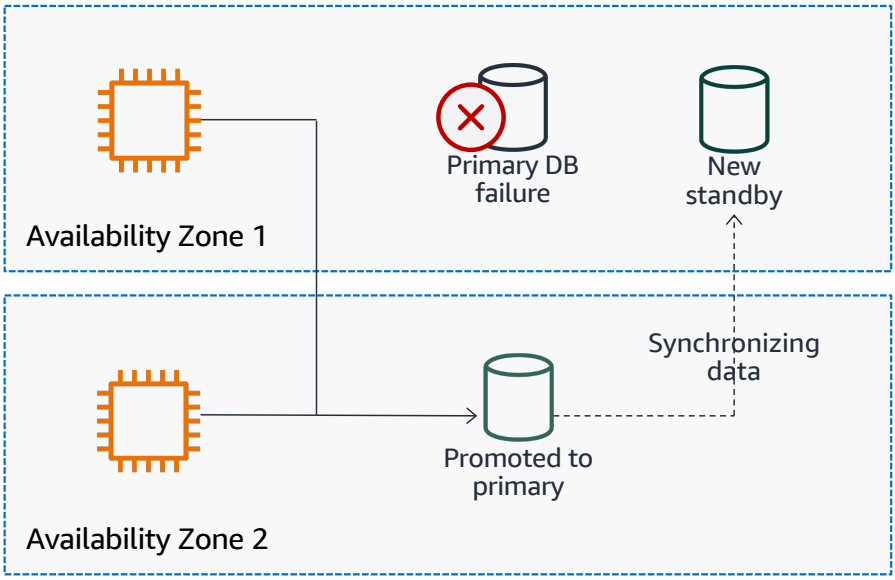
Amazon RDS Multi-AZ failover

Upon failure, the standby DB instance picks up the load.



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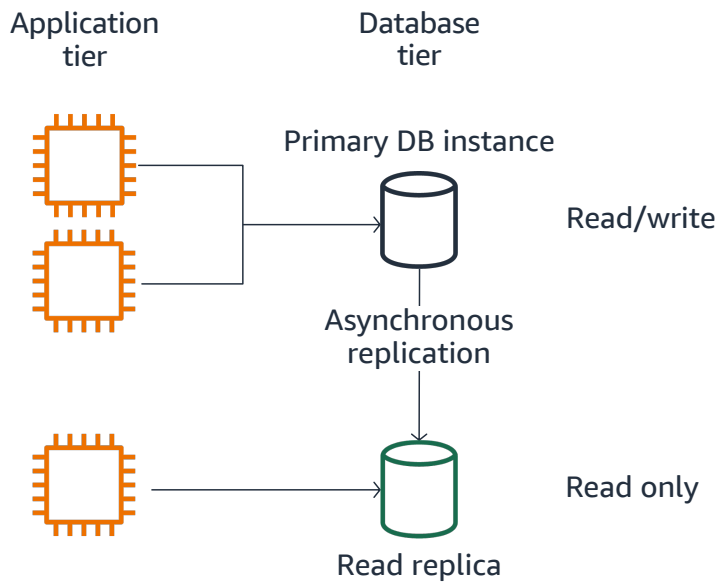
Read replicas

With read replicas, you can do the following:

- Horizontally scale for read-heavy workloads.
- Offload reporting.
- Replicate across AWS Regions.



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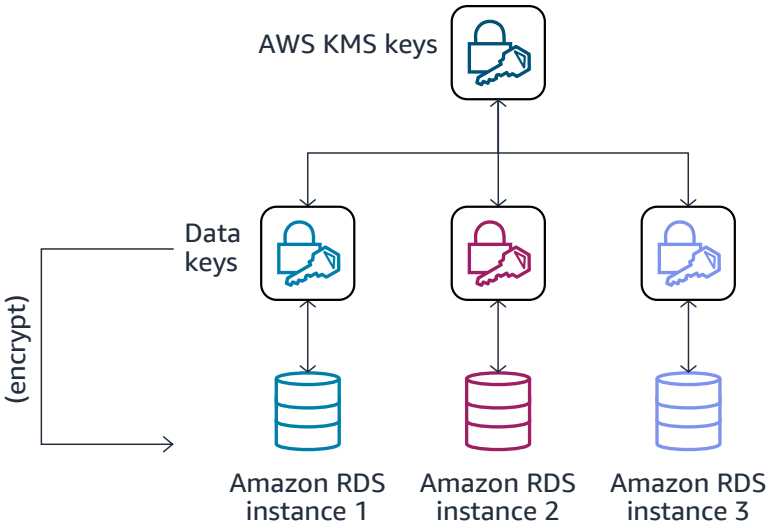
Data encryption at rest

- Encryption of data at rest is managed by AWS KMS.
- A unique data key encrypts your data.
- An AWS KMS key encrypts data keys.
- Data encryption is available for all Amazon RDS engines.



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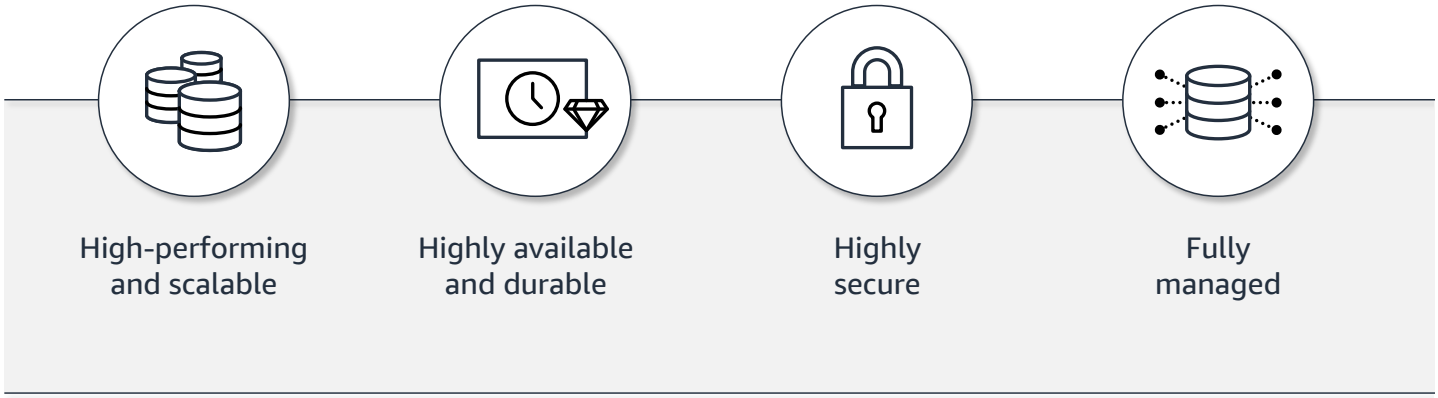
Amazon Aurora

A MySQL and PostgreSQL compatible relational database built for the cloud



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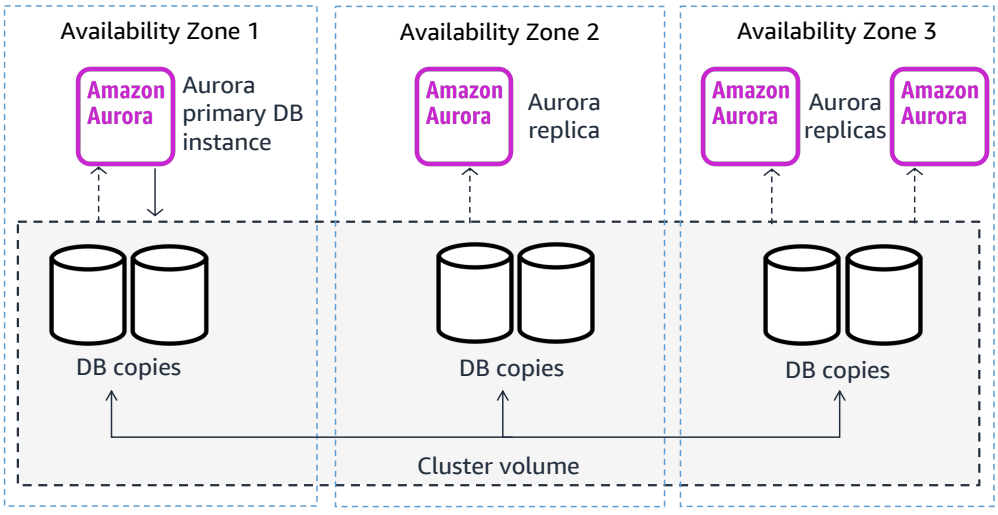
Aurora DB clusters

- A DB cluster consists of one or more DB instances and a cluster volume.
- Primary instances perform read/write operations.
- Aurora replicas are read-only.
- A cluster volume is a virtual database storage volume that spans multiple Availability Zones.



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Aurora Serverless v2 for PostgreSQL and MySQL

Scaling configuration for Aurora that automatically scales capacity up or down based on your application's needs

Start up on demand

Pay for only what you use

Scale with no impact to applications

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COIII

Amazon DynamoDB

"How can we build a scalable key-value NoSQL database?"

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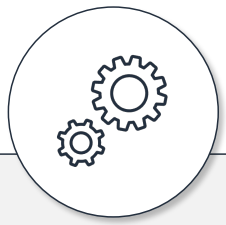
DynamoDB

A fully managed NoSQL AWS database service



Offers performance at scale

Requires no management of servers

Is enterprise ready

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Key-value data

- Is structured in key-value pairs with a flexible schema
- Is ideal for uses where the needed data can be mapped to a primary key
- Partitions data by key
- Delivers high-throughput, low-latency reads and writes

Database name: Gamers

Primary key

Attributes

GamerTag	UserId	Level	Points	TopScore	Plays
“Hammer57”	107	21	4,050	483,610	1,722
“FluffyDuffy”	285	5	1,123	10,863	43

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DynamoDB use case 1

GamerTag	UserId	TopScore	MemberSince	SubscriptionType
“Hammer57”	101	5,842	“2021-09-15:17:24:31”	“Gold”
“FluffyDuffy”	243	1,024	“2021-10-22:23:18:01”	“Platinum”
“NewPlayer”	623	687	“2021-10-22:23:22:01”	“Free”

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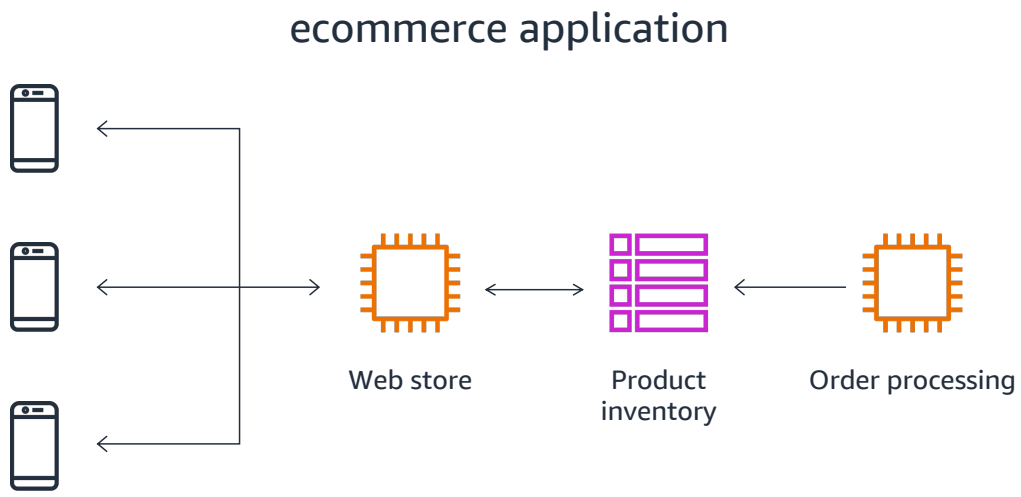
DynamoDB use case 2



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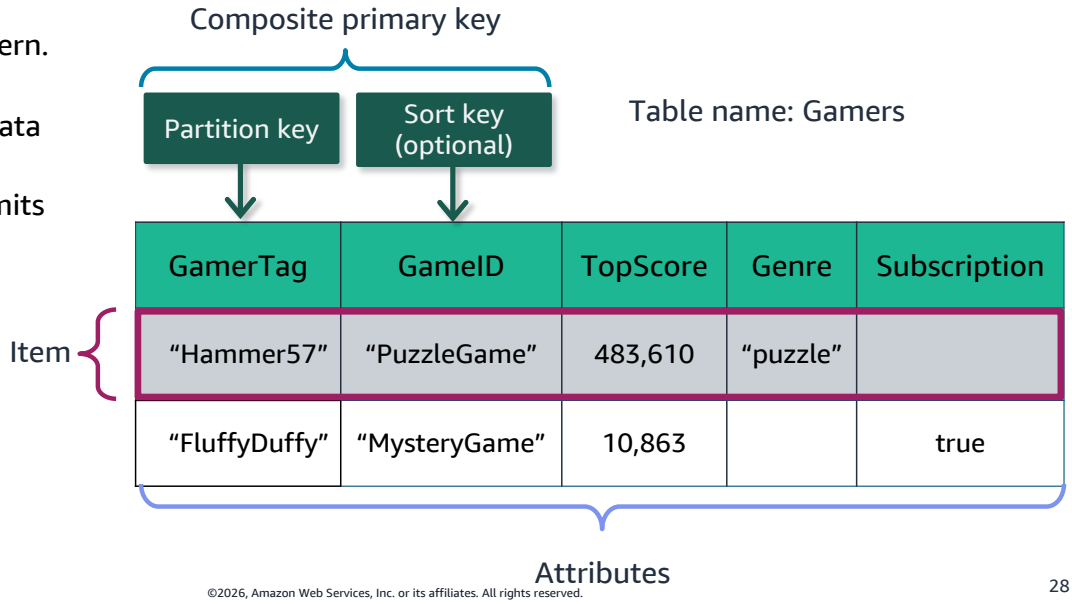
DynamoDB tables

- There is a mandatory key-value access pattern.
- The partition key determines the data distribution.
- The sort key permits rich query capabilities.



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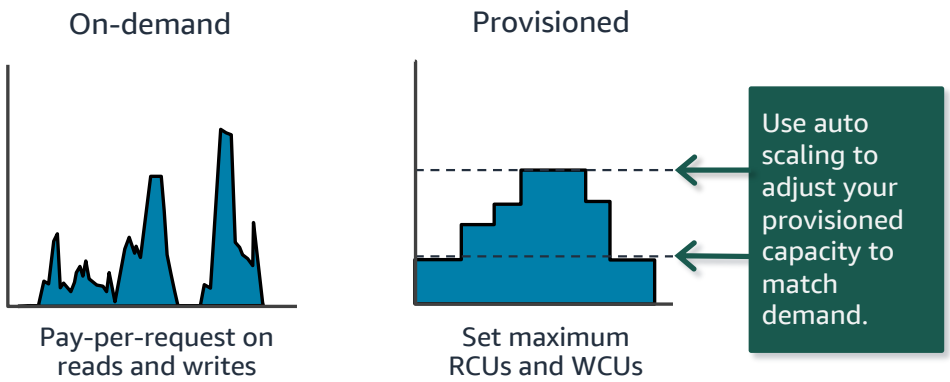
DynamoDB capacity and scaling

- DynamoDB measures read capacity in read capacity units (RCUs).
  - Read requests for up to a 4-KB item
- DynamoDB measures write capacity in write capacity units (WCUs).
  - Number of write requests per second for up to a 1-KB item

- DynamoDB has two options for managing capacity:



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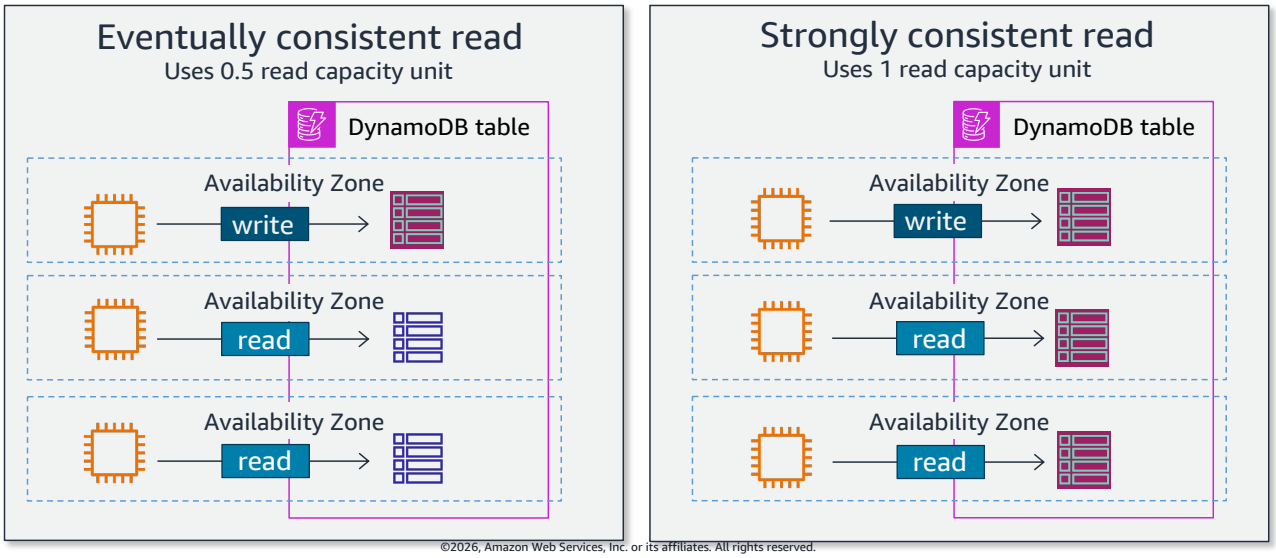
DynamoDB consistency options

DynamoDB replicates table data across three Availability Zones in a Region usually within 1 second.



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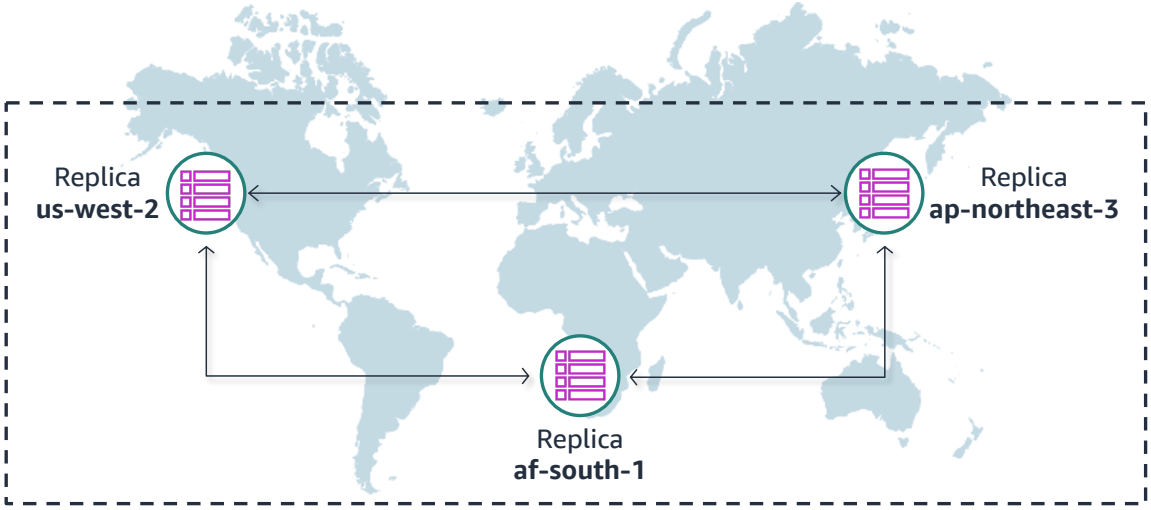
DynamoDB global tables

Global tables automate replication across Regions.



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Database caching

"How can we cache databases in the cloud to maximize performance?"

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What should you cache?

Data that requires a slow and expensive query

Frequently accessed data

Information that is relatively static

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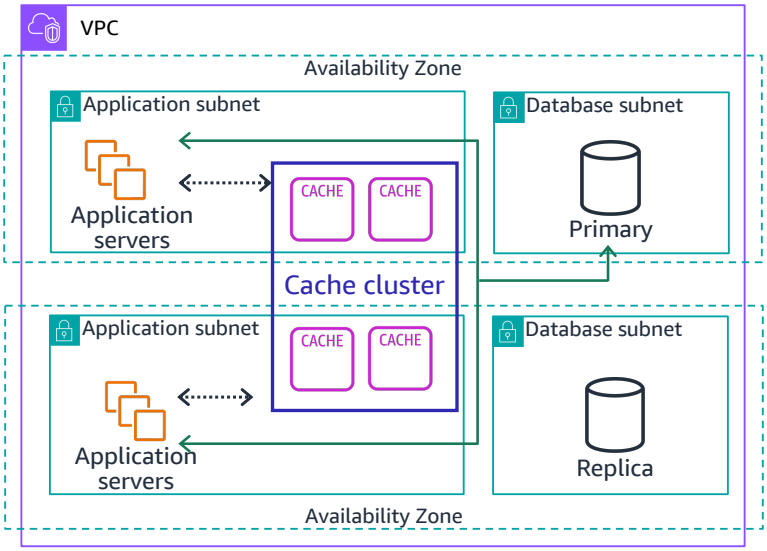
Caching architecture



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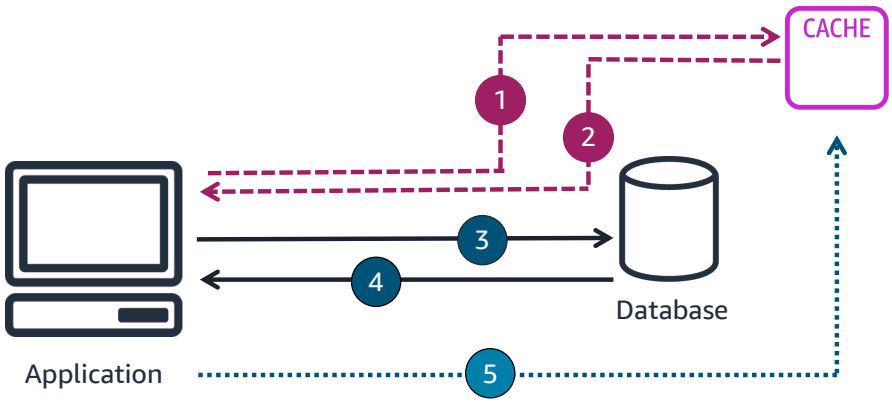
Common caching strategies: Lazy loading

1. Data request to the cache by the application
2. Cache miss
3. Missing data that the application requested from the database
4. Data that is returned from the database
5. Returned value that the application writes to the cache



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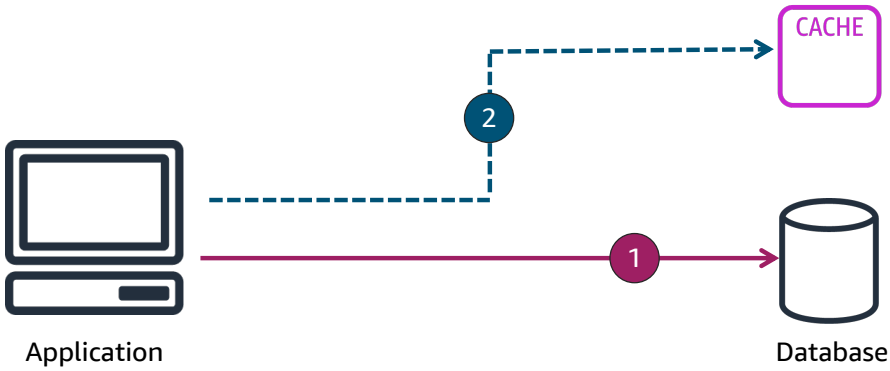
Common caching strategies: Write-through

1. The application writes data to the database.
2. The application also writes data to the cache.



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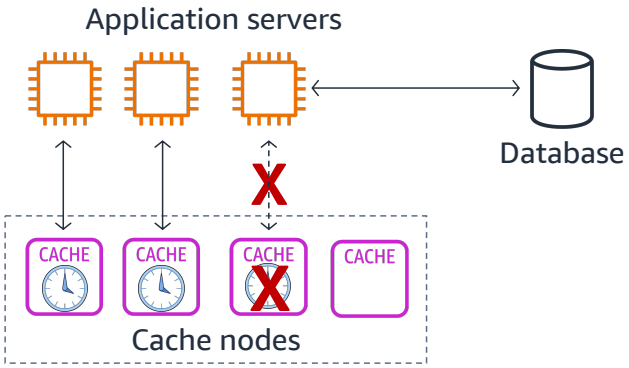
Managing your cache

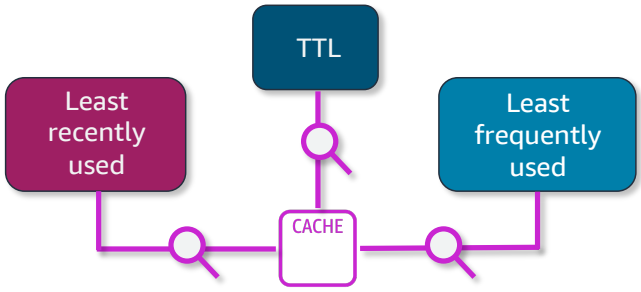
Cache validity

To minimize stale data, you can add a time to live (TTL) value to each application write.



Managing memory

When your cache memory is full, your cache evicts data based on your selected eviction policy. Eviction policies can evaluate any combination of the following:



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Amazon ElastiCache

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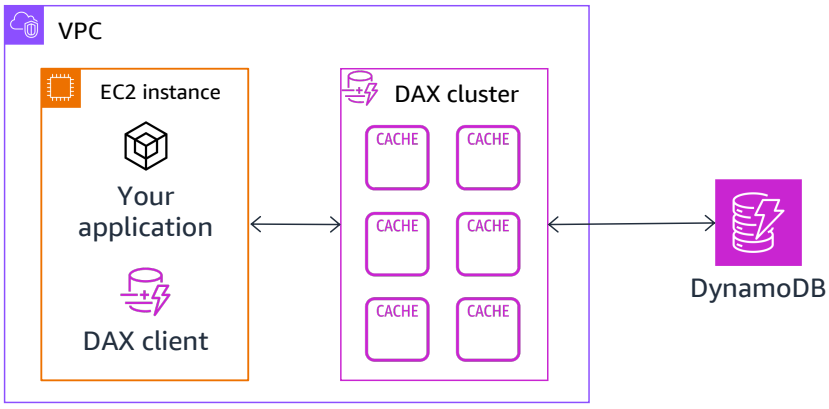
Amazon DynamoDB Accelerator (DAX)

- Is a fully managed, highly available cache for DynamoDB
- Can deliver microsecond response times
- Can scale to millions of read requests per second



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Database migration tools

"Which tools are available for migrating an existing database to the AWS Cloud?"

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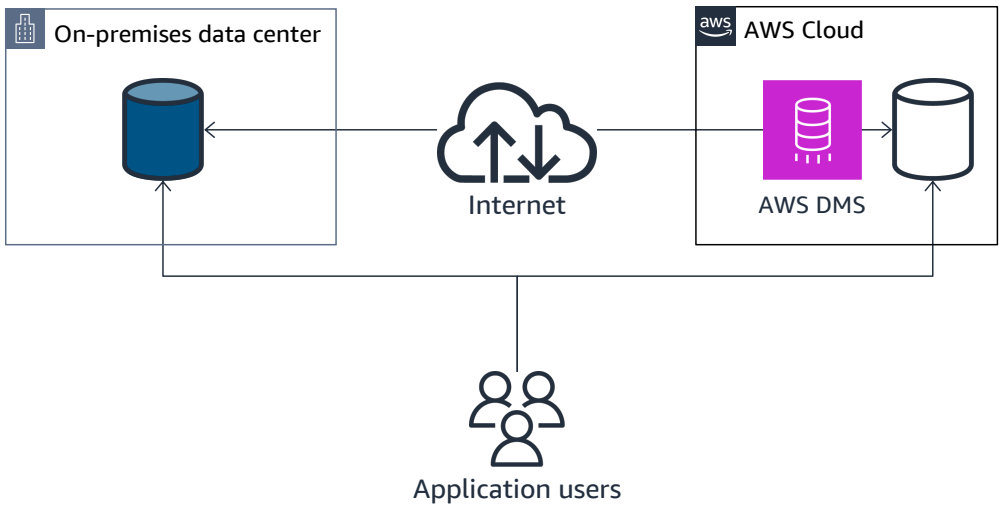
AWS Database Migration Service

AWS Database Migration Service (AWS DMS)

- Heterogeneous database migrations
- Database consolidation
- Continuous data replication
- Ability to point to a database, Amazon S3, AWS Snowball Edge, or other services



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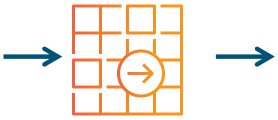
AWS Schema Conversion Tool

Source databases

- Oracle database
- Oracle data warehouse
- Azure SQL
- SQL Server
- Teradata
- IBM Netezza
- Greenplum
- HPE Vertica
- MySQL and MariaDB
- PostgreSQL
- Aurora
- IBM DB2 11.1 for Linux, Unix, and Windows (LUW)
- Apache Cassandra
- SAP Adaptive Server Enterprise (ASE)



AWS Schema Conversion Tool (AWS SCT)

Target databases on AWS

- MySQL
- PostgreSQL
- Oracle
- Amazon Redshift
- DynamoDB
- Amazon RDS for MySQL
- Aurora MySQL
- Amazon RDS for PostgreSQL
- Aurora PostgreSQL

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Review

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Present solutions

Database services manager

Consider how you would answer the following questions:

- What are the AWS database solutions?
- How can we more efficiently manage our relational databases in the cloud?
- How can we build a scalable key-value NoSQL database?
- How can we cache databases in the cloud to maximize performance?
- Which tools are available for migrating an existing database to the AWS Cloud?

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Module review

In this module, you learned about the following:

- ✓ Database services
- ✓ Amazon RDS
- ✓ DynamoDB
- ✓ Database caching
- ✓ Database migration tools

Next, you review the following:

- Capstone check-in
- Knowledge check
- Lab introduction

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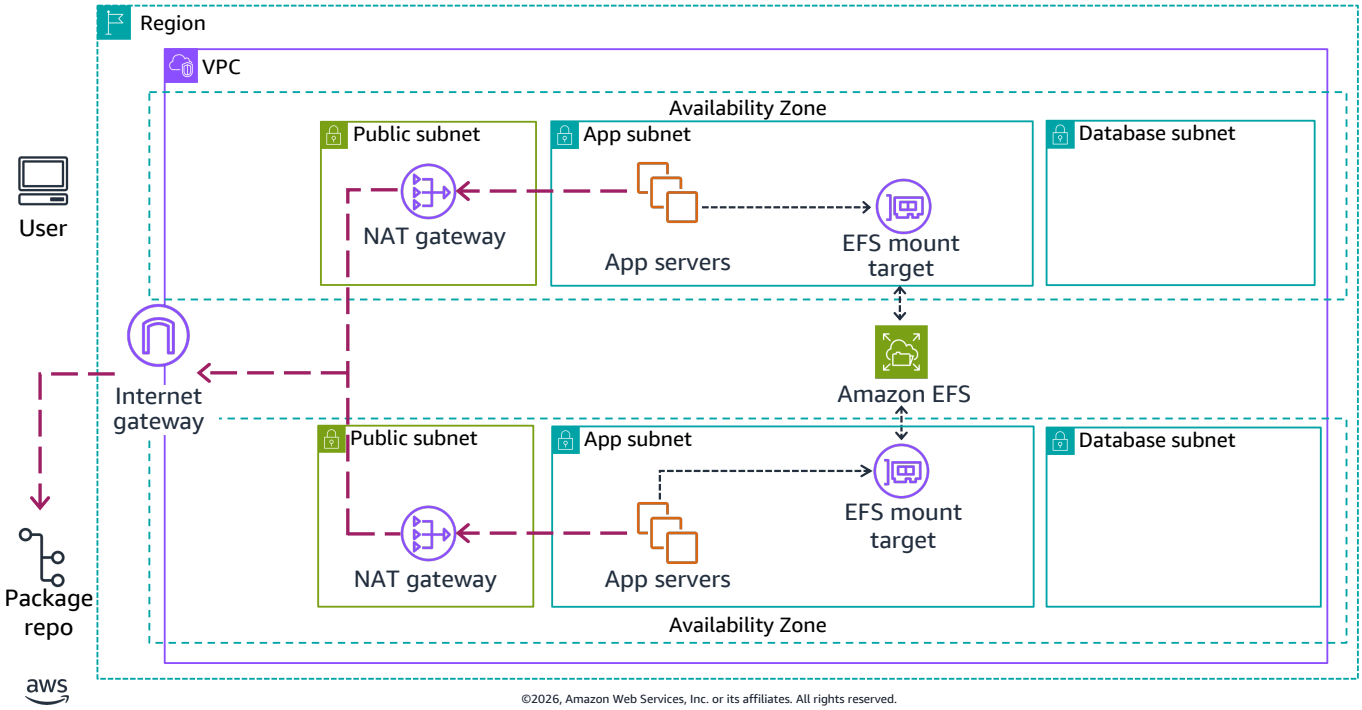
Capstone architecture previously



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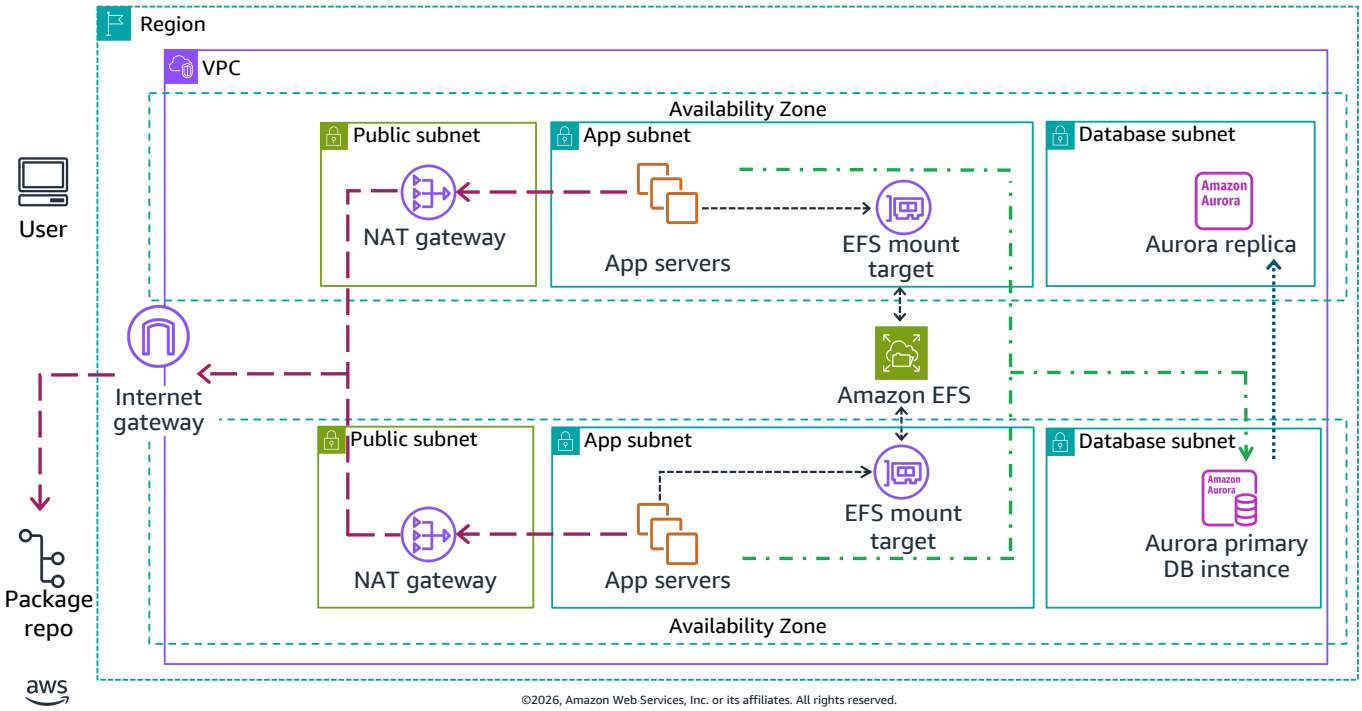
Capstone architecture check-in



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Knowledge check

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Knowledge check question 1

What is a benefit of using Amazon RDS in a Multi-AZ configuration?

Choice	Response
A	It delivers two live copies of the database running concurrently.
B	It provides automatic failover across Availability Zones.
C	It provides automatic cross-Region replication.
D	It eliminates the need for read replicas.

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Knowledge check question 1 and answer

What is a benefit of using Amazon RDS in a Multi-AZ configuration?

Choice	Response
A	It delivers two live copies of the database running concurrently.
B correct	It provides automatic failover across Availability Zones.
C	It provides automatic cross-Region replication.
D	It eliminates the need for read replicas.

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Knowledge check question 2

Which statement is true regarding Amazon DynamoDB global tables?

Choice	Response
A	Tables are updated manually or through automation tools.
B	Only two tables are active at one time.
C	Select different instance sizes to adjust performance.
D	Tables can be in different AWS Regions.

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Knowledge check question 2 and answer

Which statement is true regarding Amazon DynamoDB global tables?

Choice	Response
A	Tables are updated manually or through automation tools.
B	Only two tables are active at one time.
C	Select different instance sizes to adjust performance.
D correct	Tables can be in different AWS Regions.

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Knowledge check question 3

Which statement is true regarding an Amazon Aurora database?

Choice	Response
A	Nine copies of the data are stored across three Availability Zones.
B	Aurora has a limit of five replicas.
C	Aurora is compatible with MySQL or PostgreSQL.
D	Multi-AZ deployments are not required for high availability.

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Knowledge check question 3 and answer

Which statement is true regarding an Amazon Aurora database?

Choice	Response
A	Nine copies of the data are stored across three Availability Zones.
B	Aurora has a limit of five replicas.
C correct	Aurora is compatible with MySQL or PostgreSQL.
D	Multi-AZ deployments are not required for high availability.

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Lab 3:

Create a Database Layer in Your Amazon VPC Infrastructure

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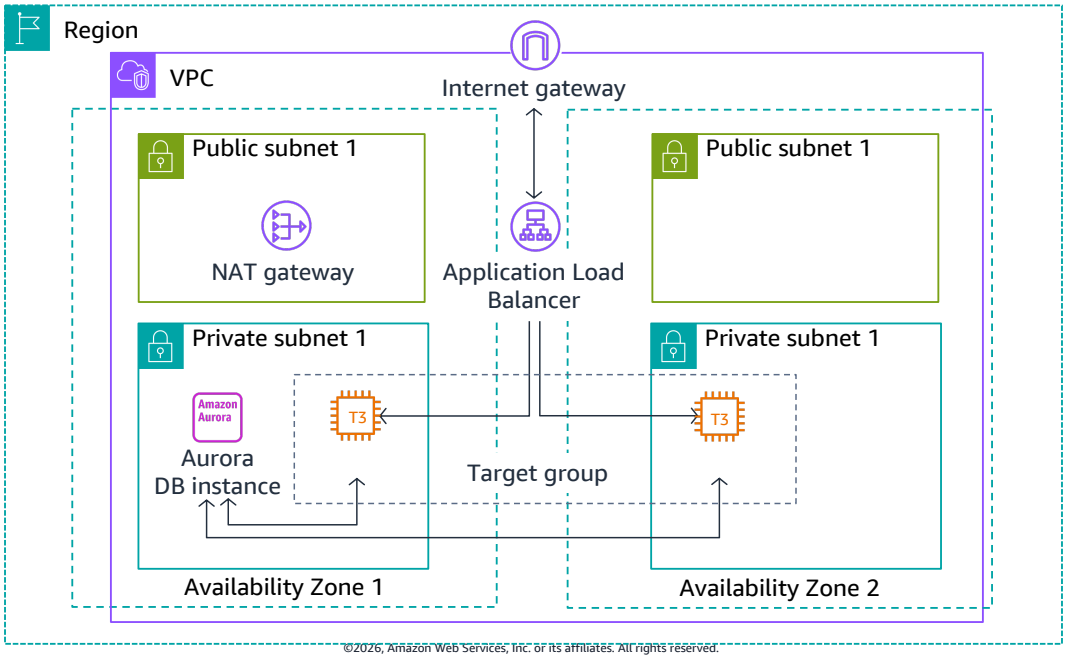
Lab 3 diagram



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Lab tasks

- Task 1: Create an Amazon RDS database
- Task 2: Create and configure an Application Load Balancer
- Task 3: Review the Amazon RDS DB instance metadata through the console
- Task 4: Test the application connectivity to the database
- Optional Task: Creating an Amazon RDS read replica in a different AWS Region

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End of Module 6

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Architecting on AWS

Module 7: Monitoring and Scaling

Lab 4

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Poll question

What factors impact your decisions about making your workloads scalable?

A. Cost

B. Usage patterns

C. Expected growth

D. Criticality of workload

E. All of the above

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Module overview

- Business request
- Monitoring
- Alarms and events
- Load balancing
- Auto scaling
- Present solutions
- Capstone check-in
- Knowledge check
- Lab 4: Configure High Availability in Your Amazon VPC

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Business request

Operations manager

The operations manager needs to know the following:

- What tools and services are available to monitor and log activity in my AWS accounts?
- How can we set thresholds and be alerted to changes in our infrastructure?
- How do we add high availability to our Amazon Elastic Compute Cloud (Amazon EC2) workloads and distribute traffic across multiple targets?
- How can we dynamically increase and decrease capacity to meet changing demand?

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Monitoring

"What tools and services are available to monitor and log activity in our AWS accounts?"

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Reasons for monitoring

Operational health

Get operational visibility and insight.

Application performance

Collect data at every layer of the performance stack.

Resource utilization

Improve resource optimization.

Security auditing

Automate and manage evidence collection, security, and integrity.

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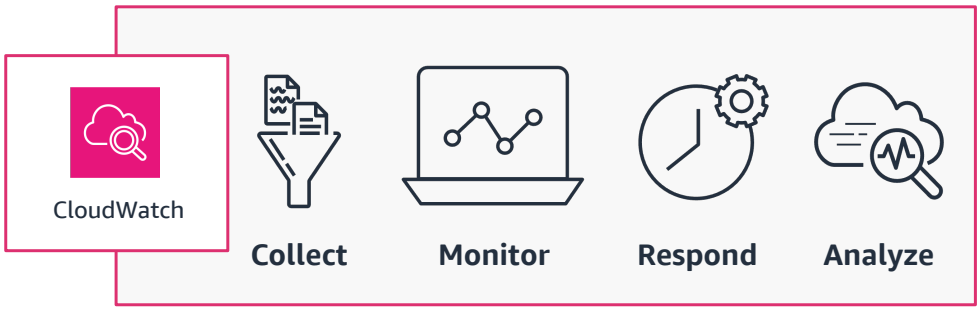
Amazon CloudWatch

- Collect near real-time metrics and logs.
- Access monitoring data in one place.
- Create alarms and send notifications.
- Initiate changes to resource capacity based on rules.
- Create and view dashboards.



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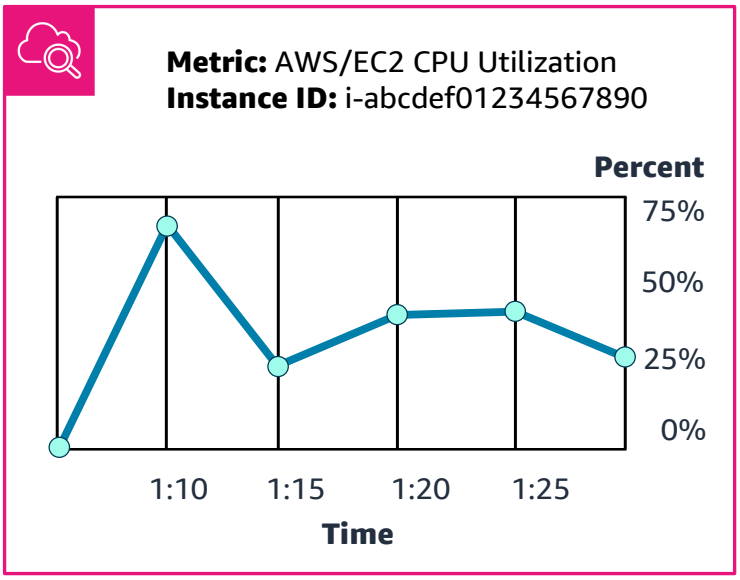
CloudWatch metrics

- Metrics are data about system performance.
- CloudWatch ingests and tracks metrics so that you can search and visualize data.



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Types of logs

Amazon CloudWatch Logs	AWS CloudTrail	VPC Flow Logs	Custom logs
Monitor apps by using log data. Store and access log files.	Track user activity and API usage.	Capture information about IP traffic to and from network interfaces.	Store custom logs that are generated from your application instances.

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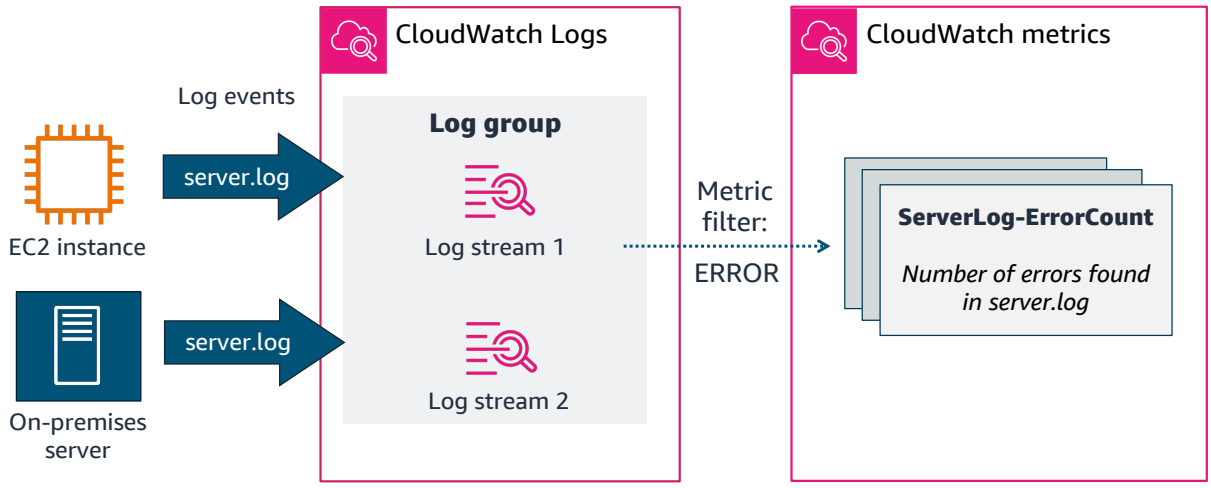
CloudWatch Logs example



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AWS CloudTrail

- Log and monitor account activity across your AWS infrastructure.
- Record API call interactions for most AWS services.
- Automatically push logs to Amazon Simple Storage Service (Amazon S3).

CloudTrail helps you understand events in your accounts.

Who **shut down** a specific instance?

What activities were **denied** due to lack of permissions?

Who **changed** a security group configuration?

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Example: CloudTrail log (1 of 4)

```
{
    "Records": [{
        "eventVersion": "1.0",
        "userIdentity": {
            "type": "IAMUser",
            "principalId": "EX_PRINCIPAL_ID",
            "arn": "arn:aws:iam::123456789012:user/Alice",
            "accountId": "123456789012",
            "accessKeyId": "EXAMPLE_KEY_ID",
            "userName": "Alice"
        },
```

Who made the request?

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Example: CloudTrail log (2 of 4)

```
"requestParameters": {
            "instancesSet": {
                "items": [{
                    "instanceId": "i-abcdefg01234567890"
                }]
            },
            "force": false
        },
```

What was the focus of the request?

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Example: CloudTrail log (3 of 4)

```
"eventTime": "2018-03-06T21:01:59Z",
        "eventSource": "ec2.amazonaws.com",
        "eventName": "StopInstances",
        "awsRegion": "us-west-2",
        "sourceIPAddress": "205.251.233.176",
        "userAgent": "ec2-api-tools 1.6.12.2",
```

When did the request occur?

What was the API call?

Where did it occur?

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Example: CloudTrail log (4 of 4)

```
"responseElements": {
    "instancesSet": {
        "items": [{
            "instanceId": "i-abcdefg01234567890",
            "currentState": {
                "code": 64,
                "name": "stopping"
            },
            "previousState": {
                "code": 16,
                "name": "running"
            }
```

What was the response?

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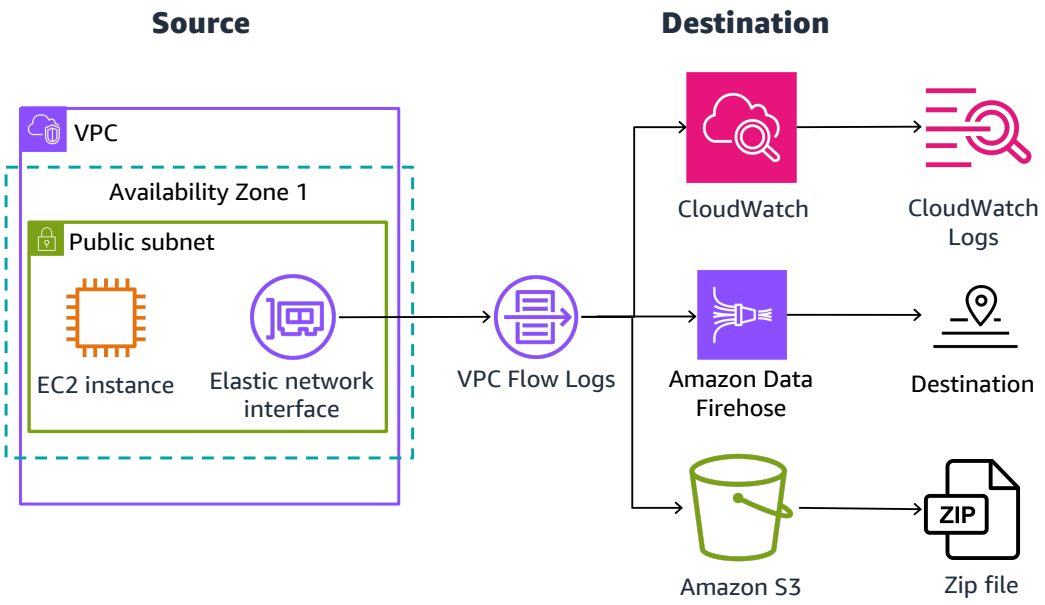
VPC Flow Logs

Capture IP traffic information that is going to and from VPC network interfaces.



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Contents of a flow log record

VPC Flow Logs
AWS account
Elastic network interface ID
Time in Unix seconds
Number of packets and bytes that are transferred
Action that is taken based on security group or network access control list (network ACL)

Version	2
Account ID	123456789010
Interface ID	eni-04b10a1942977452f
Source address	172.168.1.2
Destination address	32.68.32.56
Source port	36490
Destination port	443
Protocol	6
Packets	77
Bytes	5040
Start	1560385064
End	1560385070
Action	ACCEPT
Log status	OK

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Alarms and events

"How can we set thresholds and be alerted to changes in our infrastructure?"

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CloudWatch alarms

1. Identify the CloudWatch metric.

2. Create your alarms based on metrics.

3. Define the actions to take when your metric's threshold is exceeded.

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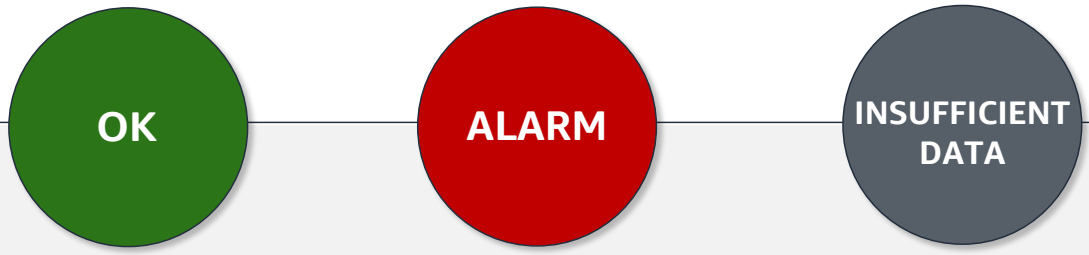
Alarm states

Test a selected metric against a specific threshold value. ALARM is not necessarily an emergency condition.



Threshold not exceeded

Threshold exceeded

Not enough information

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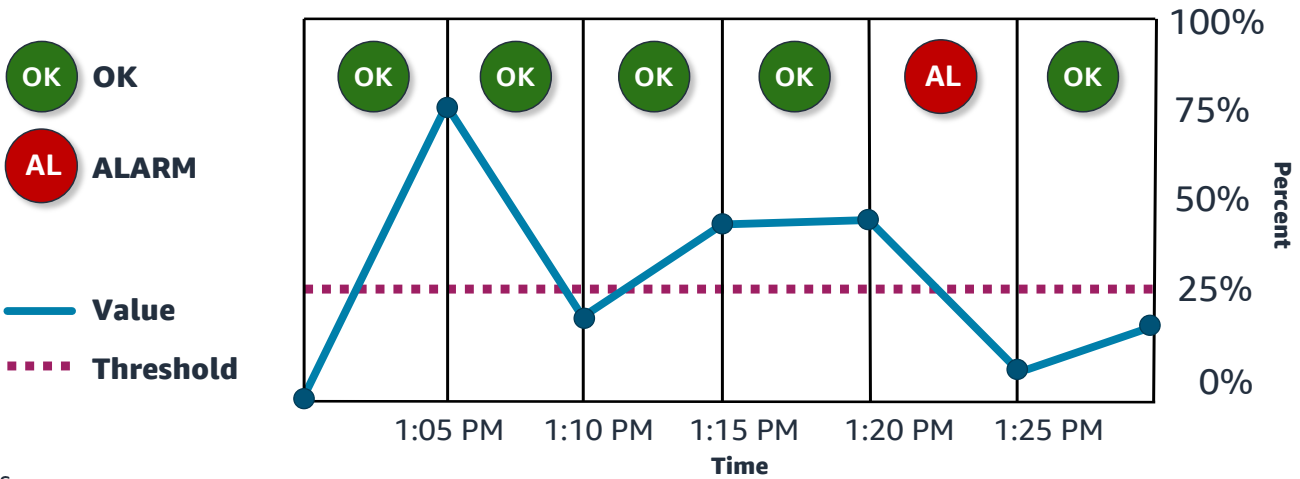
Alarm components

Metric name	Instance ID	Statistic	Period	Data points to alarm
CPUUtilization	i-abcdef012345	Average	5 minutes	2 of 2



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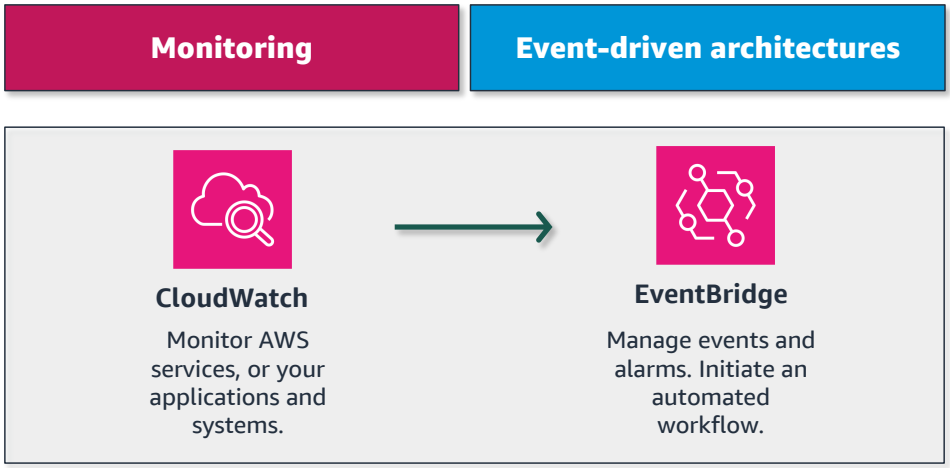
Amazon EventBridge

Amazon CloudWatch Events is now part of Amazon EventBridge.

EventBridge can do the following:

- Send messages to respond to the environment.
- Activate functions or initiate actions.
- Capture state information.



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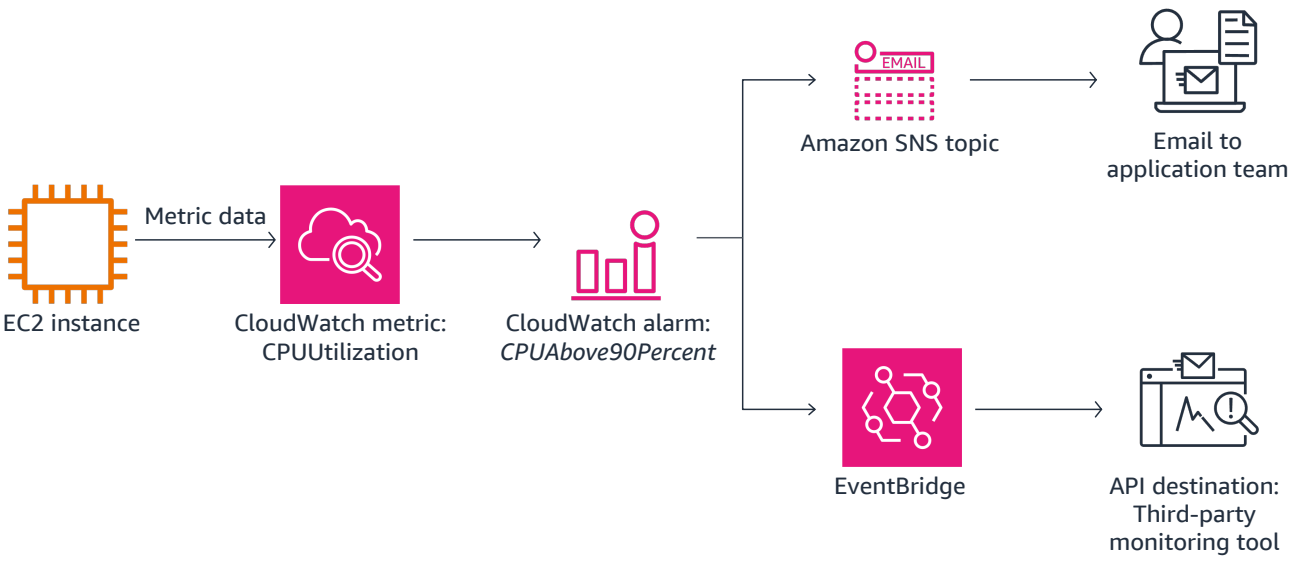
Example: CloudWatch alarm automated response



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Load balancing

"How do we add high availability to our Amazon EC2 workloads and distribute traffic across multiple targets?"

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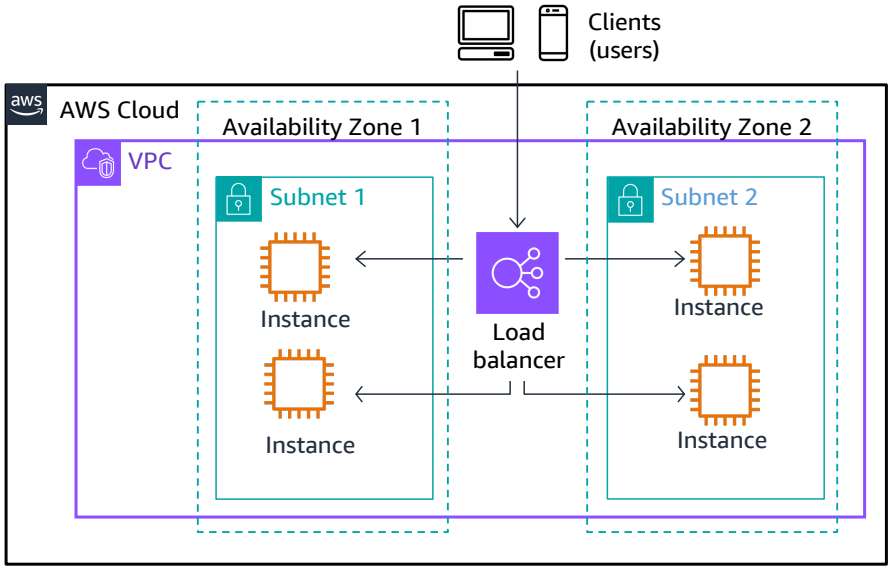
Elastic Load Balancing (ELB)

- Automatically distributes traffic across multiple targets
- Provides high availability
- Incorporates security features
- Performs health checks



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ELB load balancer types

Application Load Balancer

HTTP and HTTPS

- Flexible application management
- Advanced load balancing of traffic
- Operates at the application layer (Layer 7)

Network Load Balancer

TCP and UDP

- Extreme performance and static IP
- Load balancing of TCP traffic
- Operates at the transport layer (Layer 4)

Gateway Load Balancer

IP

- Flexible application management
- Advanced load balancing of traffic
- Operates at the network layer (Layer 3)

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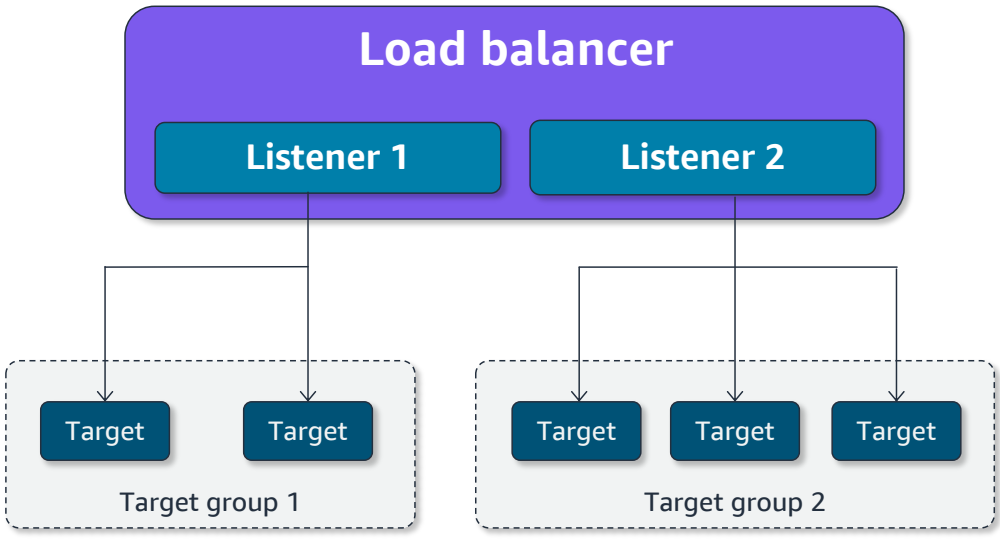
ELB load balancer components



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ELB common features

Feature	Application Load Balancer	Network Load Balancer	Gateway Load Balancer
Health checks	Yes	Yes	Yes
CloudWatch metrics	Yes	Yes	Yes
Logging	Yes	Yes	Yes
SSL offloading	Yes	Yes	
Connection draining	Yes	Yes	Yes
Preserve source IP address	Yes	Yes	Yes
Static IP address	**	Yes	
AWS Lambda functions as a target	Yes		
Redirects	Yes		
Fixed-response actions	Yes		

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Auto scaling

"How can we dynamically increase and decrease capacity to meet changing demand?"

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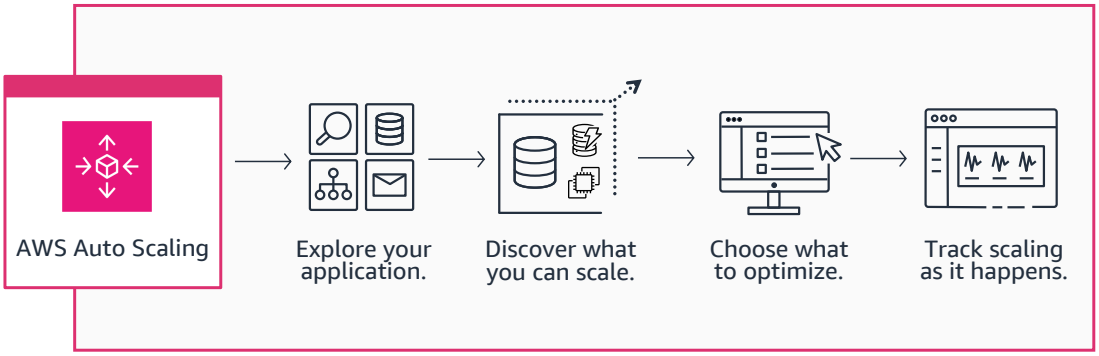
AWS Auto Scaling

Provides application scaling for multiple resources across services, in short intervals



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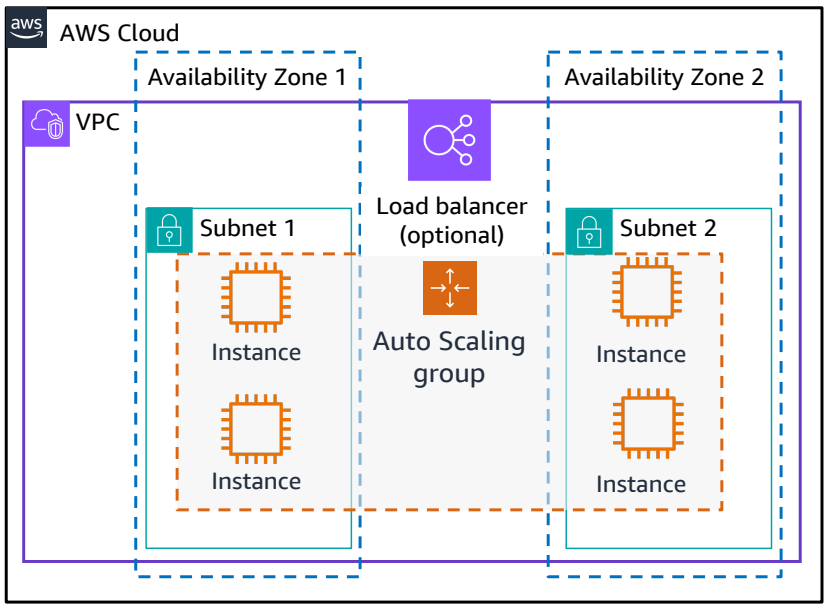
Amazon EC2 Auto Scaling

- Helps you control EC2 instances available to handle the load for your application
- Launches or terminates your AWS resources based on specified conditions
- Registers new instances with load balancers, when specified



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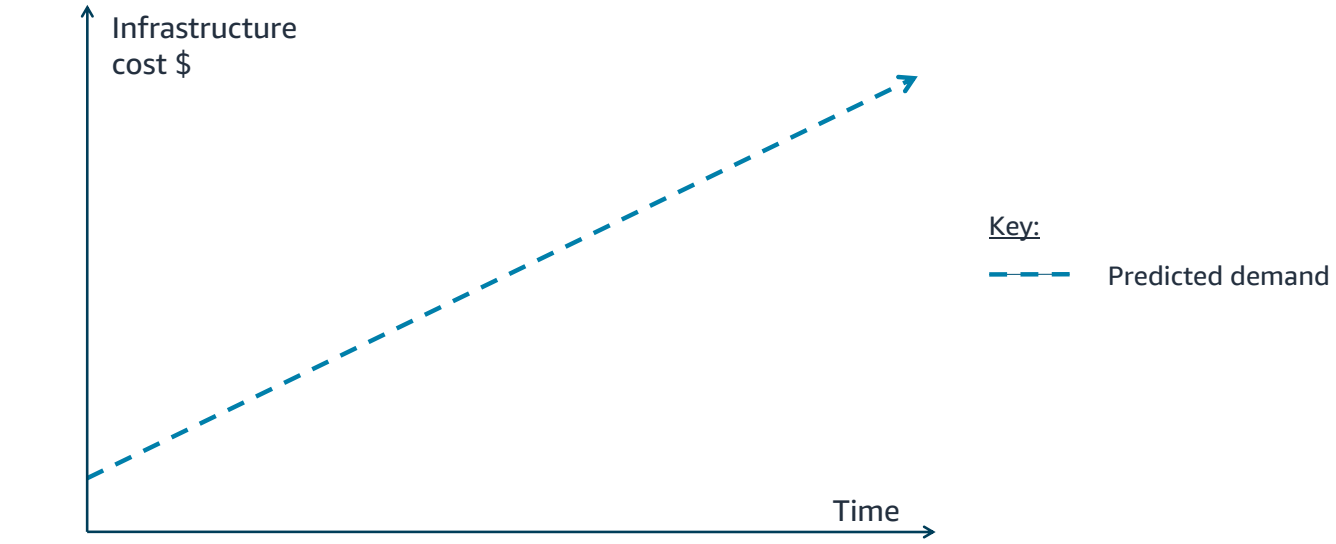
Elasticity: Scaling in and out (1 of 6)



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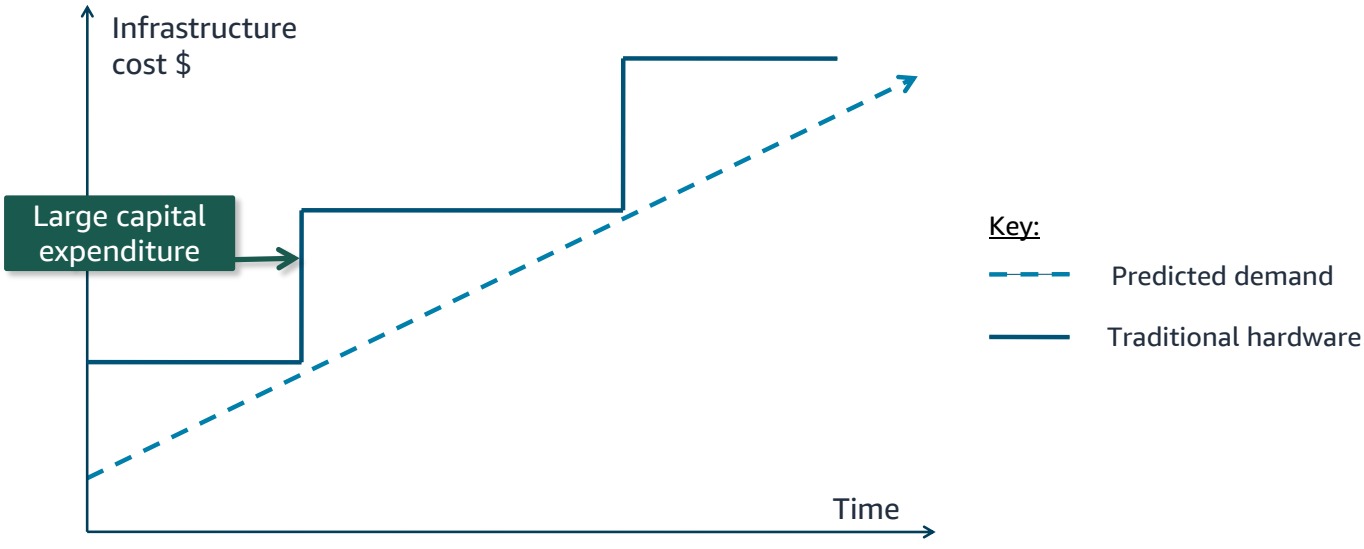
Elasticity: Scaling in and out (2 of 6)



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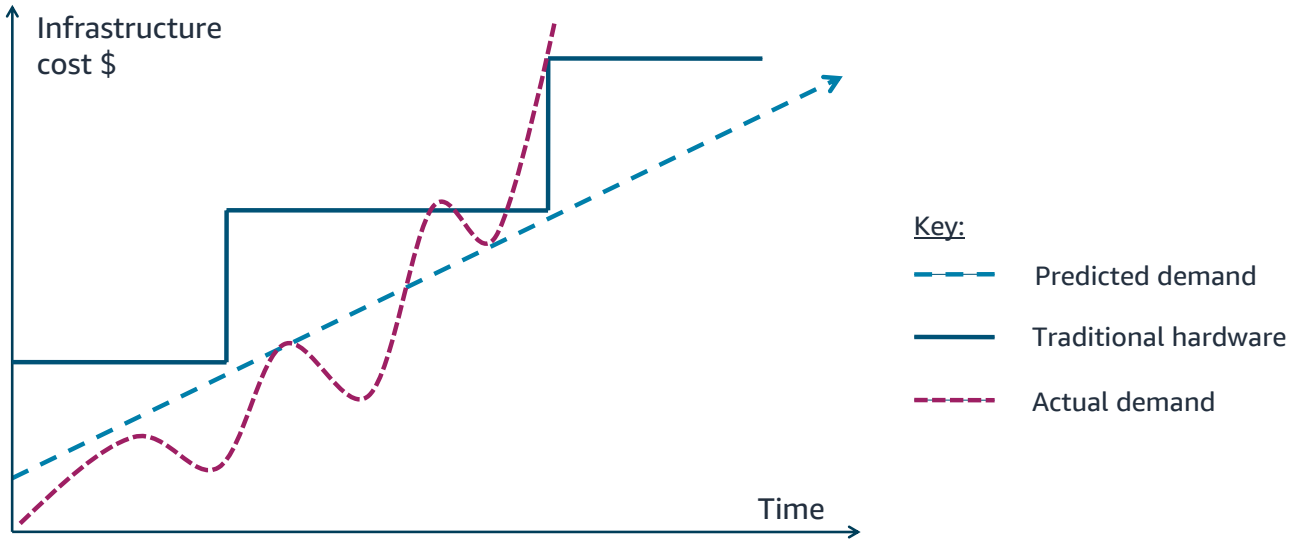
Elasticity: Scaling in and out (3 of 6)



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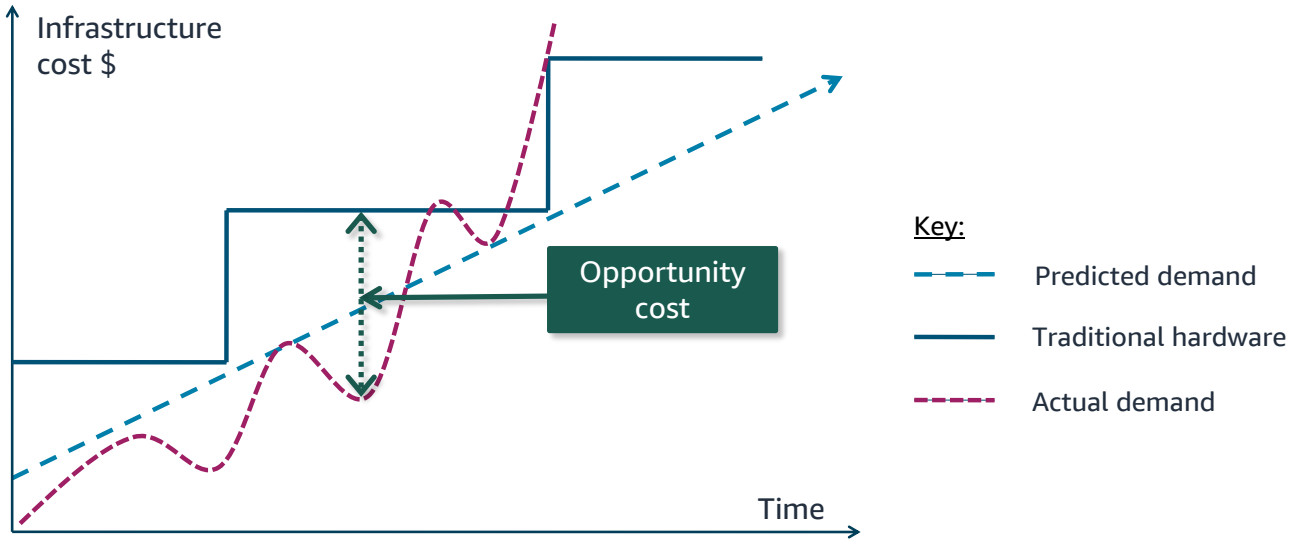
Elasticity: Scaling in and out (4 of 6)



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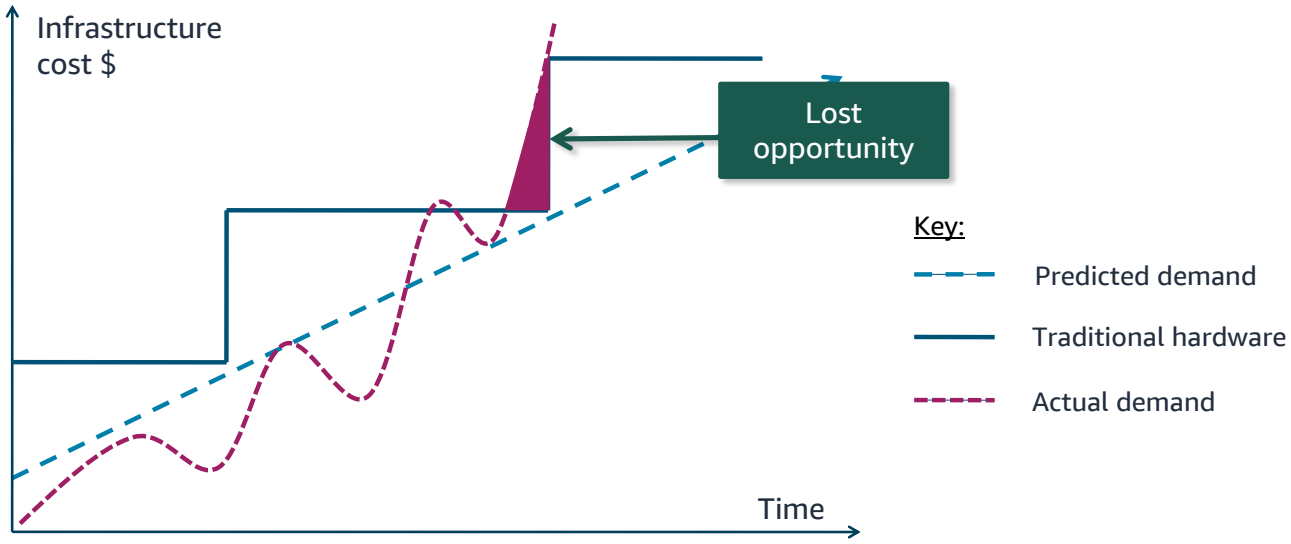
Elasticity: Scaling in and out (5 of 6)



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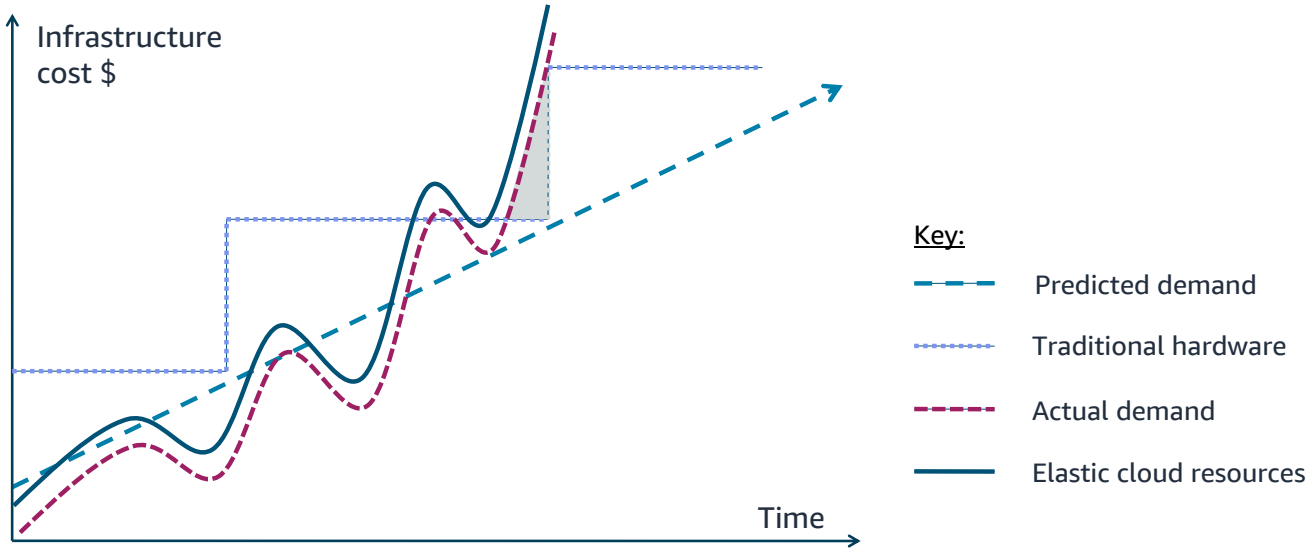
Elasticity: Scaling in and out (6 of 6)



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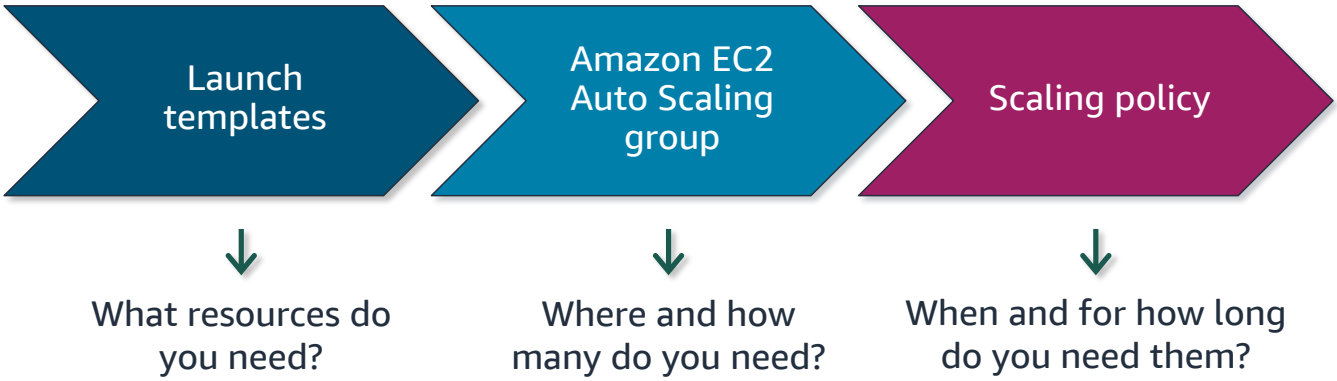
Amazon EC2 Auto Scaling components



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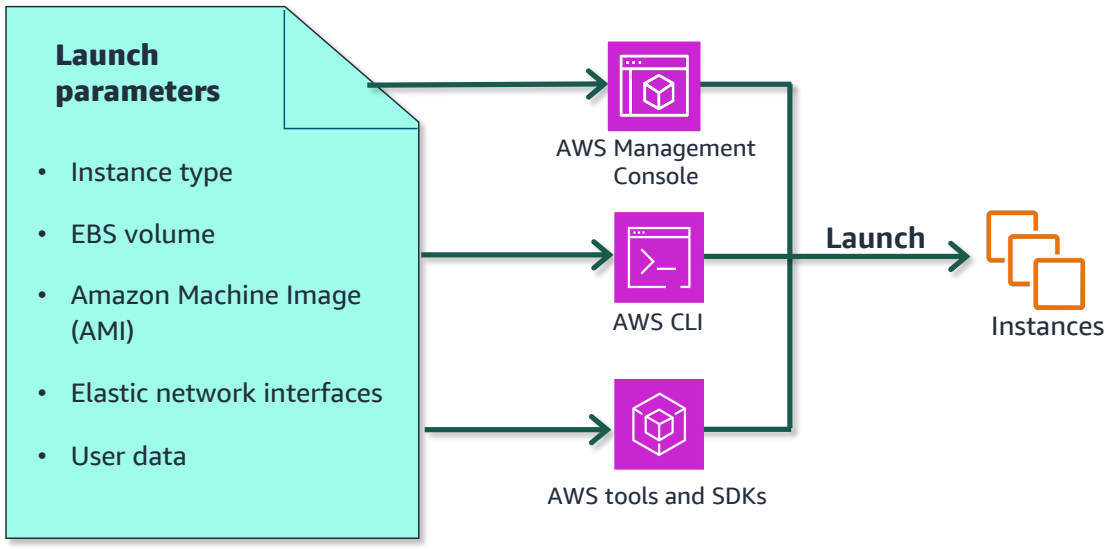
Launch template



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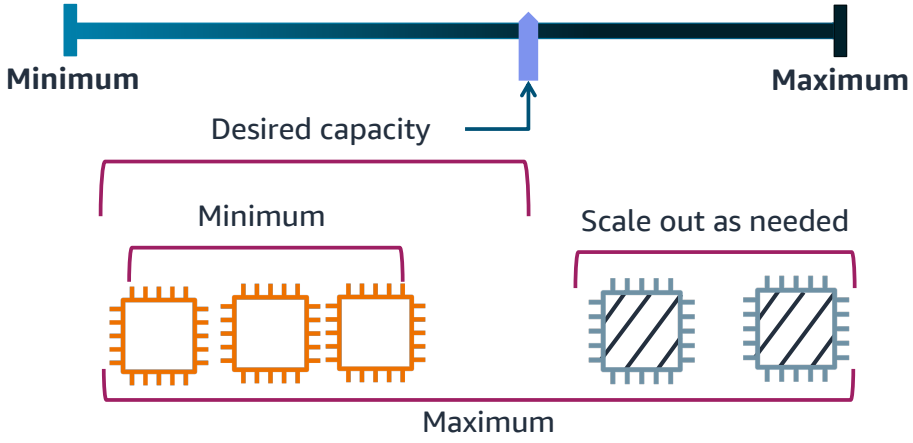
Group capacity

- Choose the VPC and subnets for your Amazon EC2 Auto Scaling group.
- Set the minimum and maximum number of instances that it can have.
- Launch or terminate instances to meet capacity demands.



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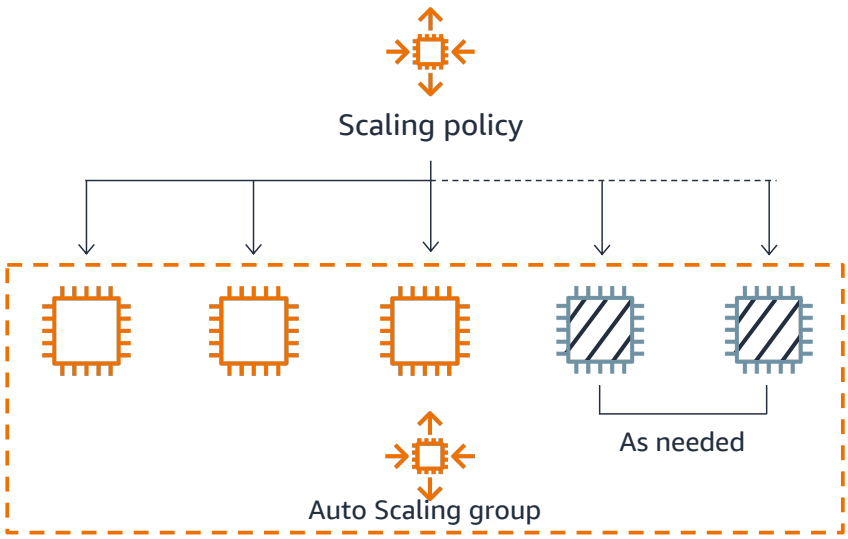
Scaling methods

- Fixed
- Manually
- Scheduled
- Dynamically
- Proactively



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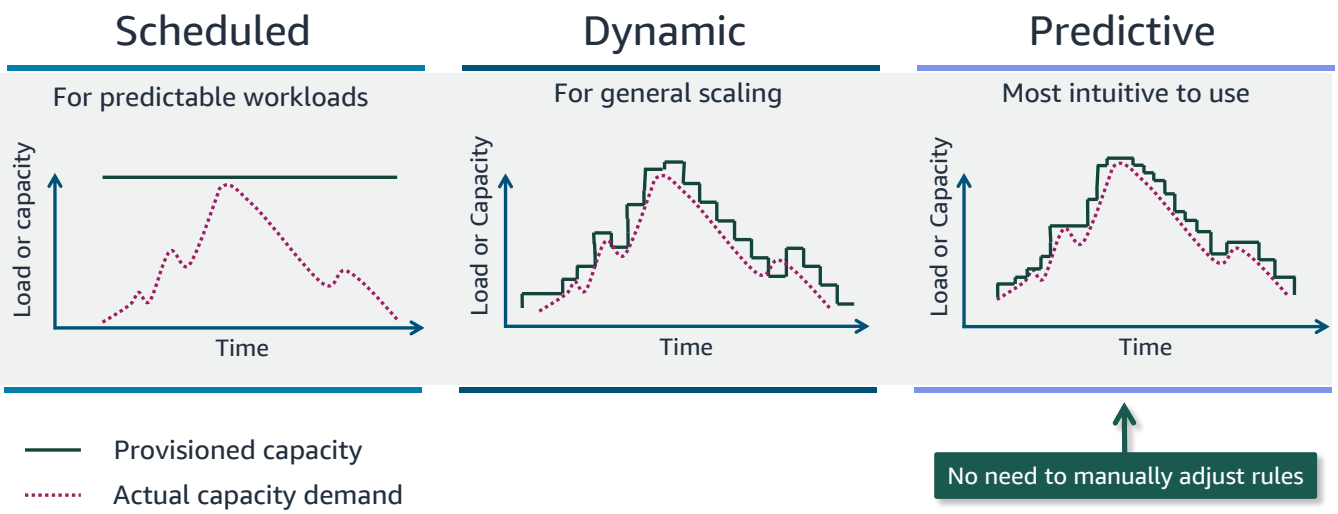
Ways to scale with Amazon EC2 Auto Scaling



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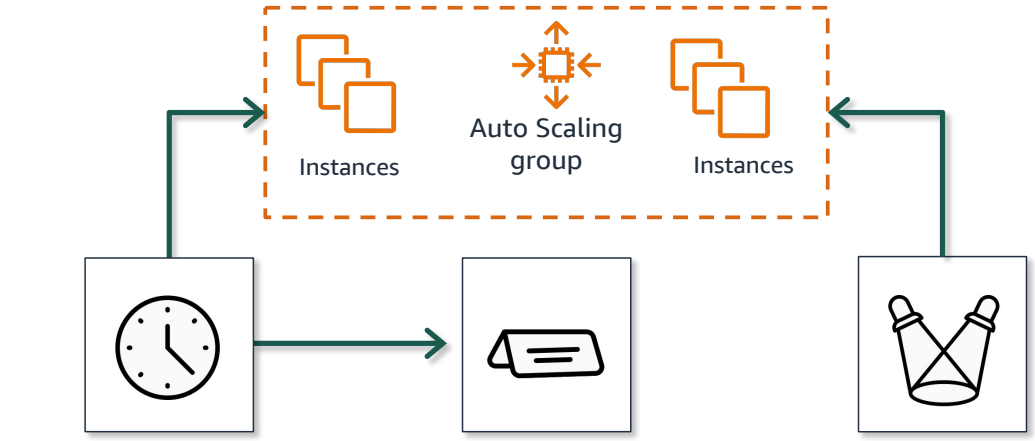
Optimize cost with Amazon EC2 Auto Scaling



On-Demand Instances
Spiky workloads, to define needs

Savings Plans
Committed and steady-state usage

Spot Instances
Fault-tolerant, flexible, stateless workloads

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Review

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Present solutions

Operations manager

Consider how you would answer the following questions:

- What tools and services are available to monitor and log activity in your AWS accounts?
- How can you set thresholds and be alerted to changes in your infrastructure?
- How do you add high availability to your Amazon EC2 workloads and distribute traffic across multiple targets?
- How can you dynamically increase and decrease capacity to meet changing demand?

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Module review

In this module, you learned about the following:

- ✓ Monitoring
- ✓ Alarms and events
- ✓ Load balancing
- ✓ Auto scaling

Next, you review the following:

- Capstone check-in
- Lab introduction
- Knowledge check

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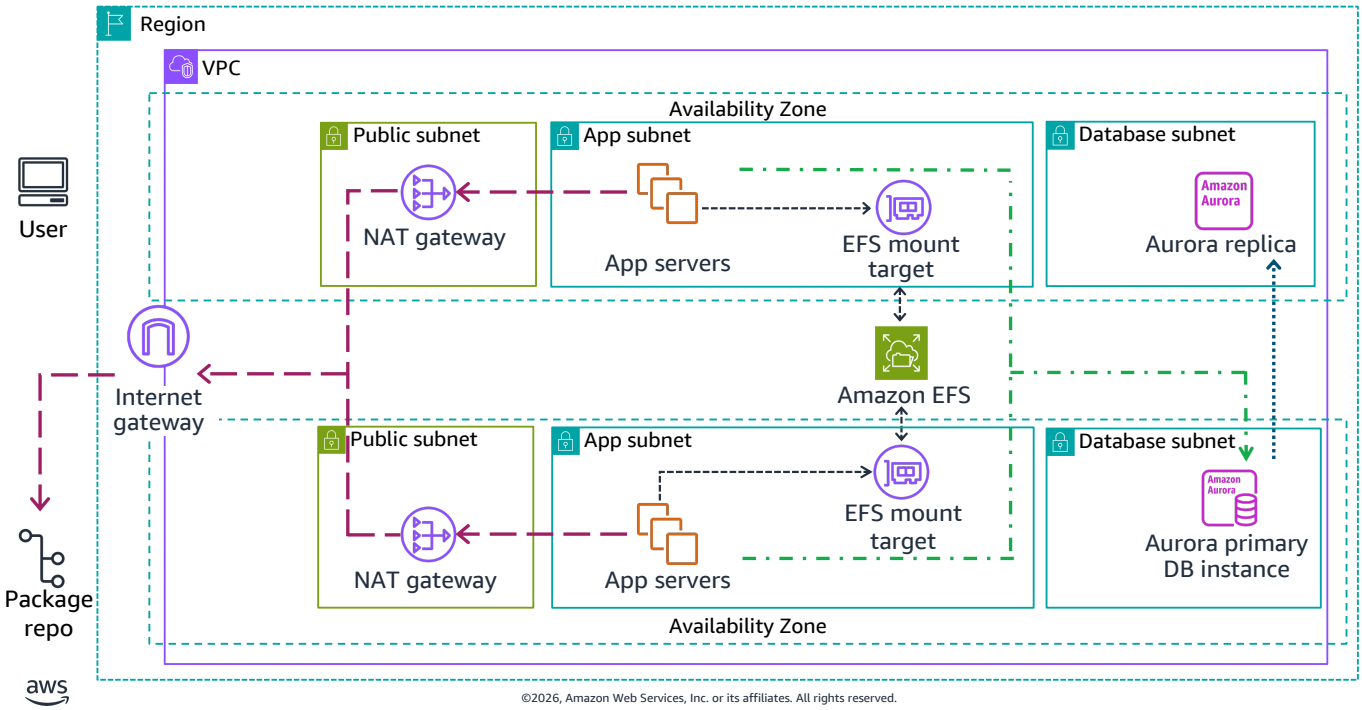
Capstone architecture previously



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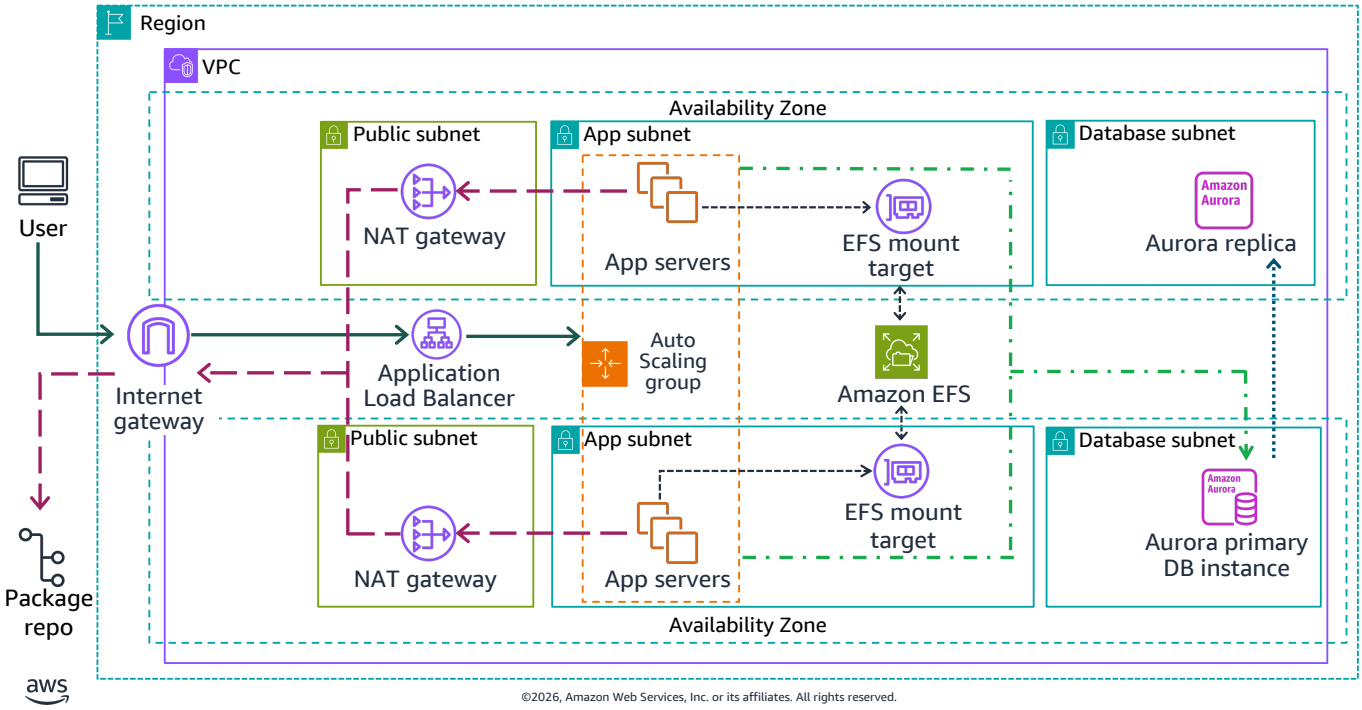
Capstone architecture final



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Knowledge check

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Knowledge check question 1

Which option is a valid target for an Application Load Balancer?

Choice	Response
A	An Amazon EC2 instance
B	An Availability Zone
C	An Amazon S3 bucket
D	A virtual private network (VPN) connection

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Knowledge check question 1 and answer

Which option is a valid target for an Application Load Balancer?

Choice	Response
A correct	An Amazon EC2 instance
B	An Availability Zone
C	An Amazon S3 bucket
D	A virtual private network (VPN) connection

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Knowledge check question 2

An application support team has an application with unpredictable traffic patterns that runs on at least two instances. They want the CPU utilization to stay at about 75 percent. Which Amazon EC2 Auto Scaling strategy should they choose?

Choice	Response
A	Scheduled
B	Dynamic
C	Predictive
D	Manual

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Knowledge check question 2 and answer

An application support team has an application with unpredictable traffic patterns that runs on at least two instances. They want the CPU utilization to stay at about 75 percent. Which Amazon EC2 Auto Scaling strategy should they choose?

Choice	Response
A	Scheduled
B correct	Dynamic
C	Predictive
D	Manual

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Knowledge check question 3

Which service can invoke actions based on data from account resources and supported third-party management services?

Choice	Response
A	Amazon CloudWatch Logs
B	Amazon EventBridge
C	AWS CloudTrail
D	Amazon EC2 Auto Scaling

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Knowledge check question 3 and answer

Which service can invoke actions based on data from account resources and supported third-party management services?

Choice	Response
A	Amazon CloudWatch Logs
B correct	Amazon EventBridge
C	AWS CloudTrail
D	Amazon EC2 Auto Scaling

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Knowledge check question 4

Which options are valid alarm states in Amazon CloudWatch? (Select TWO.)

Choice	Response
A	READY
B	ALERT
C	ALARM
D	INSUFFICIENT_DATA
E	FAILED

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Knowledge check question 4 and answer

Which options are valid alarm states in Amazon CloudWatch? (Select TWO.)

Choice	Response
A	READY
B	ALERT
C correct	ALARM
D correct	INSUFFICIENT_DATA
E	FAILED

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Knowledge check question 5

Which options are use cases for AWS CloudTrail data? (Select TWO.)

Choice	Response
A	Provide real-time observability of AWS resources.
B	Store log data as a record of account usage.
C	Log events for a particular service or application.
D	Capture root login failures.
E	Collect metric data that measures CPU utilization.

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Knowledge check question 5 and answer

Which options are use cases for AWS CloudTrail data? (Select TWO.)

Choice	Response
A	Provide real-time observability of AWS resources.
B correct	Store log data as a record of account usage.
C	Log events for a particular service or application.
D correct	Capture root login failures.
E	Collect metric data that measures CPU utilization.

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Lab 4:

Configure High Availability in Your Amazon VPC

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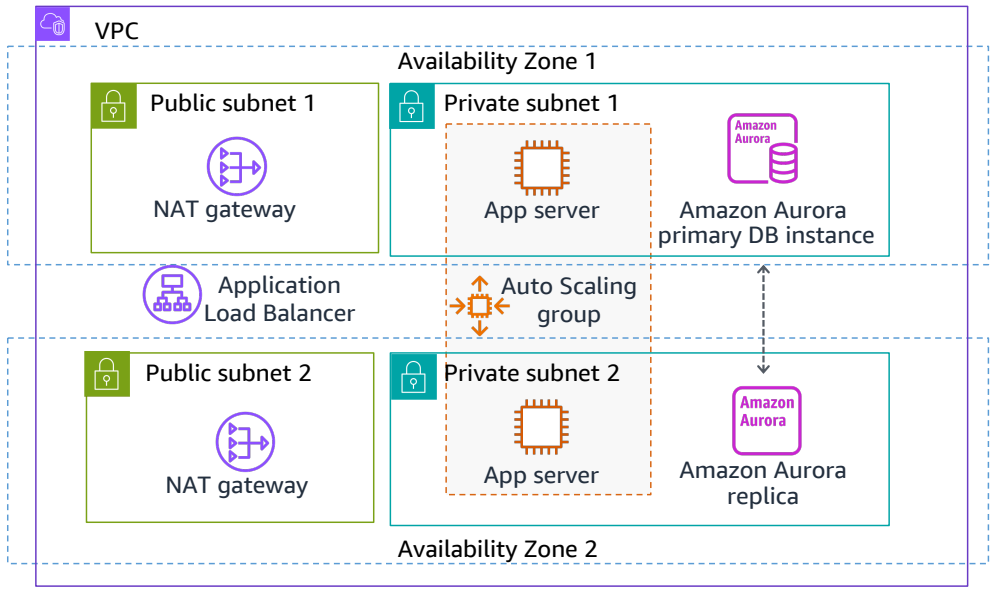
Lab 4 diagram



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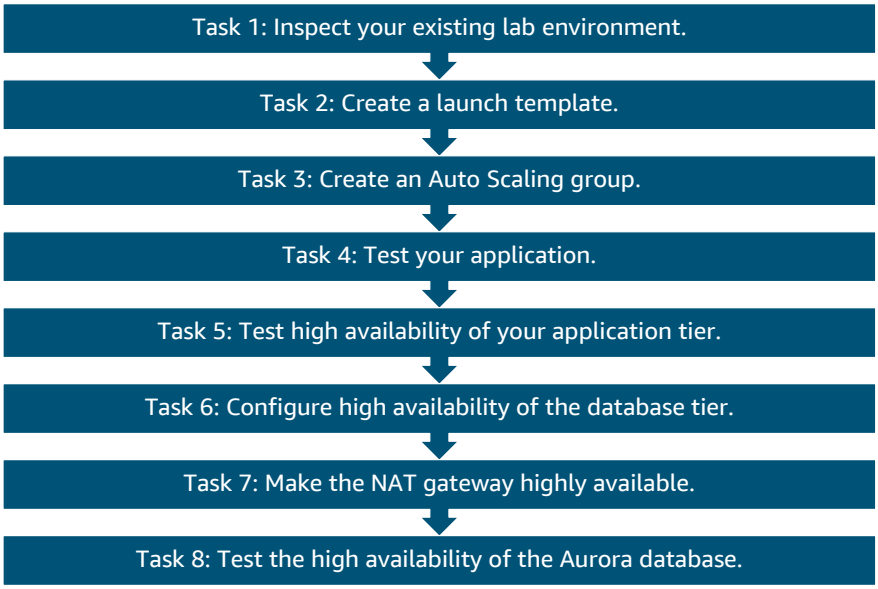
Lab tasks



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End of Module 7

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Architecting on AWS

Module 8: Automation

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Poll question

Have you used any of these resource provisioning tools?

A. AWS CloudFormation

B. AWS Elastic Beanstalk

C. Others (for example, Terraform or OpenStack Heat)

D. Not yet

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Module overview

- Business request
- AWS CloudFormation
- Infrastructure management
- Amazon Q
- Present solutions
- Knowledge check

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Business request

Chief technology officer

The chief technology officer wants to know the following:

- How can we simplify our cloud infrastructure build?
- How can we deploy, maintain, and scale applications in the cloud?

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AWS CloudFormation overview

"How can we simplify our cloud infrastructure build?"

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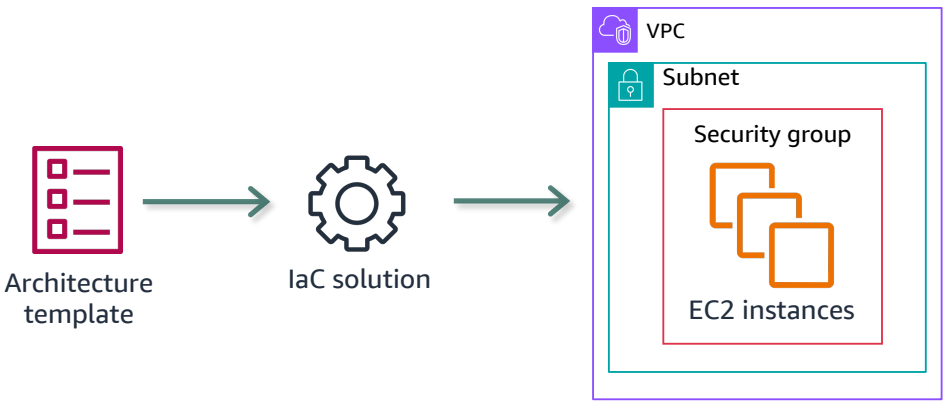
Infrastructure as code (IaC)

- Replicate, redeploy, and repurpose.
- Control versioning on infrastructure and applications.
- Detect drift.
- Roll back the service to the last good state.



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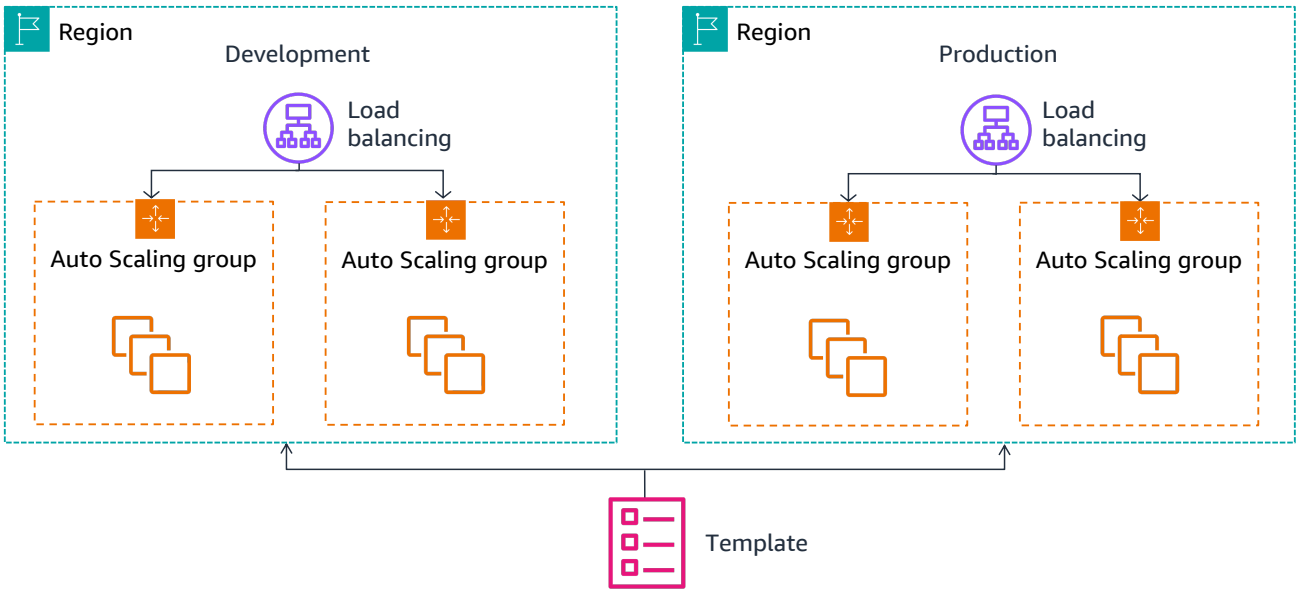
Benefits of IaC – Reusability



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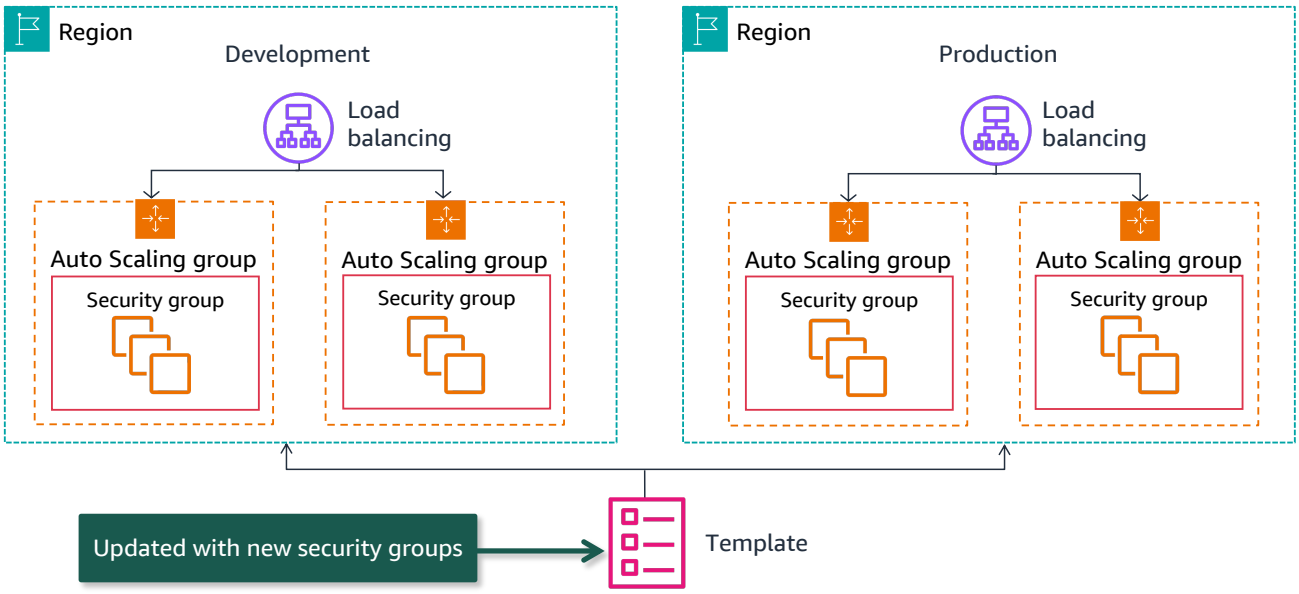
Benefits of IaC – Updates



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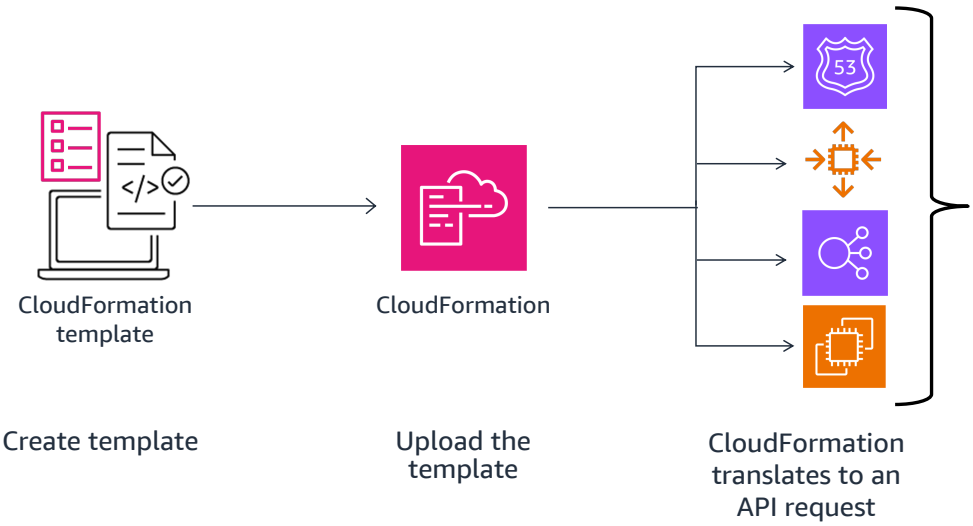
AWS CloudFormation



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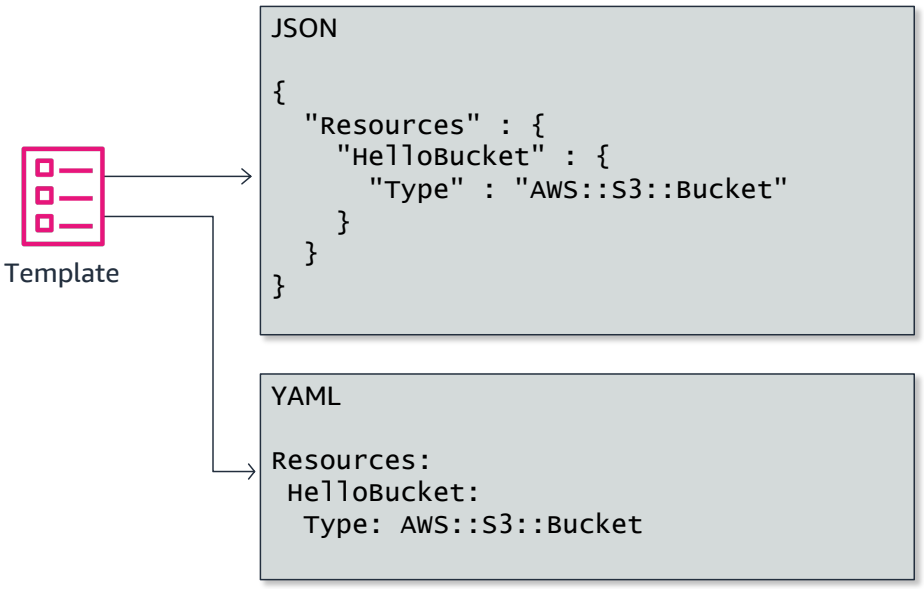
Understanding CloudFormation

- Written as JSON or YAML
- Describes the resources to be created or modified
- Treated as source code:
  - Code reviewed
  - Version controlled



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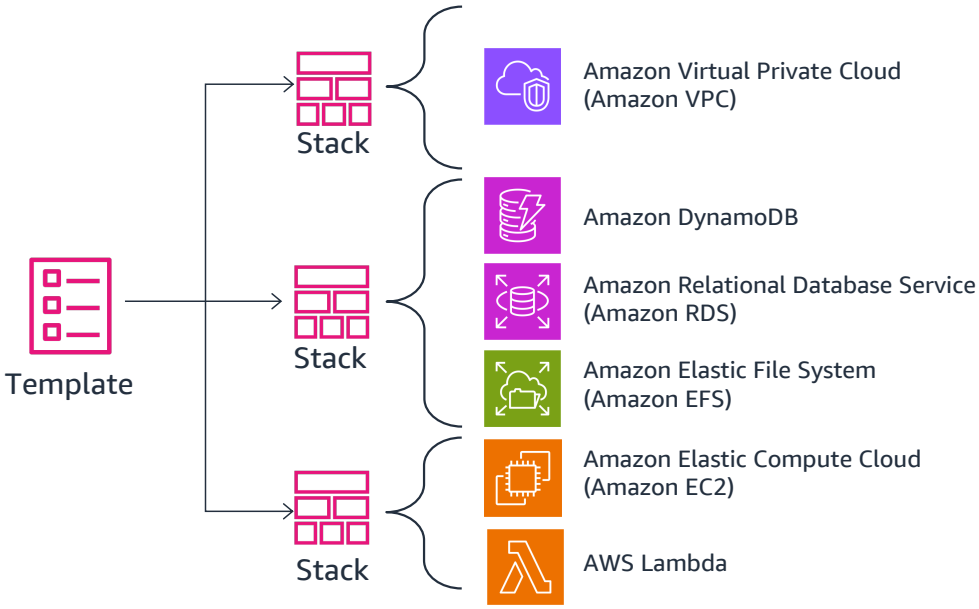
Stacks

- A collection of AWS resources that are managed as a single unit
- Can deploy and delete resources as a unit
- Can update resources and settings on running stacks
- Supports nested stacks and cross-stack references



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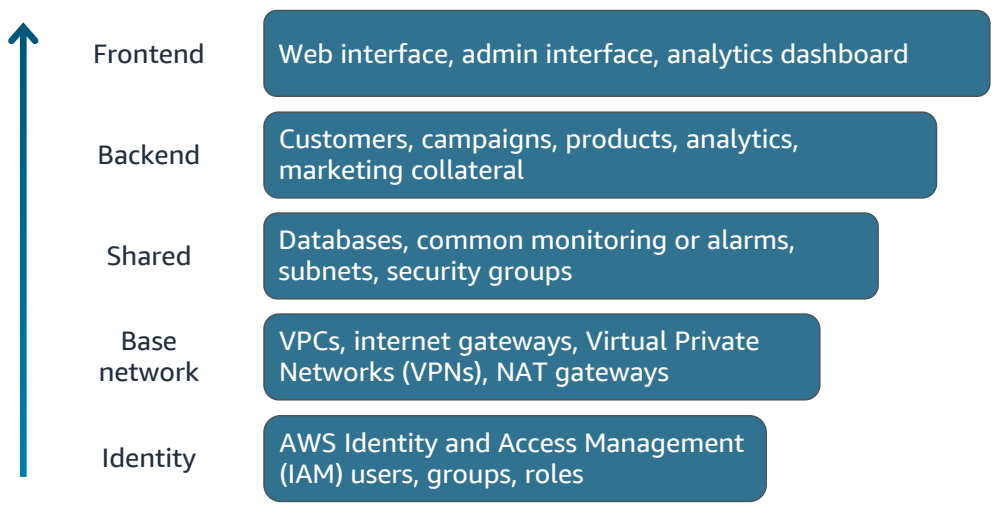
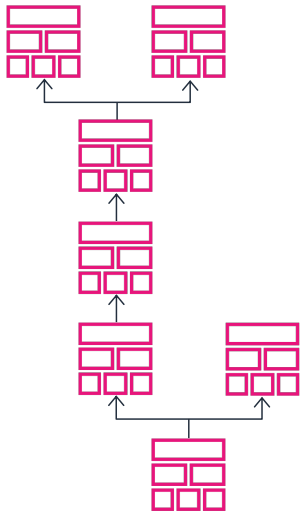
Using multiple templates

A layered architecture





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Infrastructure management

"How can we deploy, maintain, and scale applications in the cloud?"

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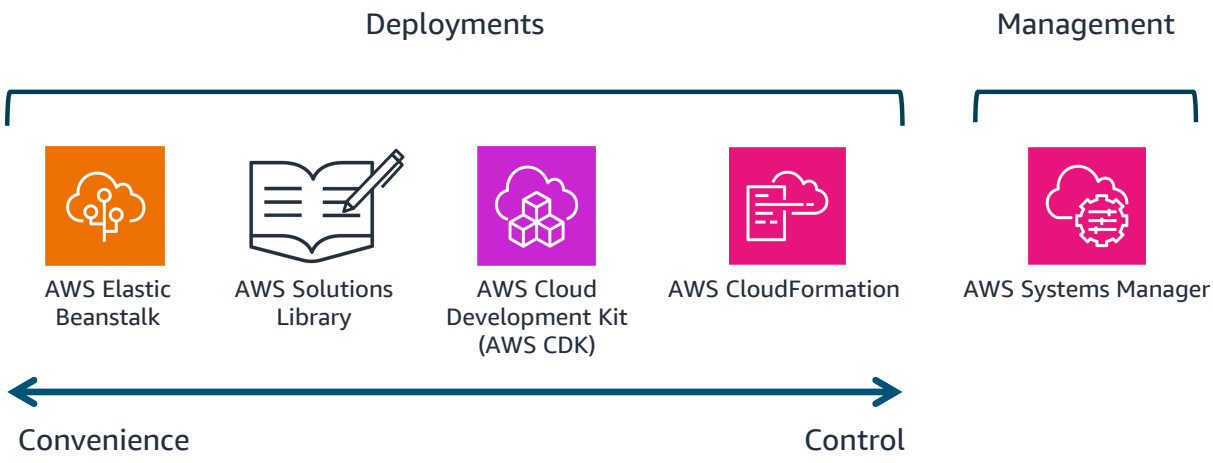
Infrastructure tools



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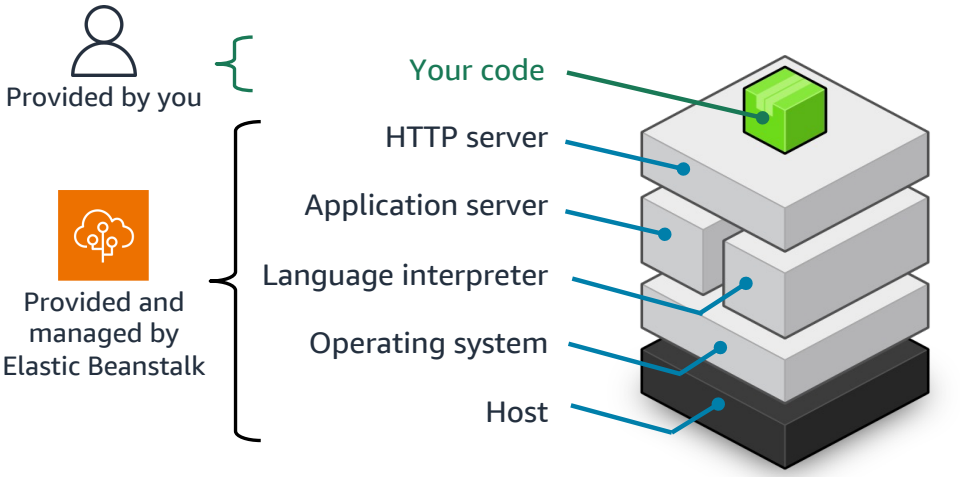
AWS Elastic Beanstalk

- Provisions and operates the infrastructure
- Manages the application stack for you
- Shows everything that is created
- Automatically scales your application up and down



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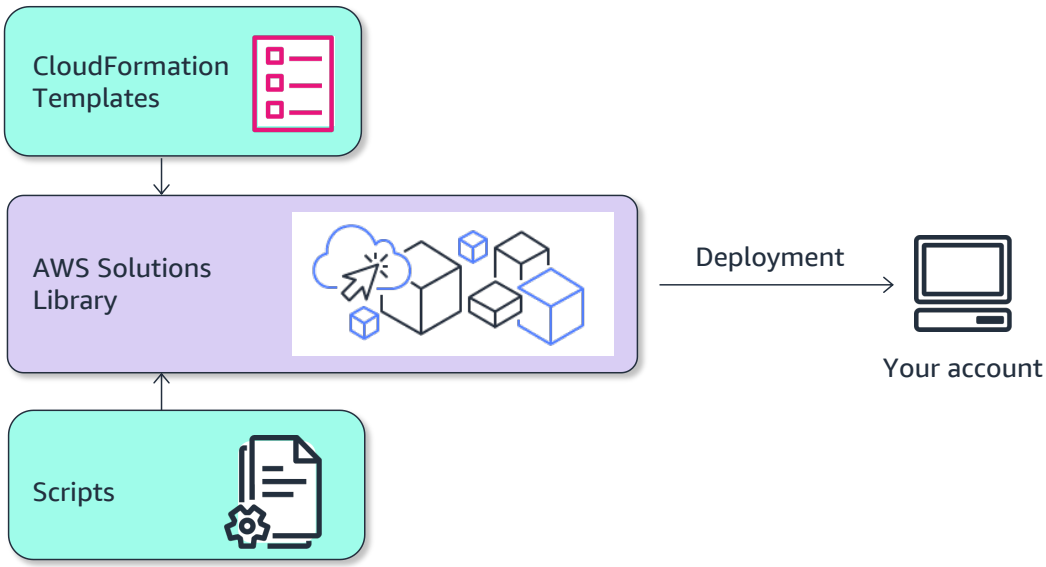
AWS Solutions Library

- Prebuilt reference architectures
- Deployment accelerator
- Solutions approved by AWS architects



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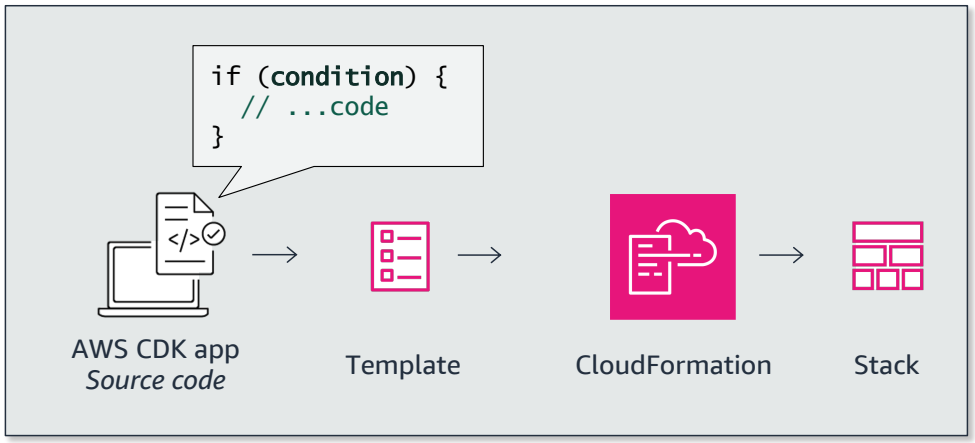
AWS CDK

- Uses any supported language to generate templates
- Supports autocomplete and in-line documentation
- Has proven defaults and reusable classes
- Provisions multiple environments



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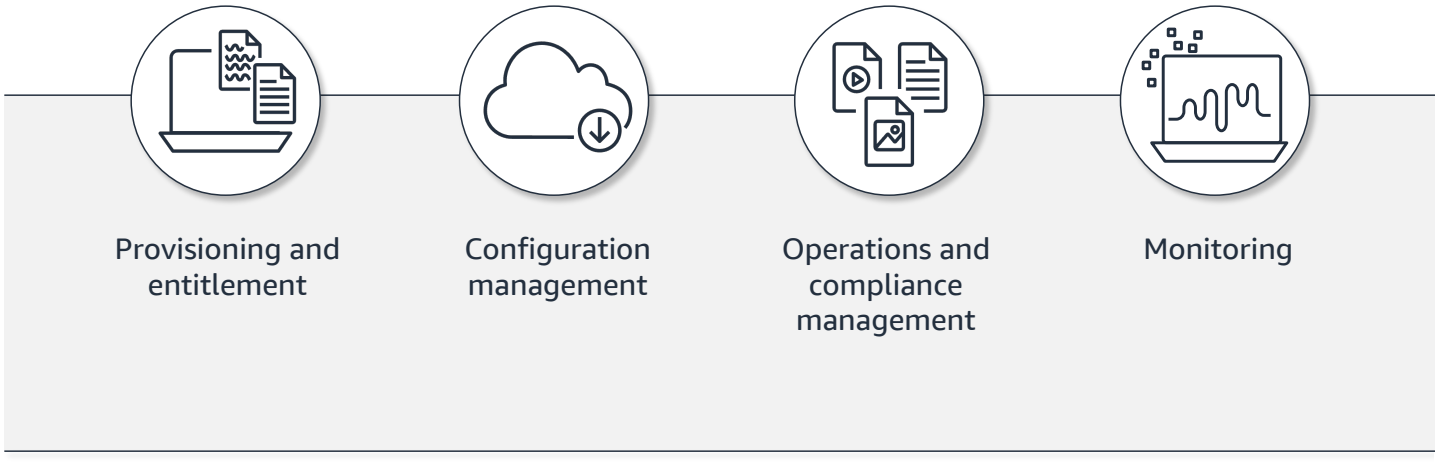
AWS Systems Manager



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Amazon Q

Build applications faster and more securely with your AI coding companion

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Amazon Q: Generative AI-powered assistant

- Secure and private by design
- Can be customized with your own data
- Provides assistants that are designed for specific use cases

Amazon Q Developer

- Designed for developers and IT professionals
- Generates code and helps you understand, build, extend, and operate AWS applications

Amazon Q Business

- Designed for business analysts and every employee
- Answers questions, generates and summarizes content, and completes non-coding tasks

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Amazon Q Developer supports you across the software development lifecycle (SDLC)

Plan	Create	Test and secure	Operate	Maintain and modernize
• Ask questions and get referenceable and contextual guidance. • Explain code with conversational coding.	• Receive in-line code recommendations for multiple languages. • Implement features through comments or code prompts. • Chat in your integrated development environment (IDE).	• Generate unit tests. • Scan project code for security vulnerabilities and get remediation suggestions.	• Troubleshoot errors. • Troubleshoot network connectivity with VPC Reachability Analyzer.	• Modernize code with Amazon Q Developer Agent for code transformation.

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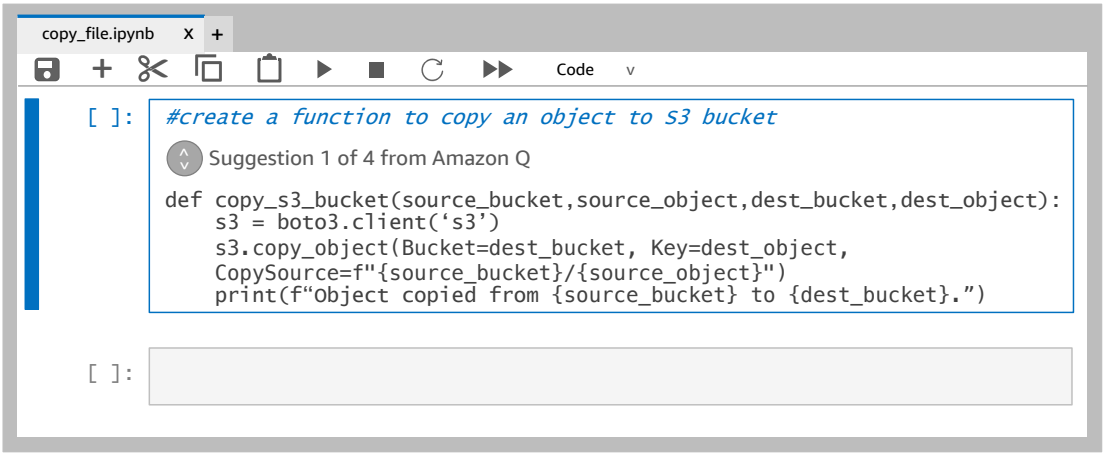
Amazon Q Developer: Code recommendations

- Recommends code based on plain English comments and prior code
- You accept or reject recommendations



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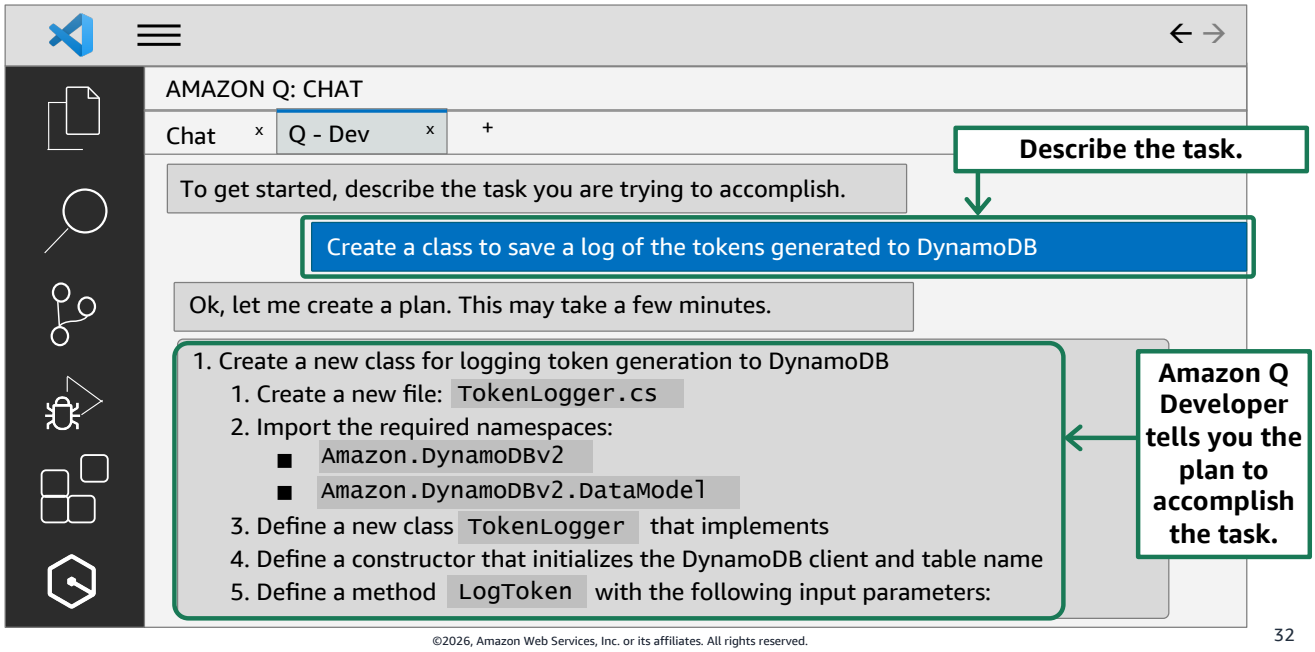
Plan and implement new functionality across files in your workspace



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Present solutions

Chief technology officer

Consider how you would answer the following:

- How can we simplify our cloud infrastructure build?
- How can we deploy, maintain, and scale applications in the cloud?

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Module review

In this module, you learned about the following:

- ✓ Amazon CloudFormation
- ✓ Amazon Q
- ✓ Infrastructure management

Next, you review the following:

Knowledge check

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Knowledge check

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Knowledge check question 1

What is an AWS CloudFormation stack?

Choice	Response
A	All of the provisioned resources that are defined in a CloudFormation template
B	All of the resources that are identified as drifted in a CloudFormation template
C	A condition in which resources are added on top of each other
D	The properties of a single resource

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Knowledge check question 1 and answer

What is an AWS CloudFormation stack?

Choice	Response
A correct	All of the provisioned resources that are defined in a CloudFormation template
B	All of the resources that are identified as drifted in a CloudFormation template
C	A condition in which resources are added on top of each other
D	The properties of a single resource

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Knowledge check question 2

Which statements are benefits of using AWS Cloud Development Kit (AWS CDK) with AWS CloudFormation? (Select TWO.)

Choice	Response
A	Developers can use common programming languages.
B	Bulk discounts are automatically applied to resource usage.
C	Developers can call preconfigured resources with proven defaults.
D	Components are limited to a single user.
E	Using AWS CDK does not require an AWS account or credentials.

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Knowledge check question 2 and answer

Which statements are benefits of using AWS Cloud Development Kit (AWS CDK) with AWS CloudFormation? (Select TWO.)

Choice	Response
A correct	Developers can use common programming languages.
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End of Module 8

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Architecting on AWS

Module 9: Containers

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Poll question

What percentage of your workloads run on containers?

A. Less than 10 percent

B. 10–50 percent

C. More than 50 percent

D. I'm not sure

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Module overview

- Business request
- Microservices
- Containers
- Container services
- Present solutions
- Knowledge check

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Business request

Compute operations manager

The compute operations manager wants to know the following:

- How can we make components of our applications more independent so that changes in one service will not affect any others?
- What are the benefits of using containers for our compute needs?
- What options do we have for managing containerized applications in the cloud?

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Microservices overview

"How can we make components of our applications more independent so that changes in one service will not affect any others?"

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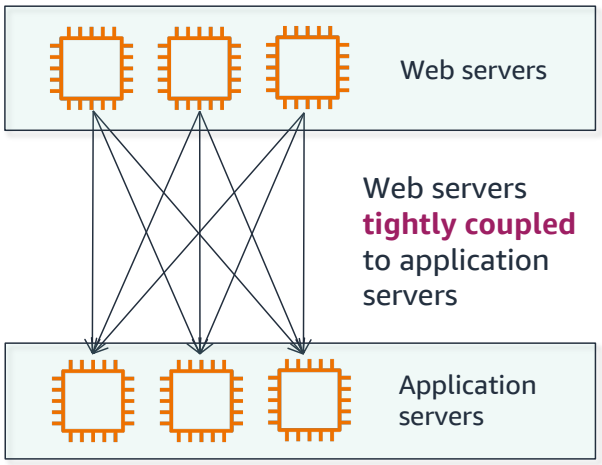
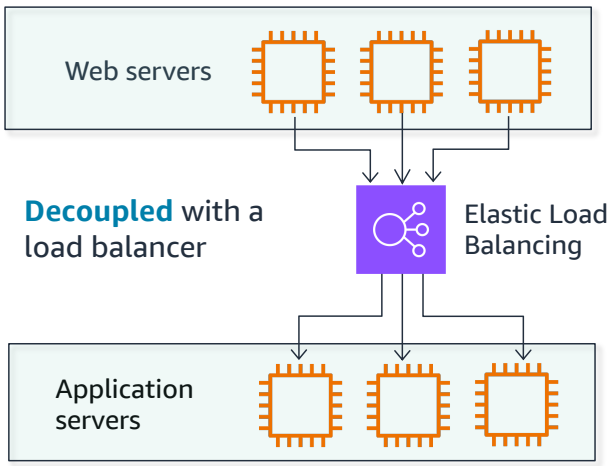
Loose coupling

Anti-pattern



Best practice



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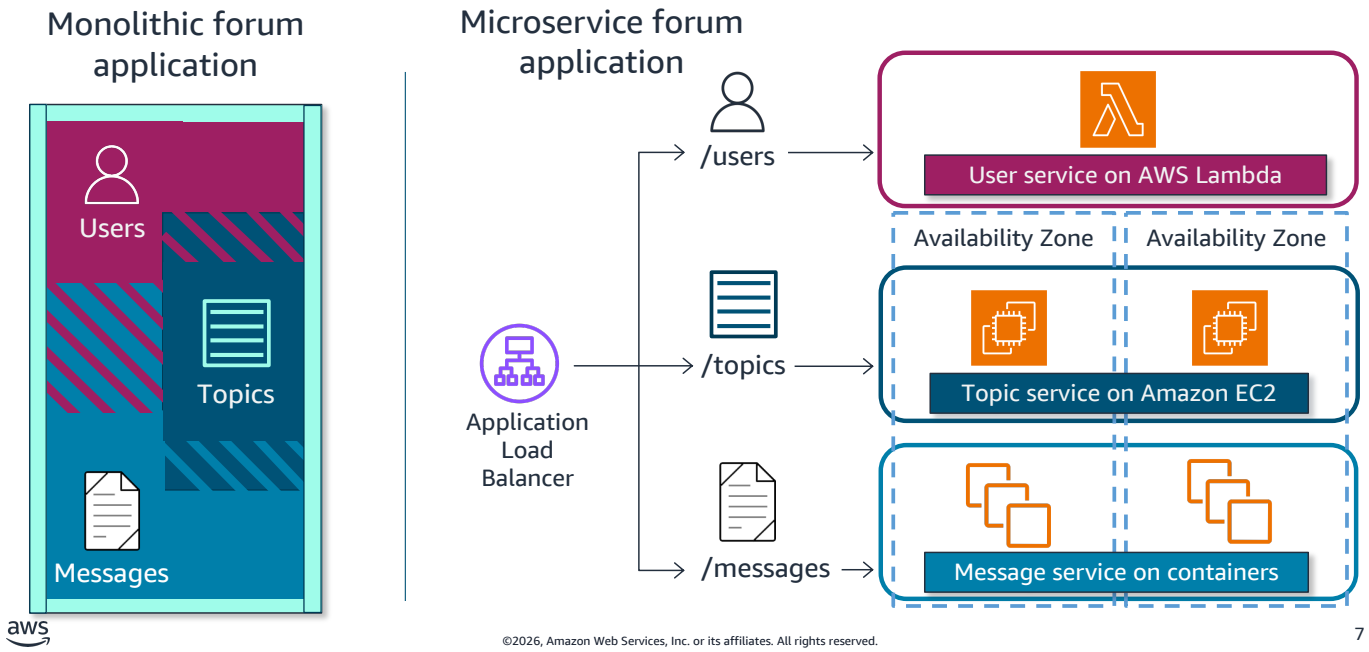
Microservices



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Containers overview

"What are the benefits of using containers for our compute needs?"

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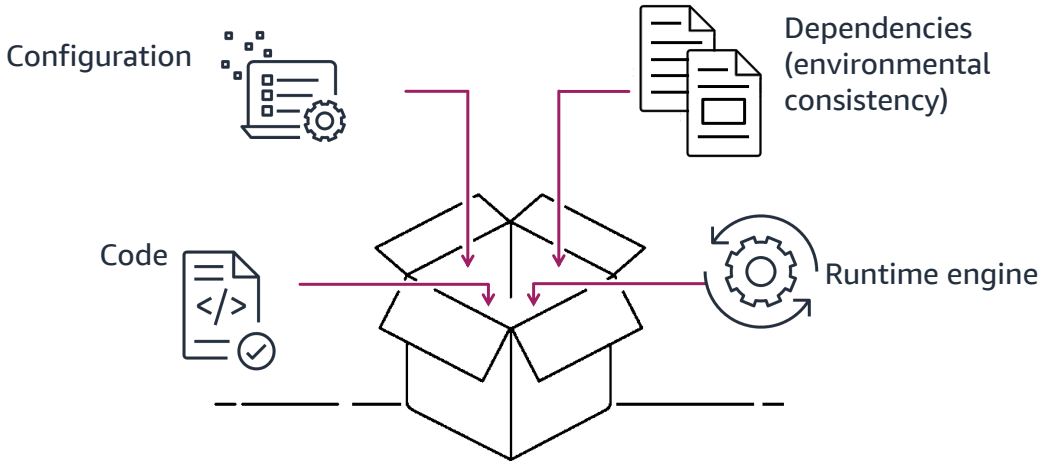
Containers

Containers are:

- Repeatable
- Self-contained environments
- Faster to spin up and down than virtual machines (VMs)
- Portable
- Scalable



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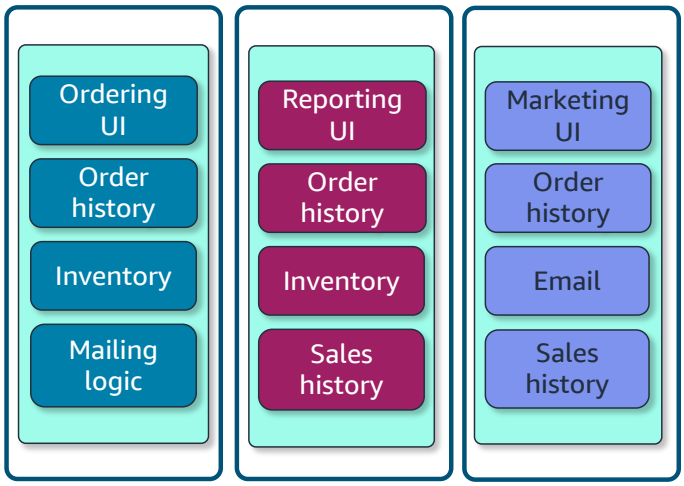
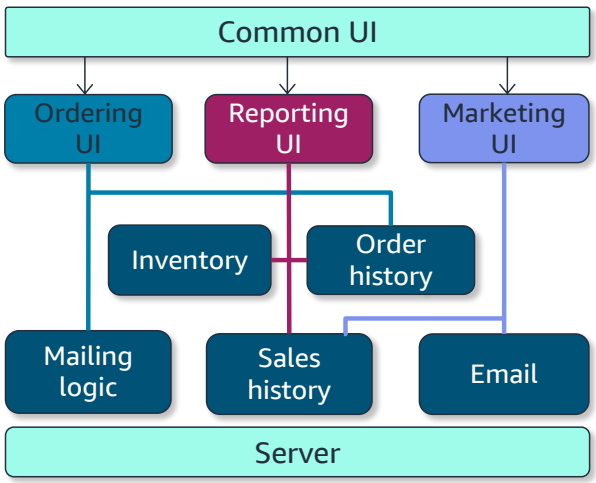
Containers and microservices

Monolithic order application



Microservice order application



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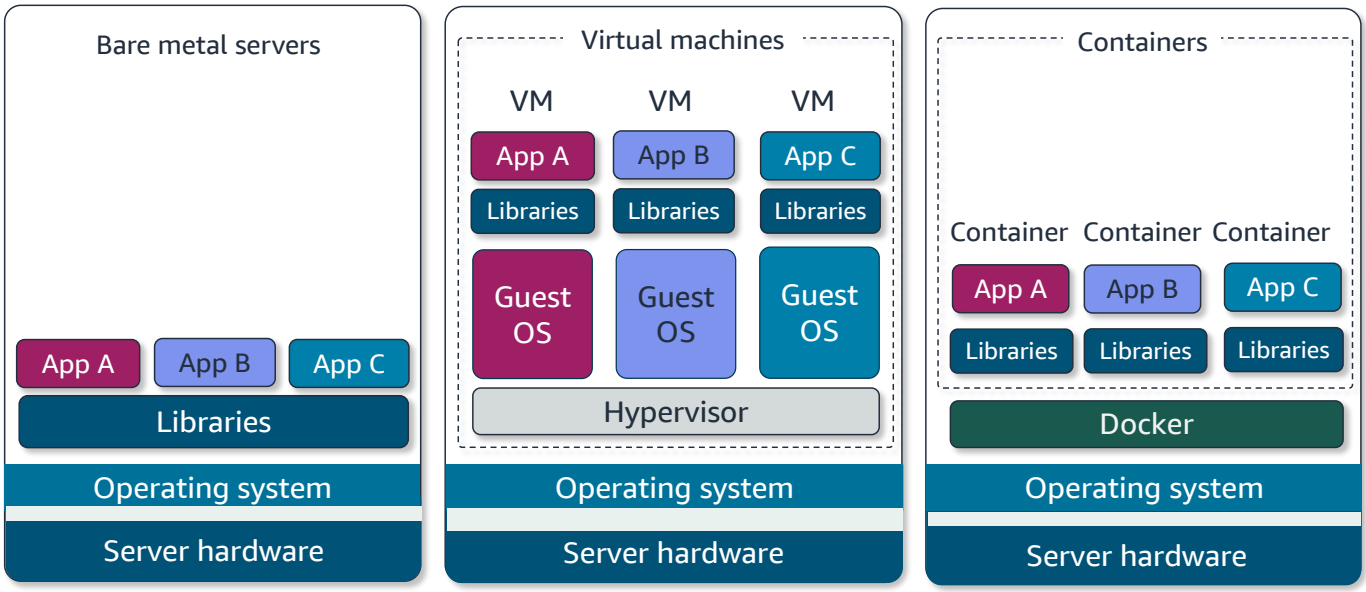
Levels of abstraction and virtualization



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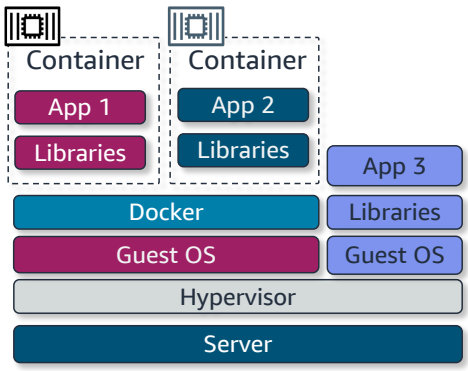
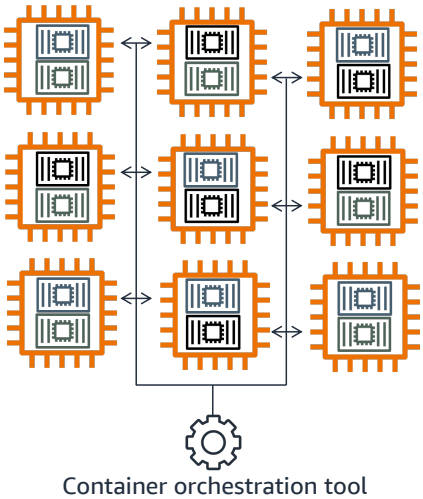
Containers on AWS

- Running containers directly on Amazon EC2 requires you to manage scaling, connectivity, and maintenance.
- Using an orchestration tool helps manage:
  - Scheduling
  - Placement
  - Networking
  - Monitoring

Containers on Amazon EC2



Containers with an orchestration tool



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Container services

"What options do we have for managing containerized applications in the cloud?"

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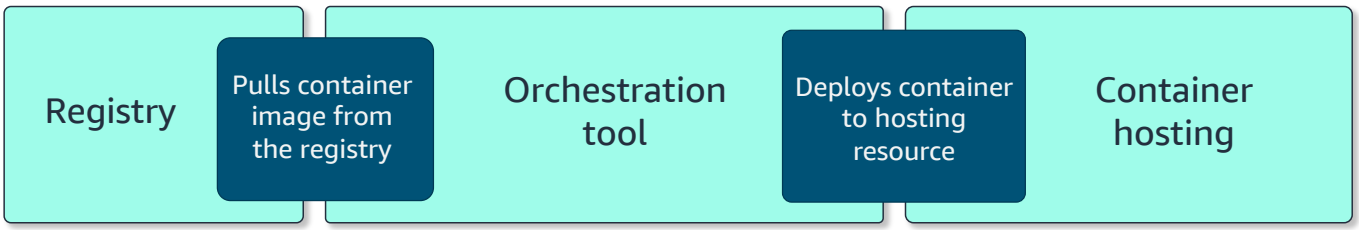
Running containers on AWS



Amazon Elastic Container Registry (Amazon ECR)

Amazon Elastic Kubernetes Service (Amazon EKS)

Amazon Elastic Container Service (Amazon ECS)

AWS Fargate

Amazon EC2

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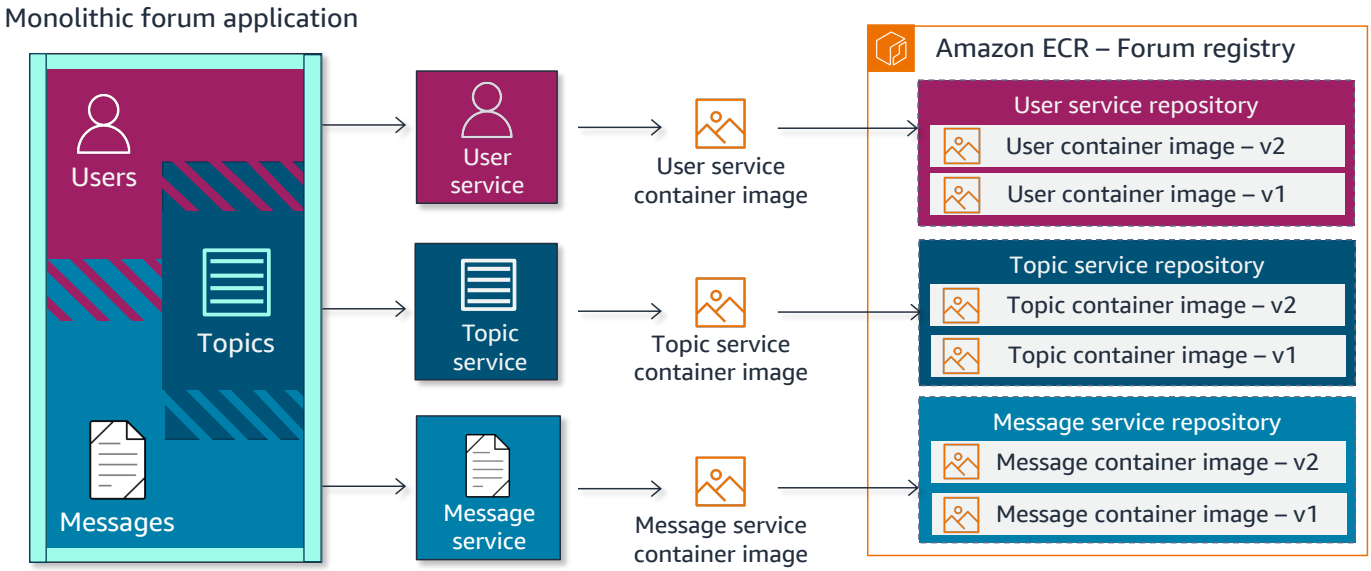
Amazon ECR



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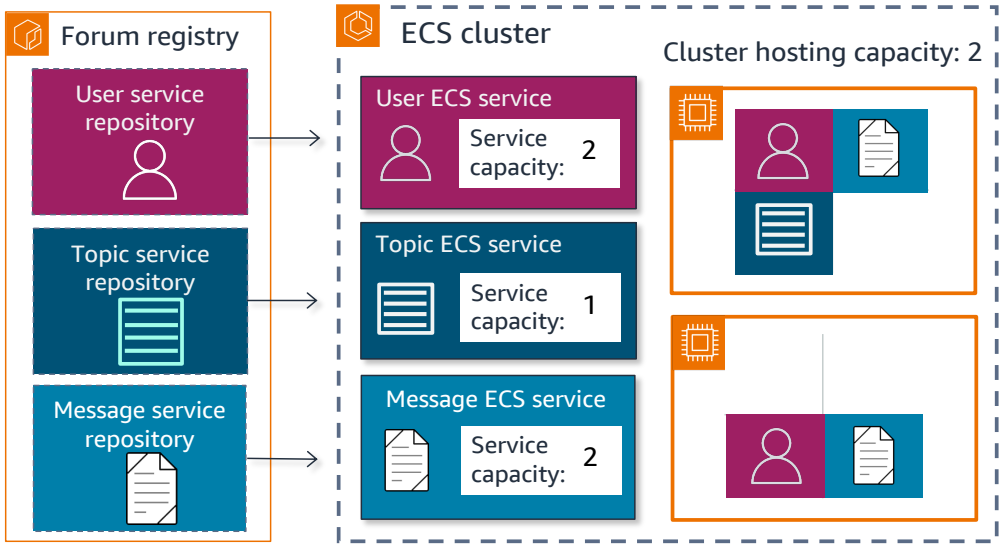
Amazon ECS orchestration

Managed container orchestration service tightly integrated with AWS:

- Pulls images from your repositories
- ECS services scale service capacity by managing container count
- ECS clusters scale hosting capacity



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Amazon ECS features

Amazon ECS

Fully managed

Service discovery

AWS integrations

Works with common development workflows

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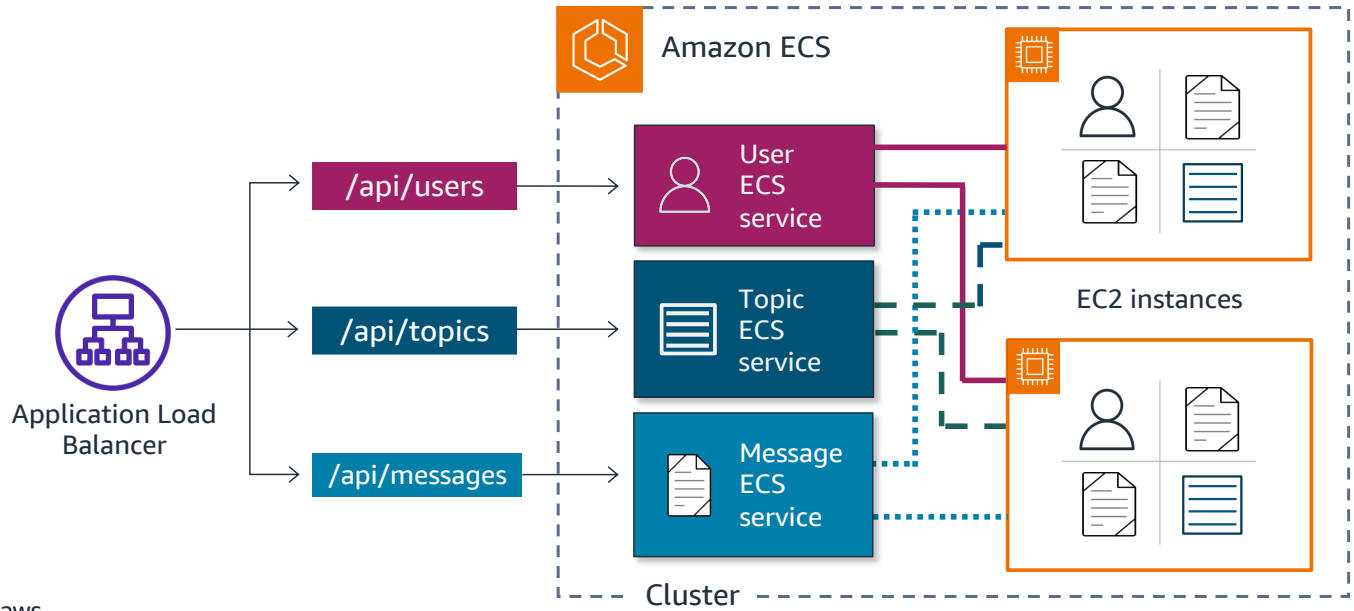
Monolithic to container-based microservices



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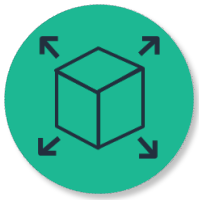
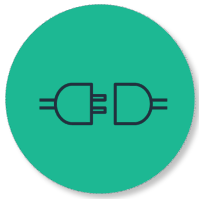
Amazon EKS



Run applications at scale

Seamlessly move applications

Run anywhere



Add new functionality

aws

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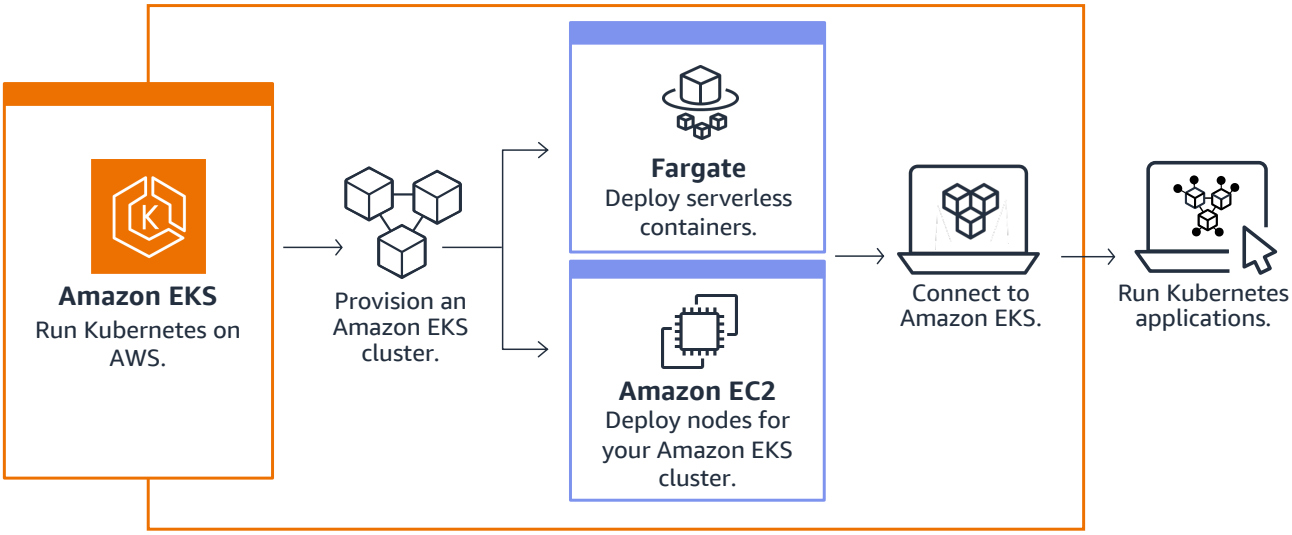
Amazon EKS solutions



aws

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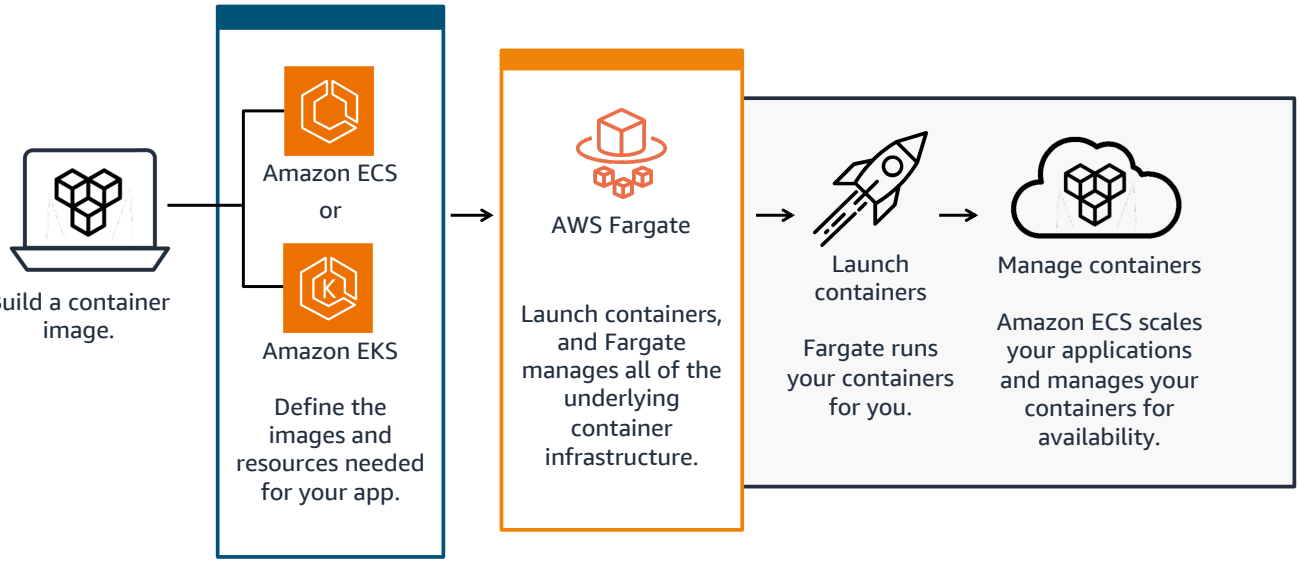
AWS Fargate



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Choosing AWS container services

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Review

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Present solutions

Compute operations manager

Consider how you would answer the following questions:

- How can we make components of our applications more independent so that changes in one service will not affect any others?
- What are the benefits of using containers for our compute needs?
- What options do we have for managing containerized applications in the cloud?

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Module review

In this module, you learned about the following:

- ✓ Microservices
- ✓ Containers
- ✓ Container services

Next, you review the following:

Knowledge check

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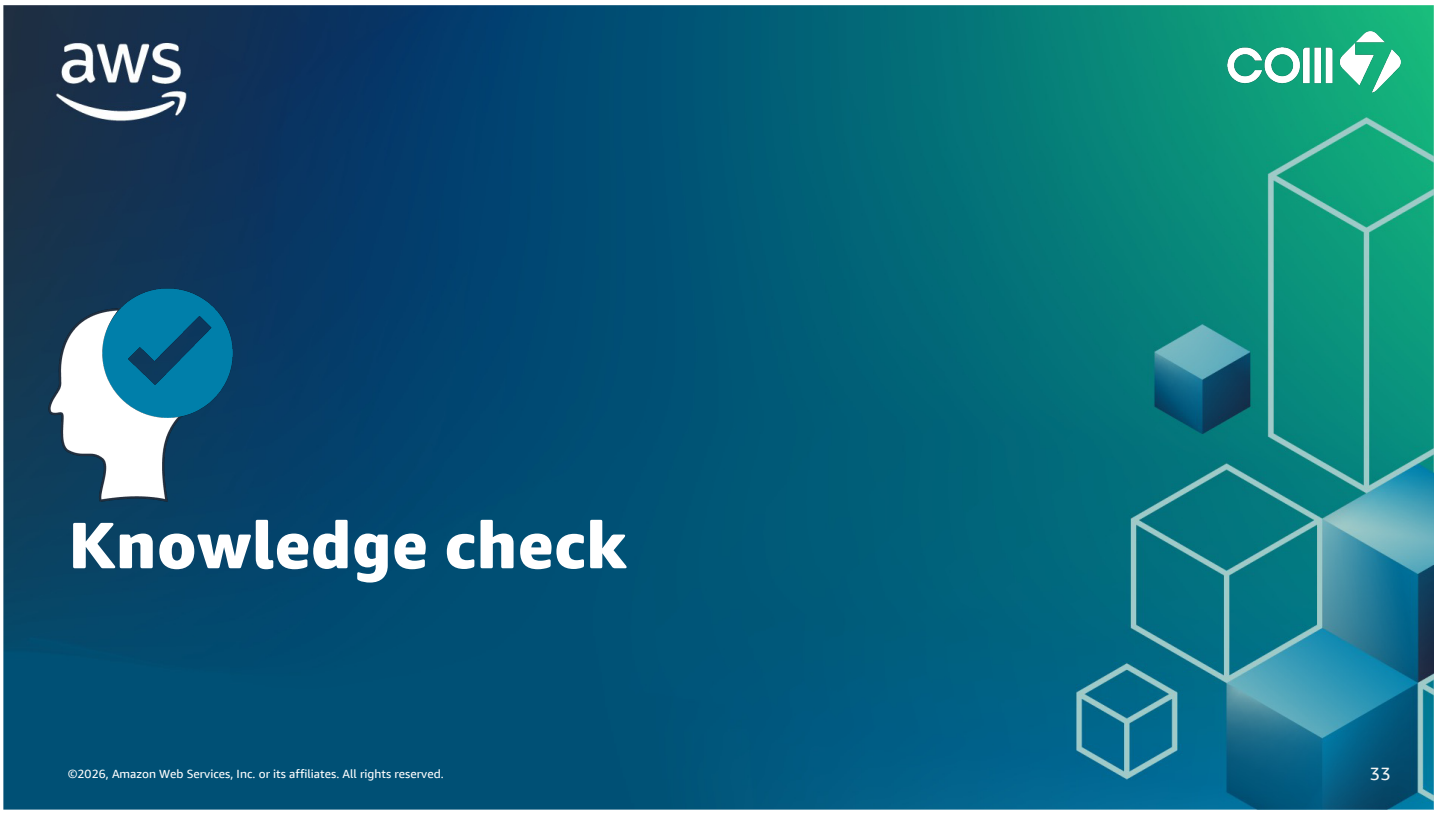

Knowledge check question 1

Which descriptions are characteristics of microservices? (Select TWO.)

Choice	Response
A	Loosely coupled
B	Redundant
C	Autonomous and independent
D	Tightly integrated and dependent
E	Interdependent components

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Knowledge check question 1 and answer

Which descriptions are characteristics of microservices? (Select TWO.)

Choice	Response
A **correct**	Loosely coupled
B	Redundant
C **correct**	Autonomous and independent
D	Tightly integrated and dependent
E	Interdependent components

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Knowledge check question 2

Which descriptions are characteristics of containers? (Select TWO.)

Choice	Response
A	Portable and scalable
B	Require a hypervisor
C	Automatic
D	Repeatable
E	Each requires its own operating system

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Knowledge check question 2 and answer

Which descriptions are characteristics of containers? (Select TWO.)

Choice	Response
A **correct**	Portable and scalable
B	Require a hypervisor
C	Automatic
D **correct**	Repeatable
E	Each requires its own operating system

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Knowledge check question 3

How are containers in Amazon Elastic Container Service (Amazon ECS) logically organized?

Choice	Response
A	In a cluster
B	In pods
C	In Amazon Elastic Block Store (Amazon EBS) volumes
D	In Amazon S3

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Knowledge check question 3 and answer

How are containers in Amazon Elastic Container Service (Amazon ECS) logically organized?

Choice	Response
A correct	In a cluster
B	In pods
C	In Amazon Elastic Block Store (Amazon EBS) volumes
D	In Amazon S3

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Knowledge check question 4

Why would developers choose to deploy their containers to AWS Fargate over Amazon EC2?

Choice	Response
A	To take control of their infrastructure
B	To avoid manual infrastructure updates
C	To optimize price for a large load
D	To manage their own patches and updates

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Knowledge check question 4 and answer

Why would developers choose to deploy their containers to AWS Fargate over Amazon EC2?

Choice	Response
A	To take control of their infrastructure
B correct	To avoid manual infrastructure updates
C	To optimize price for a large load
D	To manage their own patches and updates

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End of Module 9

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Architecting on AWS

Module 10: Networking 2

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Poll question

How many virtual private clouds (VPCs) does your organization use?

A. <20

B. 20–100

C. >100

D. I'm not sure

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Module overview

- Business request
- VPC endpoints
- VPC peering
- Hybrid networking
- AWS Transit Gateway
- Present solutions
- Knowledge check

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Business request

Network engineer

The network engineer needs to know the following:

- What can we do to keep our connections to AWS services private?
- How can we privately route traffic between our VPCs?
- What are our options to connect our on-premises network to the AWS Cloud?
- Which services can reduce the number of route tables that we need to manage our global network?

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VPC endpoints

"What can we do to keep our connections to AWS services private?"

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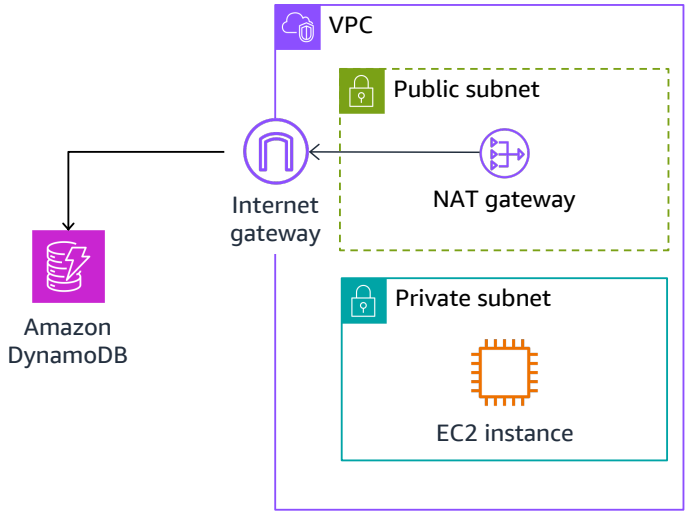
Without VPC endpoints

To consume AWS services from the instance in a private subnet you will need:

- Internet gateway
- NAT gateway
- Public IP address



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Interface VPC endpoints

- Supports access to 200+ AWS services
- Is an elastic network interface with a private IP address
- Horizontally scaled
- Redundant
- Highly available
- Powered by AWS PrivateLink

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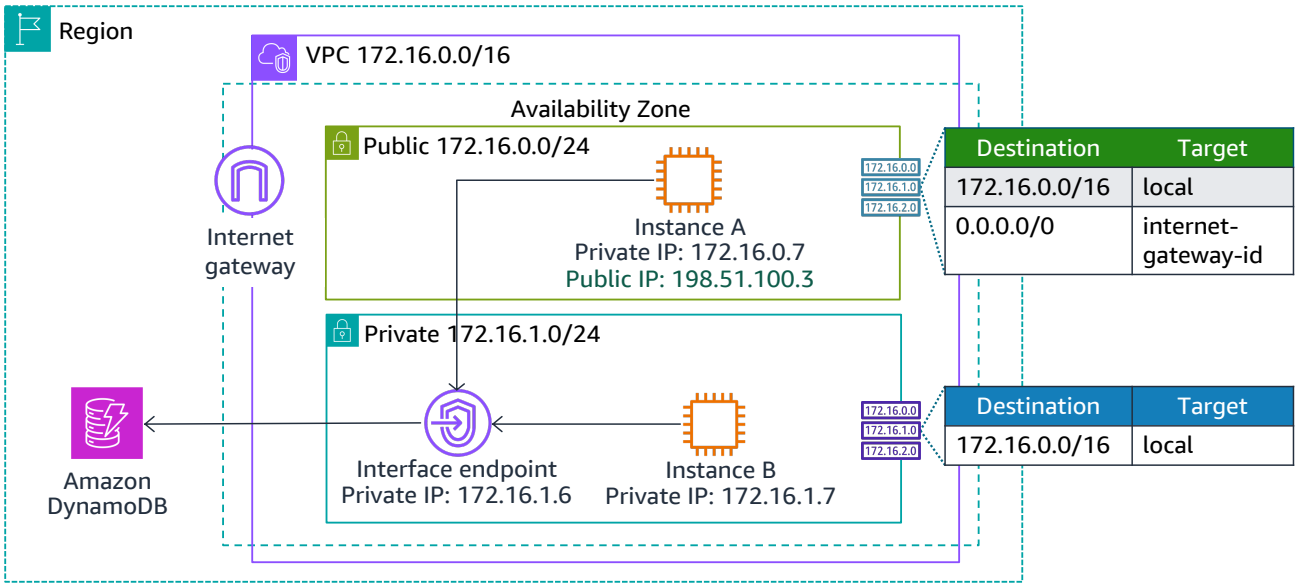
Example interface VPC endpoints



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VPC peering

"How can we privately route traffic between our VPCs?"

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VPC peering introduction

VPC peering connects networks between two VPCs.

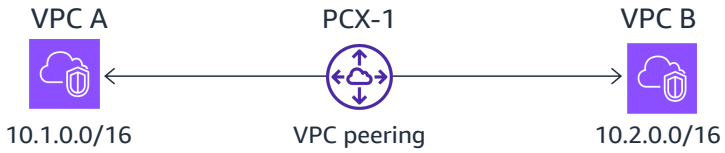
- Intra-Region and inter-Region support
- Cross-account support

Route table: VPC A

Destination	Target
10.1.0.0/16	local
10.2.0.0/16	PCX-1

Route table: VPC B

Destination	Target
10.2.0.0/16	local
10.1.0.0/16	PCX-1



Note: IP spaces cannot overlap

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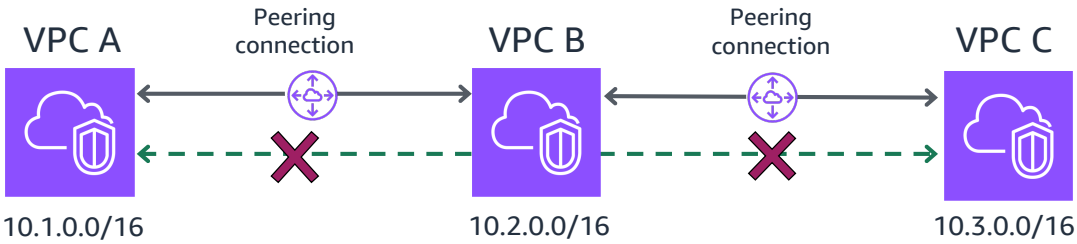
Multiple VPC peering connections



Note: No transitive peering relationships

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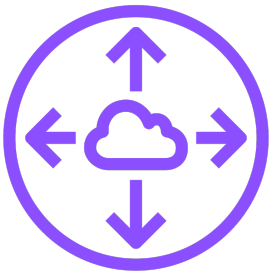
Benefits of VPC peering



- Bypasses the internet gateway or virtual private gateway
- Provides highly available connections—no single point of failure
- Avoids bandwidth bottlenecks
- Uses private IP addresses to direct traffic between VPCs

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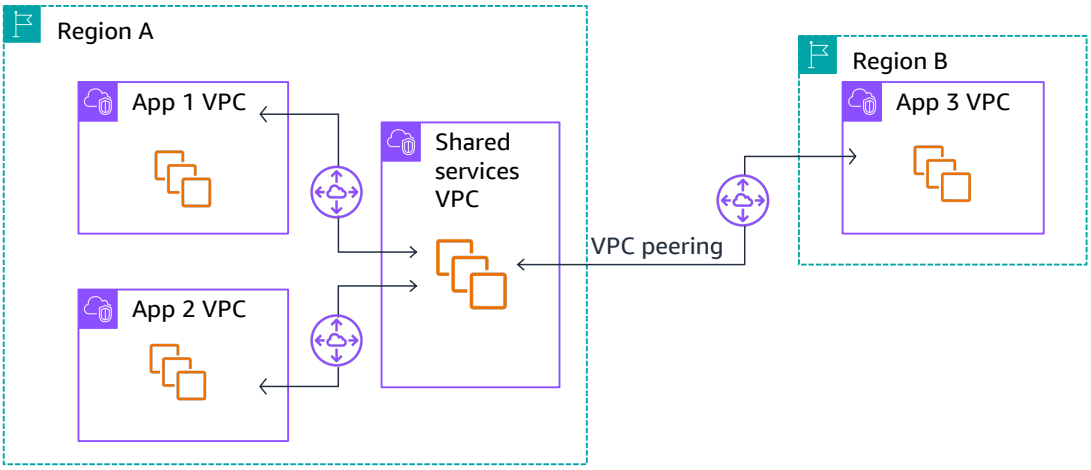
Example: VPC peering for shared services

- App VPCs have no peering with each other.
- You cannot use the shared services VPC as a transit point between app VPCs.



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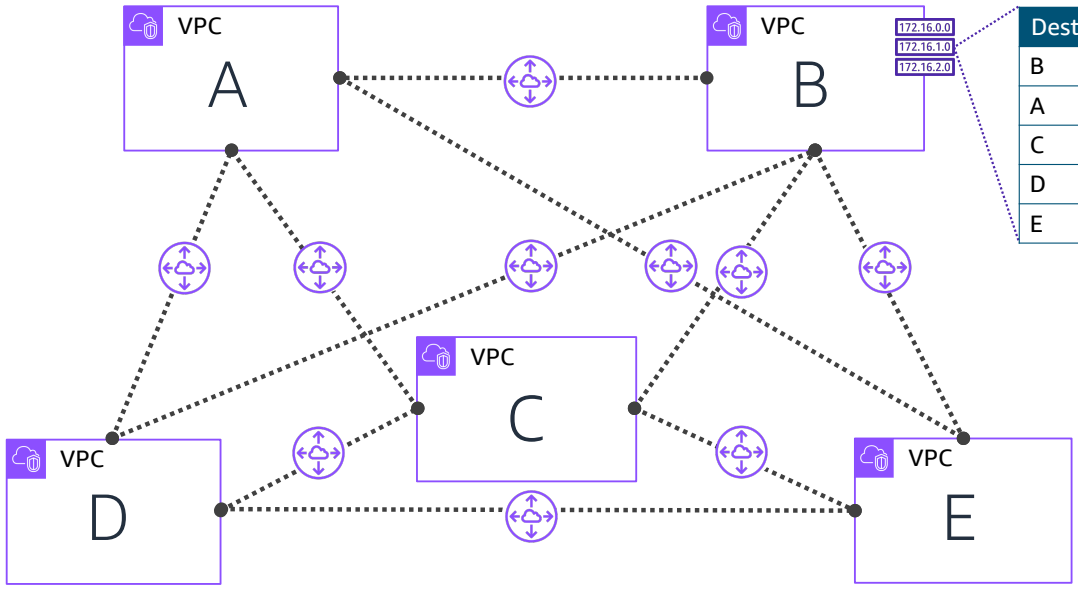
Example: Full mesh VPC peering



Destination	Target
B	Local
A	PCX-1
C	PCX-2
D	PCX-3
E	PCX-4

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Hybrid networking

"What are our options to connect our on-premises network to the AWS Cloud?"

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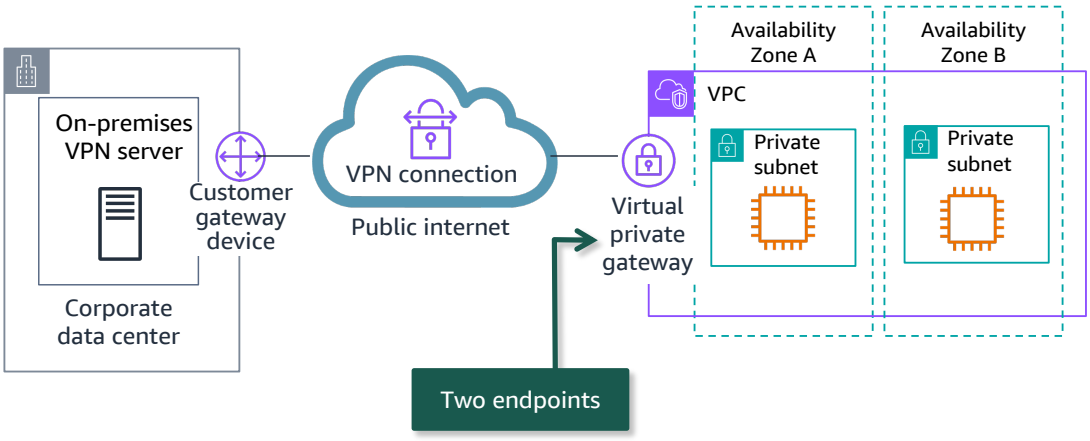
AWS Site-to-Site VPN

- Managed connection
- Static or dynamic VPN



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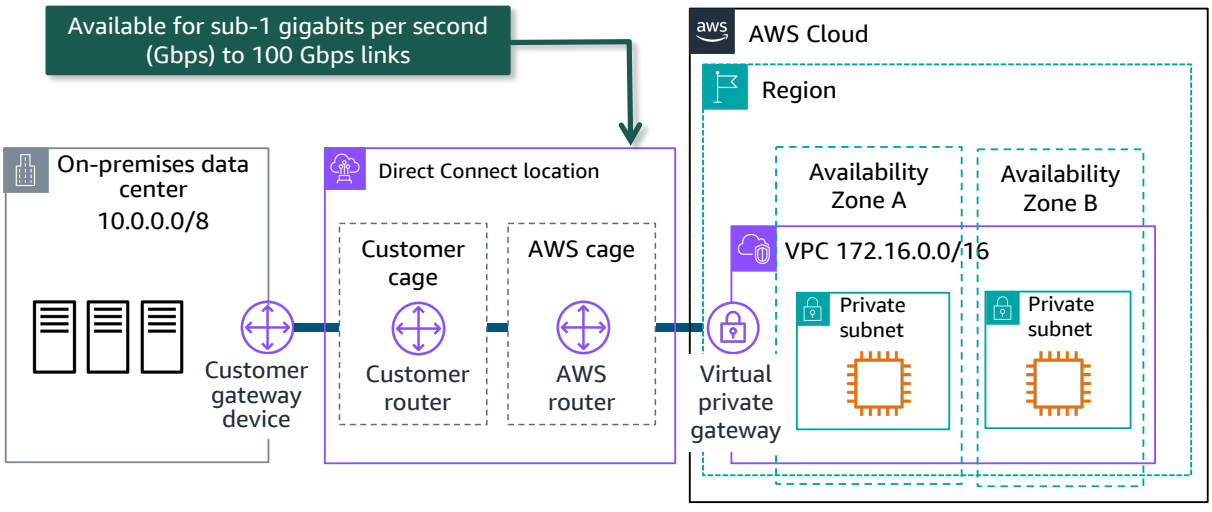
AWS Direct Connect

Create a fiber link from your data center to your AWS resources.



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AWS Site-to-Site VPN and Direct Connect pricing

Site-to-Site VPN

- Connection fee (per hour)
- Data transfer out (DTO)
  - Measured in gigabyte (GB)
  - First 100 GB are at no charge

Direct Connect

- Capacity
  - Measured in megabits per second (Mbps) or gigabits per second (Gbps)
- Port hours
  - Time that a port is provisioned for your use in the data center
- Data transfer out (DTO)
  - Measured in GB

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Choosing AWS VPN or Direct Connect

AWS Site-to-Site VPN	Direct Connect
Limited to 1.25 Gbps connection maximum	Sub-1, 1, 10, or 100 Gbps connection options
Faster to configure than Direct Connect	Requires special agreements and physical cabling to the data center
Don’t need to pay for inactive connections	Pay for port hours whether the connection is active or not
Encrypted in transit by default, but travels over public internet	Not encrypted by default, but a private, dedicated connection

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AWS Transit Gateway

"Which services can reduce the number of route tables that we need to manage our global network?"

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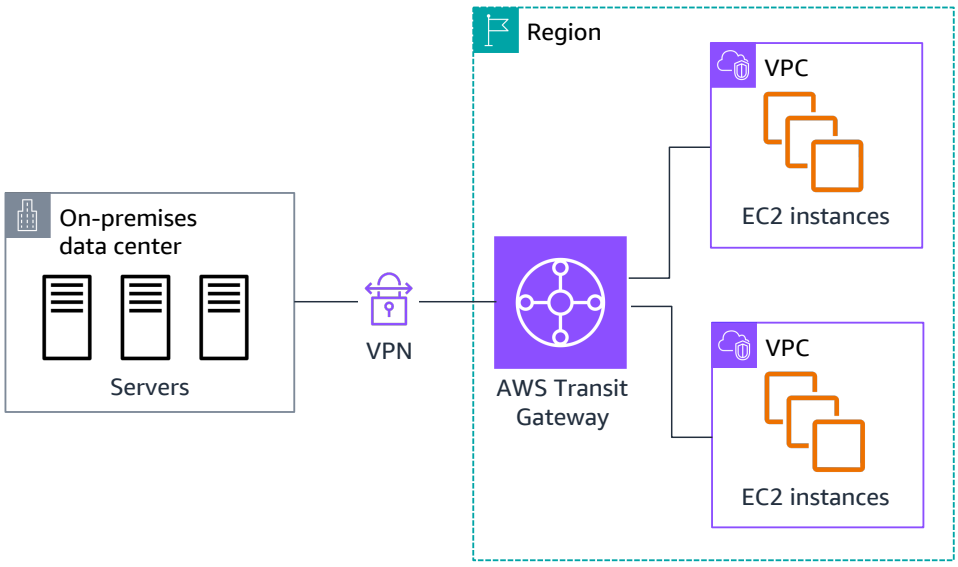
Transit Gateway

- Connects up to 5,000 VPCs and on-premises environments
- Acts as a hub for all traffic to flow through
- Permits multicast and inter-Region peering



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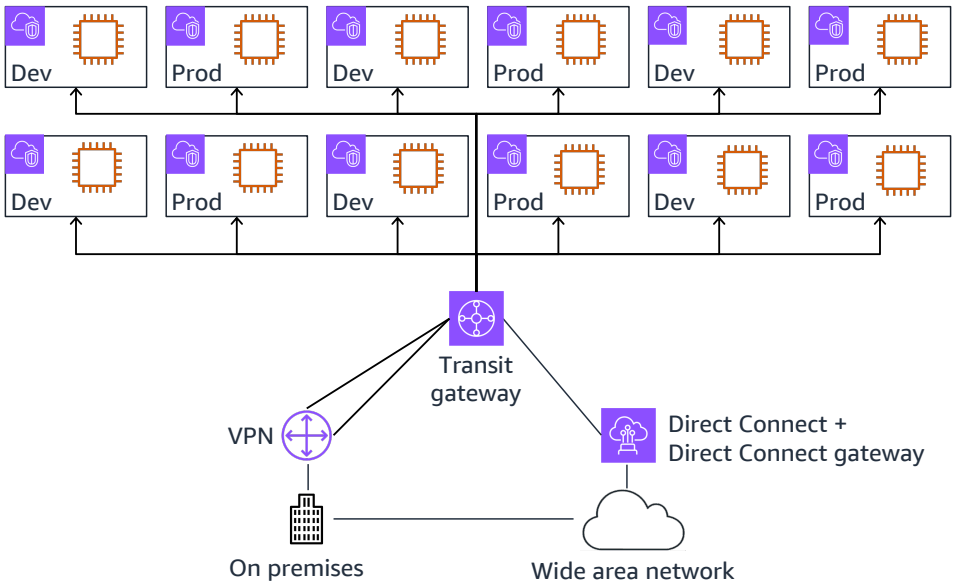
Scaling your network with Transit Gateway

- Attachment-based
- Flexible routing and segmentation
- Simplified connections
- Highly available and scalable



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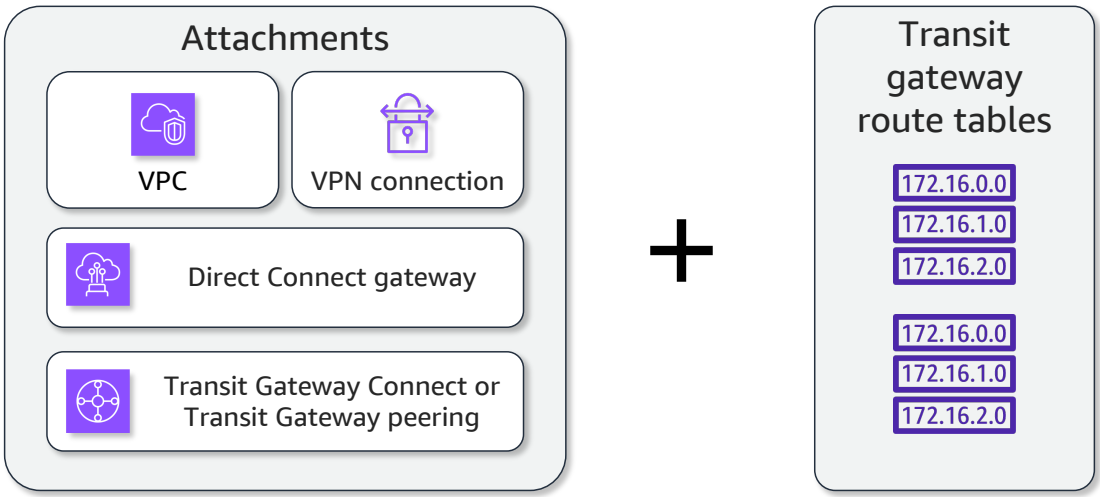
Transit Gateway components



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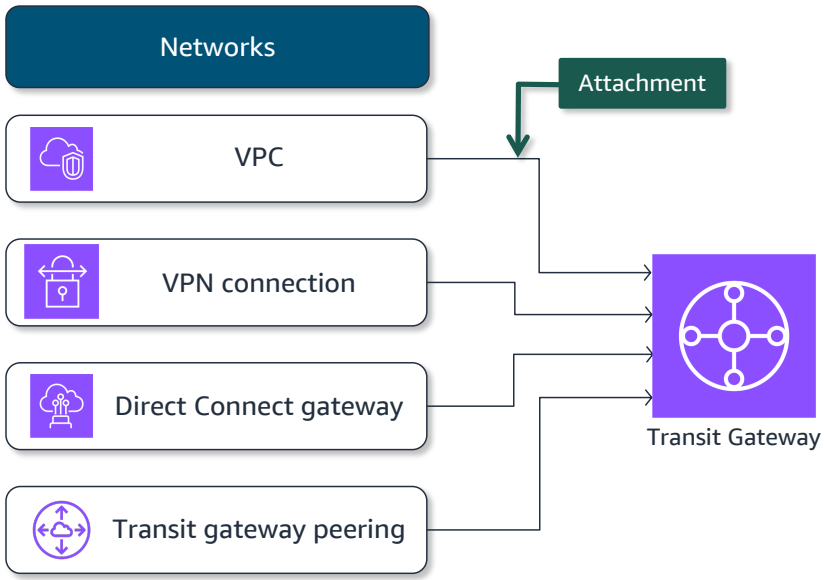
Transit Gateway setup

- Attach VPCs, VPN, Direct Connect gateway, and Transit Gateway peering connections.
- Network attachments must be in the same Region as the transit gateway.



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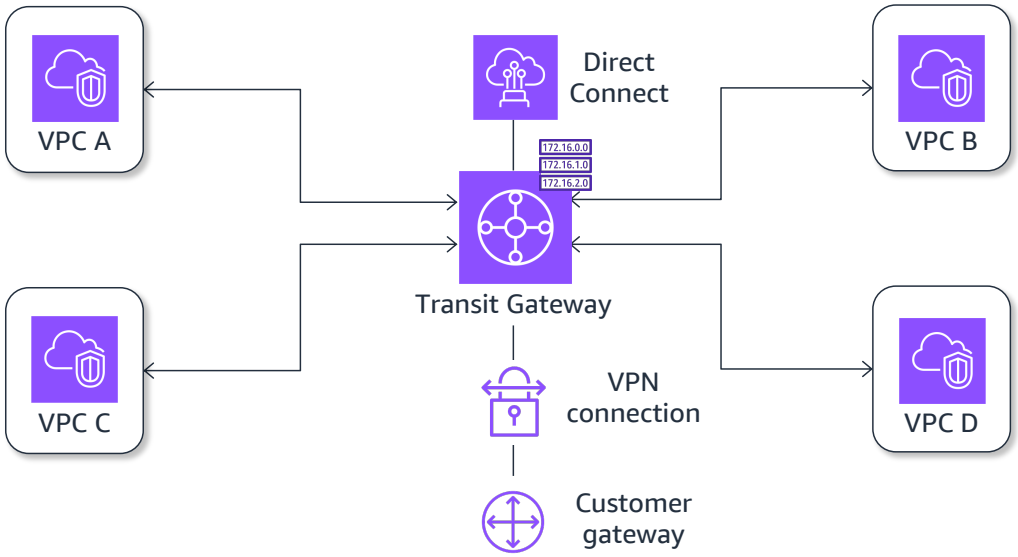
Full connectivity



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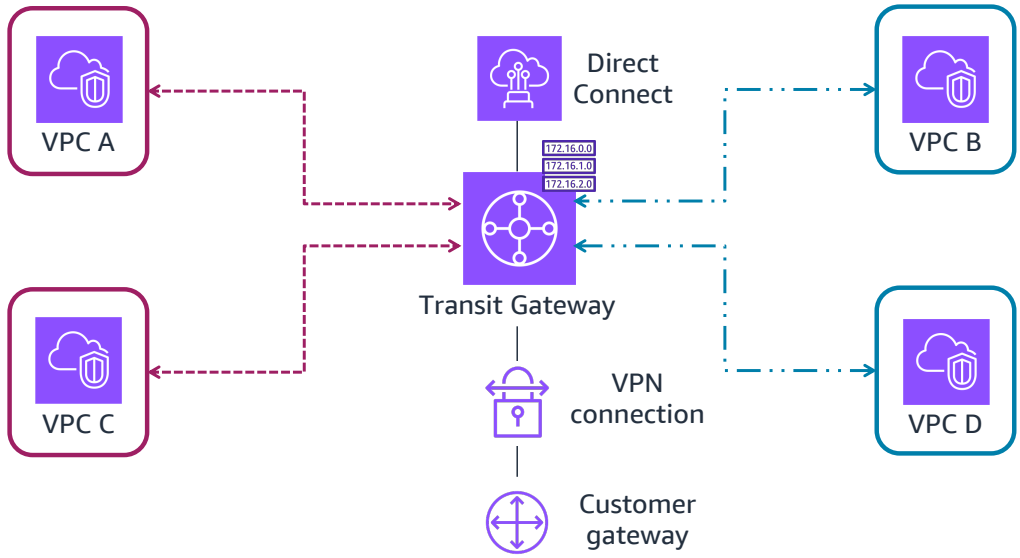
Partial connectivity



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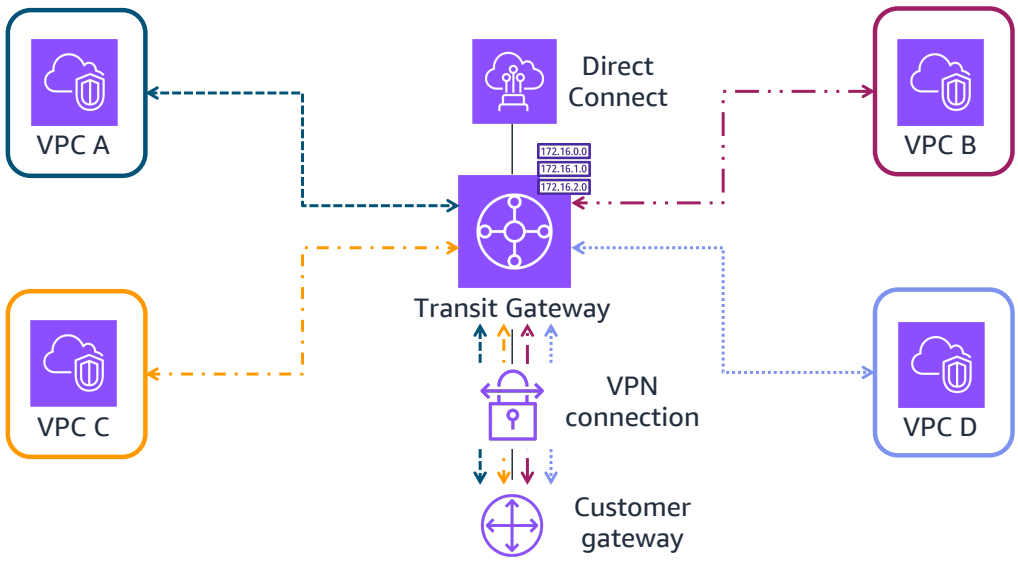
Isolation with full access from a VPN



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Review

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Present solutions

Network engineer

Consider how you would answer the following questions:

- What can we do to keep our connections to AWS services private?
- How can we privately route traffic between our VPCs?
- What are our options to connect our on-premises network to the AWS Cloud?
- Which services can reduce the number of route tables that we need to manage our global network?

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Module review

In this module, you learned about the following:

- ✓ VPC endpoints
- ✓ VPC peering
- ✓ Hybrid networking
- ✓ Transit Gateway

Next, you review the following:

Knowledge check

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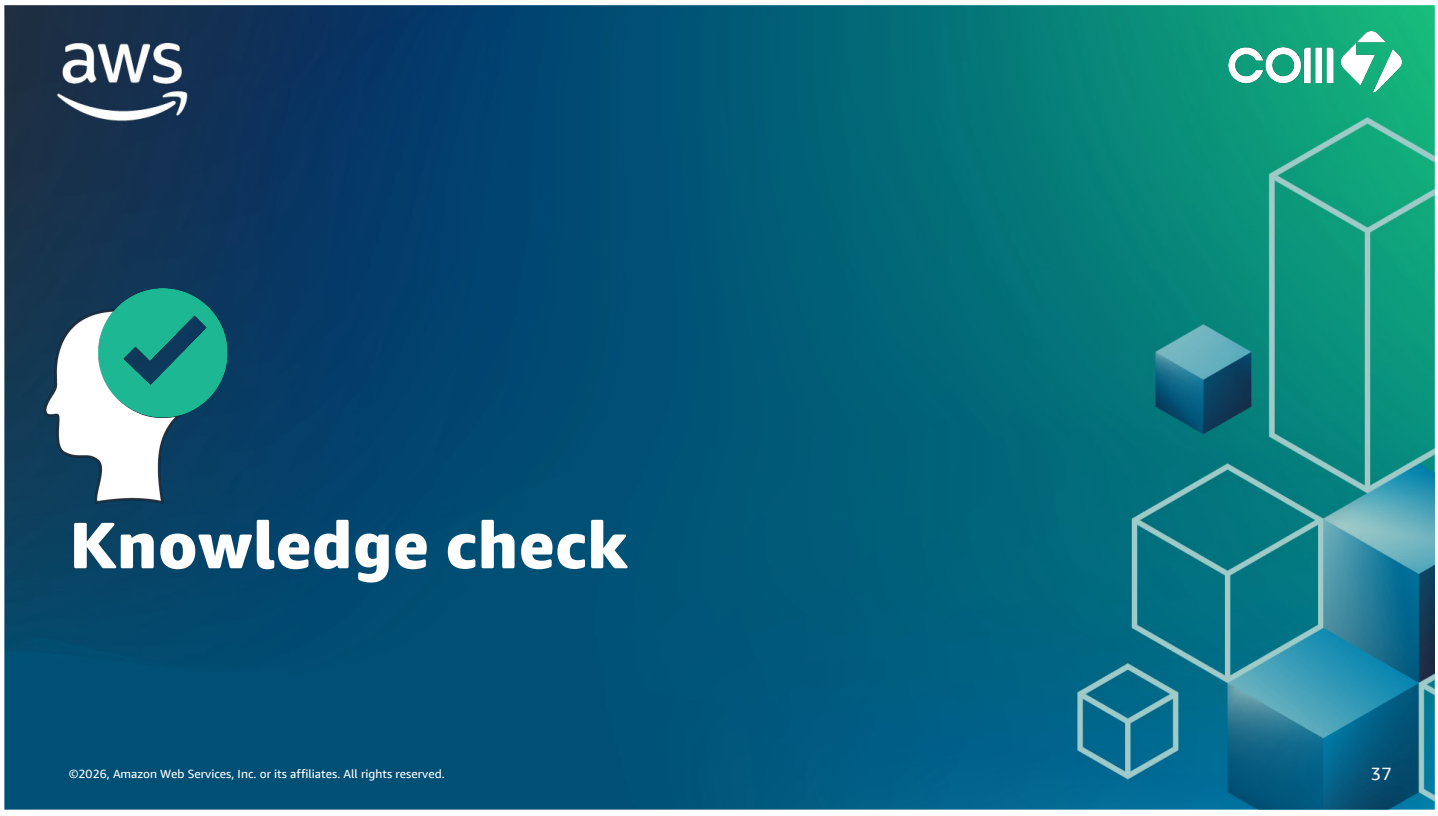

Knowledge check question 1

What is a connection to a transit gateway called?

Choice	Response
A	VPN
B	Attachment
C	Route
D	VPC

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Knowledge check question 1 and answer

What is a connection to a transit gateway called?

Choice	Response
A	VPN
B correct	Attachment
C	Route
D	VPC

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Knowledge check question 2

What are the components of an AWS Site-to-Site VPN connection? (Select TWO.)

Choice	Response
A	Customer gateway device
B	Interface endpoint
C	Virtual private gateway
D	VPC peering connection
E	Gateway endpoint

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Knowledge check question 2 and answer

What are the components of an AWS Site-to-Site VPN connection? (Select TWO.)

Choice	Response
A correct	Customer gateway device
B	Interface endpoint
C correct	Virtual private gateway
D	VPC peering connection
E	Gateway endpoint

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Knowledge check question 3

Which statements are true of VPC peering connections? (Select TWO.)

Choice	Response
A	Connections are one-to-many.
B	Connections are one-to-one.
C	Connections require a transit gateway.
D	Connections can span accounts.
E	Connections are transitive.

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Knowledge check question 3 and answer

Which statements are true of VPC peering connections? (Select TWO.)

Choice	Response
A	Connections are one-to-many.
B correct	Connections are one-to-one.
C	Connections require a transit gateway.
D correct	Connections can span accounts.
E	Connections are transitive.

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End of Module 10

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Architecting on AWS

Module 11: Serverless

Lab 5

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Poll question

Which of the following best describes your familiarity with serverless architectures?

A. I have built solutions using serverless architectures.

B. I understand serverless architectures, but have not used them.

C. I know a little bit about serverless architectures.

D. I am not familiar with serverless architectures.

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Module overview

- Business request
- What is serverless?
- Amazon API Gateway
- Amazon Simple Queue Service (Amazon SQS)
- Amazon Simple Notification Service (Amazon SNS)
- Amazon Kinesis
- AWS Step Functions
- Present solutions
- Knowledge check
- Lab 5: Build a Serverless Architecture

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Business request

Application development manager

The application development manager wants to know the following:

- How can we reduce operational overhead and optimize our resource costs?
- What is a secure way to provide APIs that use our backend services?
- How do we create a message queue for reliable service-to-service communication?
- How can we give our applications the ability to send push notifications?
- How do we ingest streaming data to power our real-time applications?
- What is an easy way to orchestrate multi-step workflows?

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What is serverless?

"How can we reduce operational overhead and optimize our resource costs?"

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Serverless

No infrastructure to provision or manage

Automatic scaling by unit of consumption

Consumption-based pricing

Built-in security; highly available compute

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AWS serverless portfolio

Compute

AWS Lambda

AWS Fargate

API proxy

Amazon API Gateway

Storage

Amazon Simple Storage Service (Amazon S3)

Database

Amazon DynamoDB

Amazon Aurora Serverless

Authentication

Amazon Cognito

Inter-process messaging

Amazon Simple Notification Service (Amazon SNS)

Amazon Simple Queue Service (Amazon SQS)

Orchestration

AWS Step Functions

Analytics

Amazon Kinesis

Amazon Athena

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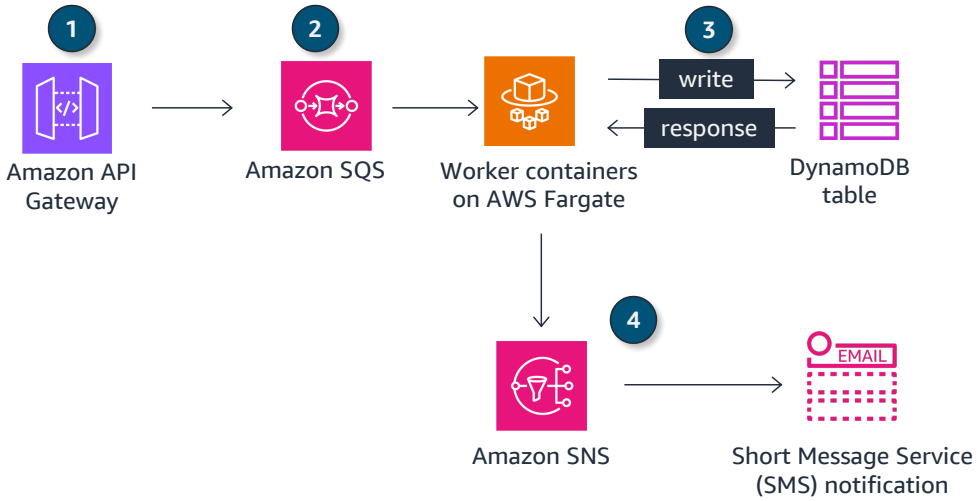
Example serverless architecture

1. POST request received
2. Request goes to a message queue to await processing by a worker service
3. Worker service processes message and writes it to Amazon DynamoDB
4. Worker service prompts the notification service to send an SMS notice to subscribed users



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Amazon API Gateway

"What is a secure way to provide APIs that use our backend services?"

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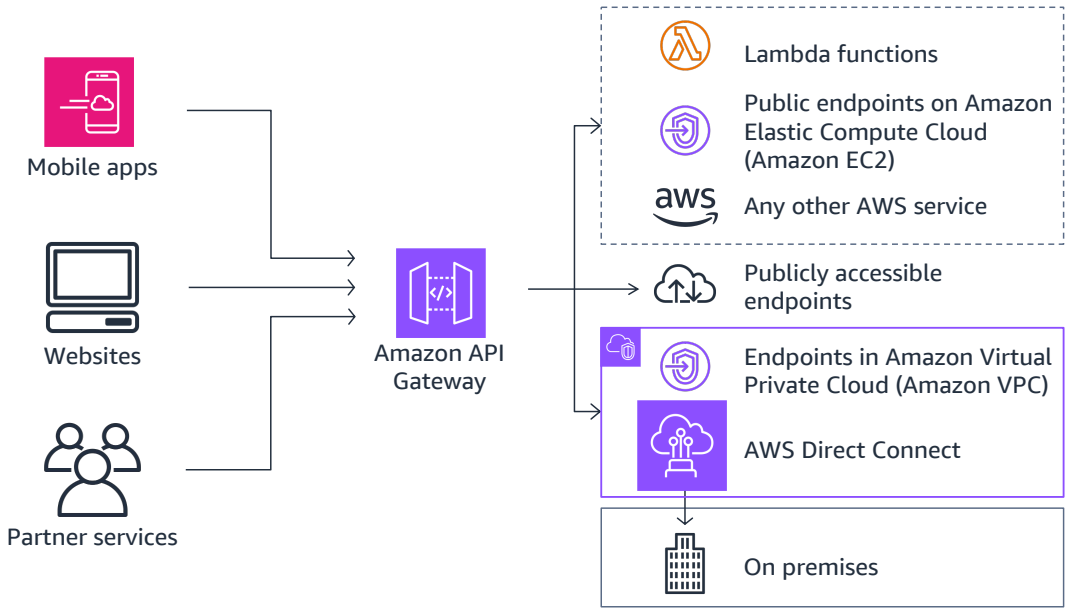
API Gateway

- Create an entry point for your applications.
- Process thousands of concurrent API calls.
- Choose internet-facing or internal only.



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API Gateway features

	Creates a unified API frontend for multiple microservices
	Provides distributed denial of service (DDoS) protection and throttling for your backend
	Authenticates and authorizes requests to a backend
	Throttles, meters, and monetizes API usage by third-party developers

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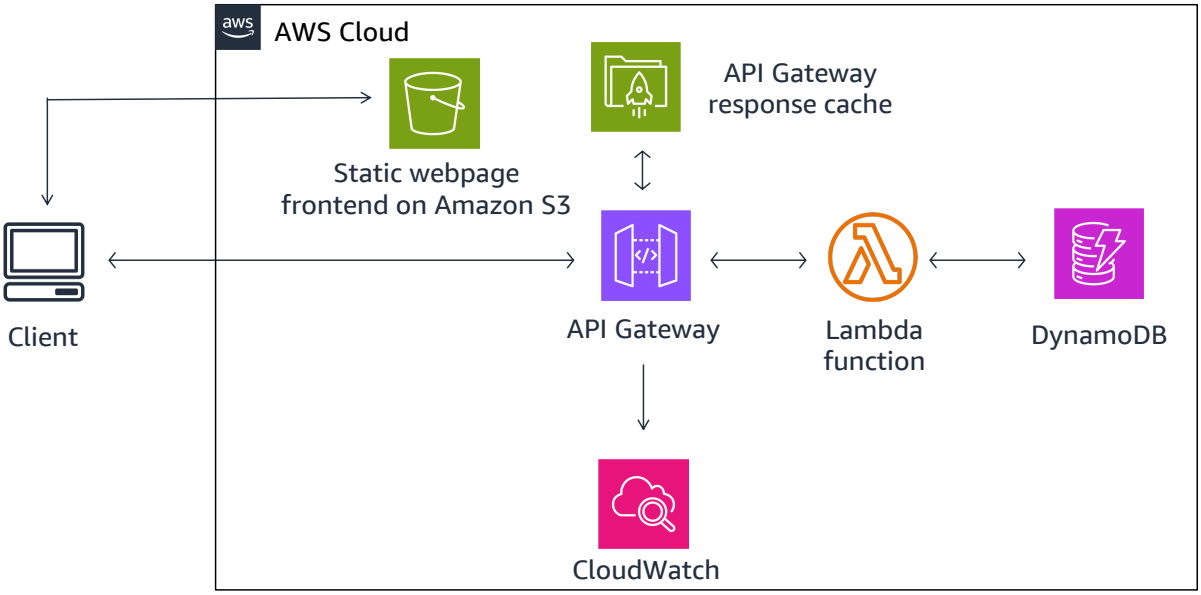
API Gateway sample architecture



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Amazon Simple Queue Service (Amazon SQS)

"How do we create a message queue for reliable service-to-service communication?"

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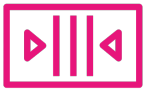
Amazon SQS

Amazon SQS



Is a fully managed message queueing service

Stores messages until they are processed and deleted

Acts as a buffer between senders and receivers

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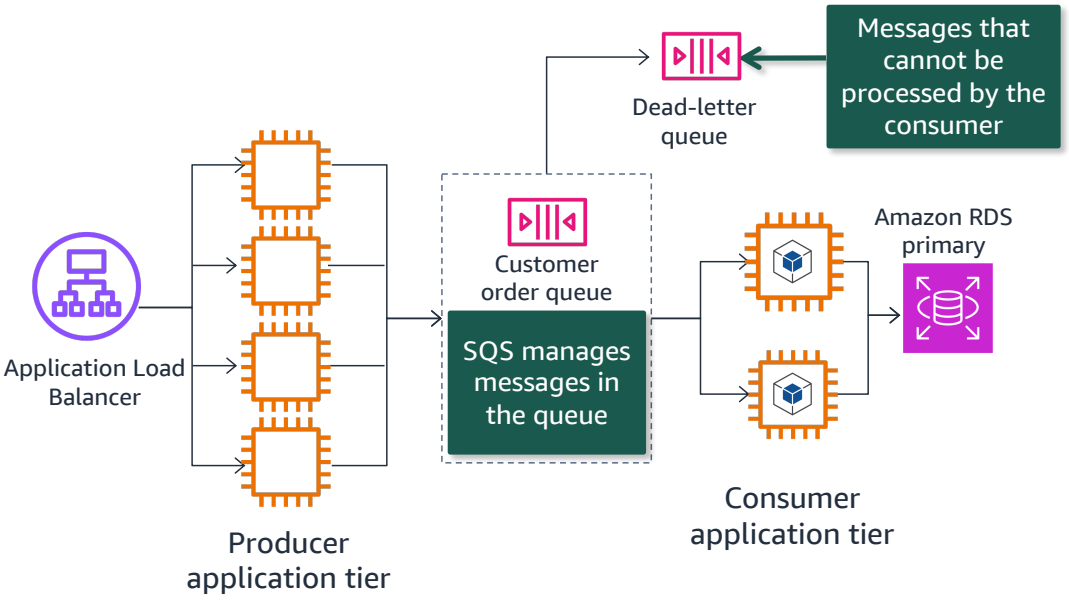
Loose coupling with Amazon SQS

- Loosely couples application components
- Uses asynchronous processing
- Creates tolerance for failed steps
- Absorbs spikes in demand



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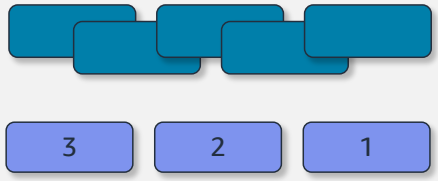
Amazon SQS use cases

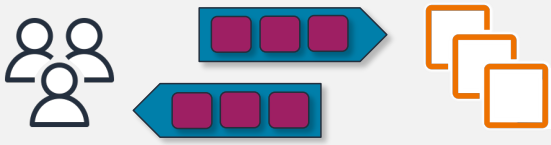
Work queues

Standard

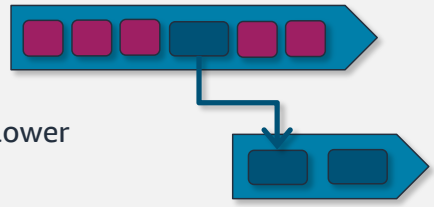
FIFO



Buffering and batch operations



Request offloading



Enqueue slower requests.

Auto scaling

Add another process to scale up the rate of messages.

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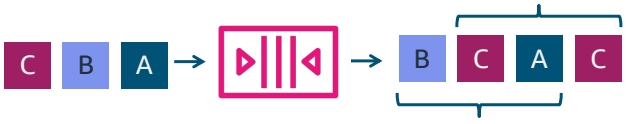
SQS queue types

Standard



Nearly unlimited API calls per second

FIFO

Limited API calls per second

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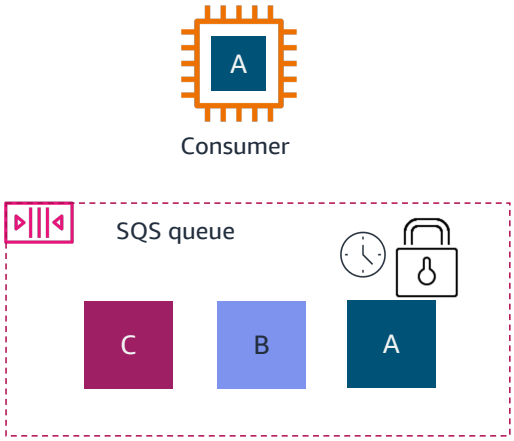
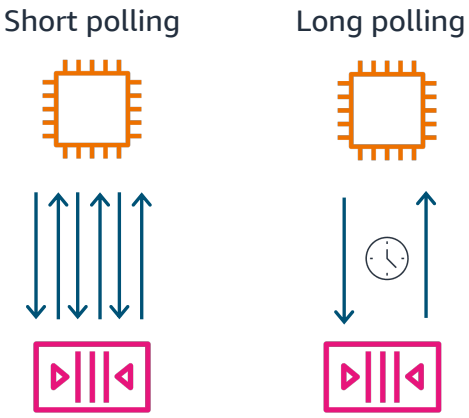
Optimizing your Amazon SQS queue configurations

Tune your visibility timeout



Choose the right polling type



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When to use message queues

✓ Service-to-service communication	✗ Selecting specific messages
✓ Asynchronous work items	✗ Large messages
✓ State change notifications	

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Amazon Simple Notification Service (Amazon SNS)

"How can I give our applications the ability to send push notifications?"

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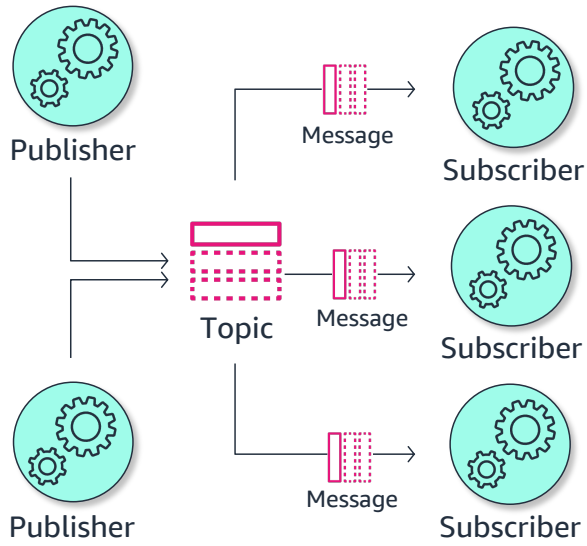
Amazon SNS



Types of subscribers

- Email/Email-JSON
- Mobile text messaging (SMS)
- Mobile push notification
- HTTP/HTTPS
- AWS Lambda
- Amazon SQS
- Amazon Data Firehose

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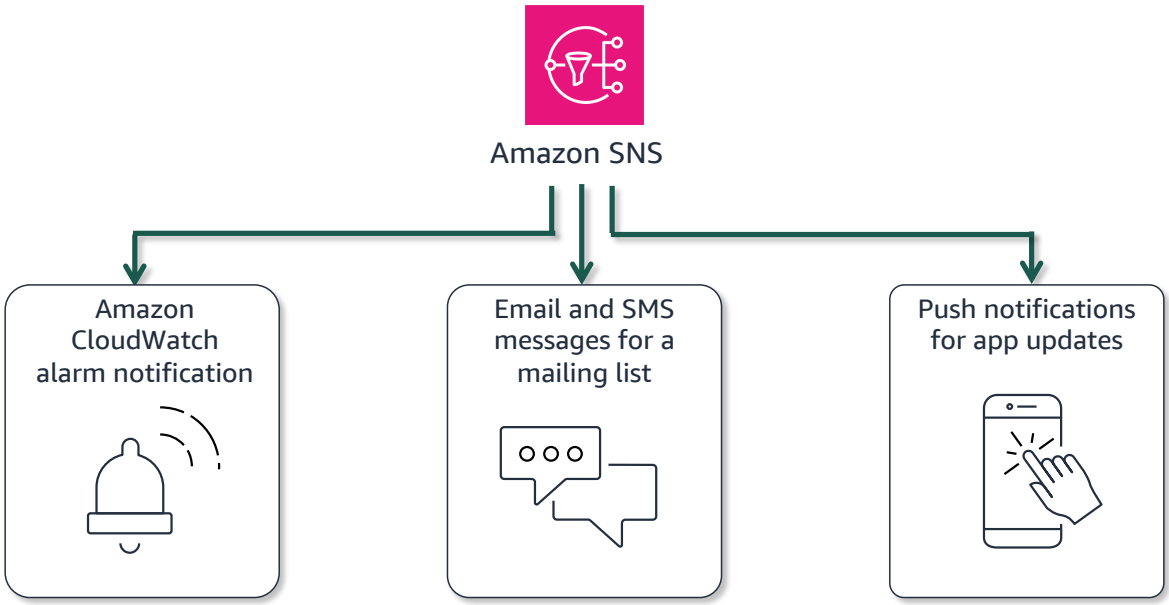
Use cases for Amazon SNS



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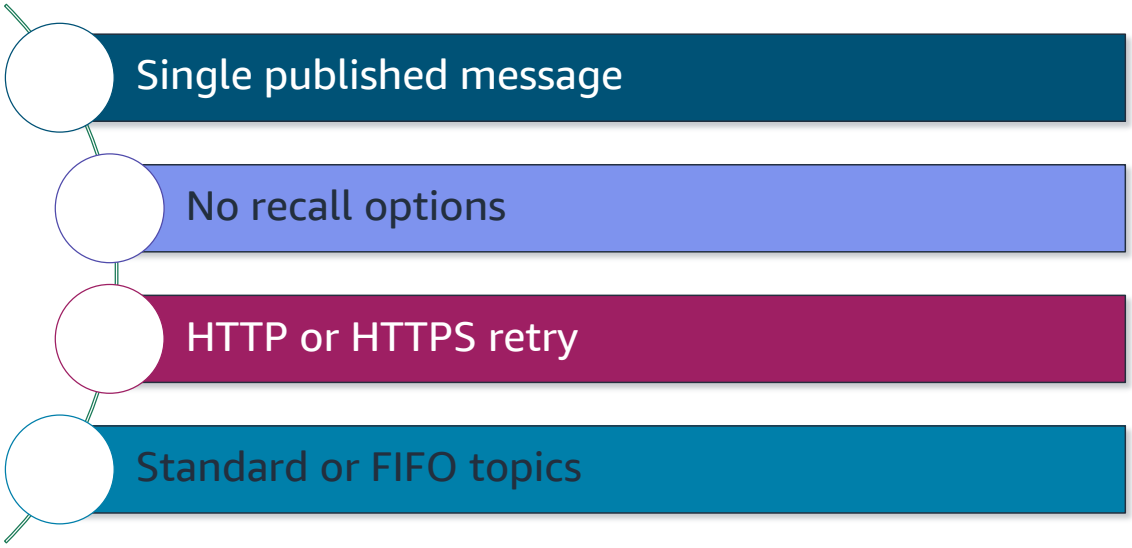
Characteristics of Amazon SNS



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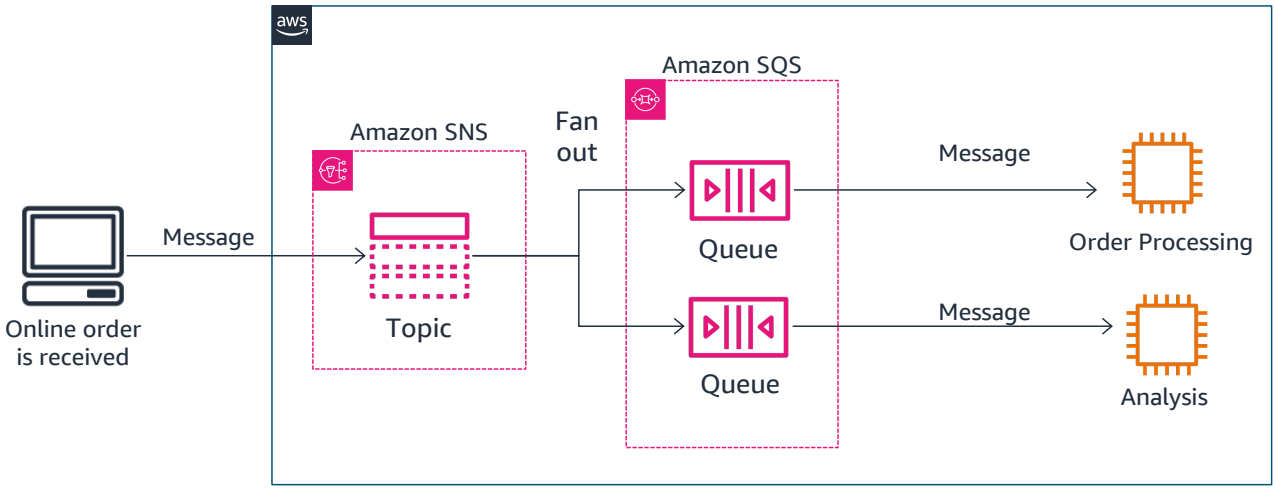
Amazon SNS publish to multiple SQS queues

Architecture example



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Amazon SNS and Amazon SQS

Features	Amazon SNS	Amazon SQS
Message persistence	No	Yes
Delivery mechanism	Push (passive)	Poll (active)
Producer and consumer	Publisher and subscriber	Send or receive
Distribution model	One to many	One to one

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Amazon Kinesis

"How do we ingest streaming data to power our real-time applications?"

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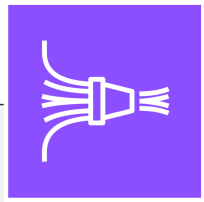
Collecting and analyzing streaming data

Amazon Kinesis Data Streams

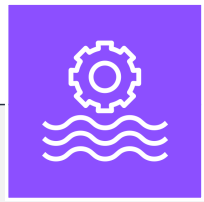
Collect and store data streams for analytics.

Amazon Data Firehose



Load data streams into AWS data stores.

Amazon Managed Service for Apache Flink



Transform and analyze streaming data in real time.

Amazon Kinesis Video Streams

Collect and store video streams for analytics.

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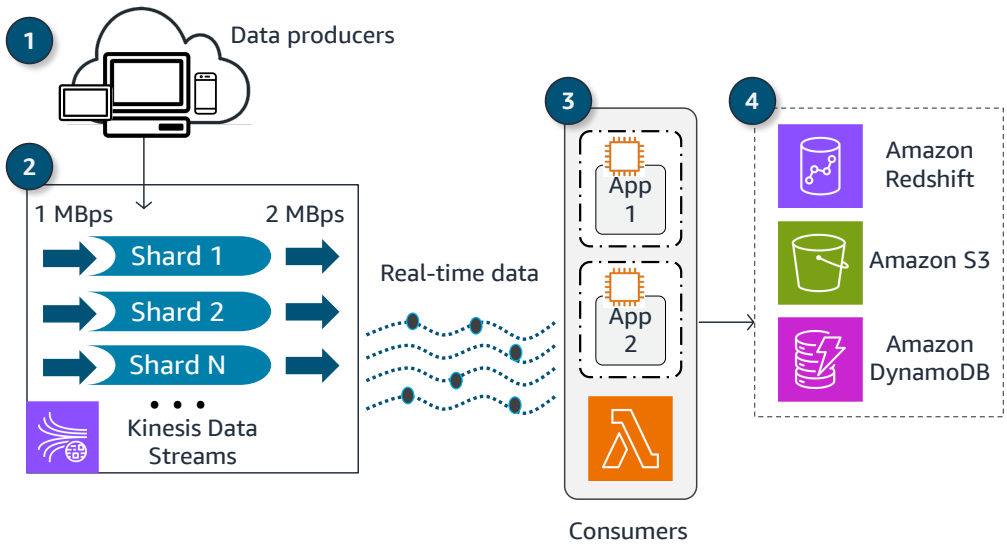
Kinesis Data Streams overview

1. Producers put data records into Kinesis Data Streams.
2. Shards hold real-time, sequenced data.
3. Consumers read from shards and process data.
4. You can store output using AWS services.



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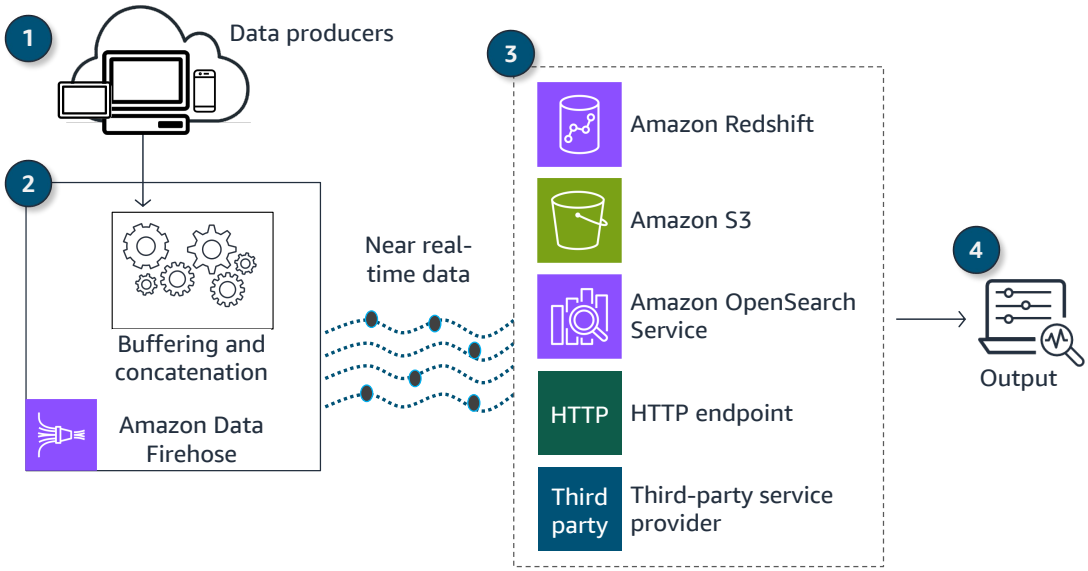
Amazon Data Firehose overview

1. Data producers send data.
2. Data can be batched and compressed before loading it into AWS.
3. Firehose writes to the destination.
4. Streaming data is processed using analytics and business intelligence.



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AWS Step Functions

"What is an easy way to orchestrate multi-step workflows?"

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Step Functions

Step Functions

- Coordinates microservices using visual workflows
- Permits you to step through the functions of your application
- Automatically initiates and tracks each step
- Provides simple error catching and logging if a step fails

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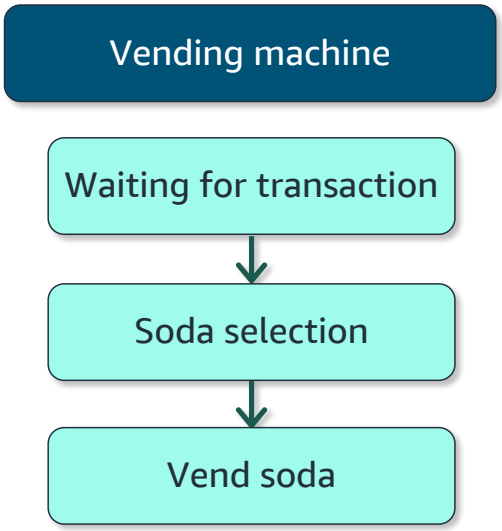
Step Functions: State machine

A *state machine* is an object that has a set number of operating conditions that depend on its previous condition to determine output.



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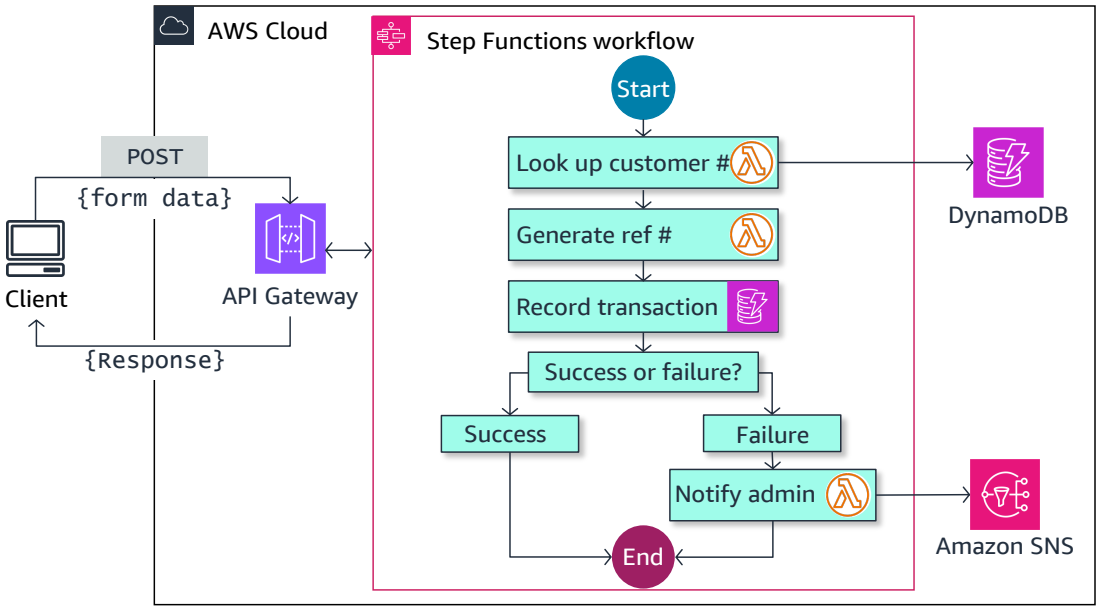
Orchestration of complex distributed workflows

Step Functions supports the following state types:

- Task
- Choice
- Fail or Succeed
- Pass
- Wait
- Parallel
- Map



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Review

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Present solutions

Application development manager

Consider how you would answer the following:

- How can we reduce operational overhead and optimize our resource costs?
- What is a secure way to provide APIs that use our backend services?
- How do we create a message queue for reliable service-to-service communication?
- How can I give our applications the ability to send push notifications?
- How do we ingest streaming data to power our real-time applications?
- What is an easy way to orchestrate multi-step workflows?

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Module review

In this module, you learned about the following:

- ✓ What is serverless?
- ✓ API Gateway
- ✓ Amazon SQS
- ✓ Amazon SNS
- ✓ Amazon Kinesis
- ✓ AWS Step Functions

Next, you review the following:

Knowledge check

Lab introduction

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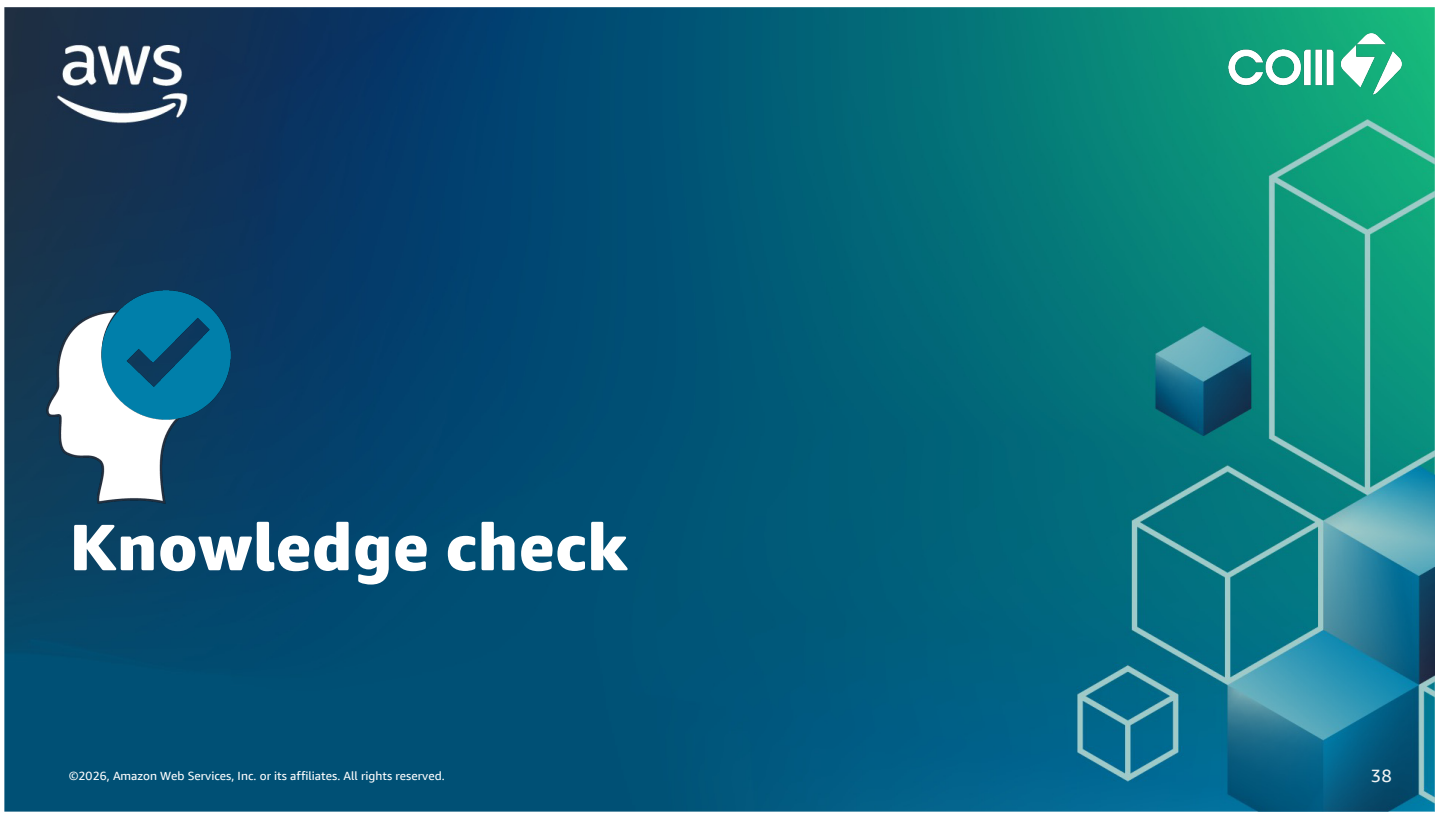

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Knowledge check question 1

Which type of Amazon SQS queue provides at-least-once delivery?

Choice	Response
A	FIFO queue
B	Standard queue
C	Dead-letter queue
D	Long polling

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Knowledge check question 1 and answer

Which type of Amazon SQS queue provides at-least-once delivery?

Choice	Response
A	FIFO queue
B correct	Standard queue
C	Dead-letter queue
D	Long polling

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Knowledge check question 2

What is an advantage of long polling compared to short polling?

Choice	Response
A	Long polling provides an immediate response from a `ReceiveMessage` call.
B	Long polling is more stable when using a single thread to poll multiple queues.
C	Long polling reduces the cost of using Amazon SQS by reducing the number of empty responses and false empty responses.
D	Long polling reduces cost by only sampling a subset of Amazon SQS servers.

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Knowledge check question 2 and answer

What is an advantage of long polling compared to short polling?

Choice	Response
A	Long polling provides an immediate response from a `ReceiveMessage` call.
B	Long polling is more stable when using a single thread to poll multiple queues.
C **correct**	Long polling reduces the cost of using Amazon SQS by reducing the number of empty responses and false empty responses.
D	Long polling reduces cost by only sampling a subset of Amazon SQS servers.

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Knowledge check question 3

What is a feature of Amazon SNS?

Choice	Response
A	Amazon SNS exchanges messages through a polling model.
B	Amazon SNS can send messages to decoupled components of a distributed application that do not process the same amount of work simultaneously.
C	Amazon SNS can push messages to multiple subscribers.
D	Amazon SNS keeps messages persistent.

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Knowledge check question 3 and answer

What is a feature of Amazon SNS?

Choice	Response
A	Amazon SNS exchanges messages through a polling model.
B	Amazon SNS can send messages to decoupled components of a distributed application that do not process the same amount of work simultaneously.
C correct	Amazon SNS can push messages to multiple subscribers.
D	Amazon SNS keeps messages persistent.

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Lab 5:

Build a Serverless Architecture

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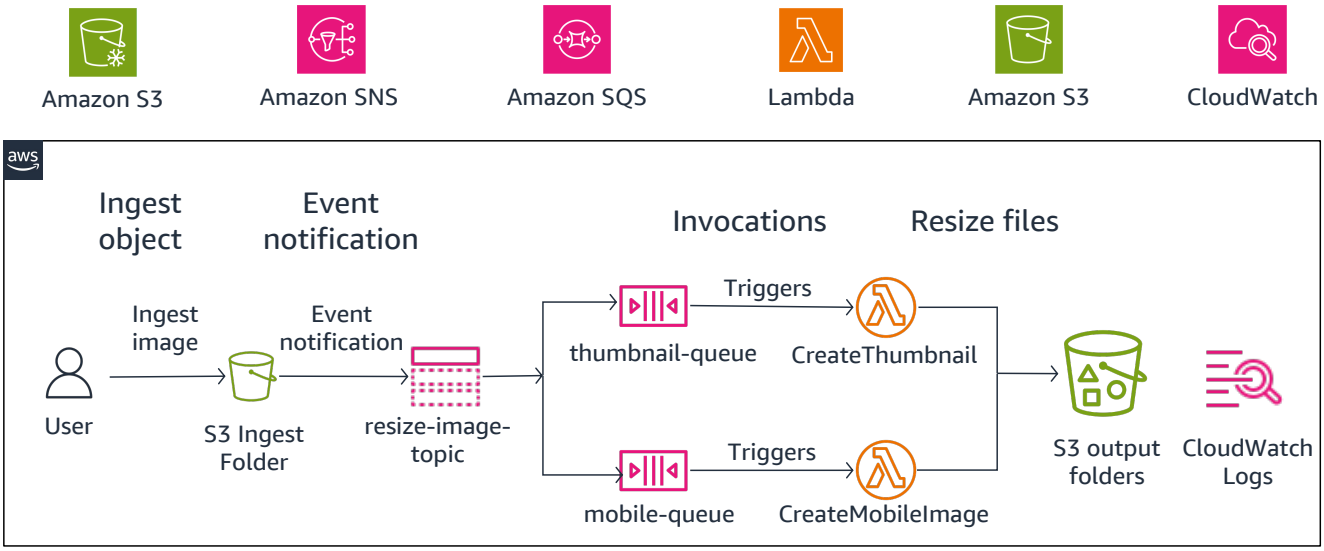
Lab 5 diagram



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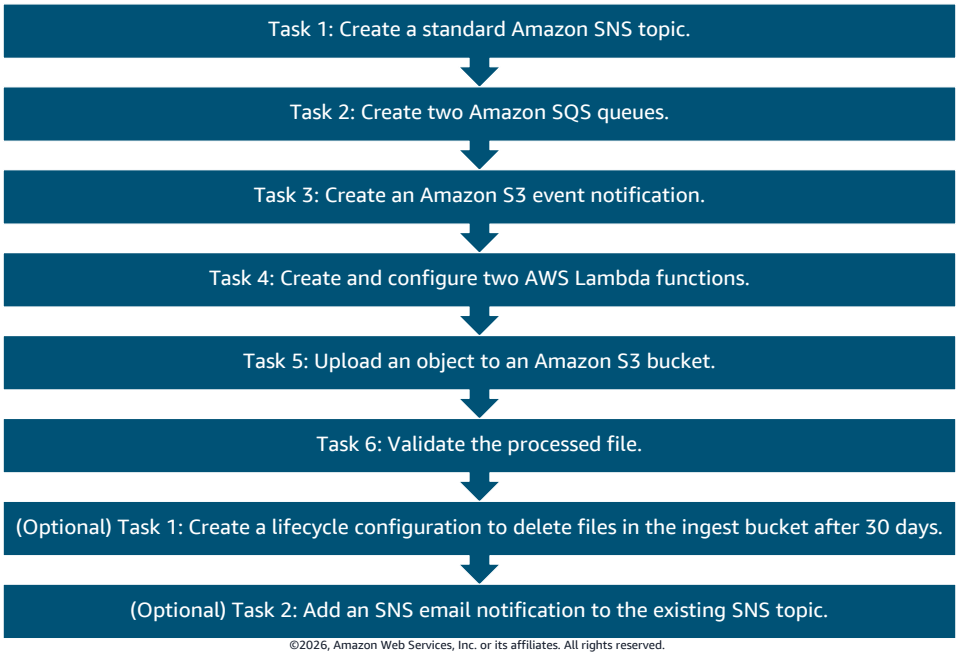
Lab tasks



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End of Module 11

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Architecting on AWS

Module 12: Edge Services

Lab 6

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Poll question

What percentage of your workloads requires ultra-low latency to devices and users?

A. 0%

B. 1–25%

C. 26–50%

D. 51–75%

E. 76–100%

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Module overview

- Business request
- Edge fundamentals
- Amazon Route 53
- Amazon CloudFront
- Distributed denial of service (DDoS) protection
- AWS Outposts
- Present solutions
- Knowledge check
- Lab 6: Configure an Amazon CloudFront Distribution with an Amazon S3 Origin

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Business request

Network engineer

The network engineer needs to know the following:

- Is there a DNS solution for AWS that is both highly available and scalable?
- What service can provide the content delivery network that we need?
- How can we protect public-facing applications?
- Does AWS support any services running on premises to meet our latency and residency requirements?

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Edge fundamentals

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AWS Cloud at the edge

Delivering the cloud anywhere customers need it

AWS Regions	Edge locations	AWS Local Zones	AWS Outposts	AWS Snow Family

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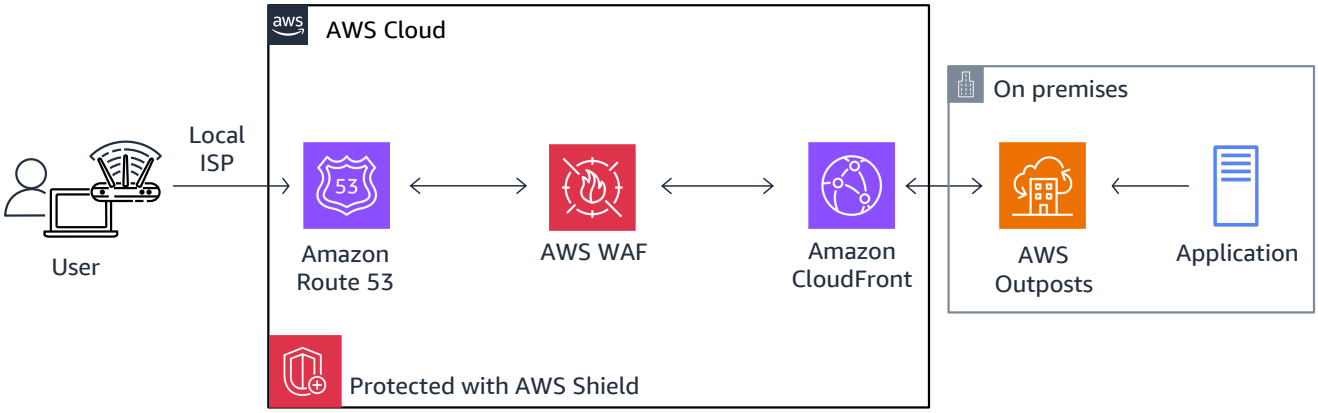
Edge services architecture

In this module, you learn about each AWS service involved in this example:



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Amazon Route 53

"Is there a DNS solution for AWS that is both highly available and scalable?"

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Route 53

Amazon Route 53

- Resolves domain names to IP addresses
- Registers or transfers a domain name
- Routes requests based on latency, health checks, and other criteria

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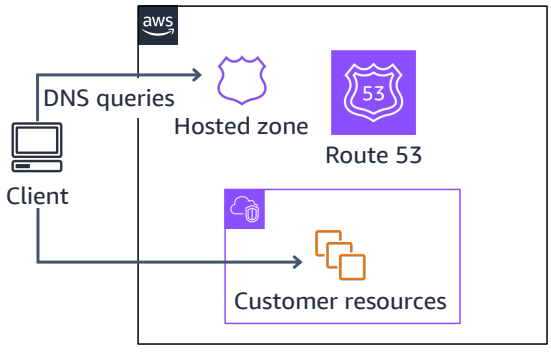
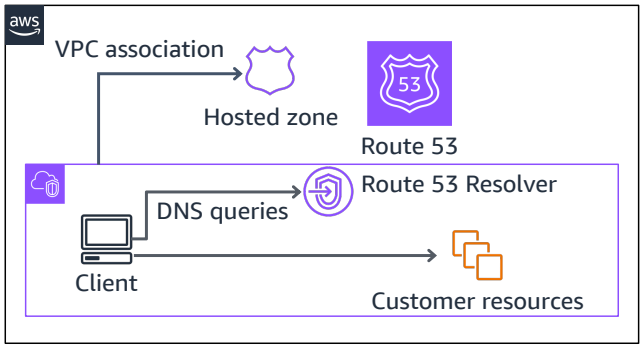
Route 53 public and private DNS

Public hosted zone



- Route to internet-facing resources
- Resolve from the internet
- Use global routing policies

Private hosted zone



- Route to VPC resources
- Resolve from inside the VPC
- Integrate with on-premises private zones using forwarding rules and endpoints

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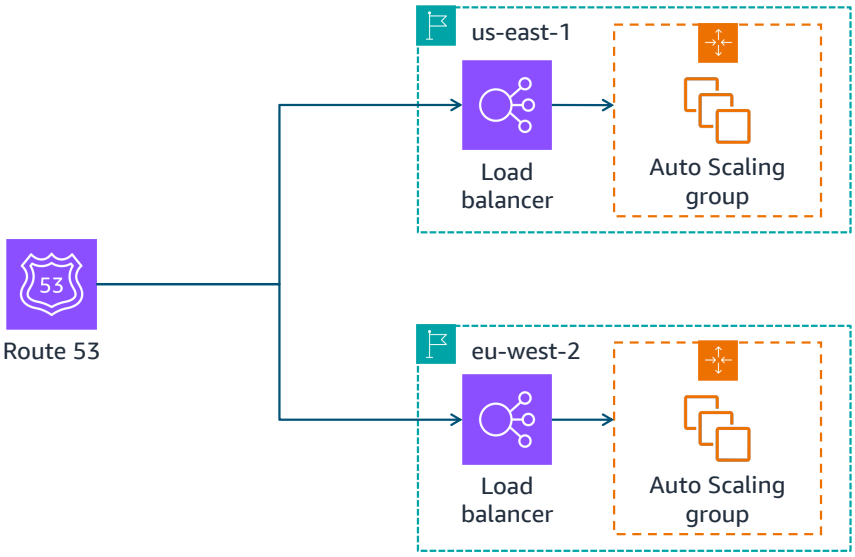
Routing policies

Routing policies:

- Simple
- Failover
- Geolocation
- Geoproximity
- Latency-based
- Multivalue answer
- Weighted



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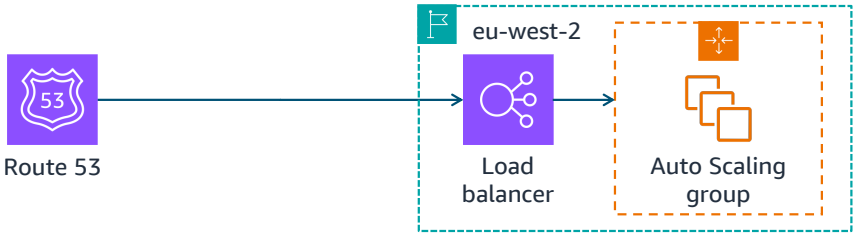
Simple routing



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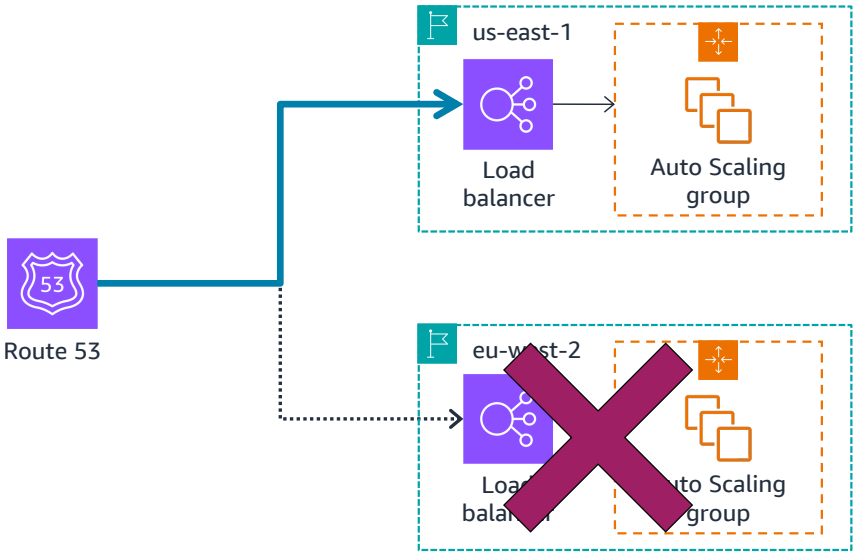
Failover routing



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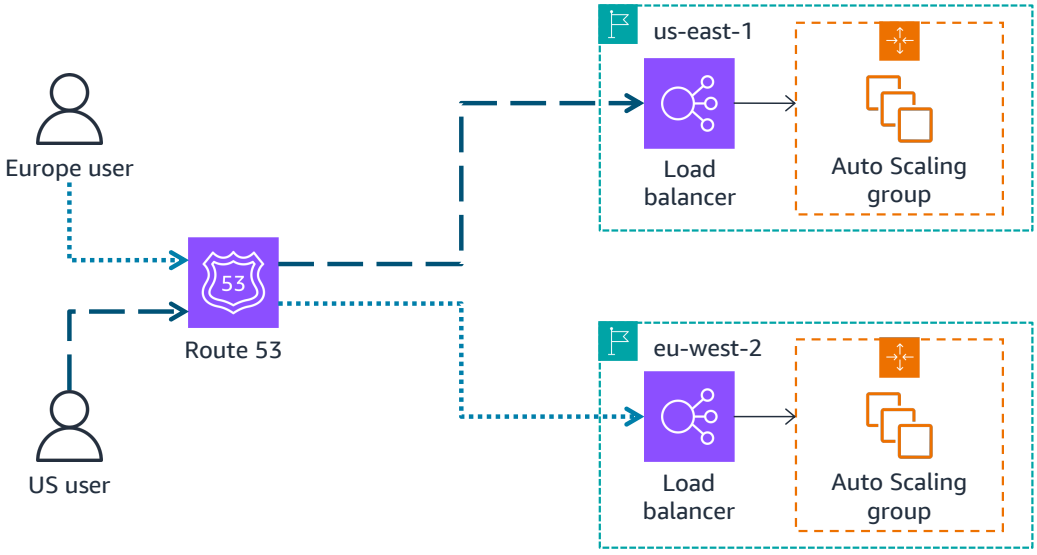
Geolocation routing



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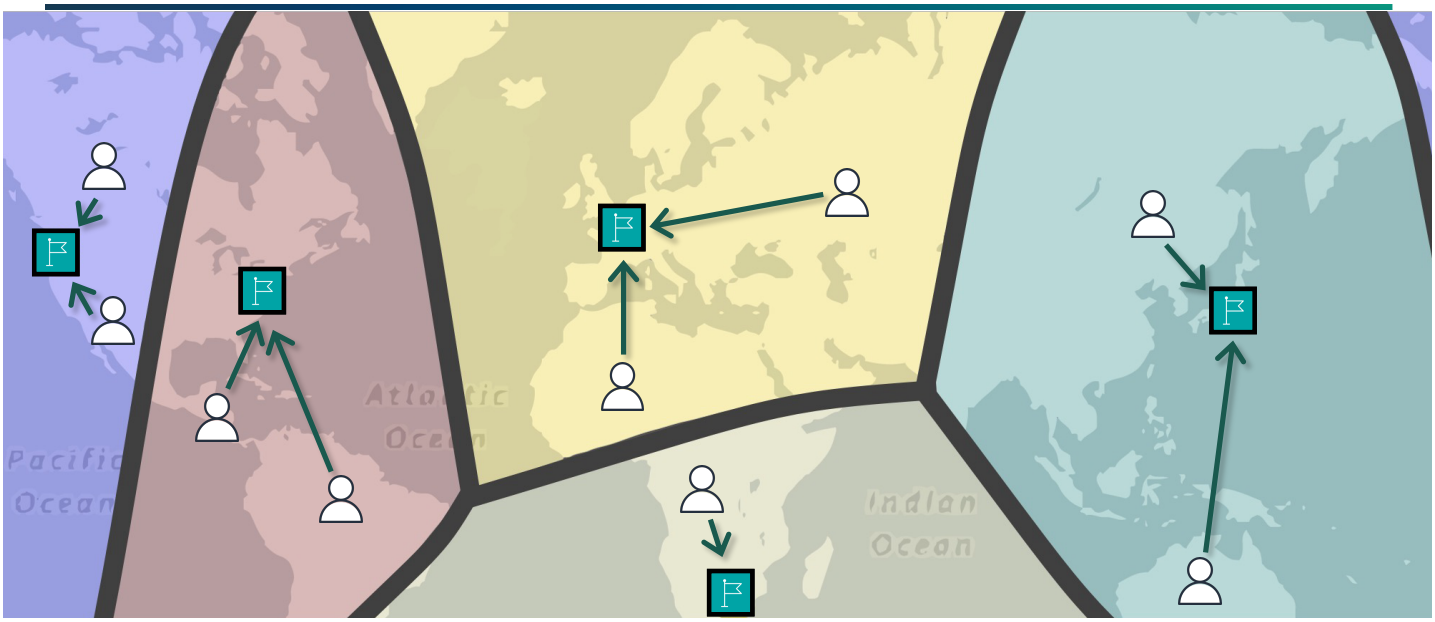
Geoproximity routing



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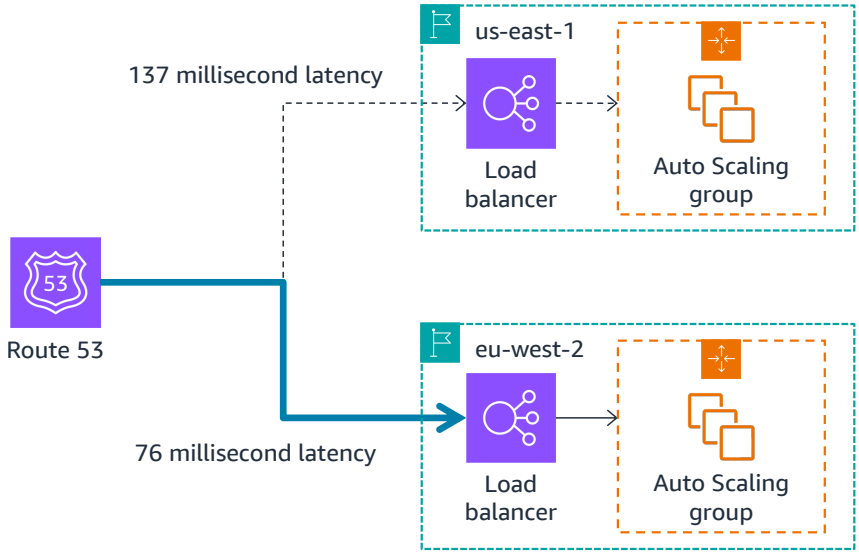
Latency-based routing



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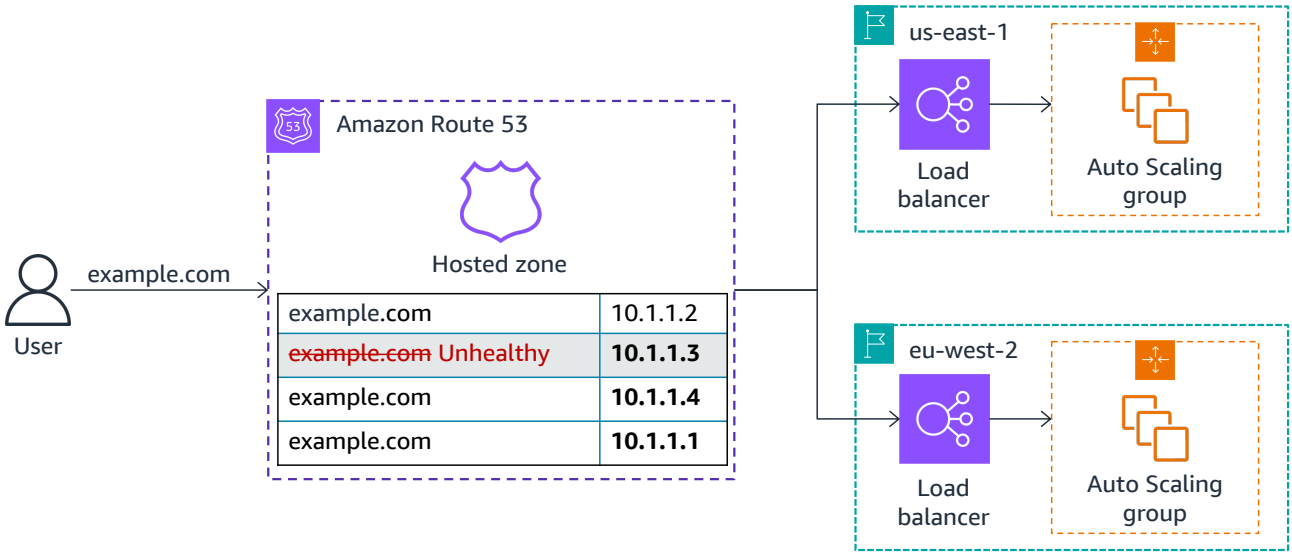
Multivalue answer routing



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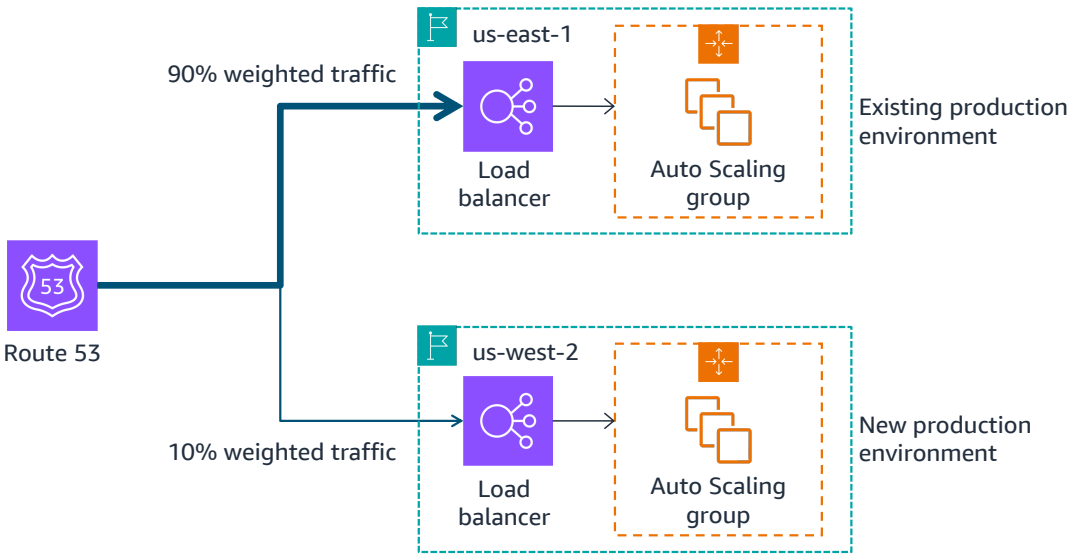
Weighted routing



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Amazon CloudFront

"What service can provide the content delivery network that we need?"

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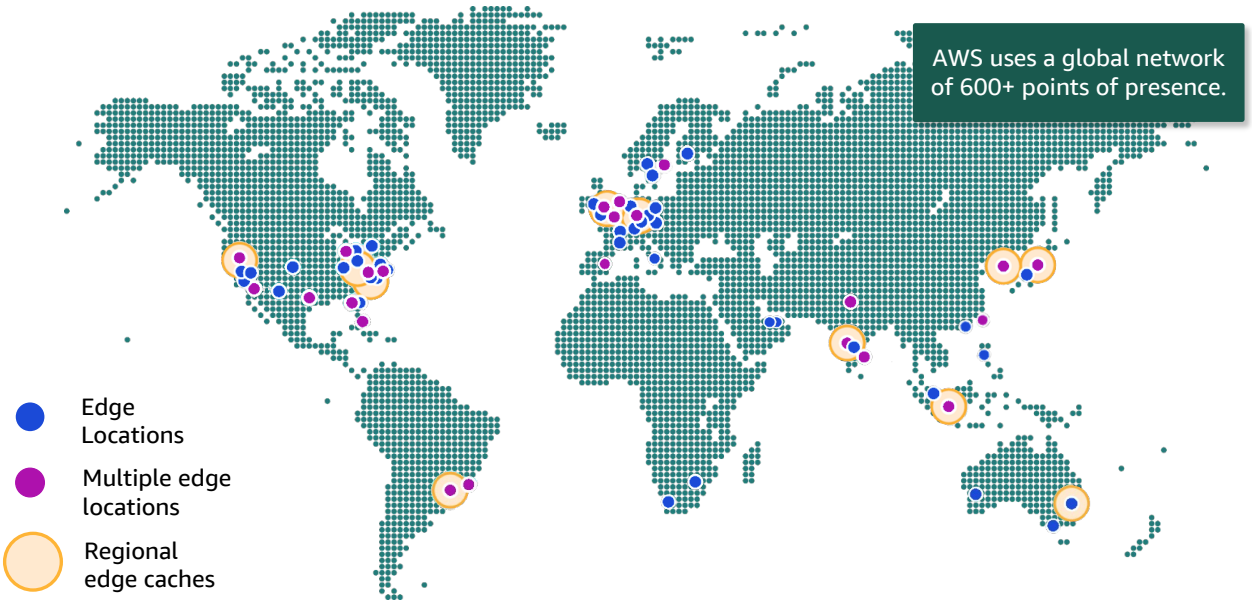
Content delivery networks



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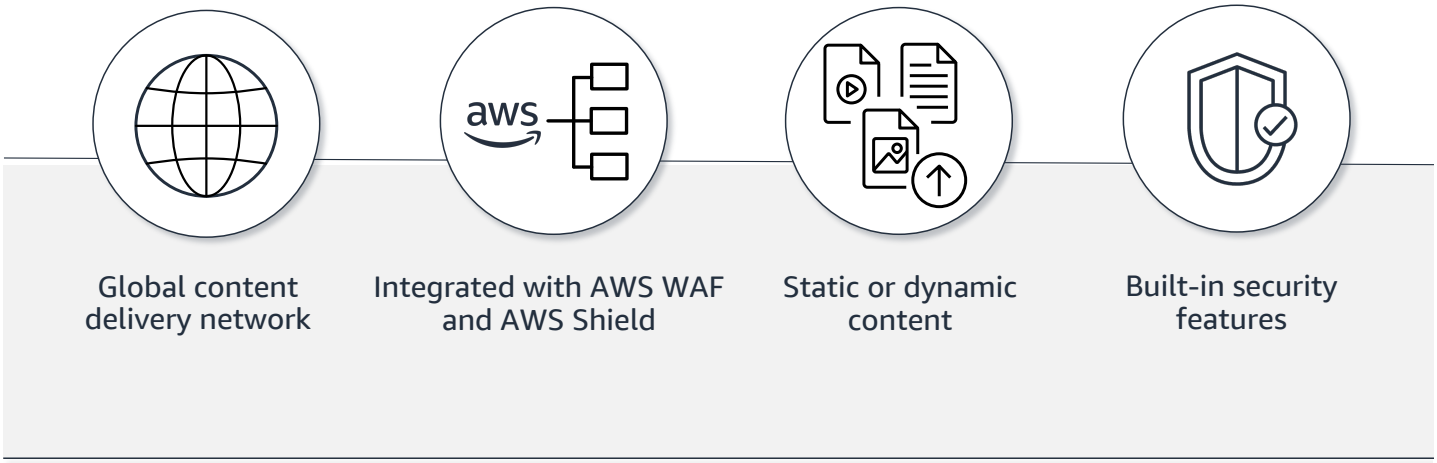
Amazon CloudFront overview



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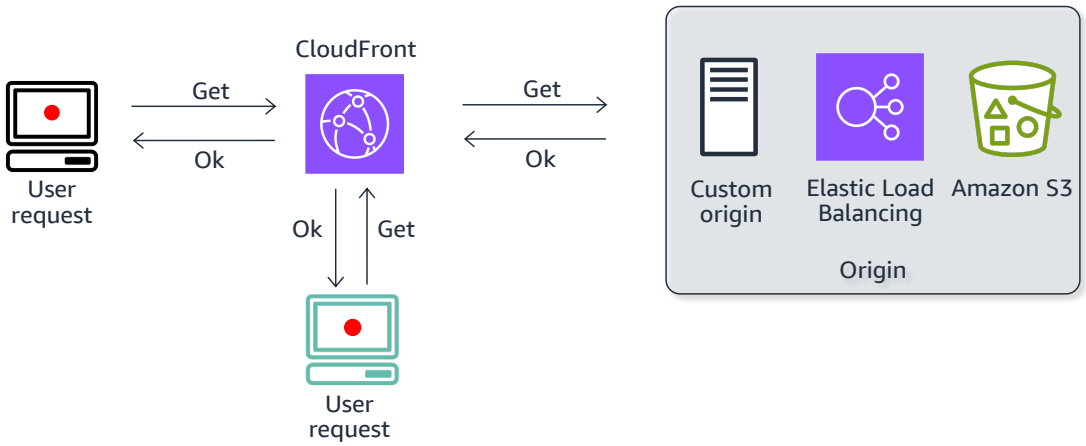
Edge caching

- Decrease latency by caching data at edge locations
- Increase security



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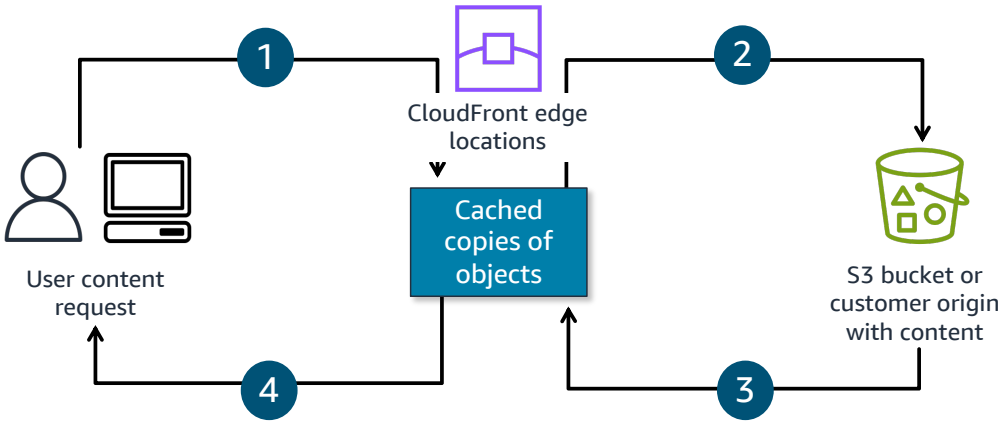
CloudFront caching steps

1. The request is routed to the optimal edge location.
2. Non-cached content is retrieved from the origin.
3. Origin content is transferred to a CloudFront edge location for caching.
4. Data is transferred to the user.



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Configuring CloudFront

1 Choose your origin

- S3 bucket
- ELB load balancer
- Custom origin
  - EC2 instance
  - On-premises server

2 Create distribution

Define cache behavior

- Path pattern
- Protocol policy
- HTTP methods
- Signed URL
- Cache policy
  - Time to live (TTL)
  - Cache key settings

(Optional)

- Associate function
- Associate AWS WAF web access control list (web ACL)
- Add custom domain name

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Improving performance

What AWS does:	What you can do:
• TCP optimization	• Choose your caching strategy
• TLS 1.3 support	• Improve your cache hit ratio
• Dynamic content placement	• Use CloudFront Origin Shield

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DDoS protection

"How can we protect public-facing applications?"

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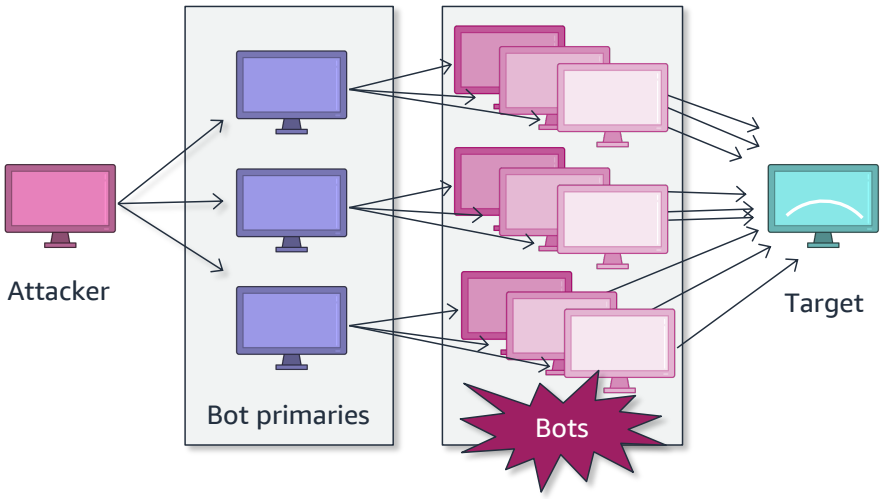
DDoS attacks

Each of the compromised hosts participates in the attack, generating a flood of requests to overwhelm the intended target.



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OSI layer attacks

DDoS attacks can be categorized by the Open Systems Interconnection (OSI) layer that they attack.

Host layer	7 Application	HTTP floods, DNS query floods	Application layer attacks
	6 Presentation	SSL abuse	Presentation layer attacks
	5 Session	N/A	
	4 Transport	SYN floods	Infrastructure layer attacks
Media layer	3 Network	UDP reflection attacks	
	2 Data link	N/A	
	1 Physical	N/A	

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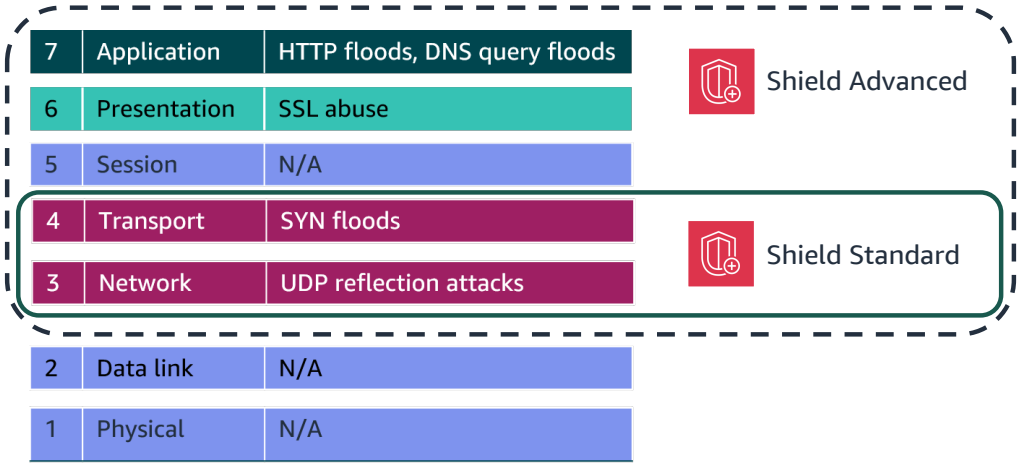
AWS Shield

Managed DDoS protection service that protects your applications on AWS

Two types of protection:

- AWS Shield Standard
- AWS Shield Advanced



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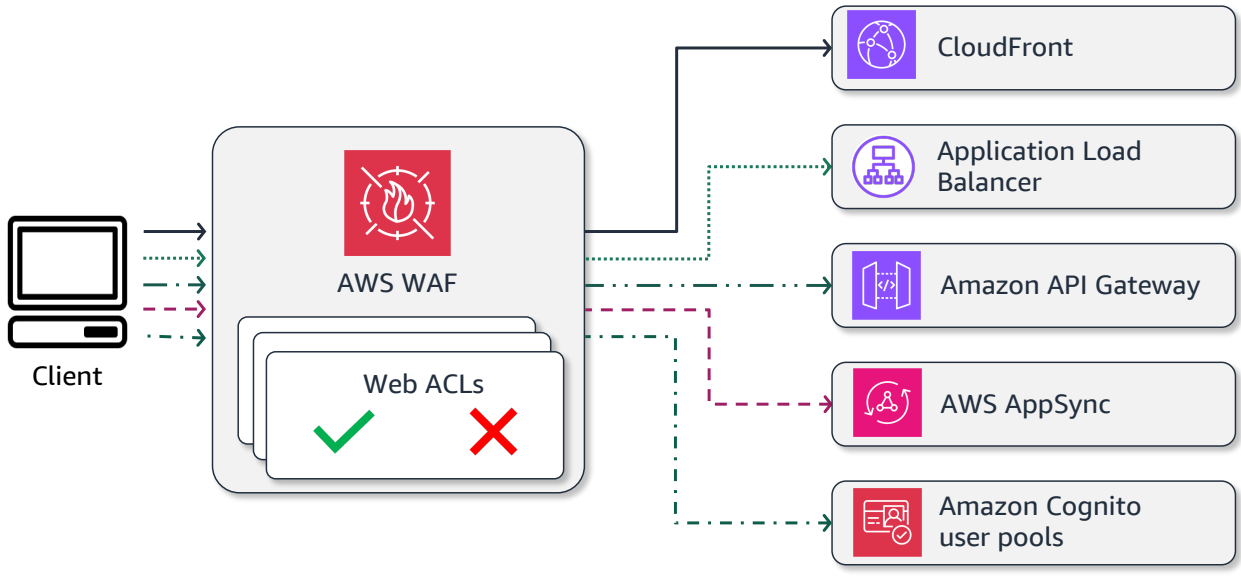
AWS WAF



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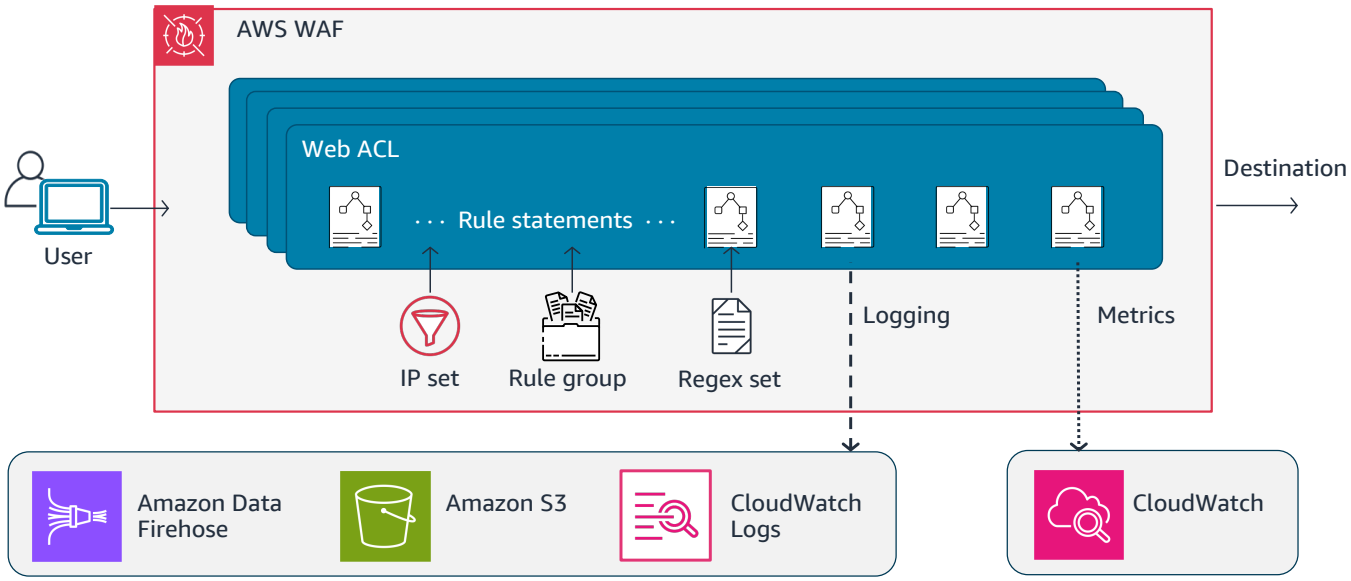
Components of access control



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Control traffic with ACL rule statements

Attack prevention

- SQL injection and cross-site scripting detection
- AWS Managed Rules for AWS WAF
- AWS Marketplace managed rule groups

Traffic filtering

- Rate limiting
- IP filtering with full Classless Inter-Domain Routing (CIDR) range support
- Geofencing by country

Pattern matching

- Regex match
- String match
- Size constraint match

Logical operation

- AND statement
- OR statement
- NOT statement

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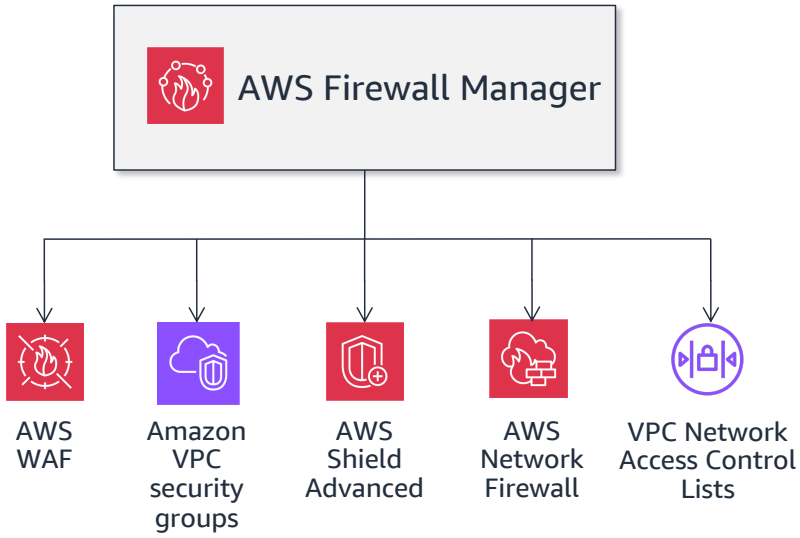
AWS Firewall Manager

- Centrally set up baseline security.
- Consistently enforce the protections.
- Seamlessly manage multiple accounts.



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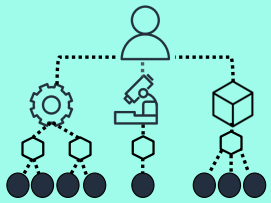
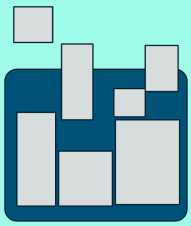
AWS Firewall Manager use cases

Large number of accounts and resources



New applications created all the time



Central organization-wide visibility into threats

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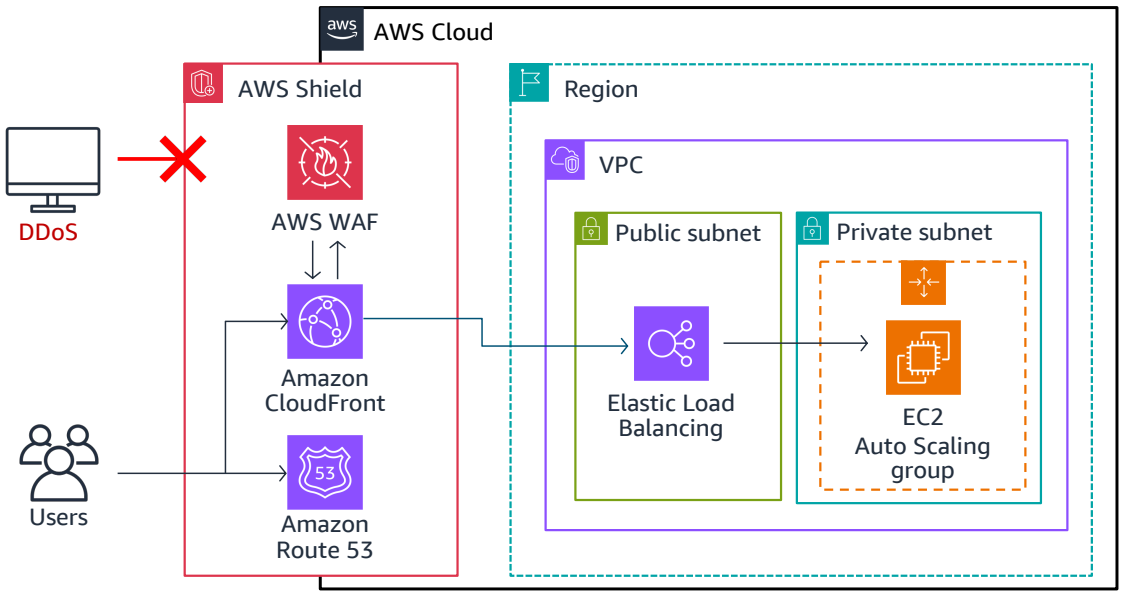
DDoS-resilient reference architecture



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AWS Outposts

"Does AWS support any services running on premises to meet our latency and residency requirements?"

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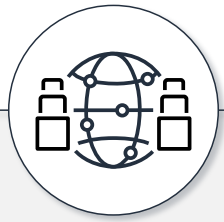
AWS Outposts family

Host AWS services on premises in your office or data center.



Meet data residency requirements or create resources that provide low latency.

Choose from a full AWS Outposts rack or a 1U or 2U-sized Outposts server.

(U = rack unit)

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Outposts rack and Outposts servers

Outposts rack*

- Scale up to 96 42U–standard racks
- Pool compute and storage capacity between multiple Outposts racks
- Get more service options

Outposts servers

- Place in your own rack
- Choose from:
  - 1U Graviton-based processor
  - 2U Intel Xeon Scalable processor

*Outposts rack requires an AWS Enterprise Support plan

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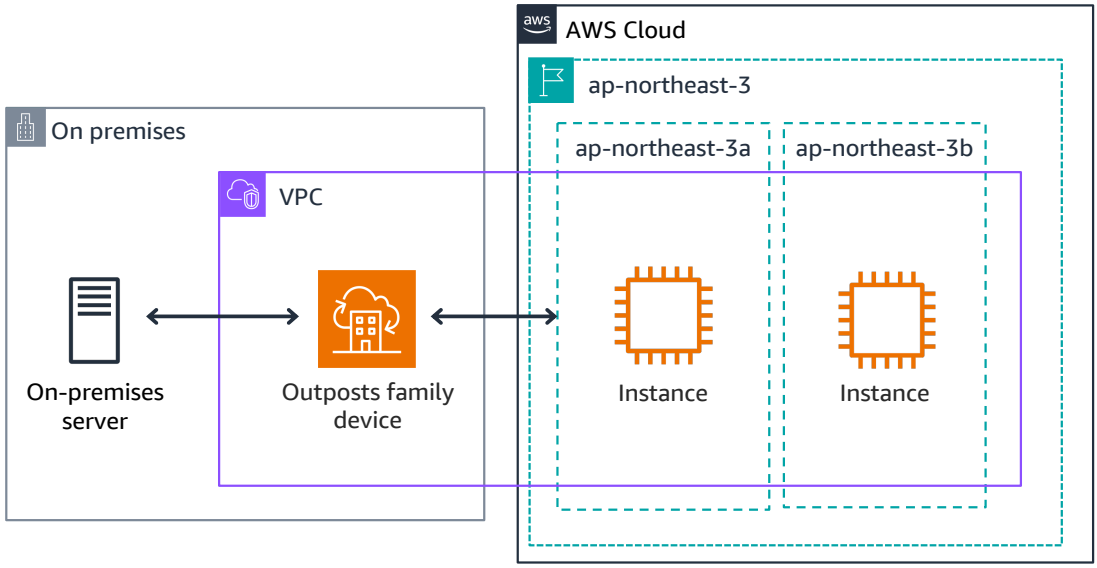
Outposts extend your VPC



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AWS resources on Outposts

AWS service availability differs between Outposts racks and Outposts 1U and 2U servers.

For more information about service availability, review the *AWS Outposts User Guide.*

Compute and storage
- Amazon EC2
- Amazon EBS
- Amazon S3

Networking
- Amazon VPC
- Application Load Balancer
- Amazon Route 53

Database
- Amazon RDS
- Amazon EMR
- Amazon ElastiCache

Containers
- Amazon Elastic Container Service (Amazon ECS)
- Amazon Elastic Kubernetes Service (Amazon EKS)

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Review

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Present solutions

Network engineer

Consider how you would answer the following:

- Is there a DNS solution for AWS that is both highly available and scalable?
- What service can provide the content delivery network that we need?
- How can we protect public-facing applications?
- Does AWS support any services running on premises to meet our latency and residency requirements?

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Module review

In this module, you learned about the following:

- ✓ Edge fundamentals
- ✓ Route 53
- ✓ Amazon CloudFront
- ✓ DDoS protection
- ✓ AWS Outposts

Next, you review the following:

- Knowledge check
- Lab introduction

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Knowledge check

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Knowledge check question 1

What are the potential benefits of implementing a CloudFront distribution? (Select TWO.)

Choice	Response
A	Increased application security
B	Automatic provisioning and renewal of SSL/TLS certificates for custom domain names
C	Automatic redundancy for all application content
D	Reduced latency for access to application content
E	On-premises data caching

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Knowledge check question 1 and answer

What are the potential benefits of implementing a CloudFront distribution? (Select TWO.)

Choice	Response
A **correct**	Increased application security
B	Automatic provisioning and renewal of SSL/TLS certificates for custom domain names
C	Automatic redundancy for all application content
D **correct**	Reduced latency for access to application content
E	On-premises data caching

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Knowledge check question 2

Which of the following services offer DDoS protection? (Select TWO.)

Choice	Response
A	AWS Outposts
B	Amazon EC2
C	Network Load Balancer
D	AWS Shield
E	AWS WAF

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Knowledge check question 2 and answer

Which of the following services offer DDoS protection? (Select TWO.)

Choice	Response
A	AWS Outposts
B	Amazon EC2
C	Network Load Balancer
D correct	AWS Shield
E correct	AWS WAF

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Knowledge check question 3

A network engineer wants to route 80 percent of web traffic to the ap-southeast-2 Region. The remaining 20 percent of traffic will be directed to the eu-west-1 Region. Which Route 53 routing policy is the best choice for this use case?

Choice	Response
A	Simple routing
B	Weighted routing
C	Geoproximity routing
D	Geolocation routing
E	Multivalue answer routing

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Knowledge check question 3 and answer

A network engineer wants to route 80 percent of web traffic to the ap-southeast-2 Region. The remaining 20 percent of traffic will be directed to the eu-west-1 Region. Which Route 53 routing policy is the best choice for this use case?

Choice	Response
A	Simple routing
B correct	Weighted routing
C	Geoproximity routing
D	Geolocation routing
E	Multivalue answer routing

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Knowledge check question 4

What is a benefit of choosing Outposts servers when compared to an Outposts rack?

Choice	Response
A	More AWS services are available on Outposts servers.
B	You can pool compute and storage capacity between multiple Outposts servers.
C	A rack-mountable server is available in a location with space constraints.
D	You don’t need to host your Outposts servers in a data center.

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Knowledge check question 4 and answer

What is a benefit of choosing Outposts servers when compared to an Outposts rack?

Choice	Response
A	More AWS services are available on Outposts servers.
B	You can pool compute and storage capacity between multiple Outposts servers.
C correct	A rack-mountable server is available in a location with space constraints.
D	You don’t need to host your Outposts servers in a data center.

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Lab 6:

Configure an Amazon CloudFront Distribution with an Amazon S3 Origin

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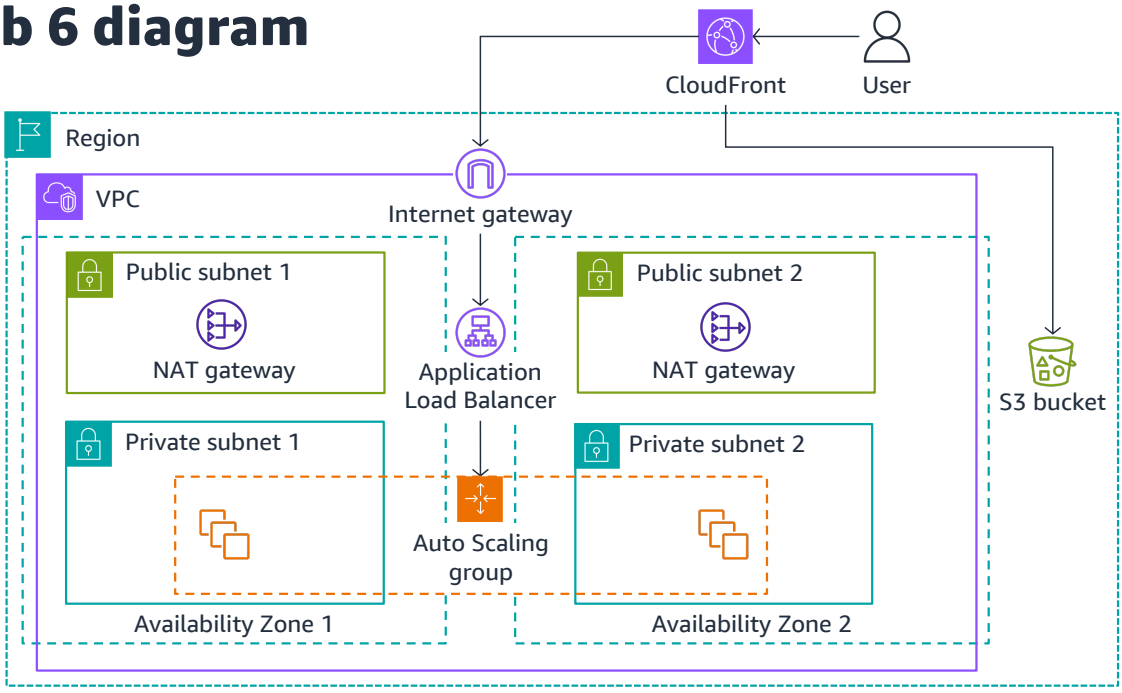
Lab 6 diagram



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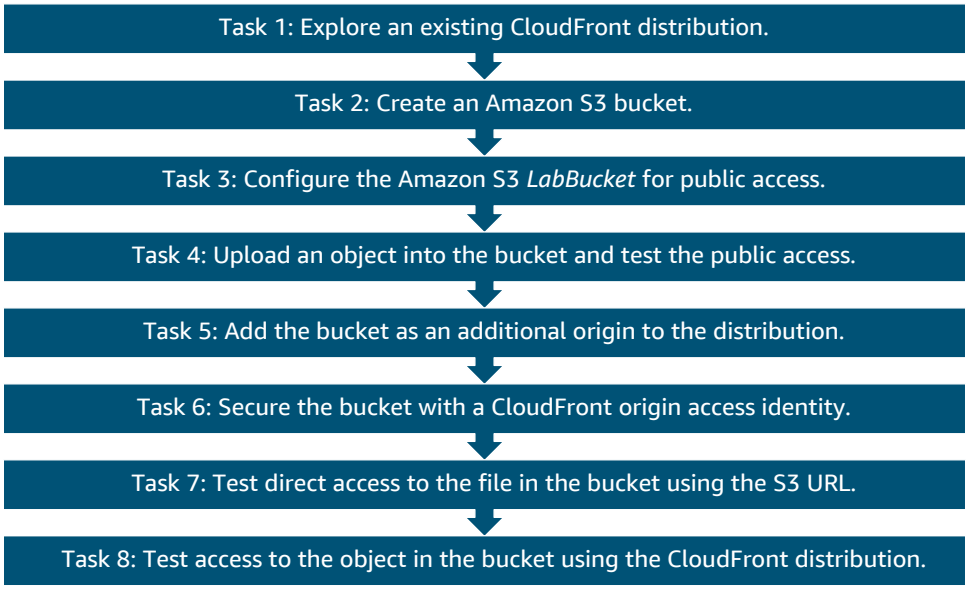
Lab tasks



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End of Module 12

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Architecting on AWS

Module 13: Backup and Recovery

Capstone lab

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Poll question

Which backup and recovery components do you work with?

A. Offsite backup storage

B. Colocation facilities

C. Tape retrievals

D. On-premises backup

E. All of the above

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Module overview

- Business request
- Disaster planning
- AWS Backup
- Recovery strategies
- Present solutions
- Knowledge check
- Capstone lab: Build an AWS Multi-Tier Architecture

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Business request

Chief technology officer

The chief technology officer has the following questions:

- What strategies can we use to protect ourselves in the event of a disaster?
- How can we centralize and automate our backup strategy?
- Which disaster recovery strategy minimizes downtime but is also cost-effective?

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Disaster planning

"What strategies can we use to protect ourselves in the event of a disaster?"

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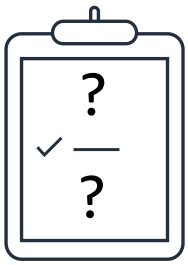
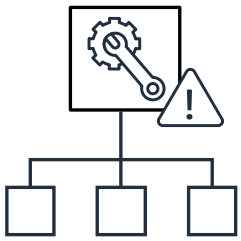
Plan for disaster recovery failure



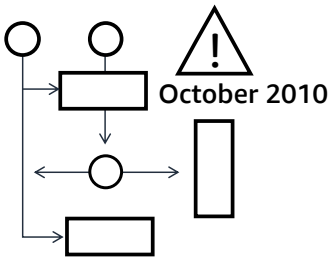
Testing

- Lack of testing



Resources

- Complex recovery path
- Lack of sufficient resources



Planning

- Outdated plans
- Changes not factored

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Availability concepts

High availability

Minimize downtime for your application.

Fault tolerance

Ensure built-in redundancy.

Backup

Make sure your data is recoverable.

Disaster recovery

Get your applications and data back after a major disaster.

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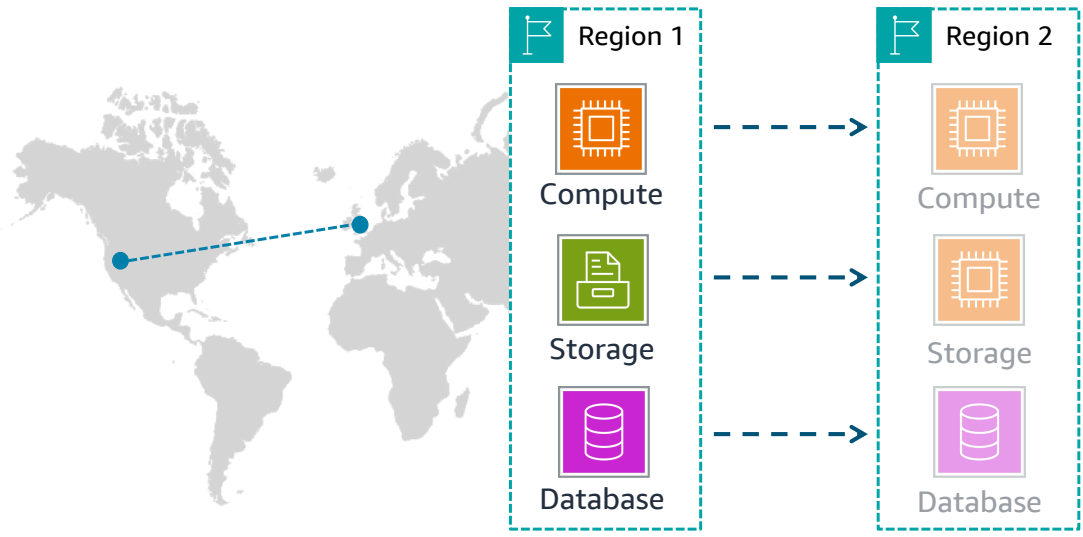
Failover and Regions

- Plan for instance failover.
- Plan for Availability Zone failover.
- Plan for Regional failover.



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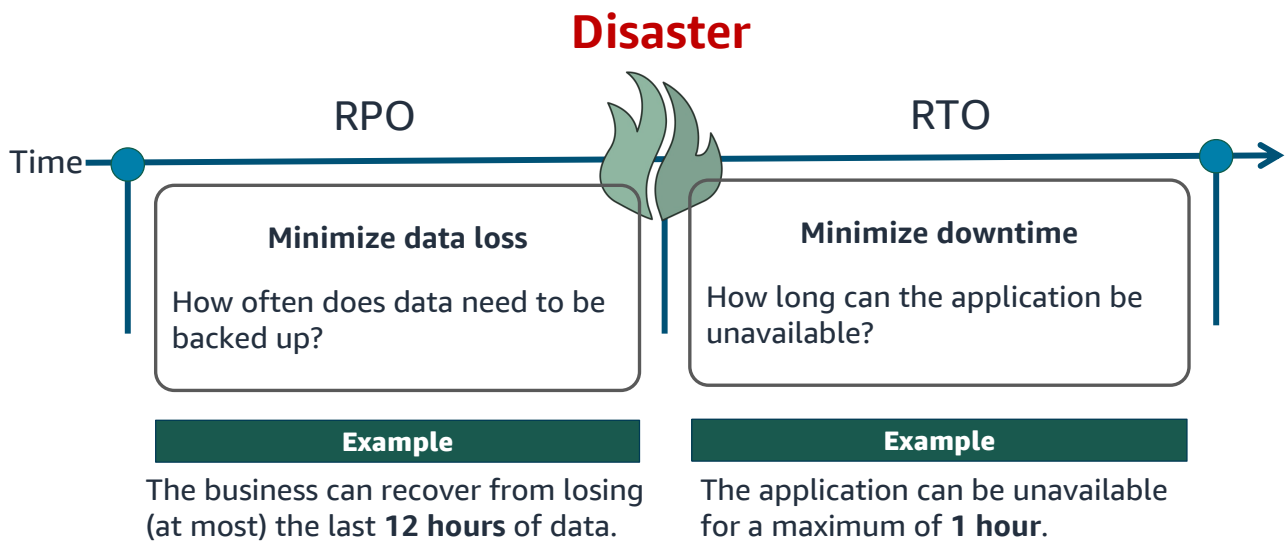
Recovery Point Objective (RPO) and Recovery Time Objective (RTO)



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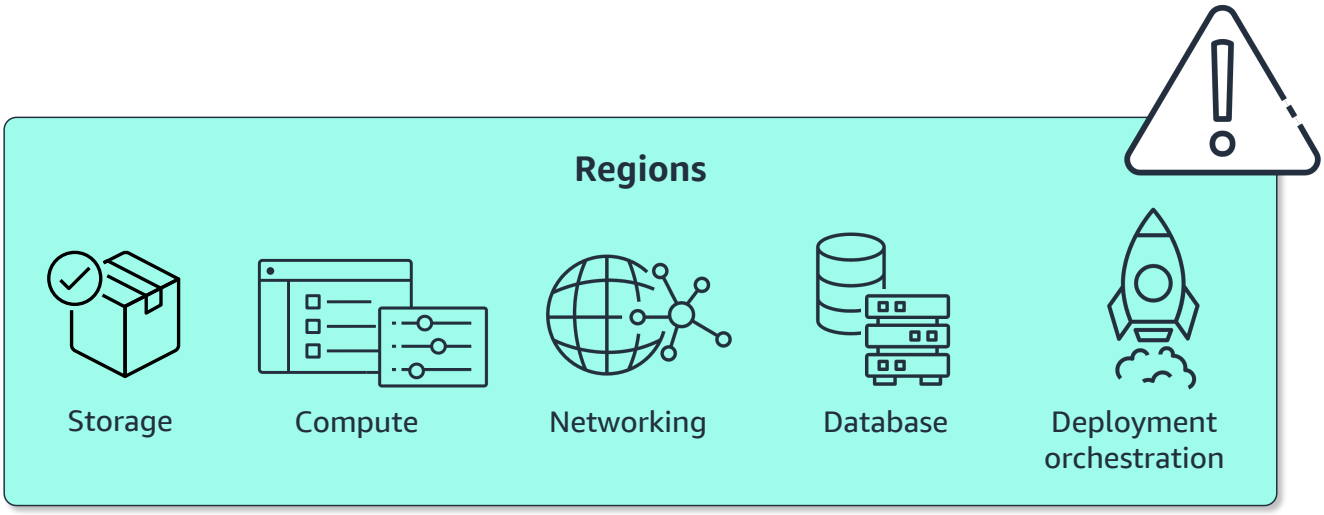
Essential AWS services and features for DR



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Duplicate your storage

Amazon S3

Replication options:
- Cross-Region
- Same-Region
- Bi-directional
- On-demand

Amazon EFS

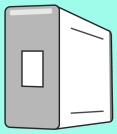
Replication options:
- Regional
- One Zone

Amazon EBS

- Create point-in-time volume snapshots.
- Copy snapshots across Regions and accounts.

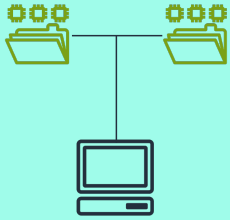
AWS Snow Family

Transfer large volumes (more than 10 TB) of data faster than high-speed internet.



AWS DataSync

Securely transfer your file or object data to, from, and between AWS storage services.



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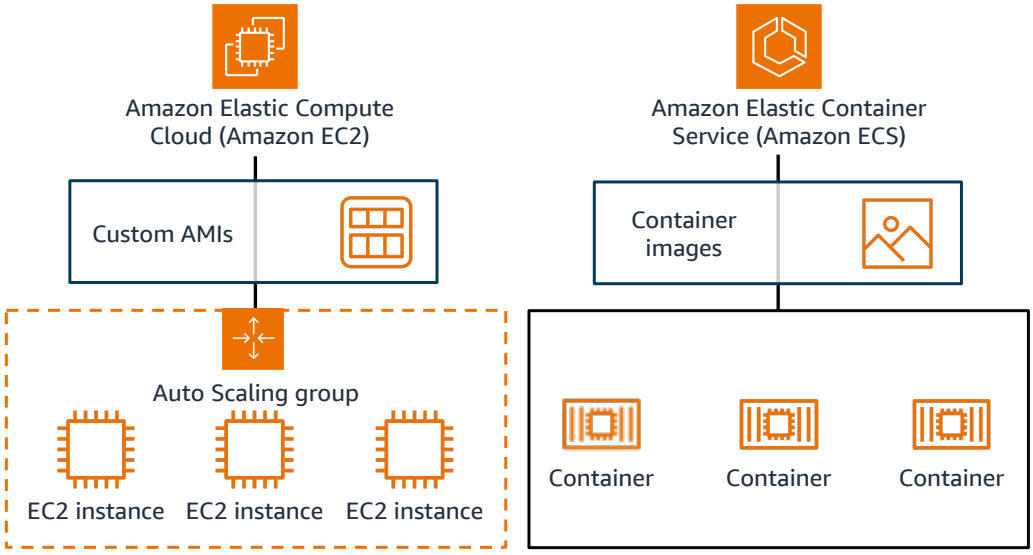
Configuring AMIs for recovery

Obtain and boot new server instances or containers within minutes.



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Failover network design

Amazon Route 53

Traffic distribution and failover

Elastic Load Balancing (ELB)

Load balancing, health checks, and failover

Amazon Virtual Private Cloud (Amazon VPC)

Extended on-premises network topology

AWS Direct Connect

Highly resilient, dedicated network connection between on-premises infrastructure and Amazon VPC

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Database backup and replicas

Amazon Relational Database Service (Amazon RDS)

- Take a snapshot of data and save it in a separate Region.
- Use Multi-AZ cluster deployment to build a resilient DR strategy by using standby instances and read replicas.
- Retain automated backups.

Amazon DynamoDB

- Back up full tables in seconds.
- Use point-in-time recovery to continuously back up tables for up to 35 days.
- Initiate backups with a single click in the console or a single API call.
- Use global tables for fast local performance for globally distributed applications.

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Templates and scripts

AWS CloudFormation

Use templates to quickly deploy collections of resources as needed.

Scripts

Use scripts to automate the provisioning of infrastructure in the cloud.

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AWS Backup

"How can we centralize and automate our backup strategy?"

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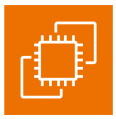
AWS Backup overview

A fully managed backup service:

- Centralizes and automates data protection
- Backs up across AWS resources

AWS Backup


Amazon EC2

Amazon Neptune

Amazon Aurora

Amazon DocumentDB

Amazon DynamoDB

Amazon RDS

Amazon Timestream

AWS Storage Gateway

Amazon EFS

Amazon EBS

Amazon FSx

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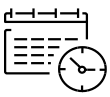
AWS Backup benefits

Simplicity

Policy-based and tag-based backup solution


Automated backup scheduling

Compliance

Monitoring and logs of centralized backup activity

Backup access policies

Encrypted backups

Cost control

Automated management of backup retention

No added cost for orchestration

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Understanding how AWS Backup works

You implement your backup strategy with AWS Backup by creating backup plans.

1. Create AWS Backup plan.

- Schedule
- Lifecycle
- Vault
- Tags for the backups

2. Assign resources.

- Select the resources that will use this plan.
  - Assign tags.
  - Use resource IDs.
- Specify the AWS Identity and Access Management (IAM) role.

3. Manage and monitor backups.

- AWS Backup automatically runs the backup plan.
- Centrally manage backups.
- Centrally monitor backup activity.

AWS Backup works with AWS Organizations to manage backup policies across AWS accounts.

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Recovery strategies

"Which disaster recovery strategy minimizes downtime but is also cost-effective?"

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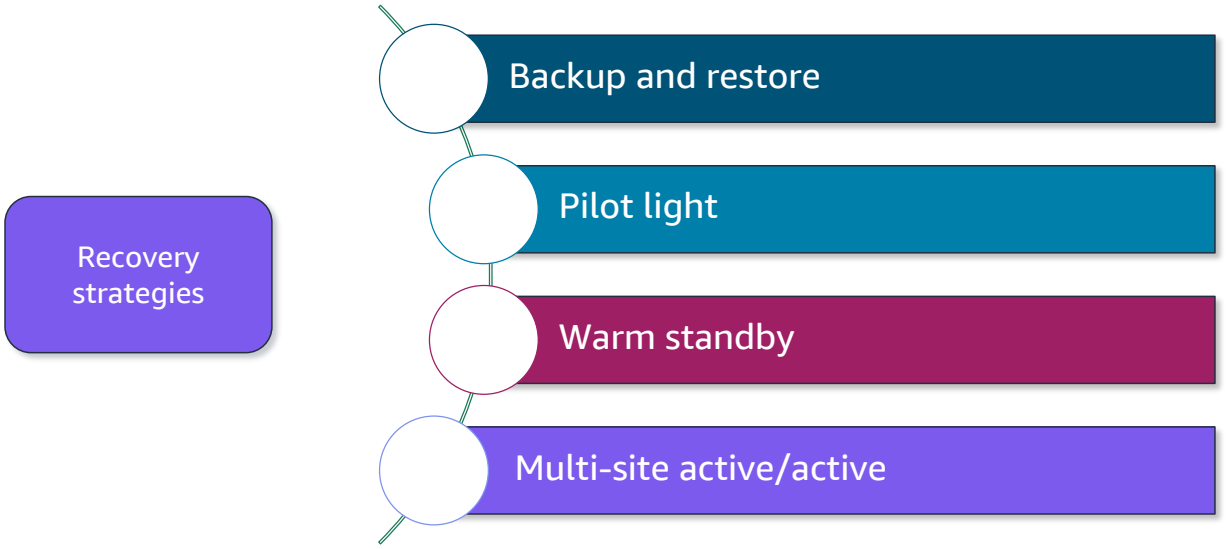
Recovery strategies overview



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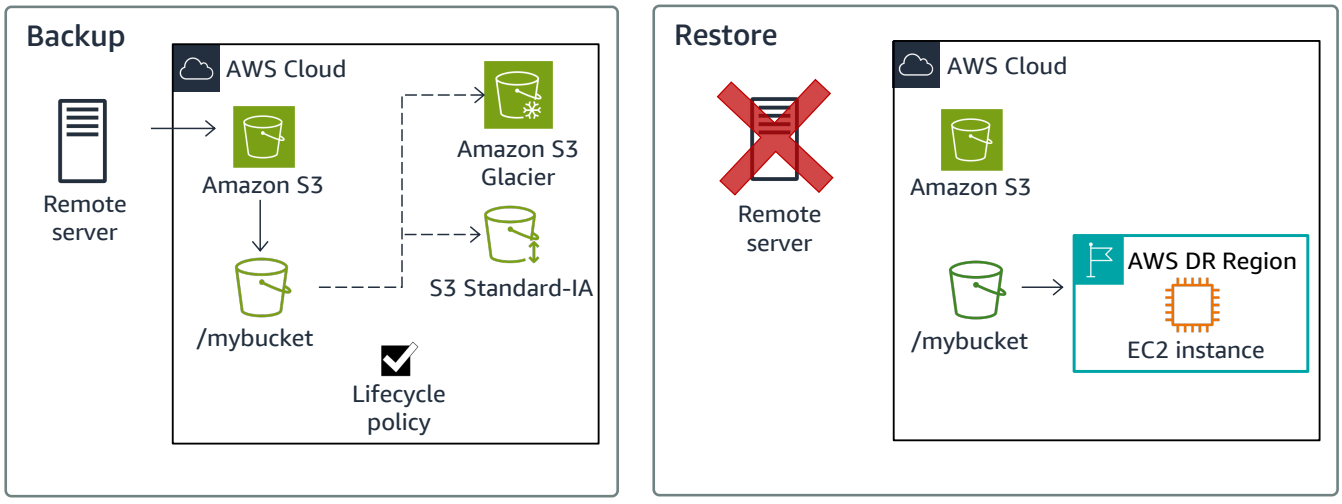
Backup and restore example



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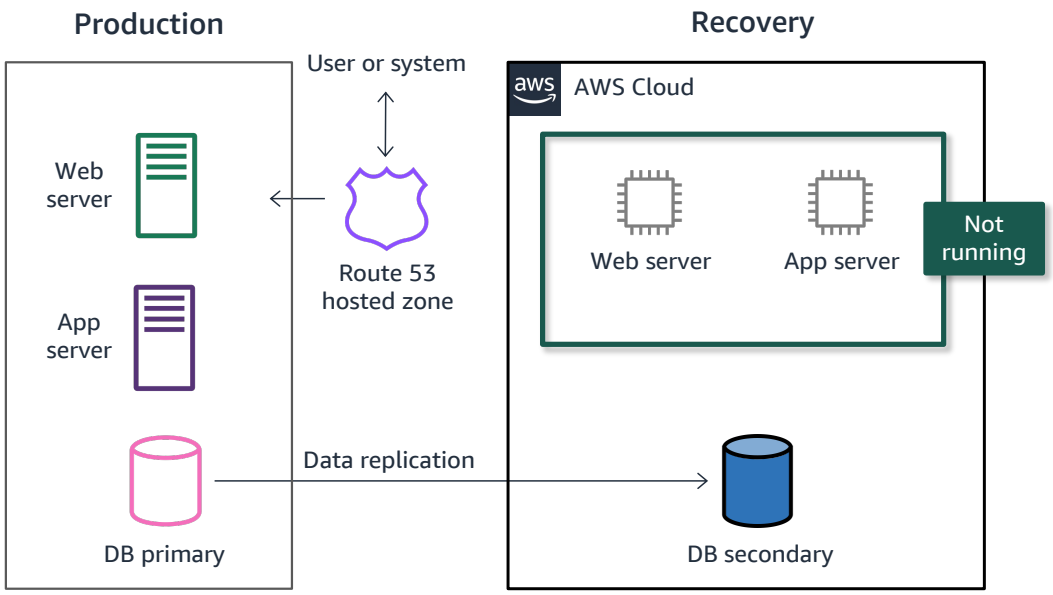
Pilot light example (1 of 2)

Very cost-effective (uses fewer 24/7 resources)



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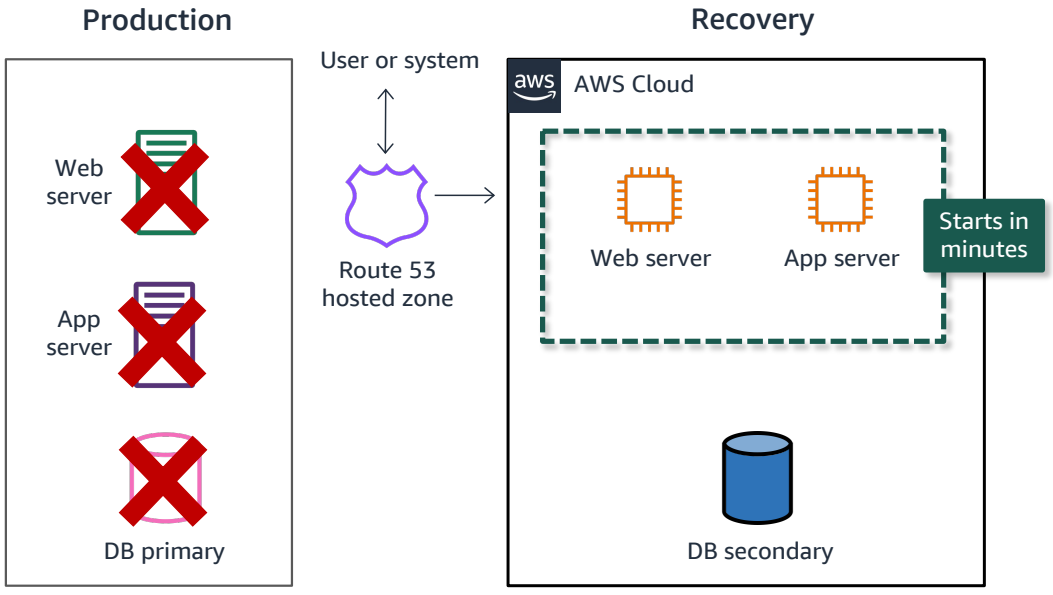
Pilot light example (2 of 2)

- Bring up resources around the replicated core dataset.
- Scale as needed to handle current production traffic.
- Switch to the new system by adjusting DNS records to point to AWS.



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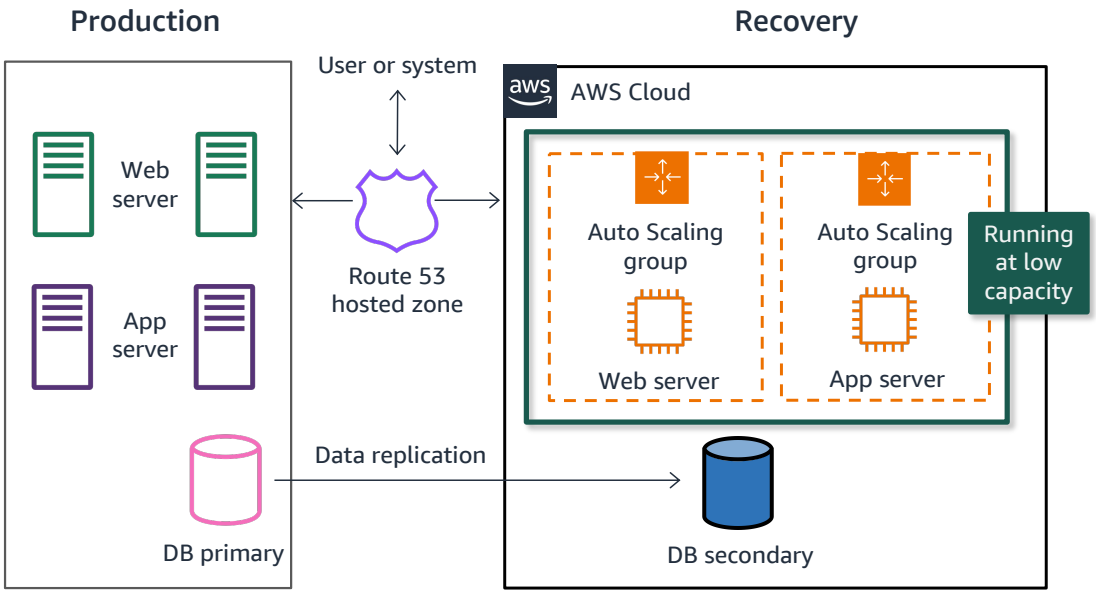
Warm standby (1 of 2)

- Is ready for production traffic
- Cannot handle a full load without scaling
- Is a low-cost option that prevents downtime



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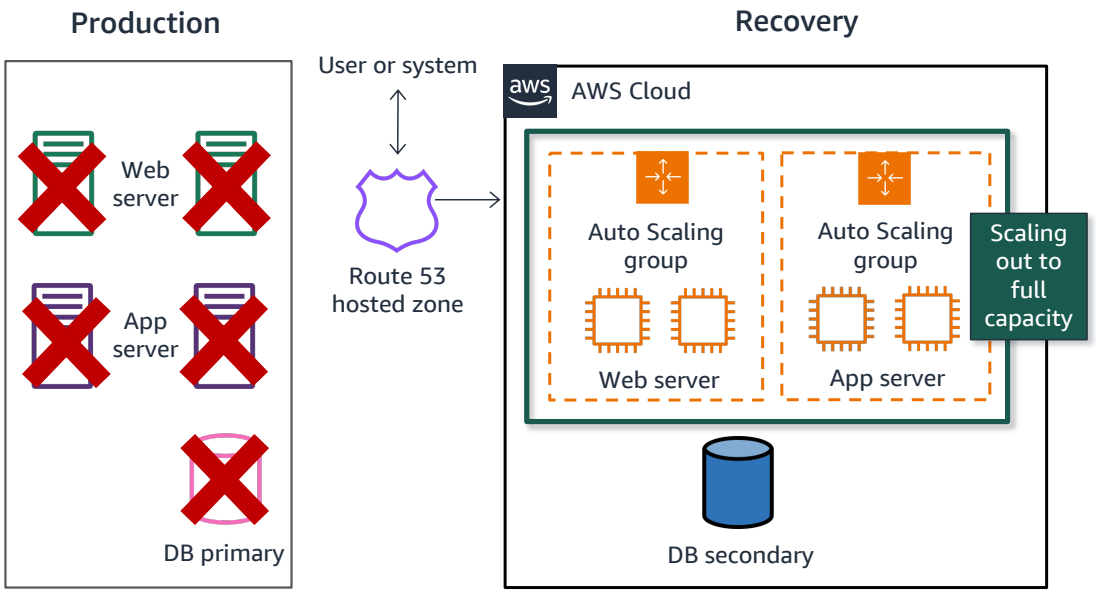
Warm standby (2 of 2)

- Immediately fail over the most critical production load.
- Adjust DNS records to point to AWS.
- Scale the system to handle a full production load.



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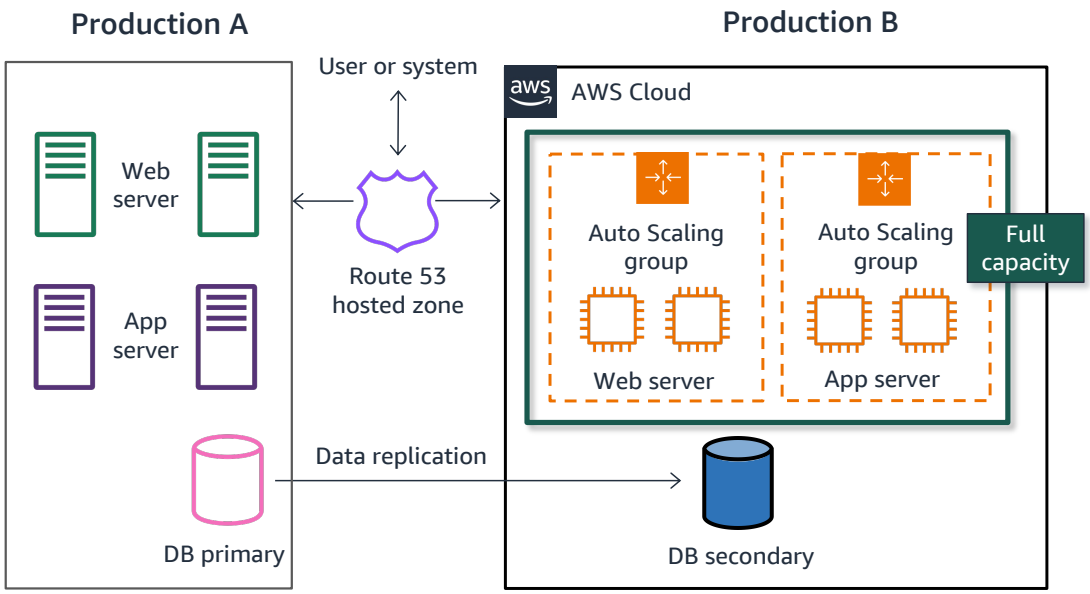
Multi-site active/active (1 of 2)

- Is ready to take the full production load
- Is similar to low-capacity standby
- Scales in or out in response to the production load



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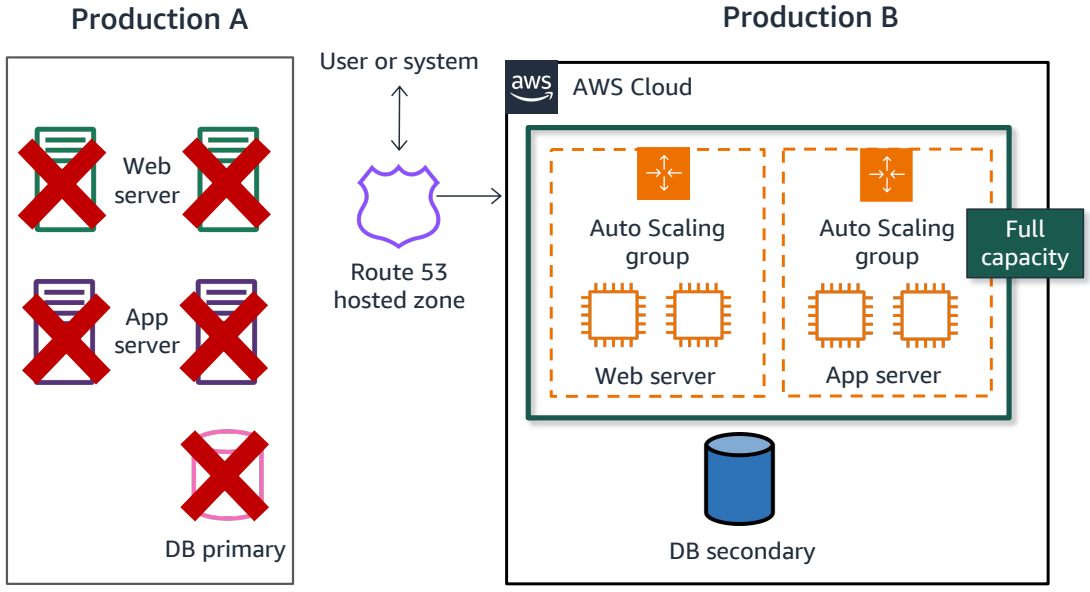
Multi-site active/active (2 of 2)

- RTO: As long as it takes to fail over
- RPO: Depends on the replication type



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Comparing common DR practices on AWS

RPO-RTO: Hours	RPO-RTO: 10s of minutes	RPO-RTO: Minutes	RPO-RTO: Real time
• Lowest priority use cases • Solutions include Amazon S3 and Storage Gateway.	• Lower RTO and RPO requirements • Core services • Scale AWS resources in response to a DR event.	• Higher priority solutions require RTO and RPO in minutes. • Business-critical services	• Highest priority • Auto-failover of your environment in AWS to a running duplicate

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Review

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Present solutions

Chief technology officer

Consider how you would answer the following:

- What strategies can we use to protect ourselves in the event of a disaster?
- How can we centralize and automate our backup strategy?
- Which disaster recovery strategy minimizes downtime but is also cost-effective?

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Module review

In this module, you learned about the following:

- ✓ Disaster planning
- ✓ AWS Backup
- ✓ Recovery strategies

Next, you will review the following:

- Knowledge check
- Capstone lab introduction

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Knowledge check

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Knowledge check question 1

Which disaster recovery model offers a Recovery Time Objective (RTO) in minutes at the lowest cost?

Choice	Response
A	Warm standby
B	Pilot light
C	Backup and restore
D	Multi-site active/active

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Knowledge check question 1 and answer

Which disaster recovery model offers a Recovery Time Objective (RTO) in minutes at the lowest cost?

Choice	Response
A	Warm standby
B correct	Pilot light
C	Backup and restore
D	Multi-site active/active

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Knowledge check question 2

Which metric defines how often data must be backed up?

Choice	Response
A	Recovery Time Objective (RTO)
B	Recovery Point Objective (RPO)
C	Available storage
D	Amount of data

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Knowledge check question 2 and answer

Which metric defines how often data must be backed up?

Choice	Response
A	Recovery Time Objective (RTO)
B correct	Recovery Point Objective (RPO)
C	Available storage
D	Amount of data

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Knowledge check question 3

Which options are features of AWS Backup? (Select THREE.)

Choice	Response
A	Encrypted backups
B	Ability to work across every AWS service
C	Ability to work across multiple services
D	Automated failover to read replicas
E	Incremental backups
F	Automated machine conversion

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Knowledge check question 3 and answers

Which options are features of AWS Backup? (Select THREE.)

Choice	Response
A correct	Encrypted backups
B	Ability to work across every AWS service
C correct	Ability to work across multiple services
D	Automated failover to read replicas
E correct	Incremental backups
F	Automated machine conversion

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Knowledge check question 4

What is the best way to make an existing Amazon RDS DB instance highly available and minimize Recovery Time Objective (RTO)?

Choice	Response
A	Run a secondary copy of the DB instance in another Region.
B	Run a Multi-AZ DB instance in the same Region.
C	Create a read replica in another Region.
D	Create a read replica in the same Region.

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Knowledge check question 4 and answer

What is the best way to make an existing Amazon RDS DB instance highly available and minimize Recovery Time Objective (RTO)?

Choice	Response
A	Run a secondary copy of the DB instance in another Region.
B correct	Run a Multi-AZ DB instance in the same Region.
C	Create a read replica in another Region.
D	Create a read replica in the same Region.

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Capstone lab

Build an AWS Multi-Tier Architecture

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Capstone summary

- Review and analyze architectural solutions based on project data, best practices, and the AWS Well-Architected Framework.
- Without specific guidance, design the architecture in a lab.

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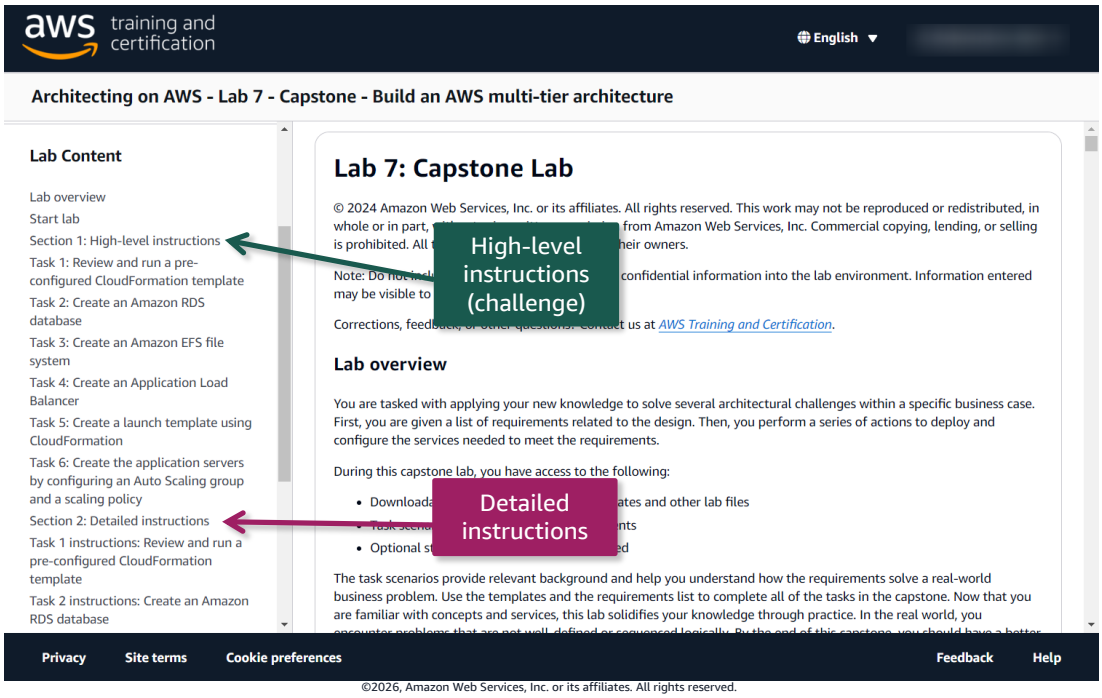
Lab navigation



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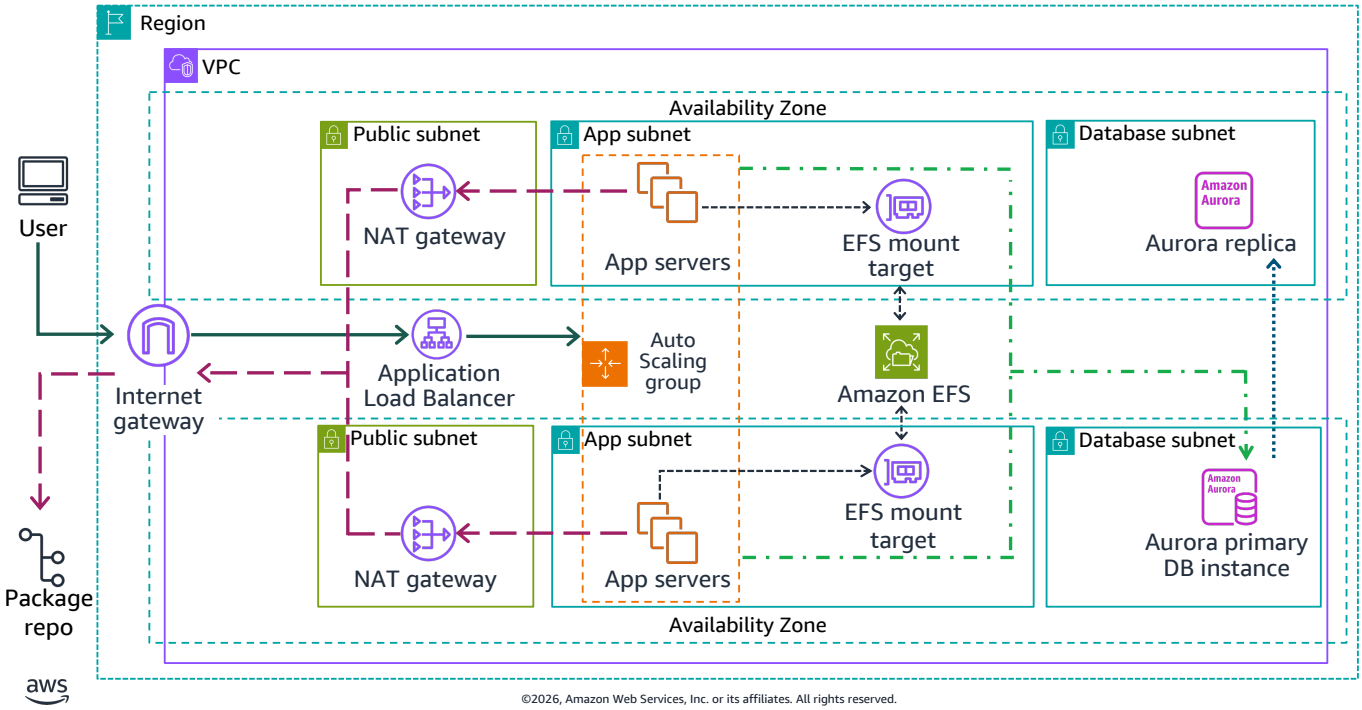
Capstone architecture



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Lab tasks

Task 1: Review and run a preconfigured CloudFormation template.

Task 2: Create an Amazon RDS database.

Task 3: Create an Amazon EFS file system.

Task 4: Create an Application Load Balancer.

Task 5: Create a launch template using CloudFormation.

Task 6: Create the application servers by configuring an Auto Scaling group and a scaling policy.

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End of Module 13

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Architecting on AWS

Course Summary

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Recap

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Course objective

- Identify Amazon Web Services (AWS) services, compare features, and explore best practices to architect resilient, secure, and highly available IT solutions on AWS.

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Day 1 summary

- Architecting Fundamentals
- Lab 1: Explore and Interact with the AWS Management Console and AWS Command Line Interface
- Account Security
- Networking 1
- Compute
- Lab 2: Build Your Amazon VPC Infrastructure

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Day 2 summary

- Storage
- Database Services
- Lab 3: Create a Database Layer in Your Amazon VPC Infrastructure
- Monitoring and Scaling
- Lab 4: Configure High Availability in Your Amazon VPC
- Automation
- Containers

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Day 3 summary

- Networking 2
- Serverless
- Lab 5: Build a Serverless Architecture
- Edge Services
- Lab 6: Configure an Amazon CloudFront Distribution with an Amazon S3 Origin
- Backup and Recovery
- Capstone lab: Build an AWS Multi-Tier Architecture

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Continuing your learning

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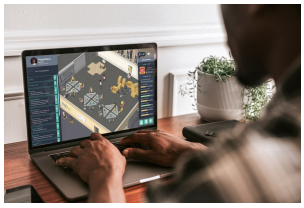
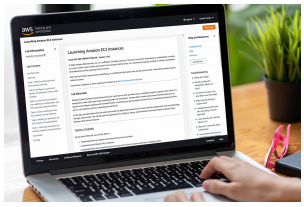
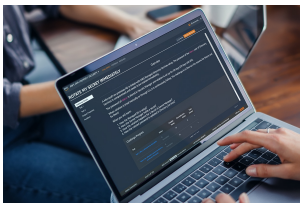
AWS Skill Builder online learning center



Game-based learning



Self-paced labs



Use case challenges

Exam preparation

Continue to deepen the skills you need, your way, with interactive training and more than 600 digital courses developed by the experts at AWS.

Get started
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Don’t miss these learning opportunities

No-charge digital training

Learn with hundreds of no-charge, self-paced digital courses on AWS fundamentals.

Classroom training

Deepen your technical skills and learn from an accredited AWS instructor.

AWS Certification

Validate your expertise with an industry-recognized credential.

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AWS Certification levels

Foundational

Knowledge-based certification for foundational understanding of AWS Cloud. No prior experience necessary.

Associate

Role-based certifications that showcase your knowledge and skills. Prior AWS Cloud or strong on-premises IT experience recommended.

Professional

Role-based certifications that validate advanced skills and knowledge. At least two years of AWS Cloud experience is recommended.

Specialty

Certifications focused on specific topics. Recommended level of experience varies.

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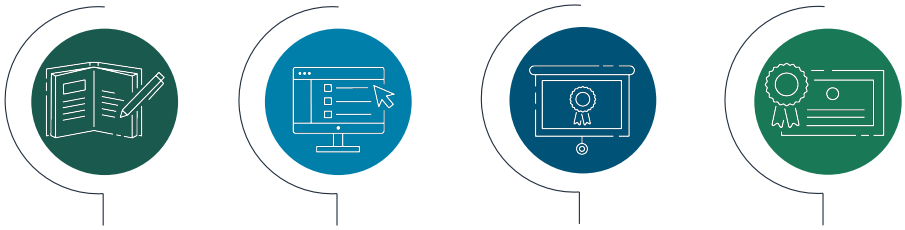
Steps to prepare for an AWS Certification exam

Approach exam day with confidence



Step 1

Get to know the exam and exam-style questions.

Step 2

Learn about exam topics in AWS Skill Builder.

Step 3

Take exam preparation training in AWS Skill Builder.

Step 4

Validate your exam readiness with official practice exams.

Explore all AWS Certification exams

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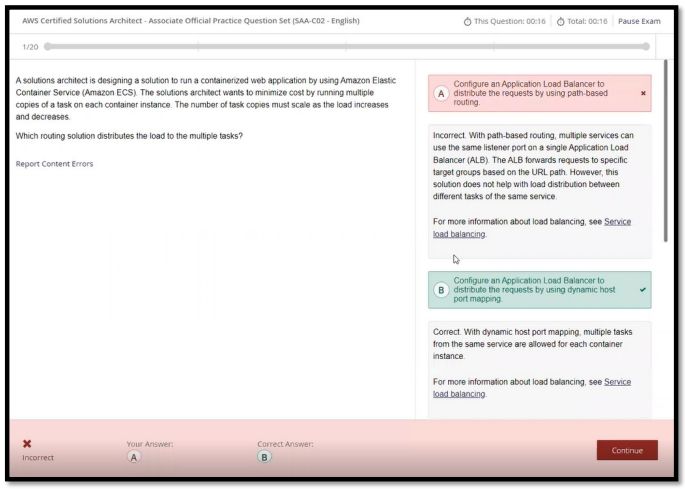
Preparing for AWS Certification: Step 1

Get to know the exam and exam-style questions

1. Review the exam guide.
2. Sign up for access to AWS Skill Builder, the AWS online learning center.
3. Enroll and take an AWS Certification Official Practice Question Set.



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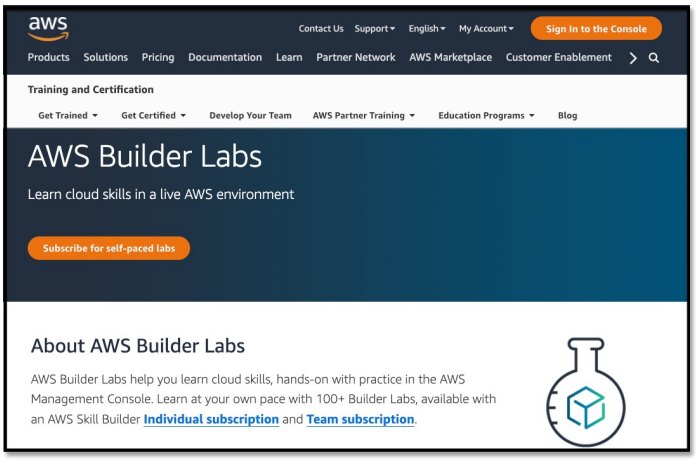
Preparing for AWS Certification: Step 2

Learn about exam topics in AWS Skill Builder

1. Identify gaps in your exam topic knowledge.
2. Enroll in self-paced digital courses you need to learn about.
3. Access AWS Builder Labs to get hands-on. Apply your skills in AWS Management Console.



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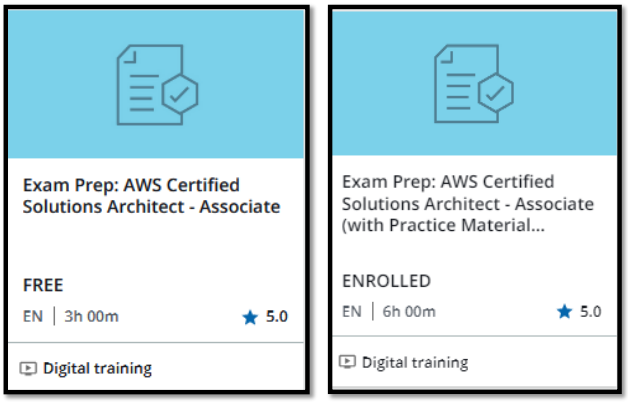
Preparing for AWS Certification: Step 3

Take exam prep training in AWS Skill Builder

1. AWS Skill Builder offers courses across all domains.
2. AWS Builder Labs contain hundreds of self-paced labs.
3. Use gaming to prepare for your AWS Certification with AWS Cloud Quest.



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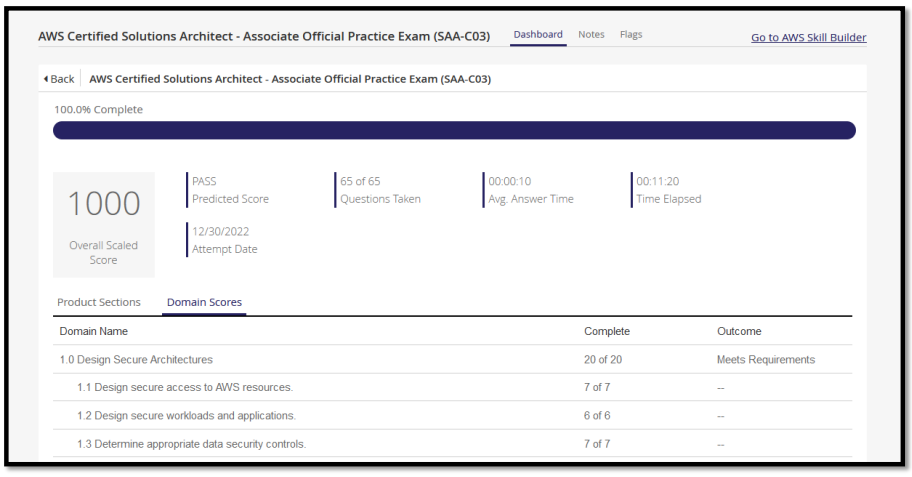
Preparing for AWS Certification: Step 4

Validate your exam readiness

Take an AWS Certification official practice exam with exam-style scoring.



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Registering for your exam

Learn about options for taking the exam

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